Note that this quadratic polynomial function takes the form $z = -(x^2 + y^2)$, so we can see that it will have a **relative** (and, in fact, absolute) **maximum** at its vertex (the critical point $(4, -1)$). We can also argue that since we are subtracting squared terms from 8, we cannot possibly obtain a function value larger than 8, and since we do get a value of 8 at the critical point $(4, -1)$, we know it will be the absolute maximum value of this function.

8) $f(x, y) = x^2 + y^2 + 2x - 6y + 6$

9) $f(x, y) = \sqrt{x^2 + y^2} + 1$

Answer

Crit. Pts.: $(0, 0)$

Extrema: f has a relative minimum of 1 at $(0, 0)$.

To justify this, consider the fact that the square root function cannot give a negative value, so this function cannot possibly return a value less than 1. Since it's value is 1 at the critical point $(0, 0)$, we know 1 must be the function's **absolute minimum** value.

In exercises 10 - 34, identify any critical points and use the Second Partials Test to determine the behavior of the function at each critical point, whether there is a maximum, minimum, saddle point, or none of these. If the Second Partials Test fails, determine the function's behavior at the point using another method and justify your answer clearly.

10) $f(x, y) = -x^3 + 4xy - 2y^2 + 1$

11) $f(x, y) = x^2 y^2$

Answer

Crit. pts.: All points on the lines $x = 0$ and $y = 0$ are critical points of this function.

Exrema: The Second Partials Test fails.

Since $x^2 y^2 > 0$ for all x and y different from zero, and $x^2 y^2 = 0$ when either x or y equals zero (or both), then the absolute minimum of 0 occurs at all points on the x- or y-axes, that is, for all points on the lines $x = 0$ and $y = 0$.

12) $f(x, y) = x^2 - 6x + y^2 + 4y - 8$

13) $f(x, y) = 2xy + 3x + 4y$

Answer

Crit. pts.: $\left(-2, -\frac{3}{2}\right)$

Exrema: f has a saddle point at $\left(-2, -\frac{3}{2}, -6\right)$.

14) $f(x, y) = 8xy(x + y) + 7$

15) $f(x, y) = x^2 + 4xy + y^2$

Answer

Crit. pts.: $(0, 0)$

Exrema: f has a saddle point at $(0, 0, 0)$.

16) $f(x, y) = x^3 + y^3 - 300x - 75y - 3$

17) $f(x, y) = 9 - x^4 y^4$

Answer

Crit. pts.: All points on the lines $x = 0$ and $y = 0$ are critical points of this function.

Extrema: The Second Partials Test fails.

Since the term $-x^4 y^4 < 0$ for all x and y different from zero, and $-x^4 y^4 = 0$ when either x or y equals zero (or both), then

this function cannot attain a value greater than 9 anywhere, but is 9 at the critical points. Thus f has an absolute maximum of 9 at all points on the x- or y-axes, that is, for all points on the lines $x=0$ and $y=0$.

18) $f(x,y)=x^2+10xy+y^2$

Answer

Crit. pts.: $(0,0)$
Extrema: f has a saddle point at $(0,0,0)$.

19) $f(x,y)=x^4+y^2+2xy+3$

Answer

Crit. pts.: $(0,0), \quad \left(-\frac{\sqrt{2}}{2},\frac{\sqrt{2}}{2}\right), \quad \left(\frac{\sqrt{2}}{2},-\frac{\sqrt{2}}{2}\right)$
Extrema: f has a saddle point at $(0,0,3)$,
f has a local minimum of 2.75 at the point $\left(-\frac{\sqrt{2}}{2},\frac{\sqrt{2}}{2}\right)$.
f has a local minimum of 2.75 at the point $\left(\frac{\sqrt{2}}{2},-\frac{\sqrt{2}}{2}\right)$.

20) $f(x,y)=7x^2y+9xy^2$

21) $f(x,y)=3x^2-2xy+y^2-8y$

Answer

Crit. pts.: $(2,6)$
Extrema: f has a relative minimum of -24 located at $(2,6)$.

22) $f(x,y)=3x^2+2xy+y^2$

23) $f(x,y)=y^2+xy+3y+2x+3$

Answer

Crit. pts.: $(1,-2)$
Extrema: f has a saddle point at $(1,-2,1)$.

24) $f(x,y)=x^2+xy+y^2-3x$

25) $f(x,y)=x^2+2y^2-x^2y$

Answer

Crit. pts.: $(0,0), \quad (-2,1), \quad (2,1)$
Extrema: f has a relative minimum of 0 at $(0,0)$ and saddle points at $(2,1,2)$ and $(-2,1,2)$.

26) $f(x,y)=x^2+y-e^y$

27) $f(x,y)=e^{-(x^2+y^2+2x)}$

Answer

Crit. pts.: $(-1,0)$
Extrema: f has a relative maximum of e located at $(-1,0)$.
See this problem illustrated in CalcPlot3D.

28) $f(x,y)=x^2+xy+y^2-x-y+1$

29) $f(x,y)=x^2y(9-x+y)$

Answer

Crit. pts.: $\left(\frac{9}{2}, -\frac{9}{4}\right)$, $(9, 0)$, and all points on the line $x = 0$
Extrema: f has a saddle point at $(9, 0, 0)$ and a relative minimum of -102.515625 at $\left(\frac{9}{2}, -\frac{9}{4}\right)$.
At the critical points on the line $x = 0$, f has neither relative extrema nor saddle points, but they do represent a sort of trough on the surface.

30) $f(x, y) = -x^2 - 5y^2 + 10x - 30y - 62$

31) $f(x, y) = 120x + 120y - xy - x^2 - y^2$

Answer

Crit. pts.: $(40, 40)$
Extrema: f has a relative maximum of 4800 located at $(40, 40)$.

32) $f(x, y) = 2x^2 + 2xy + y^2 + 2x - 3$

33) $f(x, y) = x^2 + x - 3xy + y^3 - 5$

Answer

Crit. pts.: $\left(\frac{1}{4}, \frac{1}{2}\right)$ and $(1, 1)$
Extrema: f has a saddle point at $\left(\frac{1}{4}, \frac{1}{2}, -\frac{79}{16}\right)$ and a relative minimum of -5 at $(1, 1)$.

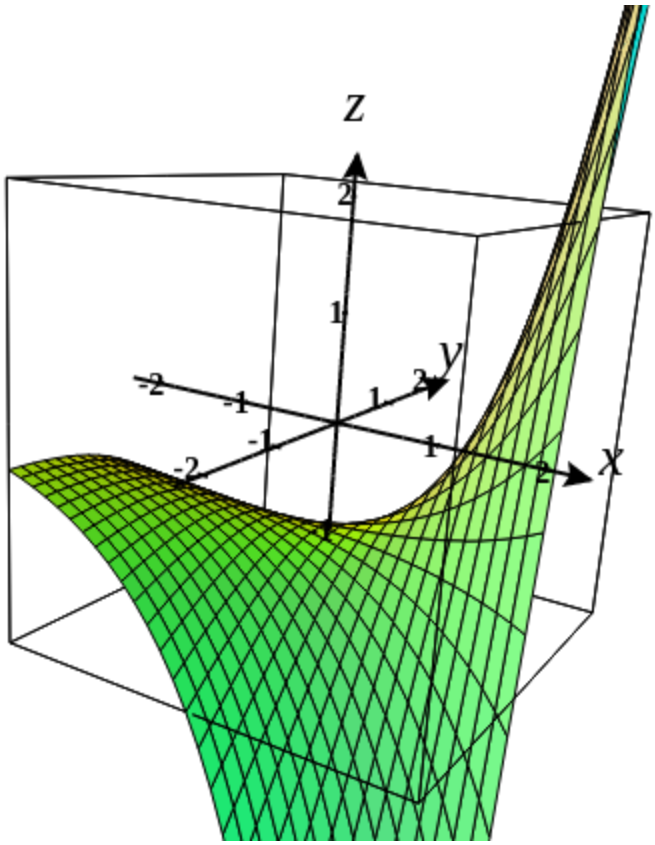
34) $f(x, y) = 2xye^{-x^2-y^2}$

In exercises 35 - 37, determine the extreme values and the saddle points. Use a CAS to graph the function.

35) [T] $f(x, y) = ye^x - e^y$

Answer

A saddle point is located at $(0, 0, -1)$.

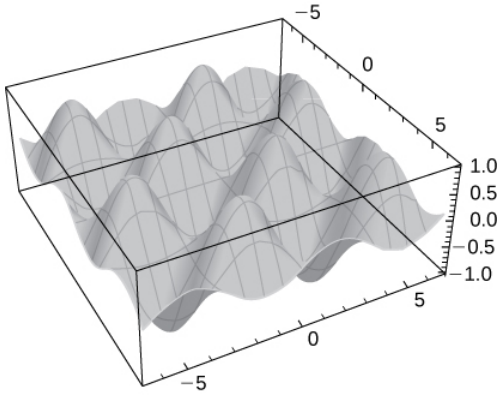


36) [T] $f(x,y) = x\sin(y)$

37) [T] $f(x,y) = \sin(x)\sin(y), \quad x \in (0, 2\pi), \quad y \in (0, 2\pi)$

Answer

There is a saddle point at (π, π), local maxima at $(\frac{\pi}{2}, \frac{\pi}{2})$ and $(\frac{3\pi}{2}, \frac{3\pi}{2})$, and local minima at $(\frac{\pi}{2}, \frac{3\pi}{2})$ and $(\frac{3\pi}{2}, \frac{\pi}{2})$.



In exercises 38 - 41, find the absolute extrema of the given function on the indicated closed and bounded set R.

38) $f(x,y) = xy - x - 3y$ where R is the triangular region with vertices $(0,0)$, $(0,4)$, and $(5,0)$.

39) Find the absolute maximum and minimum values of $f(x,y) = x^2 + y^2 - 2y + 1$ on the region $R = \{(x,y) \,|\, x^2 + y^2 \leq 4\}$.

Answer

$(0,1,0)$ is the absolute minimum and $(0,-2,9)$ is the absolute maximum.

40) $f(x,y)=x^3-3xy-y^3$ on $R=\{(x,y)\,|\,-2\le x\le 2,\ -2\le y\le 2\}$

41) $f(x,y)=\dfrac{-2y}{x^2+y^2+1}$ on $R=\{(x,y)\,|\,x^2+y^2\le 4\}$

Answer

There is an absolute minimum at $(0,1,-1)$ and an absolute maximum at $(0,-1,1)$.

42) Find three positive numbers the sum of which is 27, such that the sum of their squares is as small as possible.

43) Find the points on the surface $x^2-yz=5$ that are closest to the origin.

Answer

$\left(\sqrt{5},0,0\right),\ \left(-\sqrt{5},0,0\right)$

44) Find the maximum volume of a rectangular box with three faces in the coordinate planes and a vertex in the first octant on the line $x+y+z=1$.

45) The sum of the length and the girth (perimeter of a cross-section) of a package carried by a delivery service cannot exceed 108 in. Find the dimensions of the rectangular package of largest volume that can be sent.

Answer

18 by 36 by 18 in.

46) A cardboard box without a lid is to be made with a volume of 4 ft^3. Find the dimensions of the box that requires the least amount of cardboard.

47) Find the point on the surface $f(x,y)=x^2+y^2+10$ nearest the plane $x+2y-z=0$. Identify the point on the plane.

Answer

$\left(\frac{47}{24},\frac{47}{12},\frac{235}{24}\right)$

48) Find the point in the plane $2x-y+2z=16$ that is closest to the origin.

49) A company manufactures two types of athletic shoes: jogging shoes and cross-trainers. The total revenue from x units of jogging shoes and y units of cross-trainers is given by $R(x,y)=-5x^2-8y^2-2xy+42x+102y$, where x and y are in thousands of units. Find the values of x and y to maximize the total revenue.

Answer

$x=3$ and $y=6$

50) A shipping company handles rectangular boxes provided the sum of the length, width, and height of the box does not exceed 96 in. Find the dimensions of the box that meets this condition and has the largest volume.

51) Find the maximum volume of a cylindrical soda can such that the sum of its height and circumference is 120 cm.

Answer

$V=\dfrac{64,000}{\pi}\approx 20,372\ \text{cm}^3$

Contributors

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- Paul Seeburger (Monroe Community College) created problems 19 and 29, and added dynamic figures for problems 27 and 35.

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14.8: Lagrange Multipliers

Learning Objectives

- Use the method of Lagrange multipliers to solve optimization problems with one constraint.
- Use the method of Lagrange multipliers to solve optimization problems with two constraints.

Solving optimization problems for functions of two or more variables can be similar to solving such problems in single-variable calculus. However, techniques for dealing with multiple variables allow us to solve more varied optimization problems for which we need to deal with additional conditions or constraints. In this section, we examine one of the more common and useful methods for solving optimization problems with constraints.

Lagrange Multipliers

In the previous section, an applied situation was explored involving maximizing a profit function, subject to certain **constraints.** In that example, the constraints involved a maximum number of golf balls that could be produced and sold in 1 month (x), and a maximum number of advertising hours that could be purchased per month (y). Suppose these were combined into a single budgetary constraint, such as $20x + 4y \leq 216$, that took into account both the cost of producing the golf balls and the number of advertising hours purchased per month. The goal is still to maximize profit, but now there is a different type of constraint on the values of x and y. This constraint and the corresponding profit function

$$f(x, y) = 48x + 96y - x^2 - 2xy - 9y^2$$

is an example of an **optimization problem**, and the function $f(x, y)$ is called the **objective function**. A graph of various level curves of the function $f(x, y)$ follows.

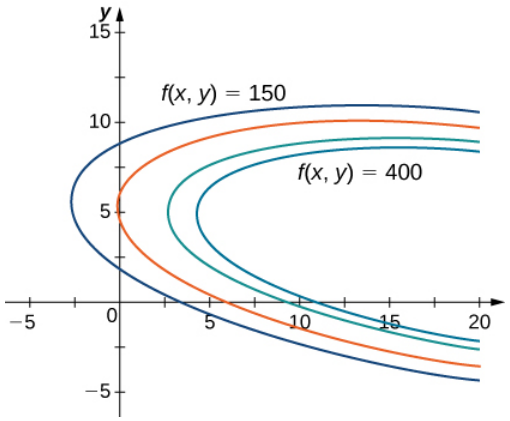


Figure 14.8.1: Graph showing level curves of the function $f(x, y) = 48x + 96y - x^2 - 2xy - 9y^2$ corresponding to $c = 150, 250, 350,$ and 400.

In Figure 14.8.1, the value c represents different profit levels (i.e., values of the function f). As the value of c increases, the curve shifts to the right. Since our goal is to maximize profit, we want to choose a curve as far to the right as possible. If there were no restrictions on the number of golf balls the company could produce or the number of units of advertising available, then we could produce as many golf balls as we want, and advertise as much as we want, and there would be not be a maximum profit for the company. Unfortunately, we have a budgetary constraint that is modeled by the inequality $20x + 4y \leq 216$. To see how this constraint interacts with the profit function, Figure 14.8.2 shows the graph of the line $20x + 4y = 216$ superimposed on the previous graph.

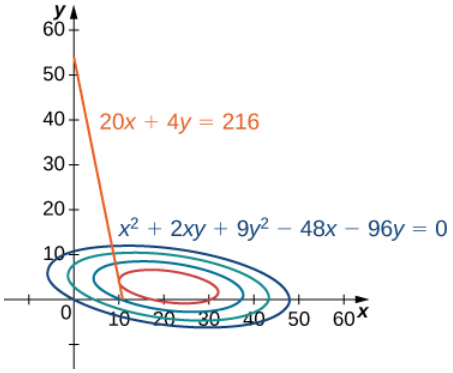


Figure 14.8.2: Graph of level curves of the function $f(x,y) = 48x + 96y - x^2 - 2xy - 9y^2$ corresponding to $c = 150, 250, 350,$ and 395. The red graph is the constraint function.

As mentioned previously, the maximum profit occurs when the level curve is as far to the right as possible. However, the level of production corresponding to this maximum profit must also satisfy the budgetary constraint, so the point at which this profit occurs must also lie on (or to the left of) the red line in Figure 14.8.2. Inspection of this graph reveals that this point exists where the line is tangent to the level curve of f. Trial and error reveals that this profit level seems to be around 395, when x and y are both just less than 5. We return to the solution of this problem later in this section. From a theoretical standpoint, at the point where the profit curve is tangent to the constraint line, the gradient of both of the functions evaluated at that point must point in the same (or opposite) direction. Recall that the gradient of a function of more than one variable is a vector. If two vectors point in the same (or opposite) directions, then one must be a constant multiple of the other. This idea is the basis of the **method of Lagrange multipliers.**

Theorem 14.8.1: Method of Lagrange Multipliers with One Constraint

Let f and g be functions of two variables with continuous partial derivatives at every point of some open set containing the smooth curve $g(x,y) = 0$. Suppose that f, when restricted to points on the curve $g(x,y) = 0$, has a local extremum at the point (x_0, y_0) and that $\vec{\nabla} g(x_0, y_0) \neq 0$. Then there is a number λ called a **Lagrange multiplier,** for which

$$\vec{\nabla} f(x_0, y_0) = \lambda \vec{\nabla} g(x_0, y_0).$$

Proof

Assume that a constrained extremum occurs at the point (x_0, y_0). Furthermore, we assume that the equation $g(x,y) = 0$ can be smoothly parameterized as

$$x = x(s) \text{ and } y = y(s)$$

where s is an arc length parameter with reference point (x_0, y_0) at $s = 0$. Therefore, the quantity $z = f(x(s), y(s))$ has a relative maximum or relative minimum at $s = 0$, and this implies that $\dfrac{dz}{ds} = 0$ at that point. From the chain rule,

$$\begin{aligned} \frac{dz}{ds} &= \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial s} \\ &= \left(\frac{\partial f}{\partial x} \hat{\mathbf{i}} + \frac{\partial f}{\partial y} \hat{\mathbf{j}} \right) \cdot \left(\frac{\partial x}{\partial s} \hat{\mathbf{i}} + \frac{\partial y}{\partial s} \hat{\mathbf{j}} \right) \\ &= 0, \end{aligned}$$

where the derivatives are all evaluated at $s = 0$. However, the first factor in the dot product is the gradient of f, and the second factor is the unit tangent vector $\vec{\mathbf{T}}(0)$ to the constraint curve. Since the point (x_0, y_0) corresponds to $s = 0$, it follows from this equation that

$$\vec{\nabla} f(x_0, y_0) \cdot \vec{\mathbf{T}}(0) = 0,$$

which implies that the gradient is either the zero vector $\vec{0}$ or it is normal to the constraint curve at a constrained relative extremum. However, the constraint curve $g(x,y) = 0$ is a level curve for the function $g(x,y)$ so that if $\vec{\nabla} g(x_0, y_0) \neq 0$ then

$\vec{\nabla} g(x_0, y_0)$ is normal to this curve at (x_0, y_0) It follows, then, that there is some scalar λ such that

$$\vec{\nabla} f(x_0, y_0) = \lambda \vec{\nabla} g(x_0, y_0)$$

□

To apply Theorem 14.8.1 to an optimization problem similar to that for the golf ball manufacturer, we need a problem-solving strategy.

Problem-Solving Strategy: Steps for Using Lagrange Multipliers

1. Determine the objective function $f(x, y)$ and the constraint function $g(x, y)$. Does the optimization problem involve maximizing or minimizing the objective function?
2. Set up a system of equations using the following template:

$$\vec{\nabla} f(x_0, y_0) = \lambda \vec{\nabla} g(x_0, y_0) \tag{14.8.1}$$
$$g(x_0, y_0) = 0 \tag{14.8.2}$$

3. Solve for x_0 and y_0.
4. The largest of the values of f at the solutions found in step 3 maximizes f; the smallest of those values minimizes f.

Example 14.8.1: Using Lagrange Multipliers

Use the method of Lagrange multipliers to find the minimum value of $f(x, y) = x^2 + 4y^2 - 2x + 8y$ subject to the constraint $x + 2y = 7$.

Solution

Let's follow the problem-solving strategy:

1. The objective function is $f(x, y) = x^2 + 4y^2 - 2x + 8y$. To determine the constraint function, we must first subtract 7 from both sides of the constraint. This gives $x + 2y - 7 = 0$. The constraint function is equal to the left-hand side, so $g(x, y) = x + 2y - 7$. The problem asks us to solve for the minimum value of f, subject to the constraint (Figure 14.8.3).

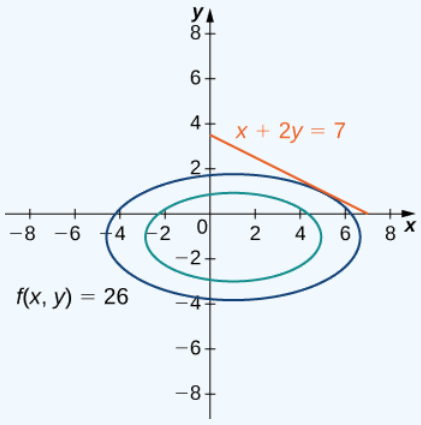


Figure 14.8.3: Graph of level curves of the function $f(x, y) = x^2 + 4y^2 - 2x + 8y$ corresponding to $c = 10$ and 26. The red graph is the constraint function.

2. We then must calculate the gradients of both f and g:

$$\vec{\nabla} f(x, y) = (2x - 2)\,\hat{\mathbf{i}} + (8y + 8)\,\hat{\mathbf{j}}$$
$$\vec{\nabla} g(x, y) = \hat{\mathbf{i}} + 2\hat{\mathbf{j}}.$$

The equation $\vec{\nabla} f(x_0, y_0) = \lambda \vec{\nabla} g(x_0, y_0)$ becomes

$$(2x_0 - 2)\,\hat{\mathbf{i}} + (8y_0 + 8)\,\hat{\mathbf{j}} = \lambda\left(\hat{\mathbf{i}} + 2\hat{\mathbf{j}}\right),$$

which can be rewritten as

$$(2x_0 - 2)\,\hat{\mathbf{i}} + (8y_0 + 8)\,\hat{\mathbf{j}} = \lambda\hat{\mathbf{i}} + 2\lambda\hat{\mathbf{j}}.$$

Next, we set the coefficients of $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ equal to each other:

$$\begin{aligned} 2x_0 - 2 &= \lambda \\ 8y_0 + 8 &= 2\lambda. \end{aligned}$$

The equation $g(x_0, y_0) = 0$ becomes $x_0 + 2y_0 - 7 = 0$. Therefore, the system of equations that needs to be solved is

$$\begin{aligned} 2x_0 - 2 &= \lambda \\ 8y_0 + 8 &= 2\lambda \\ x_0 + 2y_0 - 7 &= 0. \end{aligned}$$

3. This is a linear system of three equations in three variables. We start by solving the second equation for λ and substituting it into the first equation. This gives $\lambda = 4y_0 + 4$, so substituting this into the first equation gives

$$2x_0 - 2 = 4y_0 + 4.$$

Solving this equation for x_0 gives $x_0 = 2y_0 + 3$. We then substitute this into the third equation:

$$\begin{aligned} (2y_0 + 3) + 2y_0 - 7 &= 0 \\ 4y_0 - 4 &= 0 \\ y_0 &= 1. \end{aligned}$$

Since $x_0 = 2y_0 + 3$, this gives $x_0 = 5$.

4. Next, we evaluate $f(x, y) = x^2 + 4y^2 - 2x + 8y$ at the point $(5, 1)$,

$$f(5, 1) = 5^2 + 4(1)^2 - 2(5) + 8(1) = 27.$$

To ensure this corresponds to a minimum value on the constraint function, let's try some other points on the constraint from either side of the point $(5, 1)$, such as the intercepts of $g(x, y) = 0$, Which are $(7, 0)$ and $(0, 3.5)$.

We get $f(7, 0) = 35 > 27$ and $f(0, 3.5) = 77 > 27$.

So it appears that f has a relative minimum of 27 at $(5, 1)$, subject to the given constraint.

? Exercise 14.8.1

Use the method of Lagrange multipliers to find the maximum value of

$$f(x, y) = 9x^2 + 36xy - 4y^2 - 18x - 8y$$

subject to the constraint $3x + 4y = 32$.

Hint

Use the problem-solving strategy for the method of Lagrange multipliers.

Answer

Subject to the given constraint, f has a maximum value of 976 at the point $(8, 2)$.

Let's now return to the problem posed at the beginning of the section.

✔ Example 14.8.2: Golf Balls and Lagrange Multipliers

The golf ball manufacturer, Pro-T, has developed a profit model that depends on the number x of golf balls sold per month (measured in thousands), and the number of hours per month of advertising y, according to the function

$$z = f(x, y) = 48x + 96y - x^2 - 2xy - 9y^2,$$

where z is measured in thousands of dollars. The budgetary constraint function relating the cost of the production of thousands golf balls and advertising units is given by $20x + 4y = 216$. Find the values of x and y that maximize profit, and find the maximum profit.

Solution:

Again, we follow the problem-solving strategy:

1. The objective function is $f(x, y) = 48x + 96y - x^2 - 2xy - 9y^2$. To determine the constraint function, we first subtract 216 from both sides of the constraint, then divide both sides by 4, which gives $5x + y - 54 = 0$. The constraint function is equal to the left-hand side, so $g(x, y) = 5x + y - 54$. The problem asks us to solve for the maximum value of f, subject to this constraint.
2. So, we calculate the gradients of both f and g:

$$\vec{\nabla} f(x, y) = (48 - 2x - 2y)\hat{\mathbf{i}} + (96 - 2x - 18y)\hat{\mathbf{j}}$$
$$\vec{\nabla} g(x, y) = 5\hat{\mathbf{i}} + \hat{\mathbf{j}}.$$

   The equation $\vec{\nabla} f(x_0, y_0) = \lambda \vec{\nabla} g(x_0, y_0)$ becomes

$$(48 - 2x_0 - 2y_0)\hat{\mathbf{i}} + (96 - 2x_0 - 18y_0)\hat{\mathbf{j}} = \lambda(5\hat{\mathbf{i}} + \hat{\mathbf{j}}),$$

   which can be rewritten as

$$(48 - 2x_0 - 2y_0)\hat{\mathbf{i}} + (96 - 2x_0 - 18y_0)\hat{\mathbf{j}} = \lambda 5\hat{\mathbf{i}} + \lambda\hat{\mathbf{j}}.$$

   We then set the coefficients of $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ equal to each other:

$$48 - 2x_0 - 2y_0 = 5\lambda$$
$$96 - 2x_0 - 18y_0 = \lambda.$$

   The equation $g(x_0, y_0) = 0$ becomes $5x_0 + y_0 - 54 = 0$. Therefore, the system of equations that needs to be solved is

$$\begin{aligned} 48 - 2x_0 - 2y_0 &= 5\lambda \\ 96 - 2x_0 - 18y_0 &= \lambda \\ 5x_0 + y_0 - 54 &= 0. \end{aligned}$$

3. We use the left-hand side of the second equation to replace λ in the first equation:

$$\begin{aligned} 48 - 2x_0 - 2y_0 &= 5(96 - 2x_0 - 18y_0) \\ 48 - 2x_0 - 2y_0 &= 480 - 10x_0 - 90y_0 \\ 8x_0 &= 432 - 88y_0 \\ x_0 &= 54 - 11y_0. \end{aligned}$$

   Then we substitute this into the third equation:

$$\begin{aligned} 5(54 - 11y_0) + y_0 - 54 &= 0 \\ 270 - 55y_0 + y_0 - 54 &= 0 \\ 216 - 54y_0 &= 0 \\ y_0 &= 4. \end{aligned}$$

   Since $x_0 = 54 - 11y_0$, this gives $x_0 = 10$.
4. We then substitute $(10, 4)$ into $f(x, y) = 48x + 96y - x^2 - 2xy - 9y^2$, which gives

$$\begin{aligned} f(10, 4) &= 48(10) + 96(4) - (10)^2 - 2(10)(4) - 9(4)^2 \\ &= 480 + 384 - 100 - 80 - 144 \\ &= 540. \end{aligned}$$

Therefore the maximum profit that can be attained, subject to budgetary constraints, is $\$540,000$ with a production level of $10,000$ golf balls and 4 hours of advertising bought per month. Let's check to make sure this truly is a maximum. The endpoints of the line that defines the constraint are $(10.8, 0)$ and $(0, 54)$ Let's evaluate f at both of these points:

$$\begin{aligned} f(10.8,0) &= 48(10.8) + 96(0) - 10.8^2 - 2(10.8)(0) - 9(0^2) \\ &= 401.76 \\ f(0,54) &= 48(0) + 96(54) - 0^2 - 2(0)(54) - 9(54^2) \\ &= -21,060. \end{aligned}$$

The second value represents a loss, since no golf balls are produced. Neither of these values exceed 540, so it seems that our extremum is a maximum value of f, subject to the given constraint.

? Exercise 14.8.2: Optimizing the Cobb-Douglas function

A company has determined that its production level is given by the Cobb-Douglas function $f(x,y) = 2.5x^{0.45}y^{0.55}$ where x represents the total number of labor hours in 1 year and y represents the total capital input for the company. Suppose 1 unit of labor costs $\$40$ and 1 unit of capital costs $\$50$. Use the method of Lagrange multipliers to find the maximum value of $f(x,y) = 2.5x^{0.45}y^{0.55}$ subject to a budgetary constraint of $\$500,000$ per year.

Hint

Use the problem-solving strategy for the method of Lagrange multipliers.

Answer

Subject to the given constraint, a maximum production level of 13890 occurs with 5625 labor hours and $\$5500$ of total capital input.

In the case of an objective function with three variables and a single constraint function, it is possible to use the method of Lagrange multipliers to solve an optimization problem as well. An example of an objective function with three variables could be the **Cobb-Douglas function** in Exercise 14.8.2: $f(x,y,z) = x^{0.2}y^{0.4}z^{0.4}$, where x represents the cost of labor, y represents capital input, and z represents the cost of advertising. The method is the same as for the method with a function of two variables; the equations to be solved are

$$\begin{aligned} \vec{\nabla} f(x,y,z) &= \lambda \vec{\nabla} g(x,y,z) \\ g(x,y,z) &= 0. \end{aligned}$$

✓ Example 14.8.3: Lagrange Multipliers with a Three-Variable objective function

Maximize the function $f(x,y,z) = x^2 + y^2 + z^2$ subject to the constraint $x + y + z = 1$.

Solution

1. The objective function is $f(x,y,z) = x^2 + y^2 + z^2$. To determine the constraint function, we subtract 1 from each side of the constraint: $x + y + z - 1 = 0$ which gives the constraint function as $g(x,y,z) = x + y + z - 1$.

2. Next, we calculate $\vec{\nabla} f(x,y,z)$ and $\vec{\nabla} g(x,y,z)$:

$$\begin{aligned} \vec{\nabla} f(x,y,z) &= \langle 2x, 2y, 2z \rangle \\ \vec{\nabla} g(x,y,z) &= \langle 1,1,1 \rangle. \end{aligned}$$

This leads to the equations

$$\begin{aligned} \langle 2x_0, 2y_0, 2z_0 \rangle &= \lambda \langle 1,1,1 \rangle \\ x_0 + y_0 + z_0 - 1 &= 0 \end{aligned}$$

which can be rewritten in the following form:

$$2x_0 = \lambda$$
$$2y_0 = \lambda$$
$$2z_0 = \lambda$$
$$x_0 + y_0 + z_0 - 1 = 0.$$

3. Since each of the first three equations has λ on the right-hand side, we know that $2x_0 = 2y_0 = 2z_0$ and all three variables are equal to each other. Substituting $y_0 = x_0$ and $z_0 = x_0$ into the last equation yields $3x_0 - 1 = 0$, so $x_0 = \frac{1}{3}$ and $y_0 = \frac{1}{3}$ and $z_0 = \frac{1}{3}$ which corresponds to a critical point on the constraint curve.

4. Then, we evaluate f at the point $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$:

$$f\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right) = \left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^2 = \frac{3}{9} = \frac{1}{3}$$

Therefore, a possible extremum of the function is $\frac{1}{3}$. To verify it is a minimum, choose other points that satisfy the constraint from either side of the point we obtained above and calculate f at those points. For example,

$$f(1, 0, 0) = 1^2 + 0^2 + 0^2 = 1$$
$$f(0, -2, 3) = 0^2 + (-2)^2 + 3^2 = 13.$$

Both of these values are greater than $\frac{1}{3}$, leading us to believe the extremum is a minimum, subject to the given constraint.

? Exercise 14.8.3

Use the method of Lagrange multipliers to find the minimum value of the function

$$f(x, y, z) = x + y + z$$

subject to the constraint $x^2 + y^2 + z^2 = 1$.

Hint

Use the problem-solving strategy for the method of Lagrange multipliers with an objective function of three variables.

Answer

Evaluating f at both points we obtained, gives us,

$$f\left(\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right) = \frac{\sqrt{3}}{3} + \frac{\sqrt{3}}{3} + \frac{\sqrt{3}}{3} = \sqrt{3}$$
$$f\left(-\frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{3}\right) = -\frac{\sqrt{3}}{3} - \frac{\sqrt{3}}{3} - \frac{\sqrt{3}}{3} = -\sqrt{3}$$

Since the constraint is continuous, we compare these values and conclude that f has a relative minimum of $-\sqrt{3}$ at the point $\left(-\frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{3}\right)$, subject to the given constraint.

Problems with Two Constraints

The method of Lagrange multipliers can be applied to problems with more than one constraint. In this case the objective function, w is a function of three variables:

$$w = f(x, y, z)$$

and it is subject to two constraints:

$$g(x, y, z) = 0 \text{ and } h(x, y, z) = 0.$$

There are two Lagrange multipliers, λ_1 and λ_2, and the system of equations becomes

$$\vec{\nabla} f(x_0, y_0, z_0) = \lambda_1 \vec{\nabla} g(x_0, y_0, z_0) + \lambda_2 \vec{\nabla} h(x_0, y_0, z_0)$$
$$g(x_0, y_0, z_0) = 0$$
$$h(x_0, y_0, z_0) = 0$$

Example 14.8.4: Lagrange Multipliers with Two Constraints

Find the maximum and minimum values of the function

$$f(x, y, z) = x^2 + y^2 + z^2$$

subject to the constraints $z^2 = x^2 + y^2$ and $x + y - z + 1 = 0$.

Solution

Let's follow the problem-solving strategy:

1. The objective function is $f(x, y, z) = x^2 + y^2 + z^2$. To determine the constraint functions, we first subtract z^2 from both sides of the first constraint, which gives $x^2 + y^2 - z^2 = 0$, so $g(x, y, z) = x^2 + y^2 - z^2$. The second constraint function is $h(x, y, z) = x + y - z + 1$.
2. We then calculate the gradients of f, g, and h:

$$\vec{\nabla} f(x, y, z) = 2x\hat{\mathbf{i}} + 2y\hat{\mathbf{j}} + 2z\hat{\mathbf{k}}$$
$$\vec{\nabla} g(x, y, z) = 2x\hat{\mathbf{i}} + 2y\hat{\mathbf{j}} - 2z\hat{\mathbf{k}}$$
$$\vec{\nabla} h(x, y, z) = \hat{\mathbf{i}} + \hat{\mathbf{j}} - \hat{\mathbf{k}}.$$

The equation $\vec{\nabla} f(x_0, y_0, z_0) = \lambda_1 \vec{\nabla} g(x_0, y_0, z_0) + \lambda_2 \vec{\nabla} h(x_0, y_0, z_0)$ becomes

$$2x_0\hat{\mathbf{i}} + 2y_0\hat{\mathbf{j}} + 2z_0\hat{\mathbf{k}} = \lambda_1(2x_0\hat{\mathbf{i}} + 2y_0\hat{\mathbf{j}} - 2z_0\hat{\mathbf{k}}) + \lambda_2(\hat{\mathbf{i}} + \hat{\mathbf{j}} - \hat{\mathbf{k}}),$$

which can be rewritten as

$$2x_0\hat{\mathbf{i}} + 2y_0\hat{\mathbf{j}} + 2z_0\hat{\mathbf{k}} = (2\lambda_1 x_0 + \lambda_2)\hat{\mathbf{i}} + (2\lambda_1 y_0 + \lambda_2)\hat{\mathbf{j}} - (2\lambda_1 z_0 + \lambda_2)\hat{\mathbf{k}}.$$

Next, we set the coefficients of $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ equal to each other:

$$2x_0 = 2\lambda_1 x_0 + \lambda_2$$
$$2y_0 = 2\lambda_1 y_0 + \lambda_2$$
$$2z_0 = -2\lambda_1 z_0 - \lambda_2.$$

The two equations that arise from the constraints are $z_0^2 = x_0^2 + y_0^2$ and $x_0 + y_0 - z_0 + 1 = 0$. Combining these equations with the previous three equations gives

$$2x_0 = 2\lambda_1 x_0 + \lambda_2$$
$$2y_0 = 2\lambda_1 y_0 + \lambda_2$$
$$2z_0 = -2\lambda_1 z_0 - \lambda_2$$
$$z_0^2 = x_0^2 + y_0^2$$
$$x_0 + y_0 - z_0 + 1 = 0.$$

3. The first three equations contain the variable λ_2. Solving the third equation for λ_2 and replacing into the first and second equations reduces the number of equations to four:

$$2x_0 = 2\lambda_1 x_0 - 2\lambda_1 z_0 - 2z_0$$
$$2y_0 = 2\lambda_1 y_0 - 2\lambda_1 z_0 - 2z_0$$
$$z_0^2 = x_0^2 + y_0^2$$
$$x_0 + y_0 - z_0 + 1 = 0.$$

Next, we solve the first and second equation for λ_1. The first equation gives $\lambda_1 = \dfrac{x_0 + z_0}{x_0 - z_0}$, the second equation gives $\lambda_1 = \dfrac{y_0 + z_0}{y_0 - z_0}$. We set the right-hand side of each equation equal to each other and cross-multiply:

$$\begin{aligned}
\frac{x_0 + z_0}{x_0 - z_0} &= \frac{y_0 + z_0}{y_0 - z_0} \\
(x_0 + z_0)(y_0 - z_0) &= (x_0 - z_0)(y_0 + z_0) \\
x_0 y_0 - x_0 z_0 + y_0 z_0 - z_0^2 &= x_0 y_0 + x_0 z_0 - y_0 z_0 - z_0^2 \\
2y_0 z_0 - 2x_0 z_0 &= 0 \\
2z_0(y_0 - x_0) &= 0.
\end{aligned}$$

Therefore, either $z_0 = 0$ or $y_0 = x_0$. If $z_0 = 0$, then the first constraint becomes $0 = x_0^2 + y_0^2$. The only real solution to this equation is $x_0 = 0$ and $y_0 = 0$, which gives the ordered triple $(0, 0, 0)$. This point does not satisfy the second constraint, so it is not a solution. Next, we consider $y_0 = x_0$, which reduces the number of equations to three:

$$\begin{aligned}
y_0 &= x_0 \\
z_0^2 &= x_0^2 + y_0^2 \\
x_0 + y_0 - z_0 + 1 &= 0.
\end{aligned}$$

We substitute the first equation into the second and third equations:

$$\begin{aligned}
z_0^2 &= x_0^2 + x_0^2 \\
&= x_0 + x_0 - z_0 + 1 \qquad = 0.
\end{aligned}$$

Then, we solve the second equation for z_0, which gives $z_0 = 2x_0 + 1$. We then substitute this into the first equation,

$$\begin{aligned}
z_0^2 &= 2x_0^2 \\
(2x_0 + 1)^2 &= 2x_0^2 \\
4x_0^2 + 4x_0 + 1 &= 2x_0^2 \\
2x_0^2 + 4x_0 + 1 &= 0,
\end{aligned}$$

and use the quadratic formula to solve for x_0:

$$x_0 = \frac{-4 \pm \sqrt{4^2 - 4(2)(1)}}{2(2)} = \frac{-4 \pm \sqrt{8}}{4} = \frac{-4 \pm 2\sqrt{2}}{4} = -1 \pm \frac{\sqrt{2}}{2}.$$

Recall $y_0 = x_0$, so this solves for y_0 as well. Then, $z_0 = 2x_0 + 1$, so

$$z_0 = 2x_0 + 1 = 2\left(-1 \pm \frac{\sqrt{2}}{2}\right) + 1 = -2 + 1 \pm \sqrt{2} = -1 \pm \sqrt{2}.$$

Therefore, there are two ordered triplet solutions:

$$\left(-1 + \frac{\sqrt{2}}{2}, -1 + \frac{\sqrt{2}}{2}, -1 + \sqrt{2}\right) \text{ and } \left(-1 - \frac{\sqrt{2}}{2}, -1 - \frac{\sqrt{2}}{2}, -1 - \sqrt{2}\right).$$

4. We substitute $\left(-1 + \dfrac{\sqrt{2}}{2}, -1 + \dfrac{\sqrt{2}}{2}, -1 + \sqrt{2}\right)$ into $f(x, y, z) = x^2 + y^2 + z^2$, which gives

$$\begin{aligned}
f\left(-1 + \frac{\sqrt{2}}{2}, -1 + \frac{\sqrt{2}}{2}, -1 + \sqrt{2}\right) &= \left(-1 + \frac{\sqrt{2}}{2}\right)^2 + \left(-1 + \frac{\sqrt{2}}{2}\right)^2 + (-1 + \sqrt{2})^2 \\
&= \left(1 - \sqrt{2} + \frac{1}{2}\right) + \left(1 - \sqrt{2} + \frac{1}{2}\right) + (1 - 2\sqrt{2} + 2) \\
&= 6 - 4\sqrt{2}.
\end{aligned}$$

Then, we substitute $\left(-1 - \dfrac{\sqrt{2}}{2}, -1 + \dfrac{\sqrt{2}}{2}, -1 + \sqrt{2}\right)$ into $f(x, y, z) = x^2 + y^2 + z^2$, which gives

$$f\left(-1-\frac{\sqrt{2}}{2},-1+\frac{\sqrt{2}}{2},-1+\sqrt{2}\right)=\left(-1-\frac{\sqrt{2}}{2}\right)^2+\left(-1-\frac{\sqrt{2}}{2}\right)^2+(-1-\sqrt{2})^2$$
$$=\left(1+\sqrt{2}+\frac{1}{2}\right)+\left(1+\sqrt{2}+\frac{1}{2}\right)+(1+2\sqrt{2}+2)$$
$$=6+4\sqrt{2}.$$

$6+4\sqrt{2}$ is the maximum value and $6-4\sqrt{2}$ is the minimum value of $f(x,y,z)$, subject to the given constraints.

? Exercise 14.8.4

Use the method of Lagrange multipliers to find the minimum value of the function

$$f(x,y,z)=x^2+y^2+z^2$$

subject to the constraints $2x+y+2z=9$ and $5x+5y+7z=29$.

Hint

Use the problem-solving strategy for the method of Lagrange multipliers with two constraints.

Answer

$f(2,1,2)=9$ is a minimum value of f, subject to the given constraints.

Key Concepts

- An objective function combined with one or more constraints is an example of an optimization problem.
- To solve optimization problems, we apply the method of Lagrange multipliers using a four-step problem-solving strategy.

Key Equations

- **Method of Lagrange multipliers, one constraint**

  $\vec{\nabla}f(x_0,y_0)=\lambda\vec{\nabla}g(x_0,y_0)$

  $g(x_0,y_0)=0$

- **Method of Lagrange multipliers, two constraints**

  $\vec{\nabla}f(x_0,y_0,z_0)=\lambda_1\vec{\nabla}g(x_0,y_0,z_0)+\lambda_2\vec{\nabla}h(x_0,y_0,z_0)$

  $g(x_0,y_0,z_0)=0$

  $h(x_0,y_0,z_0)=0$

Glossary

constraint
an inequality or equation involving one or more variables that is used in an optimization problem; the constraint enforces a limit on the possible solutions for the problem

Lagrange multiplier
the constant (or constants) used in the method of Lagrange multipliers; in the case of one constant, it is represented by the variable λ

method of Lagrange multipliers
a method of solving an optimization problem subject to one or more constraints

objective function
the function that is to be maximized or minimized in an optimization problem

optimization problem

calculation of a maximum or minimum value of a function of several variables, often using Lagrange multipliers

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14.8E: Exercises for Section 14.8

In exercises 1-15, use the method of Lagrange multipliers to find the maximum and minimum values of the function subject to the given constraint.

1) Objective function: $f(x,y) = 4xy$ Constraint: $\dfrac{x^2}{9} + \dfrac{y^2}{16} = 1$

Answer

Subject to the given constraint, the function f has a relative minimum of -24 at both $\left(-\frac{3\sqrt{2}}{2}, 2\sqrt{2}\right)$ and $\left(\frac{3\sqrt{2}}{2}, -2\sqrt{2}\right)$ and a relative maximum of 24 at both $\left(\frac{3\sqrt{2}}{2}, 2\sqrt{2}\right)$ and $\left(-\frac{3\sqrt{2}}{2}, -2\sqrt{2}\right)$

2) Objective function: $f(x,y) = x^2 y$ Constraint: $x^2 + 2y^2 = 6$

3) Objective function: $f(x,y) = x^2 + y^2 + 2x - 2y + 1$ Constraint: $g(x,y) = x^2 + y^2 = 2$

Answer

Subject to the given constraint, f has a relative minimum of -1 at $(-1,1)$ and a relative maximum of 7 at $(1,-1)$.

4) Objective function: $f(x,y) = xy$ Constraint: $4x^2 + 8y^2 = 16$

5) Objective function: $f(x,y) = x^2 + y^2$ Constraint: $xy = 1$

Answer

f has a relative minimum of 2 at both $(-1,-1)$ and $(1,1)$, subject to the given constraint.

6) Objective function: $f(x,y) = x^2 - y^2$ Constraint: $x - 2y + 6 = 0$

7) Objective function: $f(x,y) = x^2 + y^2$ Constraint: $x + 2y - 5 = 0$

Answer

Subject to the given constraint, f has a relative minimum of $f(1,2) = 5$ at the point $(1,2)$.

8) Objective function: $f(x,y) = x^2 + y^2$ Constraint: $(x-1)^2 + 4y^2 = 4$

9) Objective function: $f(x,y) = 4x^3 + y^2$ Constraint: $2x^2 + y^2 = 1$

Answer

Subject to the given constraint, the function f has a relative minimum of $-\sqrt{2}$ at $\left(-\frac{\sqrt{2}}{2}, 0\right)$,
a relative minimum of $\frac{25}{27}$ at both points $\left(\frac{1}{3}, -\frac{\sqrt{7}}{3}\right)$ and $\left(\frac{1}{3}, \frac{\sqrt{7}}{3}\right)$,
a relative maximum of $\sqrt{2}$ at $\left(\frac{\sqrt{2}}{2}, 0\right)$, and a relative maximum of 1 at both points $(0,1)$ and $(0,-1)$.

Solution:

Let $g(x,y) = 2x^2 + y^2$ be the constraint function. Then:

$\vec{\nabla} f(x,y) = 12x^2\,\hat{\mathbf{i}} + 2y\hat{\mathbf{j}}$ and $\vec{\nabla} g(x,y) = 4x\,\hat{\mathbf{i}} + 2y\hat{\mathbf{j}}$

Using the Lagrange Multiplier equation,

$$\vec{\nabla} f(x,y) = \lambda \vec{\nabla} g(x,y),$$

we have:

$$12x^2\,\hat{\mathbf{i}} + 2y\hat{\mathbf{j}} = 4x\lambda\,\hat{\mathbf{i}} + 2y\lambda\hat{\mathbf{j}}$$

giving us the system of equations:

$$12x^2 = 4x\lambda, \quad 2y = 2y\lambda, \quad \text{and the constraint} \quad 2x^2 + y^2 = 1$$

Rewriting the first two equations as zero-products (by moving to one-side and factoring), we get:

$$\begin{array}{ccc} 4x(3x-\lambda)=0 & & 2y(1-\lambda)=0 \\ x=0 \quad \text{or} \quad \lambda = 3x & \text{and} & y=0 \quad \text{or} \quad \lambda = 1 \end{array}$$

Now we consider the combinations of these solutions to the above two equations and plug each of these into the constraint equation to solve for the corresponding Lagrange points.

The combination $x = 0$ and $y = 0$ produces a contradiction when placed in the constraint equation, since this point is not on the ellipse.

Taking the combination $x = 0$ and $\lambda = 1$, we put 0 in for x in the constraint and solve for y, obtaining: $y = \pm 1$. This gives us two Lagrange points: $(0, 1)$ and $(0, -1)$.

Taking the combination $\lambda = 3x$ and $y = 0$, we put 0 in for y in the constraint and solve for x, obtaining: $x = \pm\frac{\sqrt{2}}{2}$. This gives us two Lagrange points: $\left(-\frac{\sqrt{2}}{2}, 0\right)$ and $\left(\frac{\sqrt{2}}{2}, 0\right)$.

Taking the combination $\lambda = 3x$ and $\lambda = 1$, we substitute 1 into the first equation for λ, giving us $1 = 3x$ so $x = \frac{1}{3}$. Plugging this value in for x in the constraint equation and solving for y, we obtain $y = \pm\frac{\sqrt{7}}{3}$ which gives us the two Lagrange points: $\left(\frac{1}{3}, -\frac{\sqrt{7}}{3}\right)$ and $\left(\frac{1}{3}, \frac{\sqrt{7}}{3}\right)$.

Evaluating the function f at these Lagrange points, we find:

$$\begin{array}{ll} f(0,-1) = 1 & f(0,1) = 1 \\ f\left(-\frac{\sqrt{2}}{2}, 0\right) = \frac{-4(\sqrt{2})^3}{8} = -\sqrt{2} & f\left(\frac{\sqrt{2}}{2}, 0\right) = \frac{4(\sqrt{2})^3}{8} = \sqrt{2} \\ f\left(\frac{1}{3}, -\frac{\sqrt{7}}{3}\right) = \frac{25}{27} & f\left(\frac{1}{3}, \frac{\sqrt{7}}{3}\right) = \frac{25}{27} \end{array}$$

Comparing these values with where the corresponding Lagrange points lie on the constraint curve, we conclude the results stated in the answer above.

10) Objective function: $f(x, y) = 2x^2 + y^2$ Constraint: $g(x, y) = x^2 + y^2 = 1$

11) Objective function: $f(x, y, z) = x + 3y - z$ Constraint: $x^2 + y^2 + z^2 = 4$

Answer

Subject to the given constraint, f has a relative minimum of $-2\sqrt{11}$ at the point $\left(-\frac{2\sqrt{11}}{11}, -\frac{6\sqrt{11}}{11}, \frac{2\sqrt{11}}{11}\right)$ and a relative maximum of $2\sqrt{11}$ at the point $\left(\frac{2\sqrt{11}}{11}, \frac{6\sqrt{11}}{11}, -\frac{2\sqrt{11}}{11}\right)$.

12) Objective function: $f(x, y, z) = x + y + z$ Constraint: $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1$

13) Objective function: $f(x, y, z) = xyz$ Constraint: $x^2 + 2y^2 + 3z^2 = 6$

Answer

Subject to the given constraint, f has a relative minimum of $-\frac{2\sqrt{3}}{3}$ at $\left(\sqrt{2},\ 1,\ -\frac{\sqrt{6}}{3}\right)$, $\left(\sqrt{2},\ -1,\ \frac{\sqrt{6}}{3}\right)$, $\left(-\sqrt{2},\ 1,\ \frac{\sqrt{6}}{3}\right)$, and $\left(-\sqrt{2},\ -1,\ -\frac{\sqrt{6}}{3}\right)$ and a relative maximum of $\frac{2\sqrt{3}}{3}$ at $\left(\sqrt{2},\ 1,\ \frac{\sqrt{6}}{3}\right)$, $\left(\sqrt{2},\ -1,\ -\frac{\sqrt{6}}{3}\right)$, $\left(-\sqrt{2},\ -1,\ \frac{\sqrt{6}}{3}\right)$, and $\left(-\sqrt{2},\ 1,\ -\frac{\sqrt{6}}{3}\right)$.

14) Objective function: $f(x,y,z)=x^2+y^2+z^2$ Constraint: $x^4+y^4+z^4=1$

15) Objective function: $f(x,y,z)=x^2+y^2+z^2$ Constraint: $xyz=4$

Answer

Subject to the given constraint, f has a relative minimum of $6\sqrt[3]{2}$ at the points $\left(\sqrt[3]{4},\ \sqrt[3]{4},\ \sqrt[3]{4}\right)$, $\left(\sqrt[3]{4},\ -\sqrt[3]{4},\ -\sqrt[3]{4}\right)$, $\left(-\sqrt[3]{4},\ \sqrt[3]{4},\ -\sqrt[3]{4}\right)$, and $\left(-\sqrt[3]{4},\ -\sqrt[3]{4},\ \sqrt[3]{4}\right)$.
To see a 3D visualization of this problem, see: CalcPlot3D for Problem 15.

In exercises 16-21, use the method of Lagrange multipliers to find the requested extremum of the given function subject to the given constraint.

16) Maximize $f(x,y)=\sqrt{6-x^2-y^2}$ subject to the constraint, $x+y-2=0$.

17) Maximize $f(x,y)=x^2-y^2$ subject to the constraints, $g(x,y)=y-x^2=0,\quad x>0,\quad y>0$.

Answer

Subject to the given constraints, f has a relative maximum of $f\left(\frac{\sqrt{2}}{2},\ \frac{1}{2}\right)=\frac{1}{4}$ at the point $\left(\frac{\sqrt{2}}{2},\ \frac{1}{2}\right)$. If $x>0$ were not a constraint, there would have been two other Lagrange points with relative extrema of f subject to the other two constraints. These would have been $(0,0)$ and $\left(-\frac{\sqrt{2}}{2},\ \frac{1}{2}\right)$.

To verify that f really has a relative **maximum** at the point $\left(\frac{\sqrt{2}}{2},\ \frac{1}{2}\right)$, we would need to check the value of f on either side of this point on the constraint curve, $y-x^2=0$.

If $x=0.5$ which is less than $\frac{\sqrt{2}}{2}$, y would be $y=(0.5)^2=0.25$.
If $x=1$ which is greater than $\frac{\sqrt{2}}{2}$, y would be $y=(1)^2=1$.

Then we compare the value of f at the Lagrange point, $\left(\frac{\sqrt{2}}{2},\ \frac{1}{2}\right)$, with the values of f at these other points on the constraint.
We have $f(0.5,0.25)=(0.5)^2-(0.25)^2=0.25-0.0625=0.1875<\frac{1}{4}$ and $f(1,1)=(1)^2-(1)^2=0<\frac{1}{4}$.

Therefore, we can conclude that f indeed has a relative maximum of $\frac{1}{4}$ at the point $\left(\frac{\sqrt{2}}{2},\ \frac{1}{2}\right)$.

18) Maximize $U(x,y)=8x^{4/5}y^{1/5}$ subject to the constraint, $4x+2y=12$.

19) Minimize $f(x,y,z)=x^2+y^2+z^2$ subject to the constraint, $x+y+z=1$.

Answer

Subject to the given constraint, f has a relative minimum of $f\left(\frac{1}{3},\ \frac{1}{3},\ \frac{1}{3}\right)=\frac{1}{3}$ at the point $\left(\frac{1}{3},\ \frac{1}{3},\ \frac{1}{3}\right)$.

20) Minimize $f(x,y)=xy$ on the ellipse $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=1$.

21) Maximize $f(x,y,z)=2x+3y+5z$ on the sphere $x^2+y^2+z^2=19$.

Answer

Subject to the given constraint, f has a relative maximum of $19\sqrt{2}$ at the point $\left(\sqrt{2},\ \frac{3\sqrt{2}}{2},\ \frac{5\sqrt{2}}{2}\right)$.
Note that, subject to this constraint, f also has a relative minimum of $-19\sqrt{2}$ at the point $\left(-\sqrt{2},\ -\frac{3\sqrt{2}}{2},\ -\frac{5\sqrt{2}}{2}\right)$.
To see a 3D visualization of this problem, see: CalcPlot3D for Problem 21.

In exercises 22-23, use the method of Lagrange multipliers with two constraints.

22) Optimize $f(x,y,z) = yz + xy$ subject to the constraints: $xy = 1, \quad y^2 + z^2 = 1$.

Answer

maximum: $\frac{3}{2}$, minimum: $\frac{1}{2}$

23) Minimize $f(x,y,z) = x^2 + y^2 + z^2$ when $x + y + z = 9$ and $x + 2y + 3z = 20$.

Answer

minimum: $f(2,3,4) = 29$

Use the method of Lagrange multipliers to solve the following applied problems.

24) A large container in the shape of a rectangular solid must have a volume of 480 m^3. The bottom of the container costs \$5/m^2 to construct whereas the top and sides cost \$3/m^2 to construct. Use Lagrange multipliers to find the dimensions of the container of this size that has the minimum cost.

25) A rectangular box without a top (a topless box) is to be made from 12 ft^2 of cardboard. Find the maximum volume of such a box.

Answer

The maximum volume is 4 ft^3. The dimensions are $1 \times 2 \times 2$ ft.

26) Find the minimum distance from the parabola $y = x^2$ to point $(0,3)$.

27) Find the point on the line $y = 2x + 3$ that is closest to point $(4,2)$.

Answer

$(25, 195)$

29) Find the minimum distance from point $(0,1)$ to the parabola $x^2 = 4y$.

Answer

1.0 unit

30) Find the minimum and maximum distances between the ellipse $x^2 + xy + 2y^2 = 1$ and the origin.

31) Find the minimum distance from the plane $x + y + z = 1$ to point $(2,1,1)$.

Answer

$\sqrt{3}$ units

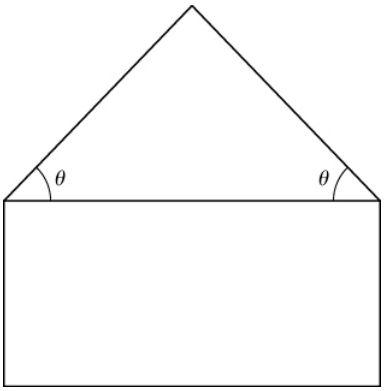
32) Find the point on the plane $4x + 3y + z = 2$ that is closest to the point $(1,-1,1)$.

33) Find the point on the surface $x^2 - 2xy + y^2 - x + y = 0$ closest to the point $(1,2,-3)$.

Answer

$\left(1, \frac{1}{2}, -3\right)$

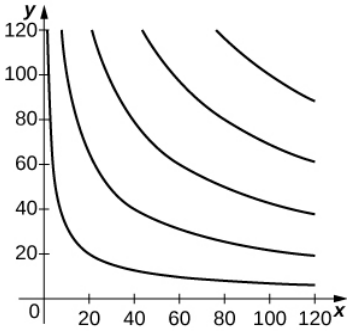
34) A pentagon is formed by placing an isosceles triangle on a rectangle, as shown in the diagram. If the perimeter of the pentagon is 10 in., find the lengths of the sides of the pentagon that will maximize the area of the pentagon.



35) **[T]** By investing x units of labor and y units of capital, a watch manufacturer can produce $P(x,y) = 50x^{0.4}y^{0.6}$ watches. Find the maximum number of watches that can be produced on a budget of $20,000 if labor costs $100/unit and capital costs $200/unit. Use a grapher like CalcPlot3D to sketch a contour plot of the function.

Answer

Roughly 3365 watches at the critical point $(80, 60)$.

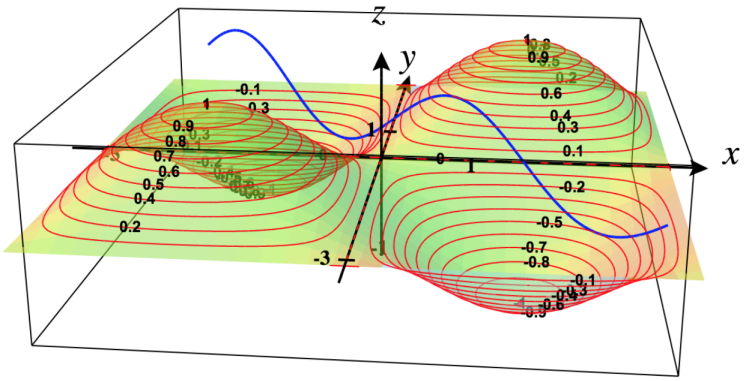


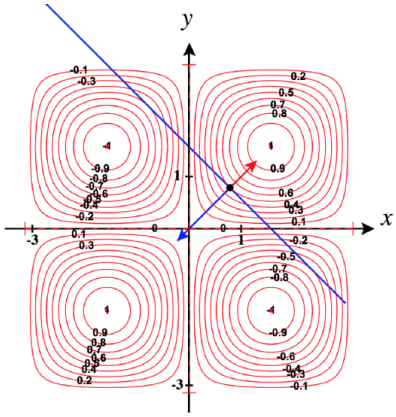
36) A rectangular solid is contained within a tetrahedron with vertices at $(1,0,0)$, $(0,1,0)$, $(0,0,1)$, and the origin. The base of the box has dimensions x and y, and the height of the box is z. If the sum of x, y, and z is 1, find the dimensions that maximizes the volume of the rectangular solid.

37) Find the maximum value of $f(x,y) = \sin x \sin y$, where x and y denote the acute angles of a right triangle. Draw the surface plot and contour plot of the function using a CAS.

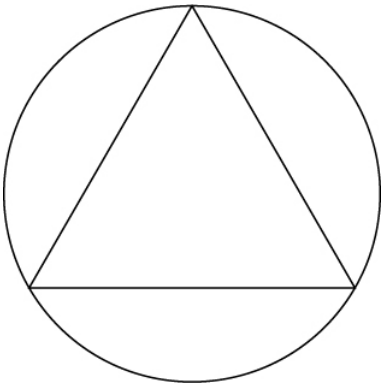
Answer

Subject to this constraint, f has a relative maximum of $\frac{1}{2}$ when $x = \frac{\pi}{4}$ and $y = \frac{\pi}{4}$.
Surface plot and contour plot for f:





38) Show that, of all the triangles inscribed in a circle of radius R (see diagram), the equilateral triangle has the largest perimeter.



Contributors

- Gilbert Strang (MIT) and Edwin “Jed” Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.
- Paul Seeburger (Monroe Community College) reordered these problems, adding problems 3 and 10 and answers for problems 15 and 17. He also added a full worked-out solution for problem 9 and a link to CalcPlot3D in problems 15, 21 and 35. He also created new images for Problem 37 and expanded the answers for many problems.

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14.R: Chapter 14 Review Exercises

For the following exercises, determine whether the statement is *true or false*. Justify your answer with a proof or a counterexample.

1. The domain of $f(x,y) = x^3 \arcsin(y)$ is $\{(x,y) \mid x \in \mathbb{R} \text{ and } -\pi \leq y \leq \pi\}$.

2. If the function $f(x,y)$ is continuous everywhere, then $f_{xy}(x,y) = f_{yx}(x,y)$.

Answer

True, by Clairaut's theorem

3. The linear approximation to the function of $f(x,y) = 5x^2 + x\tan y$ at the point $(2,\pi)$ is given by $L(x,y) = 22 + 21(x-2) + (y-\pi)$.

4. $(34, 916)$ is a critical point of $g(x,y) = 4x^3 - 2x^2y + y^2 - 2$.

Answer

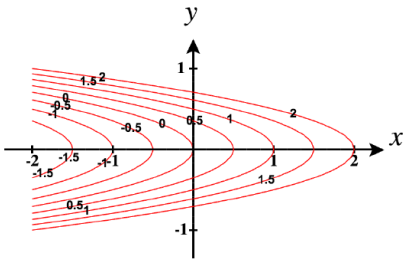
False

For the following exercises, sketch the function in one graph and, in a second, sketch several level curves.

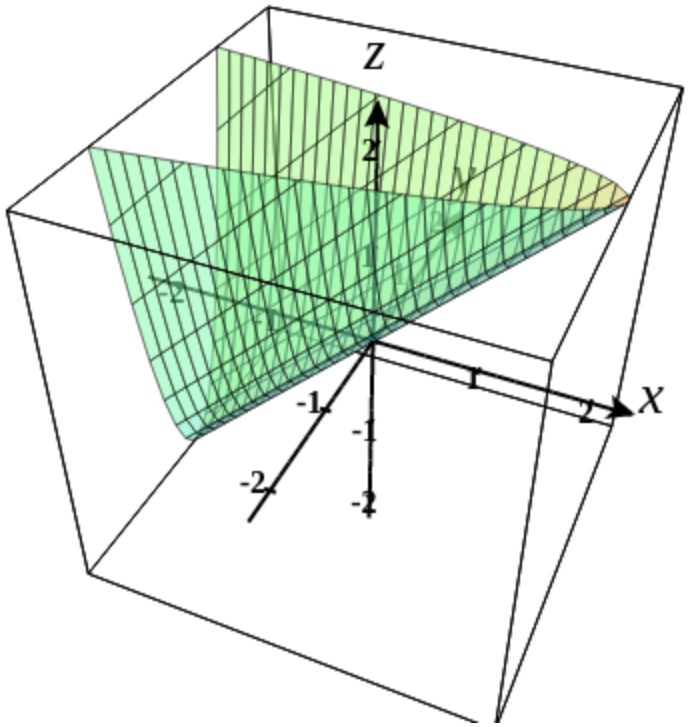
5. $f(x,y) = e^{-(x^2+2y^2)}$

6. $f(x,y) = x + 4y^2$

Answer



https://math.libretexts.org/@go/page/67640



For the following exercises, evaluate the following limits, if they exist. If they do not exist, prove it.

7. $\lim\limits_{(x,y)\to(1,1)} \dfrac{4xy}{x-2y^2}$

8. $\lim\limits_{(x,y)\to(0,0)} \dfrac{4xy}{x-2y^2}$

Answer

Does not exist.

For the following exercises, find the largest interval of continuity for the function.

9. $f(x,y) = x^3 \arcsin y$

10. $g(x,y) = \ln(4-x^2-y^2)$

Answer

Continuous at all points on the xy-plane, except where $x^2+y^2 > 4$.

For the following exercises, find all first partial derivatives.

11. $f(x,y) = x^2 - y^2$

12. $u(x,y) = x^4 - 3xy + 1$, with $x = 2t$ and $y = t^3$

Answer

$$\frac{\partial u}{\partial x} = 4x^3 - 3y,$$

$$\frac{\partial u}{\partial y} = -3x,$$

$$\frac{dx}{dt} = 2 \text{ and } \frac{dy}{dt} = 3t^2$$

$$\begin{aligned}\frac{du}{dt} &= \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} \\ &= 8x^3 - 6y - 9xt^2 \\ &= 8\big(2t\big)^3 - 6(t^3) - 9(2t)t^2 \\ &= 64t^3 - 6t^3 - 18t^3 \\ &= 40t^3\end{aligned}$$

For the following exercises, find all second partial derivatives.

13. $g(t, x) = 3t^2 - \sin(x + t)$

14. $h(x, y, z) = \dfrac{x^3 e^{2y}}{z}$

Answer

$$\begin{aligned}h_{xx}(x, y, z) &= \frac{6xe^{2y}}{z}, \\ h_{xy}(x, y, z) &= \frac{6x^2 e^{2y}}{z}, \\ h_{xz}(x, y, z) &= -\frac{3x^2 e^{2y}}{z^2}, \\ h_{yx}(x, y, z) &= \frac{6x^2 e^{2y}}{z}, \\ h_{yy}(x, y, z) &= \frac{4x^3 e^{2y}}{z}, \\ h_{yz}(x, y, z) &= -\frac{2x^3 e^{2y}}{z^2}, \\ h_{zx}(x, y, z) &= -\frac{3x^2 e^{2y}}{z^2}, \\ h_{zy}(x, y, z) &= -\frac{2x^3 e^{2y}}{z^2}, \\ h_{zz}(x, y, z) &= \frac{2x^3 e^{2y}}{z^3}\end{aligned}$$

For the following exercises, find the equation of the tangent plane to the specified surface at the given point.

15. $z = x^3 - 2y^2 + y - 1$ at point $(1, 1, -1)$

16. $z = e^x + \dfrac{2}{y}$ at point $(0, 1, 3)$

Answer

$z = x - 2y + 5$

17. Approximate $f(x, y) = e^{x^2} + \sqrt{y}$ at $(0.1, 9.1)$. Write down your linear approximation function $L(x, y)$. How accurate is the approximation to the exact answer, rounded to four digits?

18. Find the differential dz of $h(x, y) = 4x^2 + 2xy - 3y$ and approximate Δz at the point $(1, -2)$. Let $\Delta x = 0.1$ and $\Delta y = 0.01$.

Answer

$dz = 4\,dx - dy,\ dz(0.1, 0.01) = 0.39,\ \Delta z = 0.432$

19. Find the directional derivative of $f(x, y) = x^2 + 6xy - y^2$ in the direction $\vec{\mathbf{v}} = \hat{\mathbf{i}} + 4\hat{\mathbf{j}}$.

20. Find the maximal directional derivative magnitude and direction for the function $f(x, y) = x^3 + 2xy - \cos(\pi y)$ at point $(3, 0)$.

Answer

$3\sqrt{85}\langle 27, 6\rangle$

For the following exercises, find the gradient.

21. $c(x,t) = e(t-x)^2 + 3\cos t$

22. $f(x,y) = \dfrac{\sqrt{x}+y^2}{xy}$

Answer

$$\vec{\nabla} f(x,y) = -\frac{\sqrt{x}+2y^2}{2x^2y}\,\hat{\mathrm{i}} + \left(\frac{1}{x} + \frac{1}{\sqrt{x}y^2}\right)\hat{\mathrm{j}}$$

For the following exercise, find and classify the critical points.

23. $z = x^3 - xy + y^2 - 1$

For the following exercises, use Lagrange multipliers to find the maximum and minimum values for the functions with the given constraints.

24. $f(x,y) = x^2y$, subject to the constraint: $x^2 + y^2 = 4$

Answer

maximum: $\dfrac{16}{3\sqrt{3}}$, minimum: $-\dfrac{16}{3\sqrt{3}}$,

25. $f(x,y) = x^2 - y^2$, subject to the constraint: $x + 6y = 4$

26. A machinist is constructing a right circular cone out of a block of aluminum. The machine gives an error of 5% in height and 2% in radius. Find the maximum error in the volume of the cone if the machinist creates a cone of height 6 cm and radius 2 cm.

Answer

$2.3228\ \text{cm}^3$

27. A trash compactor is in the shape of a cuboid. Assume the trash compactor is filled with incompressible liquid. The length and width are decreasing at rates of 2 ft/sec and 3 ft/sec, respectively. Find the rate at which the liquid level is rising when the length is 14 ft, the width is 10 ft, and the height is 4 ft.

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CHAPTER OVERVIEW

15: Multiple Integration

In this chapter we extend the concept of a definite integral of a single variable to double and triple integrals of functions of two and three variables, respectively. We examine applications involving integration to compute volumes, masses, and centroids of more general regions. We will also see how the use of other coordinate systems (such as polar, cylindrical, and spherical coordinates) makes it simpler to compute multiple integrals over some types of regions and functions. In the preceding chapter, we discussed differential calculus with multiple independent variables. Now we examine integral calculus in multiple dimensions. Just as a partial derivative allows us to differentiate a function with respect to one variable while holding the other variables constant, we will see that an iterated integral allows us to integrate a function with respect to one variable while holding the other variables constant.

15.0: Prelude to Multiple Integration

15.1: Double Integrals over Rectangular Regions

15.1E: Exercises for Section 15.1

15.2: Double Integrals over General Regions

15.2E: Exercises for Section 15.2

15.3: Double Integrals in Polar Coordinates

15.3E: Exercises for Section 15.3

15.4: Triple Integrals

15.4E: Exercises for Section 15.4

15.5: Triple Integrals in Cylindrical and Spherical Coordinates

15.5E: Exercises for Section 15.5

15.6: Calculating Centers of Mass and Moments of Inertia

15.6E: Exercises for Section 15.6

15.7: Change of Variables in Multiple Integrals

15.7E: Exercises for Section 15.7

15.R: Chapter 15 Review Exercises

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15.0: Prelude to Multiple Integration

In this chapter we extend the concept of a definite integral of a single variable to double and triple integrals of functions of two and three variables, respectively. We examine applications involving integration to compute volumes, masses, and centroids of more general regions. We will also see how the use of other coordinate systems (such as polar, cylindrical, and spherical coordinates) makes it simpler to compute multiple integrals over some types of regions and functions. As an example, we will use polar coordinates to find the volume of structures such as l'Hemisfèric (Figure 15.0.1).

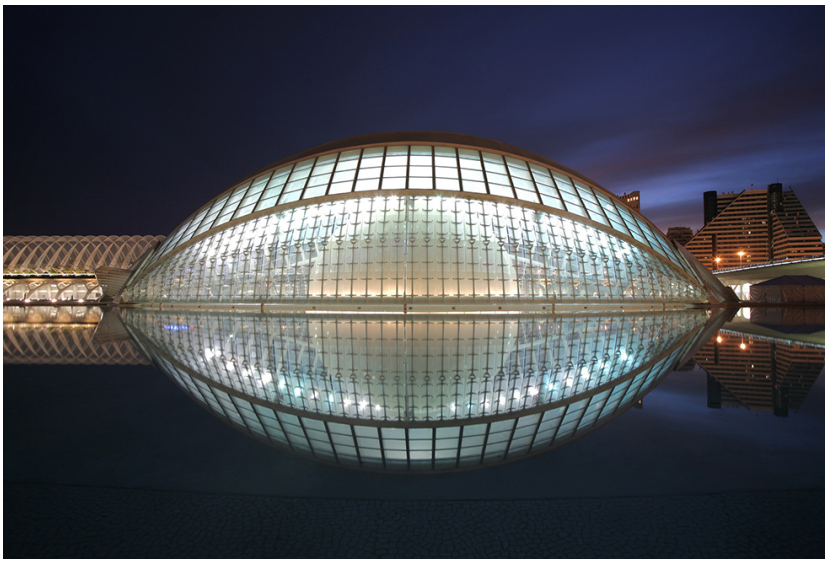


Figure 15.0.1: The City of Arts and Sciences in Valencia, Spain, has a unique structure along an axis of just two kilometers that was formerly the bed of the River Turia. The l'Hemisfèric has an IMAX cinema with three systems of modern digital projections onto a concave screen of 900 square meters. An oval roof over 100 meters long has been made to look like a huge human eye that comes alive and opens up to the world as the "Eye of Wisdom." (credit: modification of work by Javier Yaya Tur, Wikimedia Commons)

In the preceding chapter, we discussed differential calculus with multiple independent variables. Now we examine integral calculus in multiple dimensions. Just as a partial derivative allows us to differentiate a function with respect to one variable while holding the other variables constant, we will see that an iterated integral allows us to integrate a function with respect to one variable while holding the other variables constant.

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15.1: Double Integrals over Rectangular Regions

Learning Objectives

- Recognize when a function of two variables is integrable over a rectangular region.
- Recognize and use some of the properties of double integrals.
- Evaluate a double integral over a rectangular region by writing it as an iterated integral.
- Use a double integral to calculate the area of a region, volume under a surface, or average value of a function over a plane region.

In this section we investigate double integrals and show how we can use them to find the volume of a solid over a rectangular region in the xy-plane. Many of the properties of double integrals are similar to those we have already discussed for single integrals.

Volumes and Double Integrals

We begin by considering the space above a rectangular region R. Consider a continuous function $f(x,y) \geq 0$ of two variables defined on the closed rectangle R:

$$R = [a,b] \times [c,d] = \{(x,y) \in \mathbb{R}^2 | \, a \leq x \leq b, \, c \leq y \leq d\}$$

Here $[a,b] \times [c,d]$ denotes the Cartesian product of the two closed intervals $[a,b]$ and $[c,d]$. It consists of rectangular pairs (x,y) such that $a \leq x \leq b$ and $c \leq y \leq d$. The graph of f represents a surface above the xy-plane with equation $z = f(x,y)$ where z is the height of the surface at the point (x,y). Let S be the solid that lies above R and under the graph of f (Figure 15.1.1). The base of the solid is the rectangle R in the xy-plane. We want to find the volume V of the solid S.

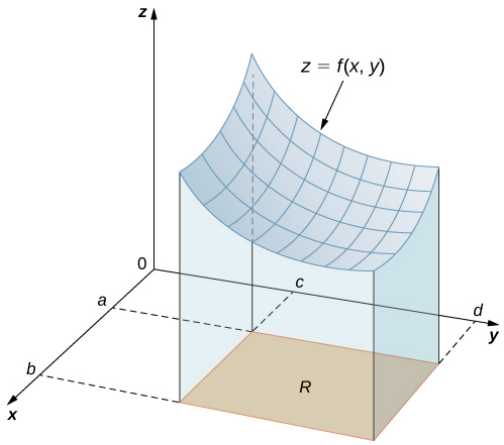


Figure 15.1.1: The graph of $f(x,y)$ *over the rectangle R in the* xy*-plane is a curved surface.*

We divide the region R into small rectangles R_{ij}, each with area ΔA and with sides Δx and Δy (Figure 15.1.2). We do this by dividing the interval $[a,b]$ into m subintervals and dividing the interval $[c,d]$ into n subintervals. Hence $\Delta x = \frac{b-a}{m}$, $\Delta y = \frac{d-c}{n}$, and $\Delta A = \Delta x \Delta y$.

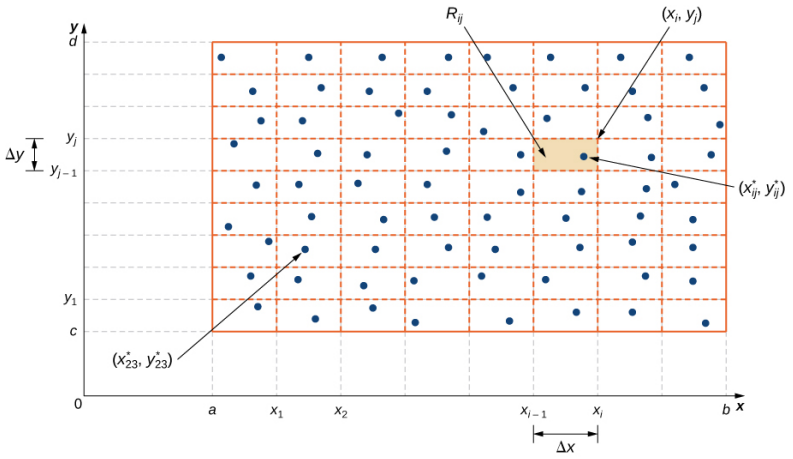


Figure 15.1.2: Rectangle R is divided into small rectangles R_{ij} each with area ΔA.

The volume of a thin rectangular box above R_{ij} is $f(x_{ij}^*, y_{ij}^*)\,\Delta A$, where (x_{ij}^*, y_{ij}^*) is an arbitrary sample point in each R_{ij} as shown in the following figure, $f(x_{ij}^*, y_{ij}^*)$ is the height of the corresponding thin rectangular box, and ΔA is the area of each rectangle R_{ij}.

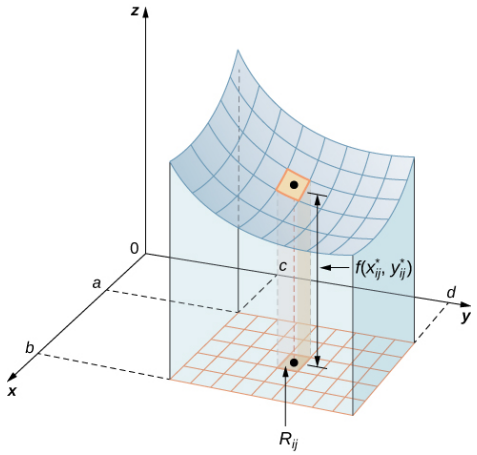


Figure 15.1.3: A thin rectangular box above R_{ij} with height $f(x_{ij}^*, y_{ij}^*)$.

Using the same idea for all the subrectangles, we obtain an approximate volume of the solid S as

$$V \approx \sum_{i=1}^{m}\sum_{j=1}^{n} f(x_{ij}^*, y_{ij}^*)\Delta A.$$

This sum is known as a **double Riemann sum** and can be used to approximate the value of the volume of the solid. Here the double sum means that for each subrectangle we evaluate the function at the chosen point, multiply by the area of each rectangle, and then add all the results.

As we have seen in the single-variable case, we obtain a better approximation to the actual volume if m and n become larger.

$$V = \lim_{m,n\to\infty} \sum_{i=1}^{m}\sum_{j=1}^{n} f(x_{ij}^*, y_{ij}^*)\Delta A$$

or

$$V = \lim_{\Delta x,\, \Delta y\to 0} \sum_{i=1}^{m}\sum_{j=1}^{n} f(x_{ij}^*, y_{ij}^*)\Delta A.$$

Note that the sum approaches a limit in either case and the limit is the volume of the solid with the base R. Now we are ready to define the double integral.

Definition: Double Integral over a Rectangular Region R

The double integral of the function $f(x, y)$ over the rectangular region R in the xy-plane is defined as

$$\iint_R f(x, y)dA = \lim_{m,n\to\infty} \sum_{i=1}^{m} \sum_{j=1}^{n} f(x_{ij}^*, y_{ij}^*)\Delta A.$$

If $f(x, y) \geq 0$, then the volume V of the solid S, which lies above R in the xy-plane and under the graph of f, is the double integral of the function $f(x, y)$ over the rectangle R. If the function is ever negative, then the double integral can be considered a "signed" volume in a manner similar to the way we defined net signed area in The Definite Integral.

Example 15.1.1: Setting up a Double Integral and Approximating It by Double Sums

Consider the function $z = f(x, y) = 3x^2 - y$ over the rectangular region $R = [0, 2] \times [0, 2]$ (Figure 15.1.4).

a. Set up a double integral for finding the value of the signed volume of the solid S that lies above R and "under" the graph of f.

b. Divide R into four squares with $m = n = 2$, and choose the sample point as the upper right corner point of each square (1,1), (2,1),(1,2), and (2,2) (Figure 15.1.4) to approximate the signed volume of the solid S that lies above R and "under" the graph of f.

c. Divide R into four squares with $m = n = 2$, and choose the sample point as the midpoint of each square: (1/2, 1/2), (3/2, 1/2), (1/2,3/2), and (3/2, 3/2) to approximate the signed volume.

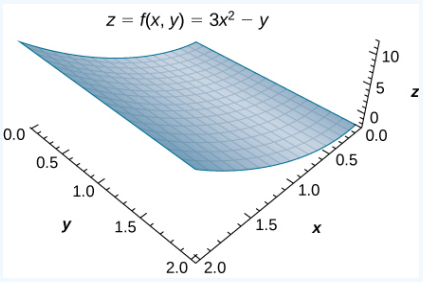


Figure 15.1.4: The function $z = f(x, y)$ graphed over the rectangular region $R = [0, 2] \times [0, 2]$.

Solution

a. As we can see, the function $z = f(x, y) = 3x^2 - y$ is above the plane. To find the signed volume of S, we need to divide the region R into small rectangles R_{ij}, each with area ΔA and with sides Δx and Δy, and choose (x_{ij}^*, y_{ij}^*) as sample points in each R_{ij}. Hence, a double integral is set up as

$$V = \iint_R (3x^2 - y)dA = \lim_{m,n\to\infty} \sum_{i=1}^{m} \sum_{j=1}^{n} [3(x_{ij}^*)^2 - y_{ij}^*]\Delta A.$$

b. Approximating the signed volume using a Riemann sum with $m = n = 2$ we have $\Delta A = \Delta x \Delta y = 1 \times 1 = 1$. Also, the sample points are (1, 1), (2, 1), (1, 2), and (2, 2) as shown in the following figure.

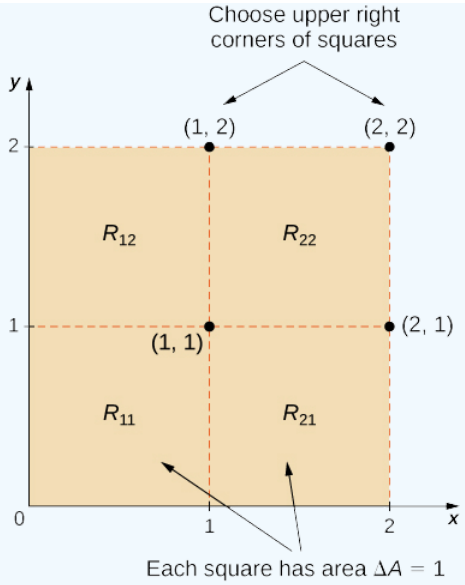


Figure 15.1.5: Subrectangles for the rectangular region $R = [0, 2] \times [0, 2]$.

Hence,

$$\begin{aligned} V &\approx \sum_{i=1}^{2}\sum_{j=1}^{2} f(x_{ij}^*, y_{ij}^*)\Delta A \\ &= \sum_{i=1}^{2}(f(x_{i1}^*, y_{i1}^*) + f(x_{i2}^*, y_{i2}^*))\Delta A \\ &= f(x_{11}^*, y_{11}^*)\Delta A + f(x_{21}^*, y_{21}^*)\Delta A + f(x_{12}^*, y_{12}^*)\Delta A + f(x_{22}^*, y_{22}^*)\Delta A \\ &= f(1,1)(1) + f(2,1)(1) + f(1,2)(1) + f(2,2)(1) \\ &= (3-1)(1) + (12-1)(1) + (3-2)(1) + (12-2)(1) \\ &= 2 + 11 + 1 + 10 = 24. \end{aligned}$$

c. Approximating the signed volume using a Riemann sum with $m = n = 2$ we have$\Delta A = \Delta x \Delta y = 1 \times 1 = 1$. In this case the sample points are (1/2, 1/2), (3/2, 1/2), (1/2, 3/2), and (3/2, 3/2).
Hence,

$$\begin{aligned} V &\approx \sum_{i=1}^{2}\sum_{j=1}^{2} f(x_{ij}^*, y_{ij}^*)\Delta A \\ &= f(x_{11}^*, y_{11}^*)\Delta A + f(x_{21}^*, y_{21}^*)\Delta A + f(x_{12}^*, y_{12}^*)\Delta A + f(x_{22}^*, y_{22}^*)\Delta A \\ &= f(1/2, 1/2)(1) + f(3/2, 1/2)(1) + f(1/2, 3/2)(1) + f(3/2, 3/2)(1) \\ &= \left(\frac{3}{4} - \frac{1}{4}\right)(1) + \left(\frac{27}{4} - \frac{1}{2}\right)(1) + \left(\frac{3}{4} - \frac{3}{2}\right)(1) + \left(\frac{27}{4} - \frac{3}{2}\right)(1) \\ &= \frac{2}{4} + \frac{25}{4} + \left(-\frac{3}{4}\right) + \frac{21}{4} = \frac{45}{4} = 11. \end{aligned}$$

Analysis

Notice that the approximate answers differ due to the choices of the sample points. In either case, we are introducing some error because we are using only a few sample points. Thus, we need to investigate how we can achieve an accurate answer.

? Exercise 15.1.1

Use the same function$z = f(x, y) = 3x^2 - y$ over the rectangular region $R = [0, 2] \times [0, 2]$.

Divide R into the same four squares with $m = n = 2$, and choose the sample points as the upper left corner point of each square (0,1), (1,1), (0,2), and (1,2) (Figure 15.1.5) to approximate the signed volume of the solid S that lies above R and "under" the graph of f.

Hint

Follow the steps of the previous example.

Answer

$$V \approx \sum_{i=1}^{2}\sum_{j=1}^{2} f(x_{ij}^*, y_{ij}^*)\,\Delta A = 0$$

Note that we developed the concept of double integral using a rectangular region R. This concept can be extended to any general region. However, when a region is not rectangular, the subrectangles may not all fit perfectly into R, particularly if the base area is curved. We examine this situation in more detail in the next section, where we study regions that are not always rectangular and subrectangles may not fit perfectly in the region R. Also, the heights may not be exact if the surface $z = f(x, y)$ is curved. However, the errors on the sides and the height where the pieces may not fit perfectly within the solid S approach 0 as m and n approach infinity. Also, the double integral of the function $z = f(x, y)$ exists provided that the function f is not too discontinuous. If the function is bounded and continuous over R except on a finite number of smooth curves, then the double integral exists and we say that f is integrable over R.

Since $\Delta A = \Delta x \Delta y = \Delta y \Delta x$, we can express dA as $dx\,dy$ or $dy\,dx$. This means that, when we are using rectangular coordinates, the double integral over a region R denoted by

$$\iint_R f(x, y)\,dA$$

can be written as

$$\iint_R f(x, y)\,dx\,dy$$

or

$$\iint_R f(x, y)\,dy\,dx.$$

Now let's list some of the properties that can be helpful to compute double integrals.

Properties of Double Integrals

The properties of double integrals are very helpful when computing them or otherwise working with them. We list here six properties of double integrals. Properties 1 and 2 are referred to as the linearity of the integral, property 3 is the additivity of the integral, property 4 is the monotonicity of the integral, and property 5 is used to find the bounds of the integral. Property 6 is used if $f(x, y)$ is a product of two functions $g(x)$ and $h(y)$.

Theorem: Properties of Double Integrals

Assume that the functions $f(x, y)$ and $g(x, y)$ are integrable over the rectangular region R; S and T are subregions of R; and assume that m and M are real numbers.

i. The sum $f(x, y) + g(x, y)$ is integrable and

$$\iint_R [f(x, y) + g(x, y)]\,dA = \iint_R f(x, y)\,dA + \iint_R g(x, y)\,dA.$$

ii. If c is a constant, then $cf(x, y)$ is integrable and

$$\iint_R cf(x, y)\,dA = c\iint_R f(x, y)\,dA.$$

iii. If $R = S \cup T$ and $S \cap T = \emptyset$ except an overlap on the boundaries, then

$$\iint_R f(x, y)\,dA = \iint_S f(x, y)\,dA + \iint_T f(x, y)\,dA.$$

iv. If $f(x,y) \geq g(x,y)$ for (x,y) in R, then

$$\iint_R f(x,y)\,dA \geq \iint_R g(x,y)\,dA.$$

v. If $m \leq f(x,y) \leq M$ and $A(R) = \text{ the area of } R$, then

$$m \cdot A(R) \leq \iint_R f(x,y)\,dA \leq M \cdot A(R).$$

vi. In the case where $f(x,y)$ can be factored as a product of a function $g(x)$ of x only and a function $h(y)$ of y only, then over the region $R = \{(x,y) \mid a \leq x \leq b,\ c \leq y \leq d\}$, the double integral can be written as

$$\iint_R f(x,y)\,dA = \left(\int_a^b g(x)\,dx\right)\left(\int_c^d h(y)\,dy\right).$$

These properties are used in the evaluation of double integrals, as we will see later. We will become skilled in using these properties once we become familiar with the computational tools of double integrals. So let's get to that now.

Iterated Integrals

So far, we have seen how to set up a double integral and how to obtain an approximate value for it. We can also imagine that evaluating double integrals by using the definition can be a very lengthy process if we choose larger values for m and n.Therefore, we need a practical and convenient technique for computing double integrals. In other words, we need to learn how to compute double integrals without employing the definition that uses limits and double sums.

The basic idea is that the evaluation becomes easier if we can break a double integral into single integrals by integrating first with respect to one variable and then with respect to the other. The key tool we need is called an iterated integral.

Definitions: Iterated Integrals

Assume a, b, c, and d are real numbers. We define an *iterated integral* for a function $f(x,y)$ over the rectangular region $R = [a,b] \times [c,d]$ as

$$\int_a^b \int_c^d f(x,y)\,dy\,dx = \int_a^b \left[\int_c^d f(x,y)\,dy\right] dx$$

or

$$\int_c^d \int_a^b f(x,y)\,dx\,dy = \int_c^d \left[\int_a^b f(x,y)\,dx\right] dy.$$

The notation $\int_a^b \left[\int_c^d f(x,y)\,dy\right] dx$ means that we integrate $f(x,y)$ with respect to y while holding x constant. Similarly, the notation $\int_c^d \left[\int_a^b f(x,y)\,dx\right] dy$ means that we integrate $f(x,y)$ with respect to x while holding y constant. The fact that double integrals can be split into iterated integrals is expressed in Fubini's theorem. Think of this theorem as an essential tool for evaluating double integrals.

Theorem: Fubini's Theorem

Suppose that $f(x,y)$ is a function of two variables that is continuous over a rectangular region $R = \{(x,y) \in \mathbb{R}^2 \mid a \leq x \leq b,\ c \leq y \leq d\}$. Then we see from Figure 15.1.6 that the double integral of f over the region equals an iterated integral,

$$\iint_R f(x,y)\,dA = \iint_R f(x,y)\,dx\,dy = \int_a^b \int_c^d f(x,y)\,dy\,dx = \int_c^d \int_a^b f(x,y)\,dx\,dy.$$

More generally, Fubini's theorem is true if f is bounded on R and f is discontinuous only on a finite number of continuous curves. In other words, f has to be integrable over R.

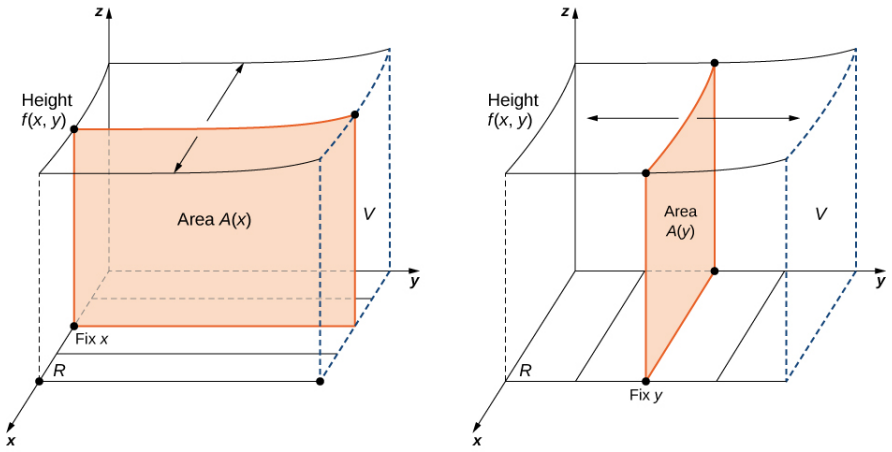


Figure 15.1.6: (a) Integrating first with respect to y and then with respect to x to find the area $A(x)$ and then the volume V; (b) integrating first with respect to x and then with respect to y to find the area $A(y)$ and then the volume V.

Example 15.1.2: Using Fubini's Theorem

Use Fubini's theorem to compute the double integral $\iint_R f(x,y)\,dA$ where $f(x,y)=x$ and $R=[0,2]\times[0,1]$.

Solution

Fubini's theorem offers an easier way to evaluate the double integral by the use of an iterated integral. Note how the boundary values of the region R become the upper and lower limits of integration.

$$\begin{aligned}\iint_R f(x,y)\,dA &= \iint_R f(x,y)\,dx\,dy \\ &= \int_{y=0}^{y=1}\int_{x=0}^{x=2} x\,dx\,dy \\ &= \int_{y=0}^{y=1}\left[\frac{x^2}{2}\Big|_{x=0}^{x=2}\right] dy \\ &= \int_{y=0}^{y=1} 2\,dy = 2y\Big|_{y=0}^{y=1} = 2\end{aligned}$$

The double integration in this example is simple enough to use Fubini's theorem directly, allowing us to convert a double integral into an iterated integral. Consequently, we are now ready to convert all double integrals to iterated integrals and demonstrate how the properties listed earlier can help us evaluate double integrals when the function $f(x,y)$ is more complex. Note that the order of integration can be changed (see Example 7).

Example 15.1.3: Illustrating Properties i and ii

Evaluate the double integral

$$\iint_R (xy-3xy^2)\,dA, \text{ where } R=\{(x,y)\,|\,0\le x\le 2,\ 1\le y\le 2\}.$$

Solution

This function has two pieces: one piece is xy and the other is $3xy^2$. Also, the second piece has a constant 3. Notice how we use properties i and ii to help evaluate the double integral.

$$\iint_R (xy - 3xy^2)\,dA = \iint_R xy\,dA + \iint_R (-3xy^2)\,dA$$ Property i: Integral of a sum is the sum of the integrals.

$$= \int_{y=1}^{y=2}\int_{x=0}^{x=2} xy\,dx\,dy - \int_{y=1}^{y=2}\int_{x=0}^{x=2} 3xy^2\,dx\,dy$$ Convert double integrals to iterated integrals.

$$= \int_{y=1}^{y=2}\left(\frac{x^2}{2}y\right)\Big|_{x=0}^{x=2} dy - 3\int_{y=1}^{y=2}\left(\frac{x^2}{2}y^2\right)\Big|_{x=0}^{x=2} dy$$ Integrate with respect to x, holding y constant.

$$= \int_{y=1}^{y=2} 2y\,dy - \int_{y=1}^{y=2} 6y^2 dy$$ Property ii: Placing the constant before the integral.

$$= 2\int_1^2 y\,dy - 6\int_1^2 y^2\,dy$$ Integrate with respect to y.

$$= 2\frac{y^2}{2}\Big|_1^2 - 6\frac{y^3}{3}\Big|_1^2$$

$$= y^2\Big|_1^2 - 2y^3\Big|_1^2$$

$$= (4-1) - 2(8-1) = 3 - 2(7)$$
$$= 3 - 14 = -11.$$

Example 15.1.4: Illustrating Property v.

Over the region $R = \{(x,y) \mid 1 \le x \le 3,\ 1 \le y \le 2\}$, we have $2 \le x^2 + y^2 \le 13$. Find a lower and an upper bound for the integral $\iint_R (x^2 + y^2)\,dA$.

Solution

For a lower bound, integrate the constant function 2 over the region R. For an upper bound, integrate the constant function 13 over the region R.

$$\int_1^2\int_1^3 2\,dx\,dy = \int_1^2 [2x\Big|_1^3]\,dy = \int_1^2 2(2)dy = 4y\Big|_1^2 = 4(2-1) = 4$$
$$\int_1^2\int_1^3 13dx\,dy = \int_1^2 [13x\Big|_1^3]\,dy = \int_1^2 13(2)\,dy = 26y\Big|_1^2 = 26(2-1) = 26.$$

Hence, we obtain $4 \le \iint_R (x^2 + y^2)\,dA \le 26$.

Example 15.1.5: Illustrating Property vi

Evaluate the integral $\iint_R e^y \cos x\,dA$ over the region $R = \{(x,y) \mid 0 \le x \le \frac{\pi}{2},\ 0 \le y \le 1\}$.

Solution

This is a great example for property vi because the function $f(x,y)$ is clearly the product of two single-variable functions e^y and $\cos x$. Thus we can split the integral into two parts and then integrate each one as a single-variable integration problem.

$$\iint_R e^y \cos x \, dA = \int_0^1 \int_0^{\pi/2} e^y \cos x \, dx \, dy$$
$$= \left(\int_0^1 e^y dy \right) \left(\int_0^{\pi/2} \cos x \, dx \right)$$
$$= (e^y \Big|_0^1)(\sin x \Big|_0^{\pi/2})$$
$$= e - 1.$$

? Exercise 15.1.2

a. Use the properties of the double integral and Fubini's theorem to evaluate the integral

$$\int_0^1 \int_{-1}^3 (3 - x + 4y) \, dy \, dx.$$

b. Show that $0 \leq \iint_R \sin \pi x \, \cos \pi y \, dA \leq \frac{1}{32}$ where $R = (0, \frac{1}{4}) \, (\frac{1}{4}, \frac{1}{2})$.

Hint

Use properties i. and ii. and evaluate the iterated integral, and then use property v.

Answer

a. 26

b. Answers may vary.

As we mentioned before, when we are using rectangular coordinates, the double integral over a region R denoted by $\iint_R f(x, y) \, dA$ can be written as $\iint_R \, f(x, y) \, dx \, dy$ or $\iint_R \, f(x, y) \, dy \, dx$. The next example shows that the results are the same regardless of which order of integration we choose.

✓ Example 15.1.6: Evaluating an Iterated Integral in Two Ways

Let's return to the function $f(x, y) = 3x^2 - y$ from Example 1, this time over the rectangular region $R = [0, 2] \times [0, 3]$. Use Fubini's theorem to evaluate $\iint_R f(x, y) \, dA$ in two different ways:

a. First integrate with respect to y and then with respect to x;
b. First integrate with respect to x and then with respect to y.

Solution

Figure 15.1.6 shows how the calculation works in two different ways.

a. First integrate with respect to y and then integrate with respect to x:

$$\iint_R f(x, y) \, dA = \int_{x=0}^{x=2} \int_{y=0}^{y=3} (3x^2 - y) \, dy \, dx$$
$$= \int_{x=0}^{x=2} \left(\int_{y=0}^{y=3} (3x^2 - y) \, dy \right) \, dx = \int_{x=0}^{x=2} \left[3x^2 y - \frac{y^2}{2} \Big|_{y=0}^{y=3} \right] \, dx$$
$$= \int_{x=0}^{x=2} \left(9x^2 - \frac{9}{2} \right) \, dx = 3x^3 - \frac{9}{2} x \Big|_{x=0}^{x=2} = 15.$$

b. First integrate with respect to x and then integrate with respect to y:

$$\begin{aligned}\iint_R f(x,y)\,dA &= \int_{y=0}^{y=3}\int_{x=0}^{x=2}(3x^2-y)\,dx\,dy \\ &= \int_{y=0}^{y=3}\left(\int_{x=0}^{x=2}(3x^2-y)\,dx\right)dy \\ &= \int_{y=0}^{y=3}\left[x^3-xy\Big|_{x=0}^{x=2}\right]dy \\ &= \int_{y=0}^{y=3}(8-2y)\,dy = 8y-y^2\Big|_{y=0}^{y=3} = 15.\end{aligned}$$

Analysis

With either order of integration, the double integral gives us an answer of 15. We might wish to interpret this answer as a volume in cubic units of the solid S below the function $f(x,y)=3x^2-y$ over the region $R=[0,2]\times[0,3]$. However, remember that the interpretation of a double integral as a (non-signed) volume works only when the integrand f is a nonnegative function over the base region R.

? Exercise 15.1.3

Evaluate

$$\int_{y=-3}^{y=2}\int_{x=3}^{x=5}(2-3x^2+y^2)\,dx\,dy.$$

Hint

Use Fubini's theorem.

Answer

$-\frac{1340}{3}$

In the next example we see that it can actually be beneficial to switch the order of integration to make the computation easier. We will come back to this idea several times in this chapter.

✓ Example 15.1.7: Switching the Order of Integration

Consider the double integral $\iint_R x\,\sin(xy)\,dA$ over the region $R=\{(x,y)\,|\,0\le x\le\pi,\ 1\le y\le 2\}$ (Figure 15.1.7).

a. Express the double integral in two different ways.
b. Analyze whether evaluating the double integral in one way is easier than the other and why.
c. Evaluate the integral.

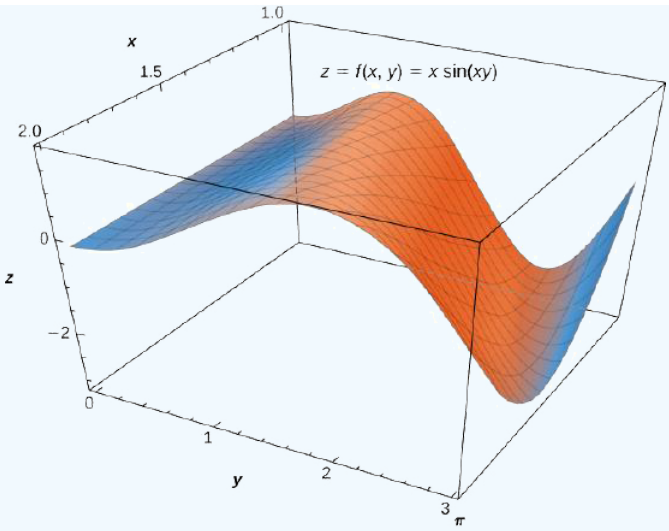


Figure 15.1.7: The function $z = f(x, y) = x\ \sin(xy)$ over the rectangular region $R = [0, \pi] \times [1, 2]$.

a. We can express $\iint_R x\ \sin(xy)\, dA$ in the following two ways: first by integrating with respect to y and then with respect to x; second by integrating with respect to x and then with respect to y.

$$\iint_R x\ \sin(xy)\, dA = \int_{x=0}^{x=\pi} \int_{y=1}^{y=2} x\ \sin(xy)\, dy\, dx$$

Integrate first with respect to y.

$$= \int_{y=1}^{y=2} \int_{x=0}^{x=\pi} x\ \sin(xy)\, dx\, dy$$

Integrate first with respect to x.

b. If we want to integrate with respect to y first and then integrate with respect to x, we see that we can use the substitution $u = xy$, which gives $du = x\, dy$. Hence the inner integral is simply $\int \sin u\, du$ and we can change the limits to be functions of x,

$$\iint_R x\ \sin(xy)\, dA = \int_{x=0}^{x=\pi} \int_{y=1}^{y=2} x\ \sin(xy)\, dy\, dx = \int_{x=0}^{x=\pi} \left[\int_{u=x}^{u=2x} \sin(u)\, du \right] dx.$$

However, integrating with respect to x first and then integrating with respect to y requires integration by parts for the inner integral, with $u = x$ and $dv = \sin(xy)dx$

Then $du = dx$ and $v = -\frac{\cos(xy)}{y}$, so

$$\iint_R x \sin(xy)\, dA = \int_{y=1}^{y=2} \int_{x=0}^{x=\pi} x \sin(xy)\, dx\, dy$$
$$= \int_{y=1}^{y=2} \left[-\frac{x\ \cos(xy)}{y} \Big|_{x=0}^{x=\pi} + \frac{1}{y} \int_{x=0}^{x=\pi} \cos(xy)\, dx \right] dy.$$

Since the evaluation is getting complicated, we will only do the computation that is easier to do, which is clearly the first method.

c. Evaluate the double integral using the easier way.

$$\iint_R x\,\sin(xy)\,dA = \int_{x=0}^{x=\pi}\int_{y=1}^{y=2} x\,\sin(xy)\,dy\,dx$$
$$= \int_{x=0}^{x=\pi}\left[\int_{u=x}^{u=2x}\sin(u)\,du\right]dx = \int_{x=0}^{x=\pi}\left[-\cos u\Big|_{u=x}^{u=2x}\right]dx$$
$$= \int_{x=0}^{x=\pi}(-\cos 2x+\cos x)\,dx$$
$$= \left(-\frac{1}{2}\sin 2x+\sin x\right)\Big|_{x=0}^{x=\pi} = 0.$$

? Exercise 15.1.4

Evaluate the integral $\iint_R xe^{xy}\,dA$ where $R=[0,1]\times[0,\ln 5]$.

Hint

Integrate with respect to y first.

Answer

$\frac{4-\ln 5}{\ln 5}$

Applications of Double Integrals

Double integrals are very useful for finding the area of a region bounded by curves of functions. We describe this situation in more detail in the next section. However, if the region is a rectangular shape, we can find its area by integrating the constant function $f(x,y)=1$ over the region R.

Definition: Area of a Region R

The area of the region R is given by

$$A(R)=\iint_R 1\,dA.$$

This definition makes sense because using $f(x,y)=1$ and evaluating the integral make it a product of length and width. Let's check this formula with an example and see how this works.

✓ Example 15.1.8: Finding Area Using a Double Integral

Find the area of the region $R=\{(x,y)\,|\,0\le x\le 3,\ 0\le y\le 2\}$ by using a double integral, that is, by integrating 1 over the region R.

Solution

The region is rectangular with length 3 and width 2, so we know that the area is 6. We get the same answer when we use a double integral:

$$A(R)=\int_0^2\int_0^3 1\,dx\,dy=\int_0^2\left[x\Big|_0^3\right]dy=\int_0^2 3dy=3\int_0^2 dy=3y\Big|_0^2=3(2)=6\text{ units}^2.$$

We have already seen how double integrals can be used to find the volume of a solid bounded above by a function $f(x,y)\ge 0$ over a region R provided $f(x,y)\ge 0$ for all (x,y) in R. Here is another example to illustrate this concept.

Example 15.1.9: Volume of an Elliptic Paraboloid

Find the volume V of the solid S that is bounded by the elliptic paraboloid $2x^2 + y^2 + z = 27$, the planes $x = 3$ and $y = 3$, and the three coordinate planes.

Solution

First notice the graph of the surface $z = 27 - 2x^2 - y^2$ in Figure 15.1.8(a) and above the square region $R_1 = [-3, 3] \times [-3, 3]$. However, we need the volume of the solid bounded by the elliptic paraboloid $2x^2 + y^2 + z = 27$, the planes $x = 3$ and $y = 3$, and the three coordinate planes.

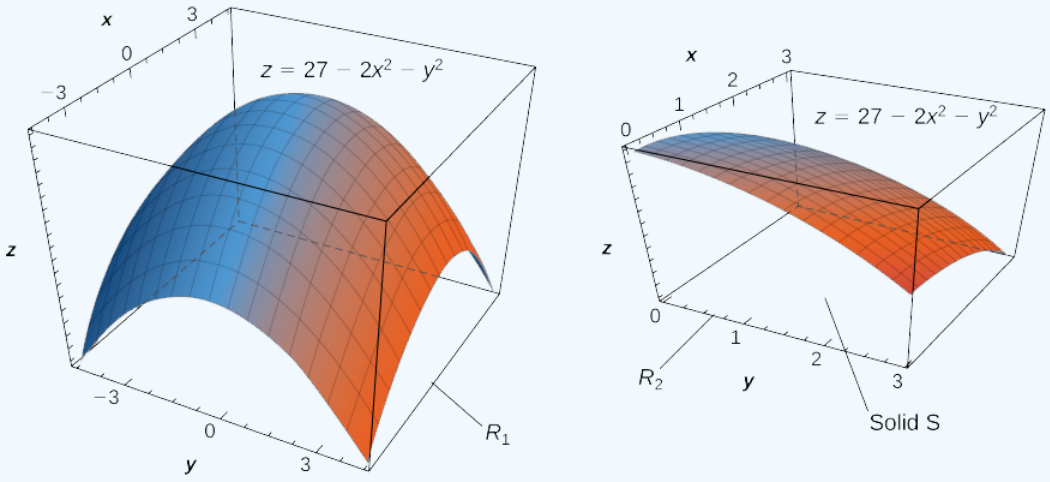


Figure 15.1.8: (a) The surface $z = 27 - 2x^2 - y^2$ above the square region $R1 = [-3, 3] \times [-3, 3]$. (b) The solid S lies under the surface $z = 27 - 2x^2 - y^2$ above the square region $R1 = [0, 3] \times [0, 3]$.

Now let's look at the graph of the surface in Figure 15.1.8(b). We determine the volume V by evaluating the double integral over R_2:

$$
\begin{aligned}
V &= \iint_R z\,dA = \iint_R (27 - 2x^2 - y^2)\,dA \\
&= \int_{y=0}^{y=3} \int_{x=0}^{x=3} (27 - 2x^2 - y^2)\,dx\,dy && \text{Convert to literal integral.} \\
&= \int_{y=0}^{y=3} \left[27x - \frac{2}{3}x^3 - y^2 x\right]\Big|_{x=0}^{x=3} dy && \text{Integrate with respect to } x. \\
&= \int_{y=0}^{y=3} (63 - 3y^2)dy = 63y - y^3\Big|_{y=0}^{y=3} = 162.
\end{aligned}
$$

Exercise 15.1.5

Find the volume of the solid bounded above by the graph of $f(x, y) = xy\sin(x^2 y)$ and below by the xy-plane on the rectangular region $R = [0, 1] \times [0, \pi]$.

Hint

Graph the function, set up the integral, and use an iterated integral.

Answer

$\frac{\pi}{2}$

Recall that we defined the average value of a function of one variable on an interval $[a, b]$ as

$$f_{ave} = \frac{1}{b-a}\int_a^b f(x)\,dx.$$

Similarly, we can define the average value of a function of two variables over a region R. The main difference is that we divide by an area instead of the width of an interval.

Definition: Average Value of a Function

The **average value of a function** of two variables over a region R is

$$F_{ave} = \frac{1}{\text{Area of } R}\iint_R f(x,y)\,dx\,dy.$$

In the next example we find the average value of a function over a rectangular region. This is a good example of obtaining useful information for an integration by making individual measurements over a grid, instead of trying to find an algebraic expression for a function.

Example 15.1.10: Calculating Average Storm Rainfall

The weather map in Figure 15.1.9 shows an unusually moist storm system associated with the remnants of Hurricane Karl, which dumped 4–8 inches (100–200 mm) of rain in some parts of the Midwest on September 22–23, 2010. The area of rainfall measured 300 miles east to west and 250 miles north to south. Estimate the average rainfall over the entire area in those two days.

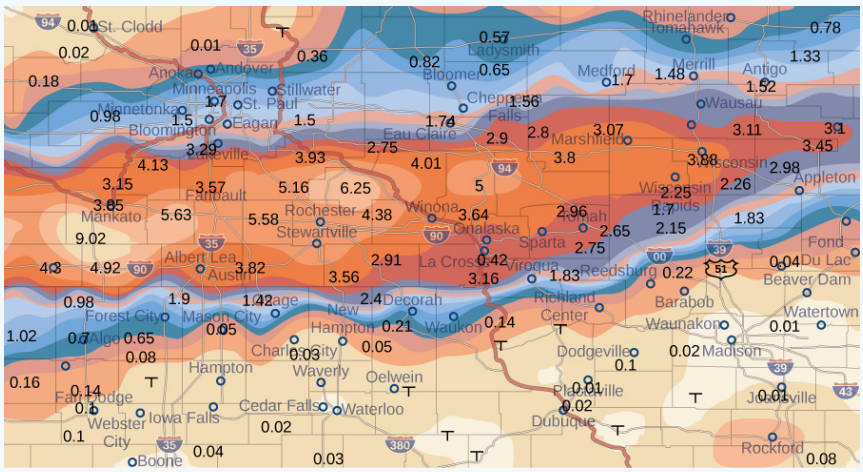


Figure 15.1.9: Effects of Hurricane Karl, which dumped 4–8 inches (100–200 mm) of rain in some parts of southwest Wisconsin, southern Minnesota, and southeast South Dakota over a span of 300 miles east to west and 250 miles north to south.

Solution

Place the origin at the southwest corner of the map so that all the values can be considered as being in the first quadrant and hence all are positive. Now divide the entire map into six rectangles ($m = 2$ and $n = 3$), as shown in Figure 15.1.9. Assume $f(x,y)$ denotes the storm rainfall in inches at a point approximately x miles to the east of the origin and y miles to the north of the origin. Let R represent the entire area of $250 \times 300 = 75000$ square miles. Then the area of each subrectangle is

$$\Delta A = \frac{1}{6}(75000) = 12500.$$

Assume $(x_{ij}*, y_{ij}*)$ are approximately the midpoints of each subrectangle R_{ij}. Note the color-coded region at each of these points, and estimate the rainfall. The rainfall at each of these points can be estimated as:

- At (x_{11}, y_{11}), the rainfall is 0.08.
- At (x_{12}, y_{12}), the rainfall is 0.08.
- At (x_{13}, y_{13}), the rainfall is 0.01.

- At (x_{21}, y_{21}), the rainfall is 1.70.
- At (x_{22}, y_{22}), the rainfall is 1.74.
- At (x_{23}, y_{23}), the rainfall is 3.00.

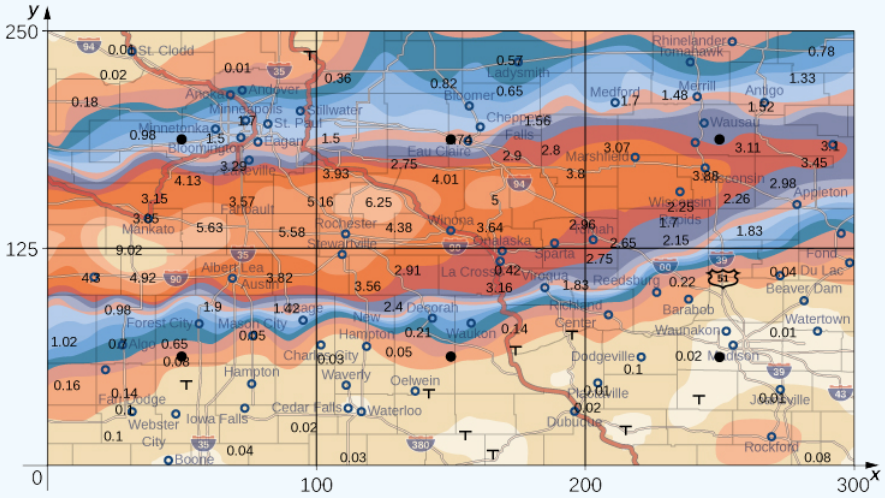


Figure 15.1.10: Storm rainfall with rectangular axes and showing the midpoints of each subrectangle.

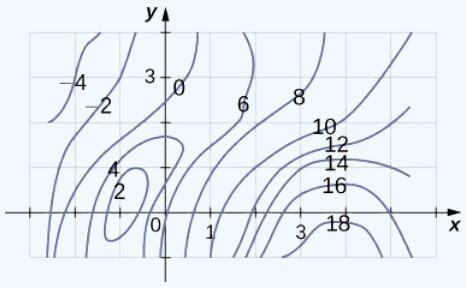
According to our definition, the average storm rainfall in the entire area during those two days was

$$
\begin{aligned}
f_{ave} &= \frac{1}{Area\, R} \\
\iint_R f(x,y)\,dx\,dy &= \frac{1}{75000}\iint_R f(x,y)\,dx\,dy \\
&\approx \frac{1}{75000}\sum_{i=1}^{3}\sum_{j=1}^{2} f(x_{ij}^*, y_{ij}^*)\Delta A \\
&= \frac{1}{75000}\Big[f(x_{11}^*, y_{11}^*)\Delta A + f(x_{12}^*, y_{12}^*)\Delta A + f(x_{13}^*, y_{13}^*)\Delta A \\
&\qquad + f(x_{21}^*, y_{21}^*)\Delta A + f(x_{22}^*, y_{22}^*)\Delta A + f(x_{23}^*, y_{23}^*)\Delta A\Big] \\
&\approx \frac{1}{75000}\Big[0.08 + 0.08 + 0.01 + 1.70 + 1.74 + 3.00\Big]\Delta A \\
&= \frac{1}{75000}\Big[0.08 + 0.08 + 0.01 + 1.70 + 1.74 + 3.00\Big]12500 \\
&= \frac{1}{6}\Big[0.08 + 0.08 + 0.01 + 1.70 + 1.74 + 3.00\Big] \\
&\approx 1.10 \text{ in.}
\end{aligned}
$$

During September 22–23, 2010 this area had an average storm rainfall of approximately 1.10 inches.

? Exercise 15.1.6

A contour map is shown for a function $f(x, y)$ on the rectangle $R = [-3, 6] \times [-1, 4]$.



a. Use the midpoint rule with $m = 3$ and $n = 2$ to estimate the value of $\iint_R f(x, y)\, dA$.

b. Estimate the average value of the function $f(x, y)$.

Hint

Divide the region into six rectangles, and use the contour lines to estimate the values for $f(x, y)$.

Answer

Answers to both parts a. and b. may vary.

Key Concepts

- We can use a double Riemann sum to approximate the volume of a solid bounded above by a function of two variables over a rectangular region. By taking the limit, this becomes a double integral representing the volume of the solid.
- Properties of double integral are useful to simplify computation and find bounds on their values.
- We can use Fubini's theorem to write and evaluate a double integral as an iterated integral.
- Double integrals are used to calculate the area of a region, the volume under a surface, and the average value of a function of two variables over a rectangular region.

Key Equations

- $$\iint_R f(x, y)\, dA = \lim_{m,n\to\infty} \sum_{i=1}^{m} \sum_{j=1}^{n} f(x_i j*, y_i j*)\, \Delta A$$
- $$\int_a^b \int_c^d f(x, y)\, dx\, dy = \int_a^b \left[\int_c^d f(x, y)\, dy \right] dx$$

  or

  $$\int_c^d \int_a^b f(x, y)\, dx\, dy = \int_c^d \left[\int_a^b f(x, y)\, dx \right] dy$$
- $$f_{ave} = \frac{1}{\text{Area of } R} \iint_R f(x, y)\, dx\, dy$$

Glossary

double integral

of the function $f(x, y)$ over the region R in the xy-plane is defined as the limit of a double Riemann sum,

$$\iint_R f(x, y)\, dA = \lim_{m,n\to\infty} \sum_{i=1}^{m} \sum_{j=1}^{n} f(x_{ij}^*, y_{ij}^*)\, \Delta A.$$

double Riemann sum

of the function $f(x, y)$ over a rectangular region R is

$$\sum_{i=1}^{m} \sum_{j=1}^{n} f(x_{ij}^*, y_{ij}^*)\, \Delta A,$$

where R is divided into smaller subrectangles R_{ij} and (x_{ij}^*, y_{ij}^*) is an arbitrary point in R_{ij}

Fubini's theorem

if $f(x, y)$ is a function of two variables that is continuous over a rectangular region $R = \{(x, y) \in \mathbb{R}^2 \mid a \le x \le b,\ c \le y \le d\}$, then the double integral of f over the region equals an iterated integral,

$$\iint_R f(x, y)\, dA = \int_a^b \int_c^d f(x, y)\, dx\, dy = \int_c^d \int_a^b f(x, y)\, dx\, dy$$

iterated integral

for a function $f(x, y)$ over the region R is

a. $\int_a^b \int_c^d f(x, y)\, dx\, dy = \int_a^b \left[\int_c^d f(x, y)\, dy \right] dx,$

b. $\int_c^d \int_a^b f(x, y)\, dx\, dy = \int_c^d \left[\int_a^b f(x, y)\, dx \right] dy,$

where a, b, c, and d are any real numbers and $R = [a, b] \times [c, d]$

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15.1E: Exercises for Section 15.1

In exercises 1 and 2, use the midpoint rule with $m = 4$ and $n = 2$ to estimate the volume of the solid bounded by the surface $z = f(x, y)$, the vertical planes $x = 1$, $x = 2$, $y = 1$, and $y = 2$, and the horizontal plane $z = 0$.

1) $f(x, y) = 4x + 2y + 8xy$

Answer

27

2) $f(x, y) = 16x^2 + \frac{y}{2}$

In exercises 3 and 4, estimate the volume of the solid under the surface $z = f(x, y)$ and above the rectangular region R by using a Riemann sum with $m = n = 2$ and the sample points to be the lower left corners of the sub-rectangles of the partition.

3) $f(x, y) = \sin x - \cos y$, $R = [0, \pi] \times [0, \pi]$

Answer

0

4) $f(x, y) = \cos x + \cos y$, $R = [0, \pi] \times [0, \frac{\pi}{2}]$

5) Use the midpoint rule with $m = n = 2$ to estimate $\iint_R f(x, y)\, dA$, where the values of the function f on $R = [8, 10] \times [9, 11]$ are given in the following table.

	y				
x	9	9.5	10	10.5	11
8	9.8	5	6.7	5	5.6
8.5	9.4	4.5	8	5.4	3.4
9	8.7	4.6	6	5.5	3.4
9.5	6.7	6	4.5	5.4	6.7
10	6.8	6.4	5.5	5.7	6.8

Answer

21.3

6) The values of the function f on the rectangle $R = [0, 2] \times [7, 9]$ are given in the following table. Estimate the double integral $\iint_R f(x, y)\, dA$ by using a Riemann sum with $m = n = 2$. Select the sample points to be the upper right corners of the sub-squares of R.

	$y_0 = 7$	$y_1 = 8$	$y_2 = 9$
$x_0 = 0$	10.22	10.21	9.85
$x_1 = 1$	6.73	9.75	9.63
$x_2 = 2$	5.62	7.83	8.21

7) The depth of a children's 4-ft by 4-ft swimming pool, measured at 1-ft intervals, is given in the following table.

a. Estimate the volume of water in the swimming pool by using a Riemann sum with $m = n = 2$. Select the sample points using the midpoint rule on $R = [0, 4] \times [0, 4]$.
b. Find the average depth of the swimming pool.

	y				
x	0	1	2	3	4
0	1	1.5	2	2.5	3
1	1	1.5	2	2.5	3
2	1	1.5	1.5	2.5	3
3	1	1	1.5	2	2.5
4	1	1	1	1.5	2

Answer

a. 28 ft^3

b. 1.75 ft.

8) The depth of a 3-ft by 3-ft hole in the ground, measured at 1-ft intervals, is given in the following table.

a. Estimate the volume of the hole by using a Riemann sum with $m = n = 3$ and the sample points to be the upper left corners of the sub-squares of R.

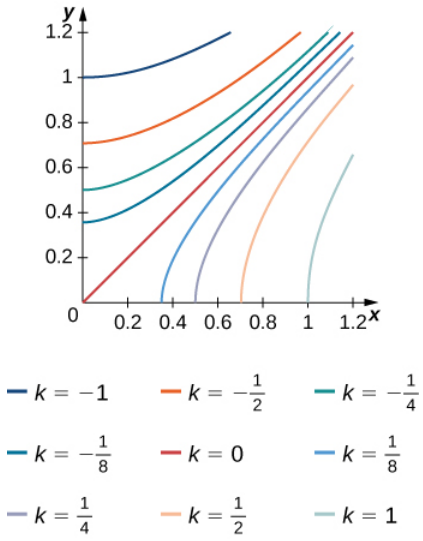
b. Find the average depth of the hole.

	y			
x	0	1	2	3
0	6	6.5	6.4	6
1	6.5	7	7.5	6.5
2	6.5	6.7	6.5	6
3	6	6.5	5	5.6

9) The level curves $f(x, y) = k$ of the function f are given in the following graph, where k is a constant.

a. Apply the midpoint rule with $m = n = 2$ to estimate the double integral $\iint_R f(x, y)\, dA$, where $R = [0.2, 1] \times [0, 0.8]$.

b. Estimate the average value of the function f on R.



Answer

a. Answers will vary somewhat, but should be close to these: $f(0.4, 0.2) \approx 0.125$, $f(0.4, 0.6) \approx -0.2$, $f(0.8, 0.2) \approx 0.6$, and $f(0.8, 0.6) \approx 0.3$.

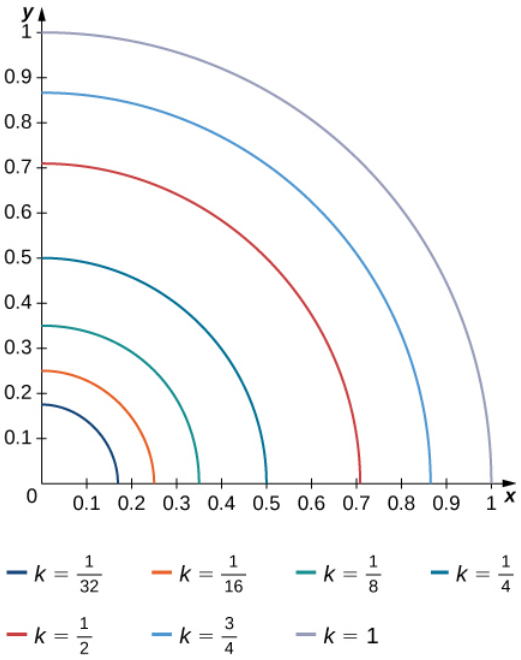
Since the area of each rectangle is $\Delta A_i = (0.4)(0.4) = 0.16 \text{ units}^2$, $\iint_R f(x,y)\,dA = \int_{0.2}^{1}\int_{0}^{0.8} f(x,y)\,dy\,dx \approx f(0.4, 0.2)(0.16) + f(0.4, 0.6)(0.16) + f(0.8, 0.2)(0.16) + f(0.8, 0.6)(0.16) = 0.132$

b. $f_{ave} = \dfrac{\int_{0.2}^{1}\int_{0}^{0.8} f(x,y)\,dy\,dx}{\text{area of } R} \approx \dfrac{\int_{0.2}^{1}\int_{0}^{0.8} f(x,y)\,dy\,dx}{0.64} \approx 0.206$

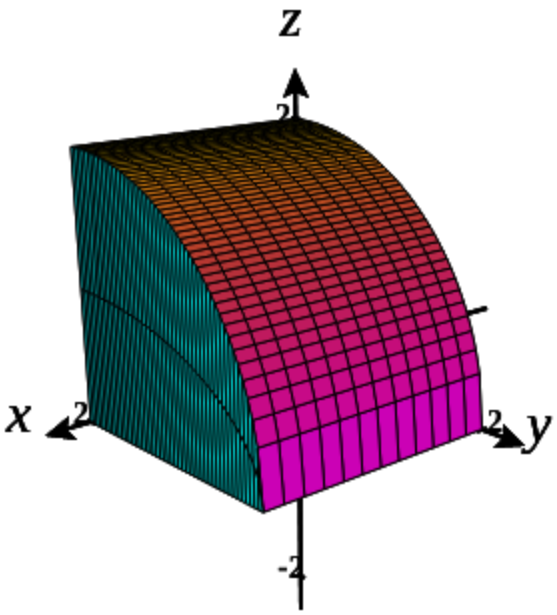
10) The level curves $f(x, y) = k$ of the function f are given in the following graph, where k is a constant.

a. Apply the midpoint rule with $m = n = 2$ to estimate the double integral $\iint_R f(x, y)\,dA$, where $R = [0.1, 0.5] \times [0.1, 0.5]$.

b. Estimate the average value of the function f on R.



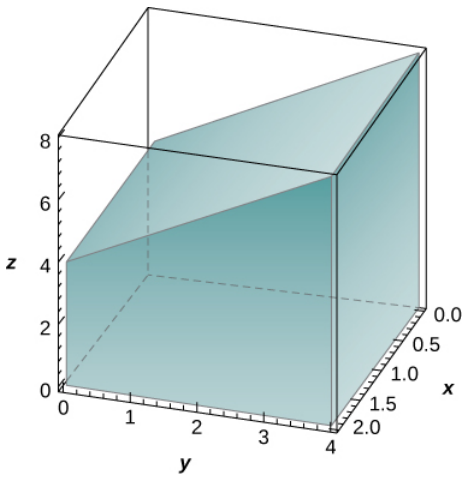
11) The solid lying under the surface $z = \sqrt{4 - y^2}$ and above the rectangular region$R = [0, 2] \times [0, 2]$ is illustrated in the following graph. Evaluate the double integral $\iint_R f(x, y)$, where $f(x, y) = \sqrt{4 - y^2}$ by finding the volume of the corresponding solid.



Answer

2π

12) The solid lying under the plane $z = y + 4$ and above the rectangular region $R = [0, 2] \times [0, 4]$ is illustrated in the following graph. Evaluate the double integral $\iint_R f(x, y)\, dA$, where $f(x, y) = y + 4$, by finding the volume of the corresponding solid.



In the exercises 13 - 20, calculate the integrals by reversing the order of integration.

13) $\int_{-1}^{1} \left(\int_{-2}^{2} (2x + 3y + 5)\, dx \right)\, dy$

Answer

40

14) $\int_0^2 \left(\int_0^1 (x + 2e^y + 3)\, dx \right)\, dy$

15) $\int_1^{27} \left(\int_1^2 (\sqrt[3]{x} + \sqrt[3]{y})\, dy \right)\, dx$

Answer

$\frac{81}{2} + 39\sqrt[3]{2}$

16) $\int_1^{16} \left(\int_1^8 (\sqrt[4]{x} + 2\sqrt[3]{y})\, dy \right)\, dx$

17) $\int_{\ln 2}^{\ln 3} \left(\int_0^1 e^{x+y}\, dy \right)\, dx$

Answer

$e - 1$

18) $\int_0^2 \left(\int_0^1 3^{x+y}\, dy \right)\, dx$

19) $\int_1^6 \left(\int_2^9 \frac{\sqrt{y}}{x^2}\, dy \right)\, dx$

Answer

$15 - \frac{10\sqrt{2}}{9}$

20) $\int_1^9 \left(\int_4^2 \frac{\sqrt{x}}{y^2}\, dy \right)\, dx$

In exercises 21 - 34, evaluate the iterated integrals by choosing the order of integration.

21) $\int_0^{\pi} \int_0^{\pi/2} \sin(2x)\cos(3y)\, dx\, dy$

Answer

0

22) $\int_{\pi/12}^{\pi/8} \int_{\pi/4}^{\pi/3} [\cot x + \tan(2y)]\, dx\, dy$

23) $\int_1^e \int_1^e \left[\frac{1}{x}\sin(\ln x) + \frac{1}{y}\cos(\ln y) \right]\, dx\, dy$

Answer

$(e-1)(1 + \sin 1 - \cos 1)$

24) $\int_1^e \int_1^e \frac{\sin(\ln x)\cos(\ln y)}{xy}\, dx\, dy$

25) $\int_1^2 \int_1^2 \left(\frac{\ln y}{x} + \frac{x}{2y+1} \right)\, dy\, dx$

Answer

$\frac{3}{4}\ln\left(\frac{5}{3}\right) + 2(\ln 2)^2 - \ln 2$

26) $\int_1^e \int_1^2 x^2 \ln(x)\, dy\, dx$

27) $\int_1^{\sqrt{3}} \int_1^2 y \arctan\left(\frac{1}{x}\right)\, dy\, dx$

Answer

$\frac{1}{8}[(2\sqrt{3}-3)\pi+6\ \ln 2]$

28) $\int_0^1 \int_0^{1/2} (\arcsin x + \arcsin y)\, dy\, dx$

29) $\int_0^1 \int_1^2 xe^{x+4y}\, dy\, dx$

Answer

$\frac{1}{4}e^4(e^4-1)$

30) $\int_1^2 \int_0^1 xe^{x-y}\, dy\, dx$

31) $\int_1^e \int_1^e \left(\frac{\ln y}{\sqrt{y}} + \frac{\ln x}{\sqrt{x}}\right) dy\, dx$

Answer

$4(e-1)(2-\sqrt{e})$

32) $\int_1^e \int_1^e \left(\frac{x\ \ln y}{\sqrt{y}} + \frac{y\ \ln x}{\sqrt{x}}\right) dy\, dx$

33) $\int_0^1 \int_1^2 \left(\frac{x}{x^2+y^2}\right) dy\, dx$

Answer

$-\frac{\pi}{4} + \ln\left(\frac{5}{4}\right) - \frac{1}{2}\ln 2 + \arctan 2$

34) $\int_0^1 \int_1^2 \frac{y}{x+y^2}\, dy\, dx$

In exercises 35 - 38, find the average value of the function over the given rectangles.

35) $f(x,y) = -x + 2y$, $R = [0,1] \times [0,1]$

Answer

$\frac{1}{2}$

36) $f(x,y) = x^4 + 2y^3$, $R = [1,2] \times [2,3]$

37) $f(x,y) = \sinh x + \sinh y$, $R = [0,1] \times [0,2]$

Answer

$\frac{1}{2}(2\ \cosh 1 + \cosh 2 - 3)$.

38) $f(x,y) = \arctan(xy)$, $R = [0,1] \times [0,1]$

39) Let f and g be two continuous functions such that $0 \le m_1 \le f(x) \le M_1$ for any $x \in [a,b]$ and $0 \le m_2 \le g(y) \le M_2$ for any$y \in [c,d]$. Show that the following inequality is true:

$$m_1 m_2 (b-a)(c-d) \le \int_a^b \int_c^d f(x)g(y)\, dydx \le M_1 M_2 (b-a)(c-d).$$

In exercises 40 - 43, use property v. of double integrals and the answer from the preceding exercise to show that the following inequalities are true.

40) $\frac{1}{e^2} \le \iint_R e^{-x^2-y^2}\, dA \le 1$, where $R = [0,1] \times [0,1]$

41) $\frac{\pi^2}{144} \le \iint_R \sin x \cos y\, dA \le \frac{\pi^2}{48}$, where $R = [\frac{\pi}{6}, \frac{\pi}{3}] \times [\frac{\pi}{6}, \frac{\pi}{3}]$

42) $0 \le \iint_R e^{-y}\ \cos x\, dA \le \frac{\pi}{2}$, where $R = [0, \frac{\pi}{2}] \times [0, \frac{\pi}{2}]$

43) $0 \le \iint_R (\ln x)(\ln y)\, dA \le (e-1)^2$, where $R = [1, e] \times [1, e]$

44) Let f and g be two continuous functions such that $0 \le m_1 \le f(x) \le M_1$ for any $x \in [a, b]$ and $0 \le m_2 \le g(y) \le M_2$ for any $y \in [c, d]$. Show that the following inequality is true:

$$(m_1 + m_2)(b-a)(c-d) \le \int_a^b \int_c^d |f(x) + g(y)|\, dy\, dx \le (M_1 + M_2)(b-a)(c-d)$$

In exercises 45 - 48, use property v. of double integrals and the answer from the preceding exercise to show that the following inequalities are true.

45) $\frac{2}{e} \le \iint_R (e^{-x^2} + e^{-y^2})\, dA \le 2$, where $R = [0, 1] \times [0, 1]$

46) $\frac{\pi^2}{36} \iint_R (\sin x + \cos y)\, dA \le \frac{\pi^2\sqrt{3}}{36}$, where $R = [\frac{\pi}{6}, \frac{\pi}{3}] \times [\frac{\pi}{6}, \frac{\pi}{3}]$

47) $\frac{\pi}{2} e^{-\pi/2} \le \iint_R (\cos x + e^{-y})\, dA \le \pi$, where $R = [0, \frac{\pi}{2}] \times [0, \frac{\pi}{2}]$

48) $\frac{1}{e} \le \iint_R (e^{-y} - \ln x)\, dA \le 2$, where $R = [0, 1] \times [0, 1]$

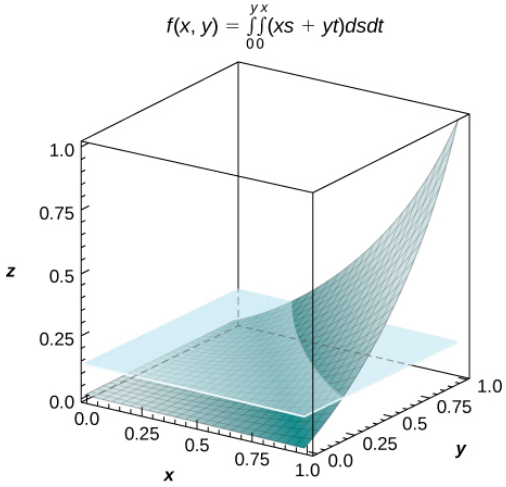
In exercises 49 - 50, the function f is given in terms of double integrals.

a. Determine the explicit form of the function f.
b. Find the volume of the solid under the surface $z = f(x, y)$ and above the region R.
c. Find the average value of the function f on R.
d. Use a computer algebra system (CAS) to plot $z = f(x, y)$ and $z = f_{ave}$ in the same system of coordinates.

49) [T] $f(x, y) = \int_0^y \int_0^x (xs + yt) ds\, dt$, where $(x, y) \in R = [0, 1] \times [0, 1]$

Answer

a. $f(x, y) = \frac{1}{2} xy(x^2 + y^2)$;
b. $V = \int_0^1 \int_0^1 f(x, y)\, dx\, dy = \frac{1}{8}$;
c. $f_{ave} = \frac{1}{8}$;
d.



50) [T] $f(x, y) = \int_0^x \int_0^y [\cos(s) + \cos(t)]\, dt\, ds$, where $(x, y) \in R = [0, 3] \times [0, 3]$

51) Show that if f and g are continuous on $[a, b]$ and $[c, d]$, respectively, then

$$\int_a^b \int_c^d |f(x) + g(y)| dy\, dx = (d-c) \int_a^b f(x)\, dx$$
$$+ \int_a^b \int_c^d g(y)\, dy\, dx = (b-a) \int_c^d g(y)\, dy + \int_c^d \int_a^b f(x)\, dx\, dy.$$

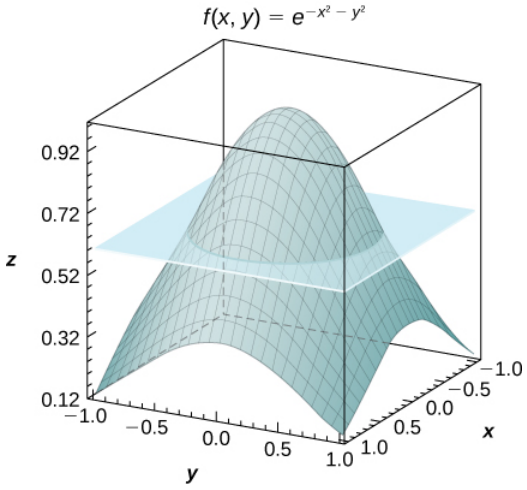
52) Show that $\int_a^b \int_c^d yf(x) + xg(y)\, dy\, dx = \frac{1}{2}(d^2 - c^2)\left(\int_a^b f(x)\, dx\right) + \frac{1}{2}(b^2 - a^2)\left(\int_c^d g(y)\, dy\right)$.

53) [T] Consider the function $f(x, y) = e^{-x^2-y^2}$, where $(x, y) \in R = [-1, 1] \times [-1, 1]$.

a. Use the midpoint rule with $m = n = 2, 4, ..., 10$ to estimate the double integral $I = \iint_R e^{-x^2-y^2} dA$. Round your answers to the nearest hundredths.
b. For $m = n = 2$, find the average value of *f* over the region *R*. Round your answer to the nearest hundredths.
c. Use a CAS to graph in the same coordinate system the solid whose volume is given by $\iint_R e^{-x^2-y^2} dA$ and the plane $z = f_{ave}$.

Answer

a. For $m = n = 2$, $I = 4e^{-0.5} \approx 2.43$
b. $f_{ave} = e^{-0.5} \simeq 0.61$;
c.



54) [T] Consider the function $f(x, y) = \sin(x^2)\ \cos(y^2)$, where $(x, y \in R = [-1, 1] \times [-1, 1]$.

a. Use the midpoint rule with $m = n = 2, 4, ..., 10$ to estimate the double integral $I = \iint_R \sin(x^2)\cos(y^2)\ dA$. Round your answers to the nearest hundredths.
b. For $m = n = 2$, find the average value of *f* over the region *R*. Round your answer to the nearest hundredths.
c. Use a CAS to graph in the same coordinate system the solid whose volume is given by $\iint_R \sin(x^2)\cos(y^2)\ dA$ and the plane $z = f_{ave}$.

In exercises 55 - 56, the functions f_n are given, where $n \geq 1$ is a natural number.

a. Find the volume of the solids S_n under the surfaces $z = f_n(x, y)$ and above the region *R*.
b. Determine the limit of the volumes of the solids S_n as n increases without bound.

55) $f(x, y) = x^n + y^n + xy,\ (x, y) \in R = [0, 1] \times [0, 1]$

Answer

a. $\frac{2}{n+1} + \frac{1}{4}$
b. $\frac{1}{4}$

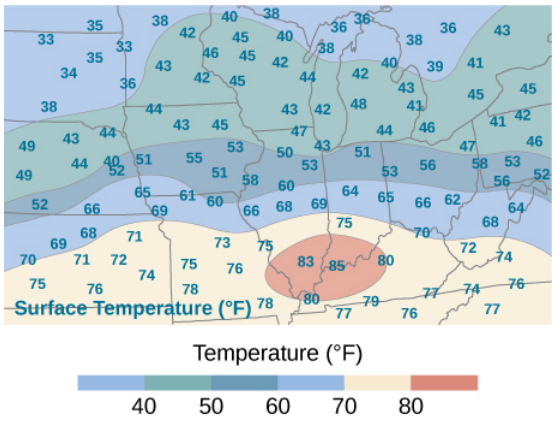
56) $f(x, y) = \frac{1}{x^n} + \frac{1}{y^n},\ (x, y) \in R = [1, 2] \times [1, 2]$

57) Show that the average value of a function f on a rectangular region $R = [a, b] \times [c, d]$ is $f_{ave} \approx \frac{1}{mn} \sum_{i=1}^{m} \sum_{j=1}^{n} f(x_{ij}^*, y_{ij}^*)$,where (x_{ij}^*, y_{ij}^*) are the sample points of the partition of *R*, where $1 \leq i \leq m$ and $1 \leq j \leq n$.

58) Use the midpoint rule with $m = n$ to show that the average value of a function f on a rectangular region $R = [a, b] \times [c, d]$ is approximated by

$$f_{ave} \approx \frac{1}{n^2} \sum_{i,j=1}^{n} f\left(\frac{1}{2}(x_{i=1} + x_i),\ \frac{1}{2}(y_{j=1} + y_j)\right).$$

59) An isotherm map is a chart connecting points having the same temperature at a given time for a given period of time. Use the preceding exercise and apply the midpoint rule with $m = n = 2$ to find the average temperature over the region given in the following figure.



Answer

56.5° F; here $f(x_1^*, y_1^*) = 71,\ f(x_2^*, y_1^*) = 72,\ f(x_2^*, y_1^*) = 40,\ f(x_2^*, y_2^*) = 43$, where x_i^* and y_j^* are the midpoints of the subintervals of the partitions of $[a, b]$ and $[c, d]$, respectively.

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15.2: Double Integrals over General Regions

Learning Objectives

- Recognize when a function of two variables is integrable over a general region.
- Evaluate a double integral by computing an iterated integral over a region bounded by two vertical lines and two functions of x, or two horizontal lines and two functions of y.
- Simplify the calculation of an iterated integral by changing the order of integration.
- Use double integrals to calculate the volume of a region between two surfaces or the area of a plane region.
- Solve problems involving double improper integrals.

Previously, we studied the concept of double integrals and examined the tools needed to compute them. We learned techniques and properties to integrate functions of two variables over rectangular regions. We also discussed several applications, such as finding the volume bounded above by a function over a rectangular region, finding area by integration, and calculating the average value of a function of two variables.

In this section we consider double integrals of functions defined over a general bounded region D on the plane. Most of the previous results hold in this situation as well, but some techniques need to be extended to cover this more general case.

General Regions of Integration

An example of a general bounded region D on a plane is shown in Figure 15.2.1. Since D is bounded on the plane, there must exist a rectangular region R on the same plane that encloses the region D that is, a rectangular region R exists such that D is a subset of $R (D \subseteq R)$.

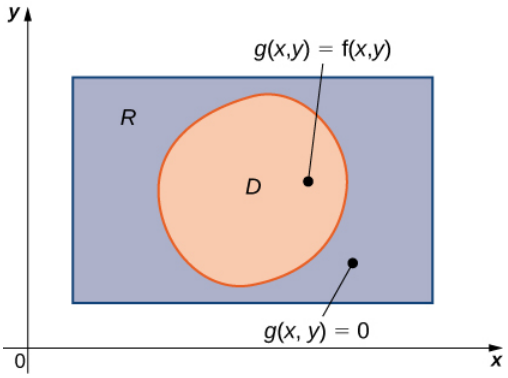


Figure 15.2.1: For a region D that is a subset of R, we can define a function $g(x, y)$ to equal $f(x, y)$ at every point in D and 0 at every point of R not in D.

Suppose $z = f(x, y)$ is defined on a general planar bounded region D as in Figure 15.2.1. In order to develop double integrals of f over D we extend the definition of the function to include all points on the rectangular region R and then use the concepts and tools from the preceding section. But how do we extend the definition of f to include all the points on R? We do this by defining a new function $g(x, y)$ on R as follows:

$$g(x,y) = \begin{cases} f(x,y), & \text{if } (x,y) \text{ is in } D \\ 0, & \text{if } (x,y) \text{ is in } R \text{ but not in } D \end{cases}$$

Note that we might have some technical difficulties if the boundary of D is complicated. So we assume the boundary to be a piecewise smooth and continuous simple closed curve. Also, since all the results developed in the section on Double Integrals over Rectangular Regions used an integrable function $f(x, y)$ we must be careful about $g(x, y)$ and verify that $g(x, y)$ is an integrable function over the rectangular region R. This happens as long as the region D is bounded by simple closed curves. For now we will concentrate on the descriptions of the regions rather than the function and extend our theory appropriately for integration.

We consider two types of planar bounded regions.

Definition: Type I and Type II regions

A region D in the (x, y)-plane is of Type I if it lies between two vertical lines and the graphs of two continuous functions $g_1(x)$ and $g_2(x)$. That is (Figure 15.2.2),

$$D = \{(x, y) \mid a \le x \le b,\ g_1(x) \le y \le g_2(x)\}.$$

A region D in the xy-plane is of Type II if it lies between two horizontal lines and the graphs of two continuous functions $h_1(y)$ and $h_2(y)$. That is (Figure 15.2.3),

$$D = \{(x, y) \mid c \le y \le d,\ h_1(y) \le x \le h_2(y)\}.$$

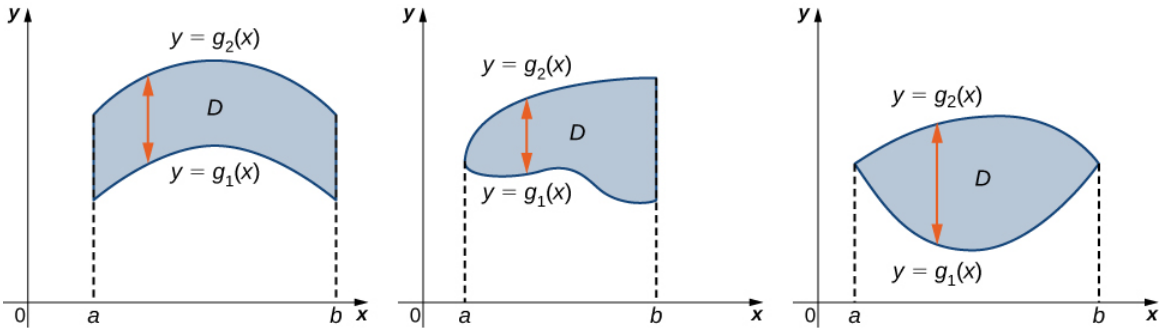


Figure 15.2.2. A Type I region lies between two vertical lines and the graphs of two functions of x.

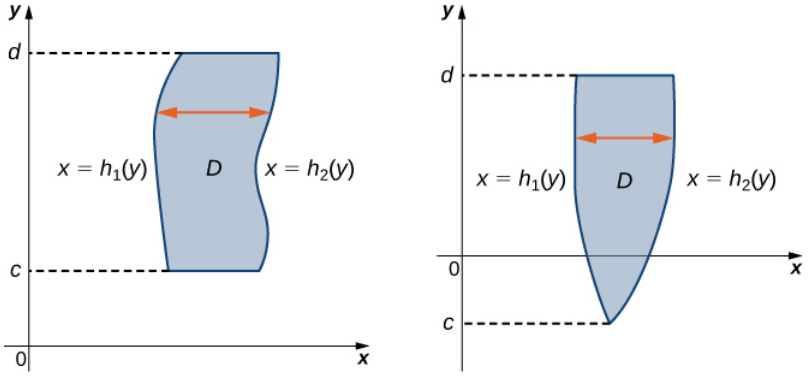


Figure 15.2.3: A Type II region lies between two horizontal lines and the graphs of two functions of y.

Example 15.2.1: Describing a Region as Type I and Also as Type II

Consider the region in the first quadrant between the functions $y = \sqrt{x}$ and $y = x^3$ (Figure 15.2.4). Describe the region first as Type I and then as Type II.

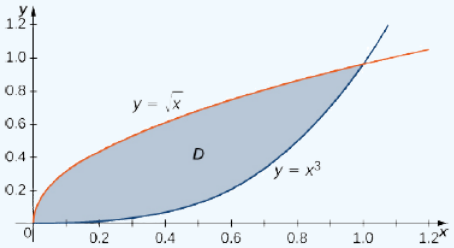


Figure 15.2.4: Region D can be described as Type I or as Type II.

When describing a region as Type I, we need to identify the function that lies above the region and the function that lies below the region. Here, region D is bounded above by $y = \sqrt{x}$ and below by $y = x^3$ in the interval for x in $[0, 1]$. Hence, as Type I, D is described as the set $\{(x, y) \mid 0 \le x \le 1,\ x^3 \le y \le \sqrt[3]{x}\}$.

However, when describing a region as Type II, we need to identify the function that lies on the left of the region and the function that lies on the right of the region. Here, the region D is bounded on the left by $x = y^2$ and on the right by $x = \sqrt[3]{y}$ in

the interval for y in $[0,1]$. Hence, as Type II, D is described as the set $\left\{(x,y)\,|\,0\le y\le 1,\ y^2\le x\le \sqrt[3]{y}\right\}$.

Exercise 15.2.1

Consider the region in the first quadrant between the functions $y=2x$ and $y=x^2$. Describe the region first as Type I and then as Type II.

Hint

Graph the functions, and draw vertical and horizontal lines.

Answer

Type I and Type II are expressed as $\left\{(x,y)\,|\,0\le x\le 2,\ x^2\le y\le 2x\right\}$ and $\left\{(x,y)|\,0\le y\le 4,\ \frac{1}{2}y\le x\le \sqrt{y}\right\}$, respectively.

Double Integrals over Non-rectangular Regions

To develop the concept and tools for evaluation of a double integral over a general, nonrectangular region, we need to first understand the region and be able to express it as Type I or Type II or a combination of both. Without understanding the regions, we will not be able to decide the limits of integrations in double integrals. As a first step, let us look at the following theorem.

Theorem: Double Integrals over Nonrectangular Regions

Suppose $g(x,y)$ is the extension to the rectangle R of the function $f(x,y)$ defined on the regions D and R as shown in Figure 15.2.1 inside R. Then $g(x,y)$ is integrable and we define the double integral of $f(x,y)$ over D by

$$\iint_D f(x,y)\,dA=\iint_R g(x,y)\,dA.$$

The right-hand side of this equation is what we have seen before, so this theorem is reasonable because R is a rectangle and $\iint_R g(x,y)dA$ has been discussed in the preceding section. Also, the equality works because the values of $g(x,y)$ are 0 for any point (x,y) that lies outside D and hence these points do not add anything to the integral. However, it is important that the rectangle R contains the region D.

As a matter of fact, if the region D is bounded by smooth curves on a plane and we are able to describe it as Type I or Type II or a mix of both, then we can use the following theorem and not have to find a rectangle R containing the region.

Theorem: Fubini's Theorem (Strong Form)

For a function $f(x,y)$ that is continuous on a region D of Type I, we have

$$\iint_D f(x,y)\,dA=\iint_D f(x,y)\,dy\,dx=\int_a^b\left[\int_{g_1(x)}^{g_2(x)} f(x,y)\,dy\right]dx.$$

Similarly, for a function $f(x,y)$ that is continuous on a region D of Type II, we have

$$\iint_D f(x,y)\,dA=\iint_D f(x,y)\,dx\,dy=\int_c^d\left[\int_{h_1(y)}^{h_2(y)} f(x,y)\,dx\right]dy.$$

The integral in each of these expressions is an iterated integral, similar to those we have seen before. Notice that, in the inner integral in the first expression, we integrate $f(x,y)$ with x being held constant and the limits of integration being $g_1(x)$ and $g_2(x)$. In the inner integral in the second expression, we integrate $f(x,y)$ with y being held constant and the limits of integration are $h_1(x)$ and $h_2(x)$.

Example 15.2.2: Evaluating an Iterated Integral over a Type I Region

Evaluate the integral $\iint_D x^2 e^{xy}\, dA$ where D is shown in Figure 15.2.5.

Solution

First construct the region as a Type I region (Figure 15.2.5). Here $D = \{(x, y) \mid 0 \le x \le 2,\ \frac{1}{2}x \le y \le 1\}$. Then we have

$$\iint_D x^2 e^{xy}\, dA = \int_{x=0}^{x=2} \int_{y=1/2x}^{y=1} x^2 e^{xy}\, dy\, dx.$$

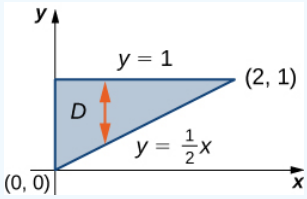


Figure 15.2.5: We can express region D as a Type I region and integrate from $y = \frac{1}{2}x$ to $y = 1$ between the lines $x = 0$ and $x = 2$.

Therefore, we have

$$\begin{aligned}
\int_{x=0}^{x=2} \int_{y=\frac{1}{2}x}^{y=1} x^2 e^{xy}\, dy\, dx &= \int_{x=0}^{x=2} \left[\int_{y=\frac{1}{2}x}^{y=1} x^2 e^{xy}\, dy \right] dx && \text{Iterated integral for a Type I region.} \\
&= \int_{x=0}^{x=2} \left[x^2 \frac{e^{xy}}{x} \right] \Big|_{y=1/2x}^{y=1} dx && \text{Integrate with respect to } y \\
&= \int_{x=0}^{x=2} \left[xe^x - xe^{x^2/2} \right] dx && \text{Integrate with respect to } x \\
&= \left[xe^x - e^x - e^{\frac{1}{2}x^2} \right] \Big|_{x=0}^{x=2} = 2.
\end{aligned}$$

In Example 15.2.2, we could have looked at the region in another way, such as $D = \{(x, y) \mid 0 \le y \le 1,\ 0 \le x \le 2y\}$ (Figure 15.2.6).

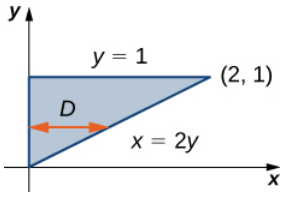


Figure 15.2.6.

This is a Type II region and the integral would then look like

$$\iint_D x^2 e^{xy}\, dA = \int_{y=0}^{y=1} \int_{x=0}^{x=2y} x^2 e^{xy}\, dx\ dy.$$

However, if we integrate first with respect to x this integral is lengthy to compute because we have to use integration by parts twice.

Example 15.2.3: Evaluating an Iterated Integral over a Type II Region

Evaluate the integral

$$\iint_D (3x^2 + y^2)\, dA$$

where $D = \{(x, y) \mid -2 \le y \le 3,\ y^2 - 3 \le x \le y + 3\}$.

Solution

Notice that D can be seen as either a Type I or a Type II region, as shown in Figure 15.2.7. However, in this case describing D as Type I is more complicated than describing it as Type II. Therefore, we use D as a Type II region for the integration.

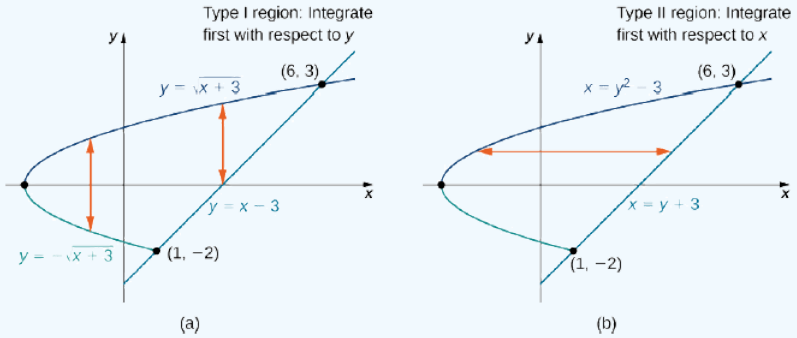


Figure 15.2.7: The region D in this example can be either (a) Type I or (b) Type II.

Choosing this order of integration, we have

$$\begin{aligned}
\iint_D (3x^2 + y^2)\, dA &= \int_{y=-2}^{y=3} \int_{x=y^2-3}^{x=y+3} (3x^2 + y^2)\, dx\ dy \\
&= \int_{y=-2}^{y=3} (x^3 + xy^2)\Big|_{y^2-3}^{y+3}\, dy && \text{Iterated integral, Type II region} \\
&= \int_{y=-2}^{y=3} \big((y+3)^3 + (y+3)y^2 - (y^2-3)^3 \\
&\quad - (y^2-3)y^2\big)\, dy \\
&= \int_{-2}^{3} (54 + 27y - 12y^2 + 2y^3 + 8y^4 && \text{Integrate with respect to } x. \\
&\quad - y^6)\, dy \\
&= \left[54y + \frac{27y^2}{2} - 4y^3 + \frac{y^4}{2} + \frac{8y^5}{5}\right. \\
&\quad \left. - \frac{y^7}{7}\right]_{-2}^{3} \\
&= \frac{2375}{7}.
\end{aligned}$$

? Exercise 15.2.2

Sketch the region D and evaluate the iterated integral

$$\iint_D xy\ dy\ dx$$

where D is the region bounded by the curves $y = \cos\ x$ and $y = \sin\ x$ in the interval $[-3\pi/4,\ \pi/4]$.

Hint

Express D as a Type I region, and integrate with respect to y first.

Answer

$\frac{\pi}{4}$

Recall from Double Integrals over Rectangular Regions the properties of double integrals. As we have seen from the examples here, all these properties are also valid for a function defined on a non-rectangular bounded region on a plane. In particular,

property 3 states:

If $R = S \cup T$ and $S \cap T = 0$ except at their boundaries, then

$$\iint_R f(x,y)\,dA = \iint_S f(x,y)\,dA + \iint_T f(x,y)\,dA.$$

Similarly, we have the following property of double integrals over a non-rectangular bounded region on a plane.

Theorem: Decomposing Regions into Smaller Regions

Suppose the region D can be expressed as $D = D_1 \cup D_2$ where D_1 and D_2 do not overlap except at their boundaries. Then

$$\iint_D f(x,y)\,dA = \iint_{D_1} f(x,y)\,dA + \iint_{D_2} f(x,y)\,dA.$$

This theorem is particularly useful for non-rectangular regions because it allows us to split a region into a union of regions of Type I and Type II. Then we can compute the double integral on each piece in a convenient way, as in the next example.

Example 15.2.4: Decomposing Regions

Express the region D shown in Figure 15.2.8 as a union of regions of Type I or Type II, and evaluate the integral

$$\iint_D (2x + 5y)\,dA.$$

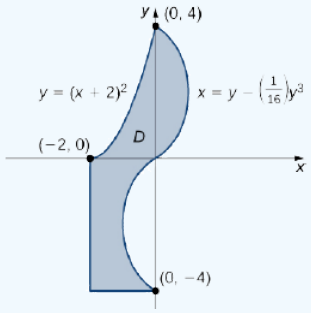


Figure 15.2.8: This region can be decomposed into a union of three regions of Type I or Type II.

Solution

The region D is not easy to decompose into any one type; it is actually a combination of different types. So we can write it as a union of three regions D_1, D_2, and D_3 where, $D_1 = \{(x,y) \mid -2 \le x \le 0,\ 0 \le y \le (x+2)^2\}$, $D_2 = \{(x,y) \mid 0 \le y \le 4,\ 0 \le x \le \left(y - \frac{1}{16}y^3\right)\}$, and $D_3 = \{(x,y) \mid -4 \le y \le 0,\ \ -2 \le x \le \left(y - \frac{1}{16}y^3\right)\}$. These regions are illustrated more clearly in Figure 15.2.9.

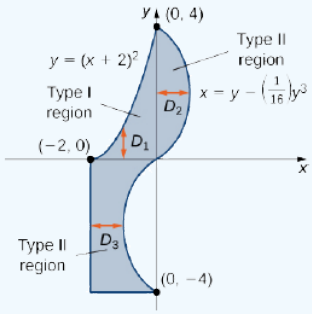


Figure 15.2.9: Breaking the region into three subregions makes it easier to set up the integration.

Here D_1 is Type I and D_2 and D_3 are both of Type II. Hence,

$$\begin{aligned}\iint_D (2x+5y)\,dA &= \iint_{D_1} (2x+5y)\,dA + \iint_{D_2} (2x+5y)\,dA + \iint_{D_3} (2x+5y)\,dA \\ &= \int_{x=-2}^{x=0}\int_{y=0}^{y=(x+2)^2} (2x+5y)\,dy\,dx + \int_{y=0}^{y=4}\int_{x=0}^{x=y-(1/16)y^3} (2x+5y)\,dx\,dy \\ &\quad + \int_{y=-4}^{y=0}\int_{x=-2}^{x=y-(1/16)y^3} (2x+5y)\,dx\,dy \\ &= \int_{x=-2}^{x=0}\left[\frac{1}{2}(2+x)^2(20+24x+5x^2)\right]dx + \int_{y=0}^{y=4}\left[\frac{1}{256}y^6 - \frac{7}{16}y^4 + 6y^2\right]dy \\ &\quad + \int_{y=-4}^{y=0}\left[\frac{1}{256}y^6 - \frac{7}{16}y^4 + 6y^2 + 10y - 4\right]dy \\ &= \frac{40}{3} + \frac{1664}{35} - \frac{1696}{35} = \frac{1304}{105}.\end{aligned}$$

Now we could redo this example using a union of two Type II regions (see Exercise 15.2.4 below).

? Exercise 15.2.3

Consider the region bounded by the curves $y = \ln x$ and $y = e^x$ in the interval $[1, 2]$. Decompose the region into smaller regions of Type II.

Hint

Sketch the region, and split it into three regions to set it up.

Answer

$$\{(x,y)\,|\,0 \le y \le 1,\ 1 \le x \le e^y\} \cup \{(x,y)\,|\,1 \le y \le e,\ 1 \le x \le 2\} \cup \{(x,y)\,|\,e \le y \le e^2,\ \ln y \le x \le 2\}$$

? Exercise 15.2.4

Redo Example 15.2.4 using a union of two Type II regions.

Hint

$$\{(x,y)\,|\,0 \le y \le 4,\ \sqrt{y} - 2 \le x \le \left(y - \frac{1}{16}y^3\right)\} \cup \{(x,y)\,|\,-4 \le y \le 0,\ -2 \le x \le \left(y - \frac{1}{16}y^3\right)\}$$

Answer

Same as in the example shown.

Changing the Order of Integration

As we have already seen when we evaluate an iterated integral, sometimes one order of integration leads to a computation that is significantly simpler than the other order of integration. Sometimes the order of integration does not matter, but it is important to learn to recognize when a change in order will simplify our work.

✓ Example 15.2.5: Changing the Order of Integration

Reverse the order of integration in the iterated integral

$$\int_{x=0}^{x=\sqrt{2}}\int_{y=0}^{y=2-x^2} xe^{x^2}\,dy\,dx.$$

Then evaluate the new iterated integral.

Solution

The region as presented is of Type I. To reverse the order of integration, we must first express the region as Type II. Refer to Figure 15.2.10.

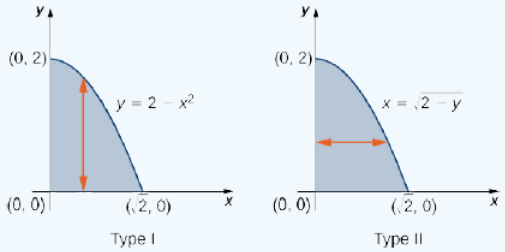


Figure 15.2.10: Converting a region from Type I to Type II.

We can see from the limits of integration that the region is bounded above by $y = 2 - x^2$ and below by $y = 0$ where x is in the interval $[0, \sqrt{2}]$. By reversing the order, we have the region bounded on the left by $x = 0$ and on the right by $x = \sqrt{2 - y}$ where y is in the interval $[0, 2]$. We solved $y = 2 - x^2$ in terms of x to obtain $x = \sqrt{2 - y}$.

Hence

$$\int_0^{\sqrt{2}} \int_0^{2-x^2} xe^{x^2}\, dy\, dx = \int_0^2 \int_0^{\sqrt{2-y}} xe^{x^2}\, dx\, dy \qquad \text{Reverse the order of integration then use substitution.}$$

$$= \int_0^2 \left[\frac{1}{2} e^{x^2} \Big|_0^{\sqrt{2-y}} \right] dy$$

$$= \int_0^2 \frac{1}{2}(e^{2-y} - 1)\, dy$$

$$= -\frac{1}{2}(e^{2-y} + y)\Big|_0^2$$

$$= \frac{1}{2}(e^2 - 3).$$

Example 15.2.6: Evaluating an Iterated Integral by Reversing the Order of Integration

Consider the iterated integral

$$\iint_R f(x, y)\, dx\, dy$$

where $z = f(x, y) = x - 2y$ over a triangular region R that has sides on $x = 0,\ y = 0$, and the line $x + y = 1$. Sketch the region, and then evaluate the iterated integral by

a. integrating first with respect to y and then
b. integrating first with respect to x.

Solution

A sketch of the region appears in Figure 15.2.11.

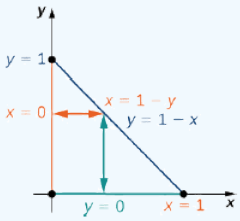


Figure 15.2.11: A triangular region R for integrating in two ways.

We can complete this integration in two different ways.

a. One way to look at it is by first integrating y from $y = 0$ to $y = 1 - x$ vertically and then integrating x from $x = 0$ to $x = 1$:

$$\iint_R f(x,y)\,dx\,dy = \int_{x=0}^{x=1}\int_{y=0}^{y=1-x}(x-2y)\,dy\,dx = \int_{x=0}^{x=1}\left(xy-y^2\right)\Big|_{y=0}^{y=1-x}dx$$

$$= \int_{x=0}^{x=1}\left[x(1-x)-(1-x)^2\right]\,dx = \int_{x=0}^{x=1}[-1+3x-2x^2]dx$$

$$= \left[-x+\frac{3}{2}x^2-\frac{2}{3}x^3\right]\Big|_{x=0}^{x=1} = -\frac{1}{6}.$$

b. The other way to do this problem is by first integrating x from $x=0$ to $x=1-y$ horizontally and then integrating y from $y=0$ to $y=1$:

$$\iint_R f(x,y)\,dx\,dy = \int_{y=0}^{y=1}\int_{x=0}^{x=1-y}(x-2y)\,dx\,dy = \int_{y=0}^{y=1}\left[\frac{1}{2}x^2-2xy\right]_{x=0}^{x=1-y}dy$$

$$= \int_{y=0}^{y=1}\left[\frac{1}{2}(1-y)^2-2y(1-y)\right]dy = \int_{y=0}^{y=1}\left[\frac{1}{2}-3y+\frac{5}{2}y^2\right]\,dy$$

$$= \left[\frac{1}{2}y-\frac{3}{2}y^2+\frac{5}{6}y^3\right]\Big|_{y=0}^{y=1} = -\frac{1}{6}.$$

? Exercise 15.2.5

Evaluate the iterated integral $\iint_D (x^2+y^2)\,dA$ over the region D in the first quadrant between the functions $y=2x$ and $y=x^2$.

Evaluate the iterated integral by integrating first with respect to y and then integrating first with resect to x.

Hint

Sketch the region and follow Example 15.2.6.

Answer

$\frac{216}{35}$

Calculating Volumes, Areas, and Average Values

We can use double integrals over general regions to compute volumes, areas, and average values. The methods are the same as those in Double Integrals over Rectangular Regions, but without the restriction to a rectangular region, we can now solve a wider variety of problems.

✓ Example 15.2.7: Finding the Volume of a Tetrahedron

Find the volume of the solid bounded by the planes $x=0,\ y=0,\ z=0$, and $2x+3y+z=6$.

Solution

The solid is a tetrahedron with the base on the xy-plane and a height $z=6-2x-3y$. The base is the region D bounded by the lines, $x=0$, $y=0$ and $2x+3y=6$ where $z=0$ (Figure 15.2.12). Note that we can consider the region D as Type I or as Type II, and we can integrate in both ways.

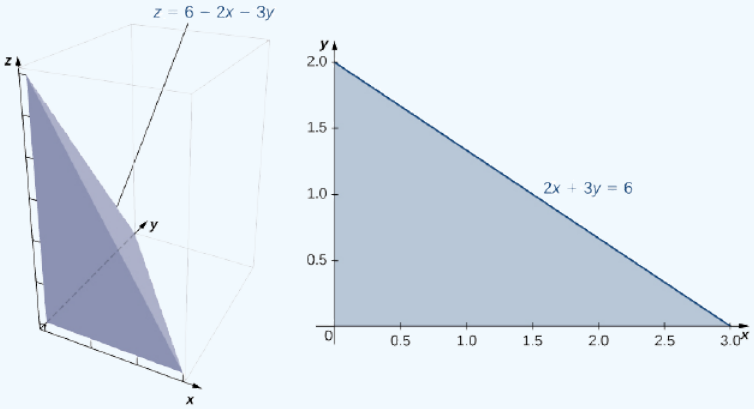


Figure 15.2.12: A tetrahedron consisting of the three coordinate planes and the plane $z = 6 - 2x - 3y$, with the base bound by $x = 0,\ y = 0$, and $2x + 3y = 6$.

First, consider D as a Type I region, and hence $D = \{(x, y) \,|\, 0 \le x \le 3,\ 0 \le y \le 2 - \frac{2}{3}x\}$.

Therefore, the volume is

$$V = \int_{x=0}^{x=3} \int_{y=0}^{y=2-(2x/3)} (6 - 2x - 3y)\, dy\, dx = \int_{x=0}^{x=3} \left[\left(6y - 2xy - \frac{3}{2}y^2 \right) \Big|_{y=0}^{y=2-(2x/3)} \right] dx$$

$$= \int_{x=0}^{x=3} \left[\frac{2}{3}(x-3)^2 \right] dx = 6.$$

Now consider D as a Type II region, so $D = \{(x, y) \,|\, 0 \le y \le 2,\ 0 \le x \le 3 - \frac{3}{2}y\}$. In this calculation, the volume is

$$V = \int_{y=0}^{y=2} \int_{x=0}^{x=3-(3y/2)} (6 - 2x - 3y)\, dx\, dy = \int_{y=0}^{y=2} \left[(6x - x^2 - 3xy) \Big|_{x=0}^{x=3-(3y/2)} \right] dy$$

$$= \int_{y=0}^{y=2} \left[\frac{9}{4}(y-2)^2 \right] dy = 6.$$

Therefore, the volume is 6 cubic units.

Exercise 15.2.6

Find the volume of the solid bounded above by $f(x, y) = 10 - 2x + y$ over the region enclosed by the curves $y = 0$ and $y = e^x$ where x is in the interval $[0, 1]$.

Hint

Sketch the region, and describe it as Type I.

Answer

$\frac{e^2}{4} + 10e - \frac{49}{4}$ cubic units

Finding the area of a rectangular region is easy, but finding the area of a non-rectangular region is not so easy. As we have seen, we can use double integrals to find a rectangular area. As a matter of fact, this comes in very handy for finding the area of a general non-rectangular region, as stated in the next definition.

Definition: Double Integrals

The area of a plane-bounded region D is defined as the double integral

$$\iint_D 1\, dA.$$

We have already seen how to find areas in terms of single integration. Here we are seeing another way of finding areas by using double integrals, which can be very useful, as we will see in the later sections of this chapter.

Example 15.2.8: Finding the Area of a Region

Find the area of the region bounded below by the curve $y = x^2$ and above by the line $y = 2x$ in the first quadrant (Figure 15.2.13).

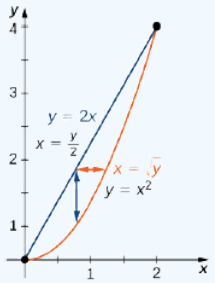


Figure 15.2.13: The region bounded by $y = x^2$ and $y = 2x$.

Solution

We just have to integrate the constant function $f(x, y) = 1$ over the region. Thus, the area A of the bounded region is $\int_{x=0}^{x=2} \int_{y=x^2}^{y=2x} dy\ dx$ or $\int_{y=0}^{y=4} \int_{x=y/2}^{x=\sqrt{y}} dx\ dy$:

$$\begin{aligned} A &= \iint_D 1\, dx\ dy \\ &= \int_{x=0}^{x=2} \int_{y=x^2}^{y=2x} 1\, dy\ dx \\ &= \int_{x=0}^{x=2} \left(y \Big|_{y=x^2}^{y=2x} \right) dx \\ &= \int_{x=0}^{x=2} (2x - x^2)\, dx \\ &= \left(x^2 - \frac{x^3}{3} \right) \Big|_0^2 = \frac{4}{3}. \end{aligned}$$

Exercise 15.2.7

Find the area of a region bounded above by the curve $y = x^3$ and below by $y = 0$ over the interval $[0, 3]$.

Hint

Sketch the region.

Answer

$\frac{81}{4}$ square units

We can also use a double integral to find the average value of a function over a general region. The definition is a direct extension of the earlier formula.

Definition: The Average Value of a Function

If $f(x, y)$ is integrable over a plane-bounded region D with positive area $A(D)$, then the average value of the function is

$$f_{ave} = \frac{1}{A(D)} \iint_D f(x, y)\, dA.$$

Note that the area is $A(D) = \iint_D 1\,dA$.

Example 15.2.9: Finding an Average Value

Find the average value of the function $f(x,y) = 7xy^2$ on the region bounded by the line $x = y$ and the curve $x = \sqrt{y}$ (Figure 15.2.14).

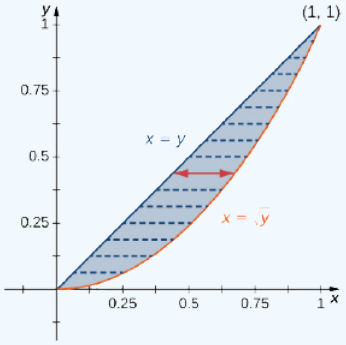


Figure 15.2.14: The region bounded by $x = y$ and $x = \sqrt{y}$.

Solution

First find the area $A(D)$ where the region D is given by the figure. We have

$$A(D) = \iint_D 1\,dA = \int_{y=0}^{y=1} \int_{x=y}^{x=\sqrt{y}} 1\,dx\,dy = \int_{y=0}^{y=1} \left[x\Big|_{x=y}^{x=\sqrt{y}}\right] dy$$
$$= \int_{y=0}^{y=1} (\sqrt{y} - y)\,dy = \frac{2}{3}y^{2/3} - \frac{y^2}{2}\Big|_0^1 = \frac{1}{6}$$

Then the average value of the given function over this region is

$$f_{ave} = \frac{1}{A(D)} \iint_D f(x,y)\,dA = \frac{1}{A(D)} \int_{y=0}^{y=1} \int_{x=y}^{x=\sqrt{y}} 7xy^2\,dx\,dy = \frac{1}{1/6} \int_{y=0}^{y=1} \left[\frac{7}{2}x^2y^2\Big|_{x=y}^{x=\sqrt{y}}\right] dy$$
$$= 6\int_{y=0}^{y=1} \left[\frac{7}{2}y^2(y - y^2)\right] dy = 6\int_{y=0}^{y=1} \left[\frac{7}{2}(y^3 - y^4)\right] dy = \frac{42}{2}\left(\frac{y^4}{4} - \frac{y^5}{5}\right)\Big|_0^1 = \frac{42}{40} = \frac{21}{20}.$$

Exercise 15.2.8

Find the average value of the function $f(x,y) = xy$ over the triangle with vertices $(0,0)$, $(1,0)$ and $(1,3)$.

Hint

Express the line joining $(0,0)$ and $(1,3)$ as a function $y = g(x)$.

Answer

$\frac{3}{4}$

Improper Double Integrals

An **improper double integral** is an integral $\iint_D f\,dA$ where either D is an unbounded region or f is an unbounded function. For example, $D = \{(x,y)\,|\,|x - y| \geq 2\}$ is an unbounded region, and the function $f(x,y) = 1/(1 - x^2 - 2y^2)$ over the ellipse $x^2 + 3y^2 \geq 1$ is an unbounded function. Hence, both of the following integrals are improper integrals:

i. $\iint_D xy\ dA$ where $D = \{(x,y) || x-y| \geq 2\}$;

ii. $\iint_D \frac{1}{1-x^2-2y^2}\ dA$ where $D = \{(x,y) | x^2+3y^2 \leq 1\}$.

In this section we would like to deal with improper integrals of functions over rectangles or simple regions such that f has only finitely many discontinuities. Not all such improper integrals can be evaluated; however, a form of Fubini's theorem does apply for some types of improper integrals.

Theorem: Fubini's Theorem for Improper Integrals

If D is a bounded rectangle or simple region in the plane defined by

$\{(x,y)\ : a \leq x \leq b,\ g(x) \leq y \leq h(x)\}$ and also by

$\{(x,y)\ : c \leq y \leq d,\ j(y) \leq x \leq k(y)\}$ and f is a nonnegative function on D with finitely many discontinuities in the interior of D then

$$\iint_D f\ dA = \int_{x=a}^{x=b}\int_{y=g(x)}^{y=h(x)} f(x,y)\,dy\,dx = \int_{y=c}^{y=d}\int_{x=j(y)}^{x=k(y)} f(x,y)\,dx\,dy$$

It is very important to note that we required that the function be nonnegative on D for the theorem to work. We consider only the case where the function has finitely many discontinuities inside D.

Example 15.2.10: Evaluating a Double Improper Integral

Consider the function $f(x,y) = \frac{e^y}{y}$ over the region $D = \{(x,y)\ : 0 \leq x \leq 1,\ x \leq y \leq \sqrt{x}\}$.

Notice that the function is nonnegative and continuous at all points on D except $(0,0)$. Use Fubini's theorem to evaluate the improper integral.

Solution

First we plot the region D (Figure 15.2.15); then we express it in another way.

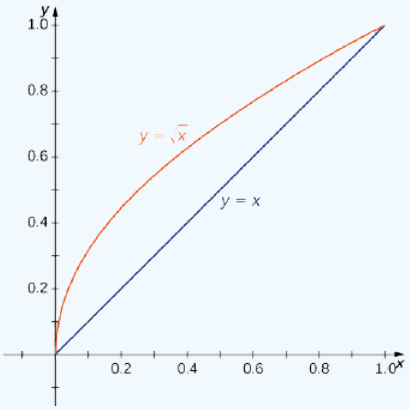


Figure 15.2.15: The function f is continuous at all points of the region D except $(0,0)$.

The other way to express the same region D is

$$D = \{(x,y)\ :\ 0 \leq y \leq 1,\ y^2 \leq x \leq y\}.$$

Thus we can use Fubini's theorem for improper integrals and evaluate the integral as

$$\int_{y=0}^{y=1}\int_{x=y^2}^{x=y} \frac{e^y}{y}\,dx\,dy.$$

Therefore, we have

$$\int_{y=0}^{y=1}\int_{x=y^2}^{x=y}\frac{e^y}{y}\,dx\,dy=\int_{y=0}^{y=1}\frac{e^y}{y}x\Big|_{x=y^2}^{x=y}dy=\int_{y=0}^{y=1}\frac{e^y}{y}(y-y^2)\,dy=\int_0^1(e^y-ye^y)\,dy=e-2.$$

As mentioned before, we also have an improper integral if the region of integration is unbounded. Suppose now that the function f is continuous in an unbounded rectangle R.

Theorem: Improper Integrals on an Unbounded Region

If R is an unbounded rectangle such as $R=\{(x,y)\,:\,a\le x\le\infty,\ c\le y\le\infty\}$, then when the limit exists, we have

$$\iint_R f(x,y)\,dA=\lim_{(b,d)\to(\infty,\infty)}\int_a^b\left(\int_c^d f(x,y)\,dy\right)dx=\lim_{(b,d)\to(\infty,\infty)}\int_c^d\left(\int_a^b f(x,y)\,dx\right)dy.$$

The following example shows how this theorem can be used in certain cases of improper integrals.

Example 15.2.11

Evaluate the integral $\iint_R xye^{-x^2-y^2}\,dA$ where R is the first quadrant of the plane.

Solution

The region R is the first quadrant of the plane, which is unbounded. So

$$\begin{aligned}
\iint_R xye^{-x^2-y^2}\,dA&=\lim_{(b,d)\to(\infty,\infty)}\int_{x=0}^{x=b}\left(\int_{y=0}^{y=d}xye^{-x^2-y^2}dy\right)dx\\
&=\lim_{(b,d)\to(\infty,\infty)}\int_{x=0}^{x=b}\left(-\frac{1}{2}x\int_{y=0}^{y=d}-2ye^{-x^2-y^2}dy\right)dx\\
&=\lim_{(b,d)\to(\infty,\infty)}\int_{x=0}^{x=b}-\frac{1}{2}x\left(e^{-x^2-y^2}\right)\Big|_{y=0}^{y=d}dx\\
&=\lim_{(b,d)\to(\infty,\infty)}\int_{x=0}^{x=b}-\frac{1}{2}x\left(e^{-x^2-d^2}-e^{-x^2}\right)dx\\
&=\lim_{(b,d)\to(\infty,\infty)}\int_{x=0}^{x=b}\frac{1}{2}xe^{-x^2}\left(1-e^{-d^2}\right)dx\\
&=\lim_{(b,d)\to(\infty,\infty)}\left(-\frac{1}{4}\right)\left(1-e^{-d^2}\right)\int_{x=0}^{x=b}-2xe^{-x^2}dx\\
&=\lim_{(b,d)\to(\infty,\infty)}\left(-\frac{1}{4}\right)\left(1-e^{-d^2}\right)e^{-x^2}\Big|_{x=0}^{x=b}dx\\
&=\lim_{(b,d)\to(\infty,\infty)}\frac{1}{4}\left(1-e^{-b^2}\right)\left(1-e^{-d^2}\right)\\
&=\lim_{(b,d)\to(\infty,\infty)}\frac{1}{4}\left(1-\cancelto{0}{e^{-b^2}}\right)\left(1-\cancelto{0}{e^{-d^2}}\right)\\
&=\frac{1}{4}
\end{aligned}$$

Thus,

$$\iint_R xye^{-x^2-y^2}\,dA$$

is convergent and the value is $\frac{1}{4}$.

Exercise 15.2.9

$$\iint_D \frac{y}{\sqrt{1-x^2-y^2}}\,dA$$

where $D = \{(x,y) : x \geq 0,\ y \geq 0,\ x^2+y^2 \leq 1\}$.

Hint

Notice that the integral is nonnegative and discontinuous on $x^2+y^2=1$. Express the region D as $D = \{(x,y) : 0 \leq x \leq 1,\ 0 \leq y \leq \sqrt{1-x^2}\}$ and integrate using the method of substitution.

Answer

$\frac{\pi}{4}$

In some situations in probability theory, we can gain insight into a problem when we are able to use double integrals over general regions. Before we go over an example with a double integral, we need to set a few definitions and become familiar with some important properties.

Definition: Joint Density Function

Consider a pair of continuous random variables X and Y such as the birthdays of two people or the number of sunny and rainy days in a month. The **joint density function** f of X and Y satisfies the probability that (X,Y) lies in a certain region D:

$$P((X,Y) \in D) = \iint_D f(x,y)\,dA.$$

Since the probabilities can never be negative and must lie between 0 and 1 the joint density function satisfies the following inequality and equation:

$$f(x,y) \geq 0 \text{ and } \iint_R f(x,y)\,dA = 1.$$

Definition: Independent Random Variables

The variables X and Y are said to be **independent random variables** if their joint density function is the product of their individual density functions:

$$f(x,y) = f_1(x)f_2(y).$$

Example 15.2.12: Application to Probability

At Sydney's Restaurant, customers must wait an average of 15 minutes for a table. From the time they are seated until they have finished their meal requires an additional 40 minutes, on average. What is the probability that a customer spends less than an hour and a half at the diner, assuming that waiting for a table and completing the meal are independent events?

Solution

Waiting times are mathematically modeled by exponential density functions, with m being the average waiting time, as

$$f(t) = \begin{cases} 0, & \text{if } t < 0 \\ \dfrac{1}{m}e^{-t/m}, & \text{if } t \geq 0. \end{cases}$$

if X and Y are random variables for 'waiting for a table' and 'completing the meal,' then the probability density functions are, respectively,

$$f_1(x) = \begin{cases} 0, & \text{if } x < 0. \\ \frac{1}{15}e^{-x/15}, & \text{if } x \geq 0. \end{cases} \quad \text{and} \quad f_2(y) = \begin{cases} 0, & \text{if } y < 0 \\ \frac{1}{40}e^{-y/40}, & \text{if } y \geq 0. \end{cases}$$

Clearly, the events are independent and hence the joint density function is the product of the individual functions

$$f(x,y) = f_1(x)f_2(y) = \begin{cases} 0, & \text{if } x < 0 \text{ or } y < 0, \\ \frac{1}{600}e^{-x/15}e^{-y/40}, & \text{if } x, y \geq 0 \end{cases}$$

We want to find the probability that the combined time $X + Y$ is less than 90 minutes. In terms of geometry, it means that the region D is in the first quadrant bounded by the line $x + y = 90$ (Figure 15.2.16).

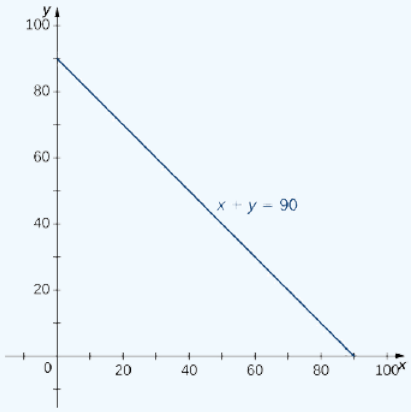


Figure 15.2.16: The region of integration for a joint probability density function.

Hence, the probability that (X, Y) is in the region D is

$$P(X + Y \leq 90) = P((X,Y) \in D) = \iint_D f(x,y)\,dA = \iint_D \frac{1}{600}e^{-x/15}e^{-y/40}\,dA.$$

Since $x + y = 90$ is the same as $y = 90 - x$, we have a region of Type I, so

$$\begin{aligned} D &= \{(x,y) \mid 0 \leq x \leq 90,\ 0 \leq y \leq 90 - x\}, \\ P(X + Y \leq 90) &= \frac{1}{600}\int_{x=0}^{x=90}\int_{y=0}^{y=90-x} e^{-x/15}e^{-y/40}dy\,dx \\ &= -\frac{1}{15}\int_{x=0}^{x=90} e^{-x/15}\int_{y=0}^{y=90-x} -\frac{1}{40}e^{-y/40}dy\,dx \\ &= -\frac{1}{15}\int_{x=0}^{x=90} e^{-x/15}e^{-y/40}\Big|_{y=0}^{y=90-x}\,dx \\ &= \quad \ldots \\ &\approx 0.8328 \end{aligned}$$

Thus, there is an 83.3% chance that a customer spends less than an hour and a half at the restaurant.

Another important application in probability that can involve improper double integrals is the calculation of expected values. First we define this concept and then show an example of a calculation.

Definition: Expected Values

In probability theory, we denote the **expected values** $E(X)$ and $E(Y)$ respectively, as the most likely outcomes of the events. The expected values $E(X)$ and $E(Y)$ are given by

$$E(X) = \iint_S x\,f(x,y)\,dA \text{ and } E(Y) = \iint_S y\,f(x,y)\,dA,$$

where S is the sample space of the random variables X and Y.

Example 15.2.13: Finding Expected Value

Find the expected time for the events ‘waiting for a table’ and ‘completing the meal’ in Example 15.2.12.

Solution

Using the first quadrant of the rectangular coordinate plane as the sample space, we have improper integrals for $E(X)$ and $E(Y)$. The expected time for a table is

$$\begin{aligned} E(X) &= \iint_S x\frac{1}{600}e^{-x/15}e^{-y/40}\,dA \\ &= \frac{1}{600}\int_{x=0}^{x=\infty}\int_{y=0}^{y=\infty} xe^{-x/15}e^{-y/40}dA \\ &= \frac{1}{600}\lim_{(a,b)\to(\infty,\infty)}\int_{x=0}^{x=a}\int_{y=0}^{y=b} xe^{-x/15}e^{-y/40}dx\;dy \\ &= \frac{1}{600}\left(\lim_{a\to\infty}\int_{x=0}^{x=a} xe^{-x/15}dx\right)\left(\lim_{b\to\infty}\int_{y=0}^{y=b} e^{-y/40}dy\right) \\ &= \frac{1}{600}\left((\lim_{a\to\infty}(-15e^{-x/15}(x+15)))\Big|_{x=0}^{x=a}\right)\left((\lim_{b\to\infty}(-40e^{-y/40}))\Big|_{y=0}^{y=b}\right) \\ &= \frac{1}{600}\left(\lim_{a\to\infty}(-15e^{-a/15}(a+15)+225)\right)\left(\lim_{b\to\infty}(-40e^{-b/40}+40)\right) \\ &= \frac{1}{600}(225)(40) = 15. \end{aligned}$$

A similar calculation shows that $E(Y) = 40$. This means that the expected values of the two random events are the average waiting time and the average dining time, respectively.

Exercise 15.2.10

The joint density function for two random variables X and Y is given by

$$f(x,y) = \begin{cases} \frac{1}{16250}(x^2+y^2), & \text{if } 0\le x\le 15,\ 0\le y\le 10 \\ 0, & \text{otherwise} \end{cases}$$

Find the probability that X is at most 10 and Y is at least 5.

Hint

Compute the probability

$$P(X\le 10,\ Y\ge 5) = \int_{x=0}^{x=10}\int_{y=5}^{y=10}\frac{1}{16250}(x^2+y^2)dy\;dx.$$

Answer

$P(X\le 10,\ Y\ge 5) = \frac{11}{39}\approx 0.28205$

Key Concepts

- A general bounded region D on the plane is a region that can be enclosed inside a rectangular region. We can use this idea to define a double integral over a general bounded region.
- To evaluate an iterated integral of a function over a general nonrectangular region, we sketch the region and express it as a Type I or as a Type II region or as a union of several Type I or Type II regions that overlap only on their boundaries.
- We can use double integrals to find volumes, areas, and average values of a function over general regions, similarly to calculations over rectangular regions.
- We can use Fubini’s theorem for improper integrals to evaluate some types of improper integrals.

Key Equations

- **Iterated integral over a Type I region**

$$\iint_D f(x,y)\,dA = \iint_D f(x,y)\,dy\,dx = \int_a^b \left[\int_{g_1(x)}^{g_2(x)} f(x,y)\,dy\right] dx$$

- **Iterated integral over a Type II region**

$$\iint_D f(x,y)\,dA = \iint_D (x,y)\,dx\,dy = \int_c^d \left[\int_{h_1(y)}^{h_2(y)} f(x,y)\,dx\right] dy$$

Glossary

improper double integral

a double integral over an unbounded region or of an unbounded function

Type I

a region D in the xy- plane is Type I if it lies between two vertical lines and the graphs of two continuous functions $g_1(x)$ and $g_2(x)$

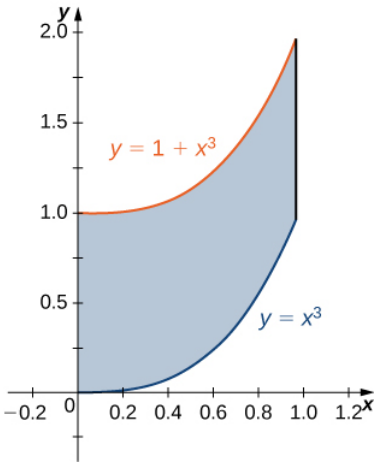
Type II

a region D in the xy-plane is Type II if it lies between two horizontal lines and the graphs of two continuous functions $h_1(y)$ and $h_2(h)$

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15.2E: Exercises for Section 15.2

1) The region D bounded by $y = x^3,\ y = x^3 + 1,\ x = 0$, and $x = 1$ as given in the following figure.



a. Classify this region as vertically simple (Type I) or horizontally simple (Type II).

Type:

Type I but not Type II

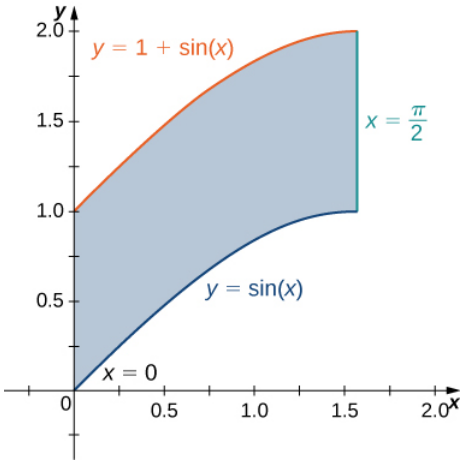
b. Find the area of the region D.

c. Find the average value of the function $f(x, y) = 3xy$ on the region graphed in the previous exercise.

Answer

$\frac{27}{20}$

2) The region D bounded by $y = \sin x,\ y = 1 + \sin x,\ x = 0$, and $x = \frac{\pi}{2}$ as given in the following figure.



a. Classify this region as vertically simple (Type I) or horizontally simple (Type II).

Type:

Type I but not Type II

b. Find the area of the region D.

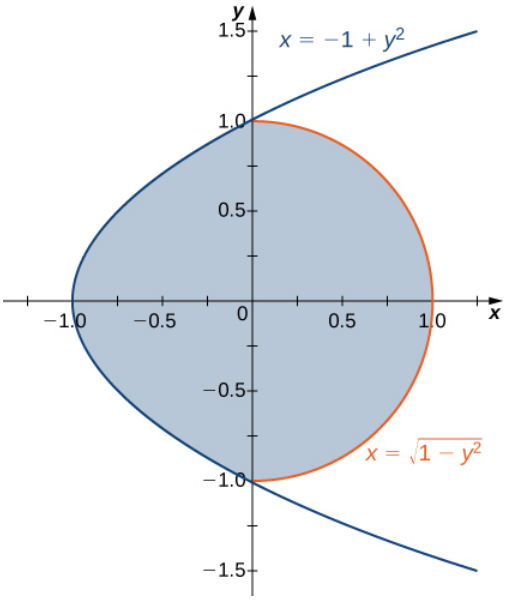
Answer

https://math.libretexts.org/@go/page/67163

$\frac{\pi}{2}$ units^2

c. Find the average value of the function $f(x, y) = \cos x$ on the region D.

3) The region D bounded by $x = y^2 - 1$ and $x = \sqrt{1 - y^2}$ as given in the following figure.



a. Classify this region as vertically simple (Type I) or horizontally simple (Type II).

Type:

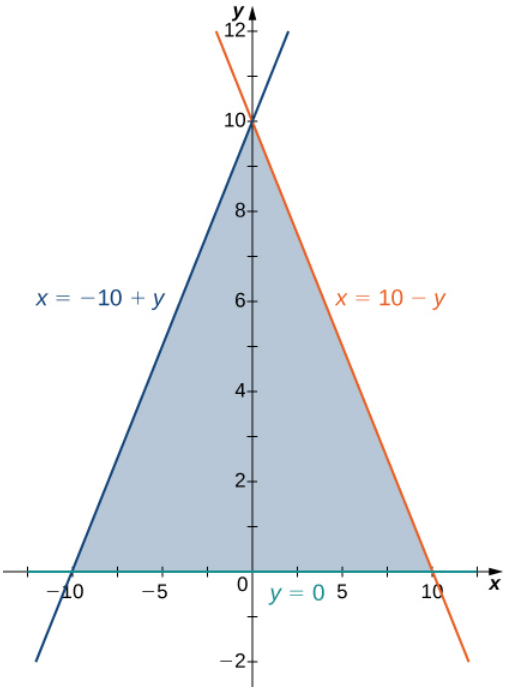
Type II but not Type I

b. Find the volume of the solid under the graph of the function $f(x, y) = xy + 1$ and above the region D.

Answer

$\frac{1}{6}(8 + 3\pi)$ units^3

4) The region D bounded by $y = 0,\ x = -10 + y$, and $x = 10 - y$ as given in the following figure.



a. Classify this region as vertically simple (Type I) or horizontally simple (Type II).

Type:

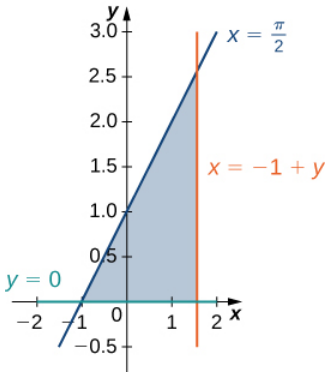
Type II but not Type I

b. Find the volume of the solid under the graph of the function $f(x, y) = x + y$ and above the region in the figure from the previous exercise.

Answer

$\frac{1000}{3}$ units^3

5) The region D bounded by $y = 0,\ x = y - 1,\ x = \frac{\pi}{2}$ as given in the following figure.

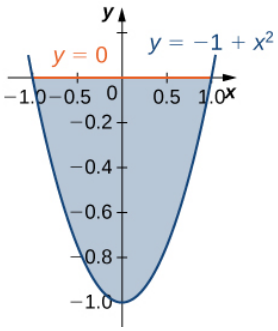


Classify this region as vertically simple (Type I) or horizontally simple (Type II).

Type:

Type I and Type II

6) The region D bounded by $y = 0$ and $y = x^2 - 1$ as given in the following figure.



Classify this region as vertically simple (Type I) or horizontally simple (Type II).

Type:

Type I and Type II

7) Let D be the region bounded by the curves of equations $y = \cos x$ and $y = 4 - x^2$ and the x-axis. Explain why D is neither of Type I nor II.

Answer

The region D is not of Type I: it does not lie between two vertical lines and the graphs of two continuous functions $g_1(x)$ and $g_2(x)$. The region is not of Type II: it does not lie between two horizontal lines and the graphs of two continuous functions $h_1(y)$ and $h_2(y)$.

8) Let D be the region bounded by the curves of equations $y = x,\ y = -x$ and $y = 2 - x^2$. Explain why D is neither of Type I nor II.

In exercises 9 - 14, evaluate the double integral $\iint_D f(x,y)\,dA$ over the region D.

9) $f(x,y) = 1$ and

$D = \{(x,y) | \ 0 \le x \le \frac{\pi}{2},\ \sin x \le y \le 1 + \sin x\}$

Answer

$\frac{\pi}{2}$

10) $f(x,y) = 2$ and

$D = \{(x,y) | \ 0 \le y \le 1,\ y - 1 \le x \le \arccos y\}$

11) $f(x,y) = xy$ and

$D = \{(x,y) | \ -1 \le y \le 1,\ y^2 - 1 \le x \le \sqrt{1 - y^2}\}$

Answer

0

12) $f(x,y) = \sin y$ and D is the triangular region with vertices $(0,0),\ (0,3)$, and $(3,0)$

13) $f(x,y) = -x + 1$ and D is the triangular region with vertices $(0,0),\ (0,2)$, and $(2,2)$

Answer

$\frac{2}{3}$

14) $f(x,y) = 2x + 4y$ and

$D = \{(x,y) | \ 0 \le x \le 1,\ x^3 \le y \le x^3 + 1\}$

In exercises 15 - 20, evaluate the iterated integrals.

15) $\int_0^1 \int_{2\sqrt{x}}^{2\sqrt{x}+1} (xy+1)\,dy\,dx$

Answer

$\frac{41}{20}$

16) $\int_0^3 \int_{2x}^{3x} (x+y^2)\,dy\,dx$

17) $\int_1^2 \int_{-u^2-1}^{-u} (8uv)\,dv\,du$

Answer

-63

18) $\int_e^{e^2} \int_{\ln u}^{2} (v+\ln u)\,dv\,du$

19) $\int_0^{1/2} \int_{-\sqrt{1-4y^2}}^{\sqrt{1-4y^2}} 4\,dx\,dy$

Answer

π

20) $\int_0^1 \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} (2x+4y^3)\,dx\,dy$

21) Let D be the region bounded by $y=1-x^2,\ y=4-x^2$, and the x- and y-axes.

a. Show that $\iint_D x\,dA = \int_0^1 \int_{1-x^2}^{4-x^2} x\,dy\,dx + \int_1^2 \int_0^{4-x^2} x\,dy\,dx$ by dividing the region D into two regions of Type I.

b. Evaluate the integral $\iint_D x\,dA$.

22) Let D be the region bounded by $y=1,\ y=x,\ y=\ln x$, and the x-axis.

a. Show that $\iint_D y^2\,dA = \int_{-1}^0 \int_{-x}^{2-x^2} y^2 dy\,dx + \int_0^1 \int_x^{2-x^2} y^2 dy\,dx$ by dividing the region D into two regions of Type I, where $D = \{(x,y)\,|\,y \geq x, y \geq -x,\ y \leq 2-x^2\}$.

b. Evaluate the integral $\iint_D y^2\,dA$.

23) Let D be the region bounded by $y=x^2$, $y=x+2$, and $y=-x$.

a. Show that $\iint_D x\,dA = \int_0^1 \int_{-y}^{\sqrt{y}} x\,dx\,dy + \int_1^4 \int_{y-2}^{\sqrt{y}} x\,dx\,dy$ by dividing the region D into two regions of Type II, where $D = \{(x,y)\,|\,y \geq x^2,\ y \geq -x,\ y \leq x+2\}$.

b. Evaluate the integral $\iint_D x\,dA$.

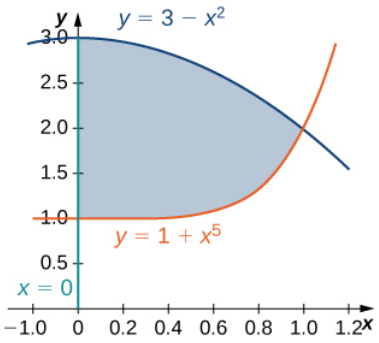
Answer

a. Answers may vary;

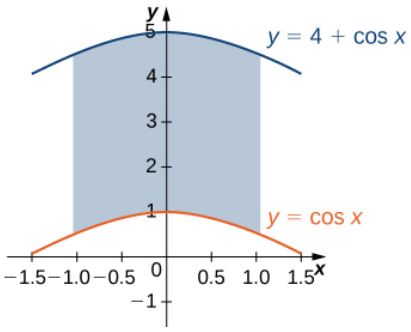
b. $\frac{7}{3}$

24) The region D bounded by $x=0, y=x^5+1$, and $y=3-x^2$ is shown in the following figure. Find the area $A(D)$ of the region D.

https://math.libretexts.org/@go/page/67163



25) The region D bounded by $y = \cos x,\ y = 4 + \cos x$, and $x = \pm\frac{\pi}{3}$ is shown in the following figure. Find the area $A(D)$ of the region D.



Answer

$\frac{8\pi}{3}$

26) Find the area $A(D)$ of the region $D = \{(x,y) | \, y \geq 1 - x^2, y \leq 4 - x^2,\ y \geq 0,\ x \geq 0\}$.

27) Let D be the region bounded by $y = 1,\ y = x,\ y = \ln x$, and the x-axis. Find the area $A(D)$ of the region D.

Answer

$\left(e - \frac{3}{2}\right)$ units^2

28) Find the average value of the function $f(x,y) = \sin y$ on the triangular region with vertices $(0,0),\ (0,3)$, and $(3,0)$.

29) Find the average value of the function $f(x,y) = -x + 1$ on the triangular region with vertices $(0,0),\ (0,2)$, and $(2,2)$.

Answer

The average value of f on this triangular region is $\frac{1}{3}$.

In exercises 30 - 33, change the order of integration and evaluate the integral.

30) $\int_{-1}^{\pi/2} \int_{0}^{x+1} \sin x \, dy \, dx$

31) $\int_{0}^{1} \int_{x-1}^{1-x} x \, dy \, dx$

Answer

$$\int_0^1 \int_{x-1}^{1-x} x \ dy \ dx = \int_{-1}^0 \int_0^{y+1} x \ dx \ dy + \int_0^1 \int_0^{1-y} x \ dx \ dy = \frac{1}{3}$$

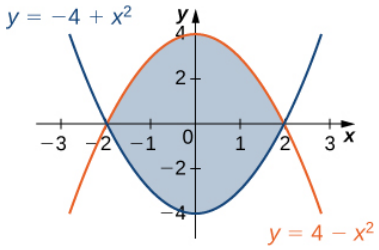
32) $\int_{-1}^{0} \int_{-\sqrt{y+1}}^{\sqrt{y+1}} y^2 dx \ dy$

33) $\int_{-1/2}^{1/2} \int_{-\sqrt{y^2+1}}^{\sqrt{y^2+1}} y \ dx \ dy$

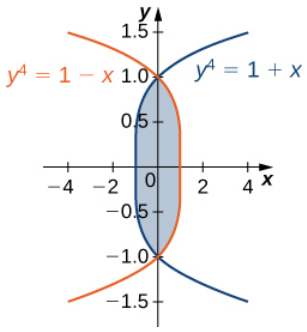
Answer

$$\int_{-1/2}^{1/2}\int_{-\sqrt{y^2+1}}^{\sqrt{y^2+1}} y\,dx\,dy = \int_{1}^{2}\int_{-\sqrt{x^2-1}}^{\sqrt{x^2-1}} y\,dy\,dx = 0$$

34) The region D is shown in the following figure. Evaluate the double integral $\iint_D (x^2+y)\,dA$ by using the easier order of integration.



35) The region D is shown in the following figure. Evaluate the double integral $\iint_D (x^2-y^2)\,dA$ by using the easier order of integration.



Answer

$$\iint_D (x^2-y^2)dA = \int_{-1}^{1}\int_{y^4-1}^{1-y^4} (x^2-y^2)dx\,dy = \frac{464}{4095}$$

36) Find the volume of the solid under the surface $z = 2x + y^2$ and above the region bounded by $y = x^5$ and $y = x$.

37) Find the volume of the solid under the plane $z = 3x + y$ and above the region determined by $y = x^7$ and $y = x$.

Answer

$\frac{4}{5}$ units^3

38) Find the volume of the solid under the plane $z = 3x + y$ and above the region bounded by $x = \tan y,\ x = -\tan y$, and $x = 1$.

39) Find the volume of the solid under the surface $z = x^3$ and above the plane region bounded by $x = \sin y,\ x = -\sin y$, and $x = 1$.

Answer

$\frac{5\pi}{32}$ units^3

40) Let g be a positive, increasing, and differentiable function on the interval $[a, b]$. Show that the volume of the solid under the surface $z = g'(x)$ and above the region bounded by $y = 0,\ y = g(x),\ x = a$, and $x = b$ is given by $\frac{1}{2}(g^2(b) - g^2(a))$.

41) Let g be a positive, increasing, and differentiable function on the interval $[a, b]$ and let k be a positive real number. Show that the volume of the solid under the surface $z = g'(x)$ and above the region bounded by $y = g(x),\ y = g(x) + k,\ x = a$, and $x = b$ is given by $k(g(b) - g(a))$.

42) Find the volume of the solid situated in the first octant and determined by the planes $z = 2,\ z = 0,\ x + y = 1,\ x = 0$, and $y = 0$.

43) Find the volume of the solid situated in the first octant and bounded by the planes $x + 2y = 1$, $x = 0,\ z = 4$, and $z = 0$.

Answer

1 units^3

44) Find the volume of the solid bounded by the planes $x + y = 1,\ x - y = 1,\ x = 0,\ z = 0$, and $z = 10$.

45) Find the volume of the solid bounded by the planes $x + y = 1,\ x - y = 1,\ x + y = -1\ x - y = -1,\ z = 1$, and $z = 0$

Answer

2 units^3

46) Let S_1 and S_2 be the solids situated in the first octant under the planes $x + y + z = 1$ and $x + y + 2z = 1$ respectively, and let S be the solid situated between $S_1,\ S_2,\ x = 0$, and $y = 0$.

a. Find the volume of the solid S_1.
b. Find the volume of the solid S_2.
c. Find the volume of the solid S by subtracting the volumes of the solids S_1 and S_2.

47) Let S_1 and S_2 be the solids situated in the first octant under the planes $2x + 2y + z = 2$ and $x + y + z = 1$ respectively, and let S be the solid situated between $S_1,\ S_2,\ x = 0$, and $y = 0$.

a. Find the volume of the solid S_1.
b. Find the volume of the solid S_2.
c. Find the volume of the solid S by subtracting the volumes of the solids S_1 and S_2.

Answer

a. $\frac{1}{3} \text{ units}^3$
b. $\frac{1}{6} \text{ units}^3$
c. $\frac{1}{6} \text{ units}^3$

48) Let S_1 and S_2 be the solids situated in the first octant under the plane $x + y + z = 2$ and under the sphere $x^2 + y^2 + z^2 = 4$, respectively. If the volume of the solid S_2 is $\frac{4\pi}{3}$ determine the volume of the solid S situated between S_1 and S_2 by subtracting the volumes of these solids.

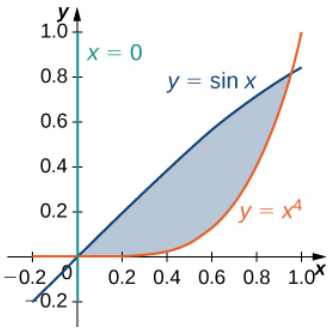
49) Let S_1 and S_2 be the solids situated in the first octant under the plane $x + y + z = 2$ and bounded by the cylinder $x^2 + y^2 = 4$, respectively.

a. Find the volume of the solid S_1.
b. Find the volume of the solid S_2.
c. Find the volume of the solid S situated between S_1 and S_2 by subtracting the volumes of the solids S_1 and S_2.

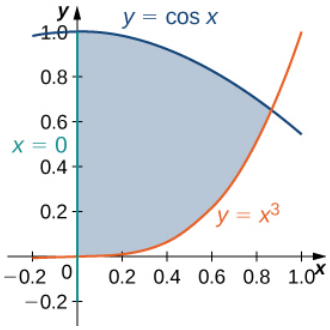
Answer

a. $\frac{4}{3} \text{ units}^3$
b. $2\pi \text{ units}^3$
c. $\frac{6\pi - 4}{3} \text{ units}^3$

50) [T] The following figure shows the region D bounded by the curves $y = \sin x,\ x = 0$, and $y = x^4$. Use a graphing calculator or CAS to find the x-coordinates of the intersection points of the curves and to determine the area of the region D. Round your answers to six decimal places.



51) [T] The region D bounded by the curves $y = \cos x,\ x = 0$, and $y = x^3$ is shown in the following figure. Use a graphing calculator or CAS to find the x-coordinates of the intersection points of the curves and to determine the area of the region D. Round your answers to six decimal places.



Answer

0 and 0.865474; $A(D) = 0.621135 \text{ units}^3$

52) Suppose that (X, Y) is the outcome of an experiment that must occur in a particular region S in the xy-plane. In this context, the region S is called the sample space of the experiment and X and Y are random variables. If D is a region included in S, then the probability of (X, Y) being in D is defined as $P[(X, Y) \in D] = \iint_D p(x, y)dx\ dy$, where $p(x, y)$ is the joint probability density of the experiment. Here, $p(x, y)$ is a nonnegative function for which $\iint_S p(x, y)dx\ dy = 1$. Assume that a point (X, Y) is chosen arbitrarily in the square $[0, 3] \times [0, 3]$ with the probability density

$$p(x, y) = \frac{1}{9}(x, y) \in [0, 3] \times [0, 3],$$

$$p(x, y) = 0 \text{ otherwise}$$

Find the probability that the point (X, Y) is inside the unit square and interpret the result.

53) Consider X and Y two random variables of probability densities $p_1(x)$ and $p_2(x)$, respectively. The random variables X and Y are said to be independent if their joint density function is given by $p_(x, y) = p_1(x)p_2(y)$. At a drive-thru restaurant, customers spend, on average, 3 minutes placing their orders and an additional 5 minutes paying for and picking up their meals. Assume that placing the order and paying for/picking up the meal are two independent events X and Y. If the waiting times are modeled by the exponential probability densities

$$p_1(x) = \frac{1}{3}e^{-x/3}\ x \geq 0,$$

$$p_1(x) = 0 \text{ otherwise}$$

$$p_2(y) = \frac{1}{5}e^{-y/5}\ y \geq 0$$

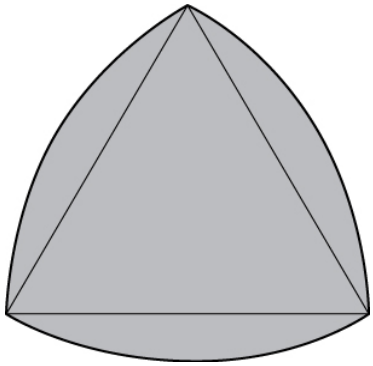
$$p_2(y) = 0 \text{ otherwise}$$

respectively, the probability that a customer will spend less than 6 minutes in the drive-thru line is given by $P[X+Y \leq 6] = \iint_D p(x,y)dx\ dy$, where $D = \{(x,y)|x \geq 0,\ y \geq 0,\ x+y \leq 6\}$. Find $P[X+Y \leq 6]$ and interpret the result.

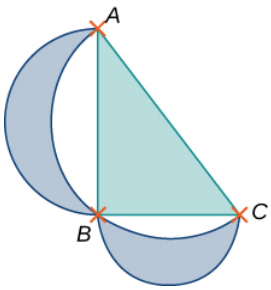
Answer

$P[X+Y \leq 6] = 1 + \frac{3}{2e^2} - \frac{5}{e^{6/5}} \approx 0.45$; there is a 45% chance that a customer will spend 6 minutes in the drive-thru line.

54) [T] The Reuleaux triangle consists of an equilateral triangle and three regions, each of them bounded by a side of the triangle and an arc of a circle of radius *s* centered at the opposite vertex of the triangle. Show that the area of the Reuleaux triangle in the following figure of side length s is $\frac{s^2}{2}(\pi - \sqrt{3})$.



55) [T] Show that the area of the lunes of Alhazen, the two blue lunes in the following figure, is the same as the area of the right triangle ABC. The outer boundaries of the lunes are semicircles of diameters AB and AC respectively, and the inner boundaries are formed by the circumcircle of the triangle ABC.



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15.3: Double Integrals in Polar Coordinates

Learning Objectives

- Recognize the format of a double integral over a polar rectangular region.
- Evaluate a double integral in polar coordinates by using an iterated integral.
- Recognize the format of a double integral over a general polar region.
- Use double integrals in polar coordinates to calculate areas and volumes.

Double integrals are sometimes much easier to evaluate if we change rectangular coordinates to polar coordinates. However, before we describe how to make this change, we need to establish the concept of a double integral in a polar rectangular region.

Polar Rectangular Regions of Integration

When we defined the double integral for a continuous function in rectangular coordinates—say, g over a region R in the xy-plane—we divided R into subrectangles with sides parallel to the coordinate axes. These sides have either constant x-values and/or constant y-values. In polar coordinates, the shape we work with is a polar rectangle, whose sides have constant r-values and/or constant θ-values. This means we can describe a polar rectangle as in Figure 15.3.1a, with $R = \{(r, \theta) \mid a \le r \le b,\ \alpha \le \theta \le \beta\}$.

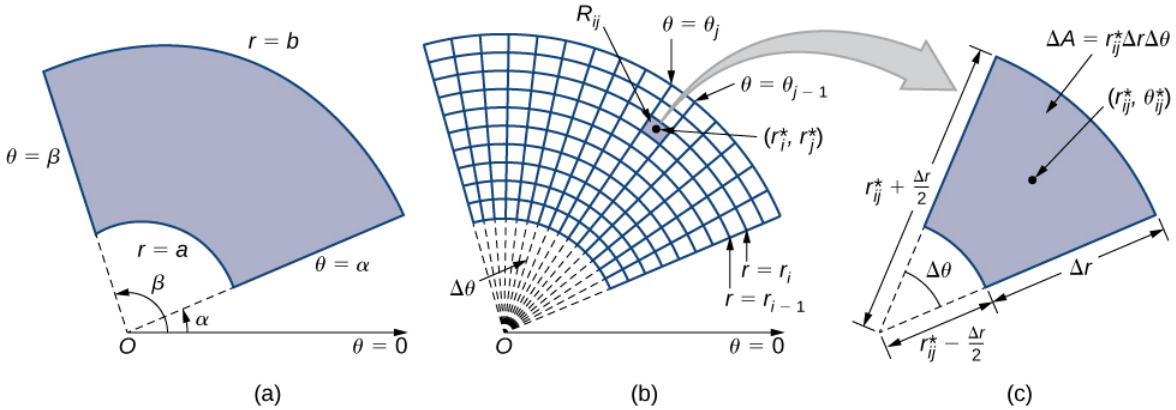


Figure 15.3.1: (a) A polar rectangle R (b) divided into subrectangles R_{ij} (c) Close-up of a subrectangle.

In this section, we are looking to integrate over polar rectangles. Consider a function $f(r, \theta)$ over a polar rectangle R. We divide the interval $[a, b]$ into m subintervals $[r_{i-1}, r_i]$ of length $\Delta r = (b - a)/m$ and divide the interval $[\alpha, \beta]$ into n subintervals $[\theta_{i-1}, \theta_i]$ of width $\Delta\theta = (\beta - \alpha)/n$. This means that the circles $r = r_i$ and rays $\theta = \theta_i$ for $1 \le i \le m$ and $1 \le j \le n$ divide the polar rectangle R into smaller polar subrectangles R_{ij} (Figure 15.3.1b).

As before, we need to find the area ΔA of the polar subrectangle R_{ij} and the "polar" volume of the thin box above R_{ij}. Recall that, in a circle of radius r the length s of an arc subtended by a central angle of θ radians is $s = r\theta$. Notice that the polar rectangle R_{ij} looks a lot like a trapezoid with parallel sides $r_{i-1}\Delta\theta$ and $r_i\Delta\theta$ and with a width Δr. Hence the area of the polar subrectangle R_{ij} is

$$\Delta A = \frac{1}{2}\Delta r(r_{i-1}\Delta\theta + r_i\Delta\theta).$$

Simplifying and letting

$$r_{ij}^* = \frac{1}{2}(r_{i-1} + r_i)$$

we have $\Delta A = r_{ij}^*\Delta r\Delta\theta$.

Therefore, the polar volume of the thin box above R_{ij} (Figure 15.3.2) is

$$f(r_{ij}^*, \theta_{ij}^*)\Delta A = f(r_{ij}^*, \theta_{ij}^*)r_{ij}^*\Delta r\Delta\theta.$$

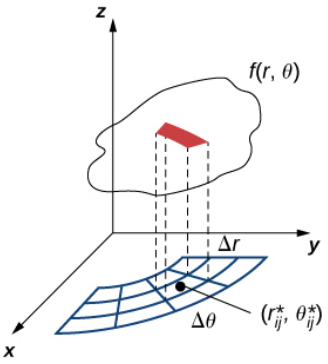


Figure 15.3.2: Finding the volume of the thin box above polar rectangle R_{ij}.

Using the same idea for all the subrectangles and summing the volumes of the rectangular boxes, we obtain a double Riemann sum as

$$\sum_{i=1}^{m}\sum_{j=1}^{n} f(r_{ij}^*, \theta_{ij}^*) r_{ij}^* \Delta r \Delta\theta.$$

As we have seen before, we obtain a better approximation to the polar volume of the solid above the region R when we let m and n become larger. Hence, we define the polar volume as the limit of the double Riemann sum,

$$V = \lim_{m,n\to\infty} \sum_{i=1}^{m}\sum_{j=1}^{n} f(r_{ij}^*, \theta_{ij}^*) r_{ij}^* \Delta r \Delta\theta.$$

This becomes the expression for the double integral.

Definition: The double integral in polar coordinates

The double integral of the function $f(r, \theta)$ over the polar rectangular region R in the $r\theta$-plane is defined as

$$\iint_R f(r,\theta) dA = \lim_{m,n\to\infty} \sum_{i=1}^{m}\sum_{j=1}^{n} f(r_{ij}^*, \theta_{ij}^*) \Delta A \tag{15.3.1}$$

$$= \lim_{m,n\to\infty} \sum_{i=1}^{m}\sum_{j=1}^{n} f(r_{ij}^*, \theta_{ij}^*) r_{ij}^* \Delta r \Delta\theta. \tag{15.3.2}$$

Again, just as in section on Double Integrals over Rectangular Regions, the double integral over a polar rectangular region can be expressed as an iterated integral in polar coordinates. Hence,

$$\iint_R f(r,\theta)\, dA = \iint_R f(r,\theta)\, r\, dr\, d\theta = \int_{\theta=\alpha}^{\theta=\beta} \int_{r=a}^{r=b} f(r,\theta)\, r\, dr\, d\theta.$$

Notice that the expression for dA is replaced by $r\, dr\, d\theta$ when working in polar coordinates. Another way to look at the polar double integral is to change the double integral in rectangular coordinates by substitution. When the function f is given in terms of x and y using $x = r\, \cos\, \theta$, $y = r\, \sin\, \theta$, and $dA = r\, dr\, d\theta$ changes it to

$$\iint_R f(x,y)\, dA = \iint_R f(r\, \cos\, \theta,\, r\, \sin\, \theta)\, r\, dr\, d\theta.$$

Note that all the properties listed in section on Double Integrals over Rectangular Regions for the double integral in rectangular coordinates hold true for the double integral in polar coordinates as well, so we can use them without hesitation.

Example 15.3.1A: Sketching a Polar Rectangular Region

Sketch the polar rectangular region

$$R = \{(r,\theta) \,|\, 1 \le r \le 3, 0 \le \theta \le \pi\}.$$

Solution

As we can see from Figure 15.3.3, $r = 1$ and $r = 3$ are circles of radius 1 and 3 and $0 \le \theta \le \pi$ covers the entire top half of the plane. Hence the region R looks like a semicircular band.

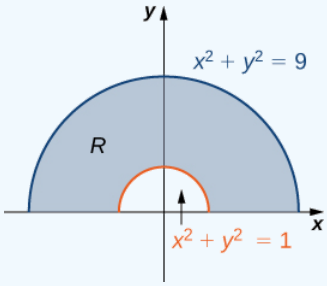


Figure 15.3.3: The polar region R lies between two semicircles.

Now that we have sketched a polar rectangular region, let us demonstrate how to evaluate a double integral over this region by using polar coordinates.

Example 15.3.1B: Evaluating a Double Integral over a Polar Rectangular Region

Evaluate the integral $\iint_R 3x\,dA$ over the region $R = \{(r,\theta)\,|\,1 \le r \le 2,\ 0 \le \theta \le \pi\}$.

Solution

First we sketch a figure similar to Figure 15.3.3, but with outer radius $r = 2$. From the figure we can see that we have

$$\begin{aligned}\iint_R 3x\,dA &= \int_{\theta=0}^{\theta=\pi}\int_{r=1}^{r=2} 3r\cos\theta\, r\,dr\,d\theta \quad \text{Use an integral with correct limits of integration.}\\ &= \int_{\theta=0}^{\theta=\pi}\cos\theta\left[r^3\big|_{r=1}^{r=2}\right]d\theta \quad \text{Integrate first with respect to } r.\\ &= \int_{\theta=0}^{\theta=\pi} 7\cos\theta\,d\theta\\ &= 7\sin\theta\Big|_{\theta=0}^{\theta=\pi} = 0.\end{aligned}$$

Exercise 15.3.1

Sketch the region $D = \{(r,\theta)|1 \le r \le 2,\ -\frac{\pi}{2} \le \theta \le \frac{\pi}{2}\}$, and evaluate $\iint_R x\,dA$.

Hint

Follow the steps in Example 15.3.1A.

Answer

$\frac{14}{3}$

Example 15.3.2A: Evaluating a Double Integral by Converting from Rectangular Coordinates

Evaluate the integral

$$\iint_R (1 - x^2 - y^2)\,dA$$

where R is the unit circle on the xy-plane.

Solution

The region R is a unit circle, so we can describe it as $R = \{(r,\theta)\,|\,0 \le r \le 1,\ 0 \le \theta \le 2\pi\}$.

Using the conversion $x = r\cos\theta$, $y = r\sin\theta$, and $dA = r\,dr\,d\theta$, we have

$$\begin{aligned}\iint_R (1 - x^2 - y^2)\,dA &= \int_0^{2\pi}\int_0^1 (1 - r^2)\,r\,dr\,d\theta \\ &= \int_0^{2\pi}\int_0^1 (r - r^3)\,dr\,d\theta \\ &= \int_0^{2\pi}\left[\frac{r^2}{2} - \frac{r^4}{4}\right]_0^1 d\theta \\ &= \int_0^{2\pi}\frac{1}{4}\,d\theta = \frac{\pi}{2}.\end{aligned}$$

Example 15.3.2B: Evaluating a Double Integral by Converting from Rectangular Coordinates

Evaluate the integral

$$\iint_R (x + y)\,dA$$

where $R = \{(x, y) \mid 1 \le x^2 + y^2 \le 4,\ x \le 0\}$.

Solution

We can see that R is an annular region that can be converted to polar coordinates and described as $R = \{(r, \theta) \mid 1 \le r \le 2,\ \frac{\pi}{2} \le \theta \le \frac{3\pi}{2}\}$ (see the following graph).

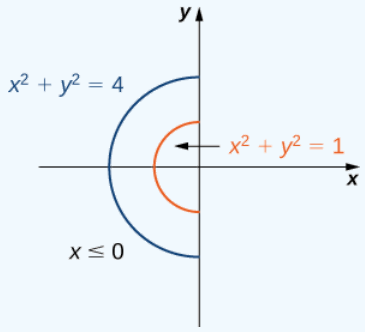


Figure 15.3.4: The annular region of integration R.

Hence, using the conversion $x = r\cos\theta$, $y = r\sin\theta$, and $dA = r\,dr\,d\theta$, we have

$$\begin{aligned}\iint_R (x + y)\,dA &= \int_{\theta=\pi/2}^{\theta=3\pi/2}\int_{r=1}^{r=2} (r\cos\theta + r\sin\theta) r\,dr\,d\theta \\ &= \left(\int_{r=1}^{r=2} r^2\,dr\right)\left(\int_{\pi/2}^{3\pi/2} (\cos\theta + \sin\theta)\,d\theta\right) \\ &= \left[\frac{r^3}{3}\right]_1^2 [\sin\theta - \cos\theta]\Big|_{\pi/2}^{3\pi/2} \\ &= -\frac{14}{3}.\end{aligned}$$

Exercise 15.3.2

Evaluate the integral

$$\iint_R (4 - x^2 - y^2)\,dA$$

where R is the circle of radius 2 on the xy-plane.

Hint

Follow the steps in the previous example.

Answer

8π

General Polar Regions of Integration

To evaluate the double integral of a continuous function by iterated integrals over general polar regions, we consider two types of regions, analogous to Type I and Type II as discussed for rectangular coordinates in section on Double Integrals over General Regions. It is more common to write polar equations as $r = f(\theta)$ than $\theta = f(r)$, so we describe a general polar region as $R = \{(r, \theta) \mid \alpha \leq \theta \leq \beta,\ h_1(\theta) \leq r \leq h_2(\theta)\}$ (Figure 15.3.5).

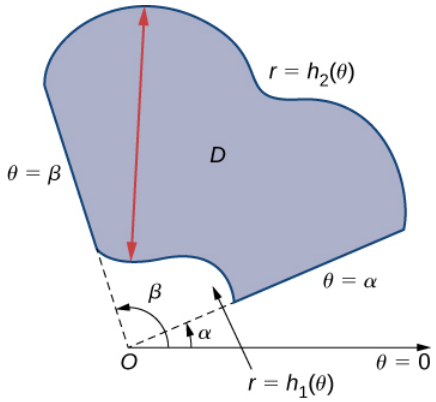


Figure 15.3.5: A general polar region between $\alpha \leq \theta \leq \beta$ and $h_1(\theta) \leq r \leq h_2(\theta)$.

Theorem: Double Integrals over General Polar Regions

If $f(r, \theta)$ is continuous on a general polar region D as described above, then

$$\iint_D f(r, \theta)\, r\, dr\, d\theta = \int_{\theta=\alpha}^{\theta=\beta} \int_{r=h_1(\theta)}^{r=h_2(\theta)} f(r, \theta)\, r\, dr\, d\theta.$$

Example 15.3.3: Evaluating a Double Integral over a General Polar Region

Evaluate the integral

$$\iint_D r^2 \sin\theta\, r\, dr\, d\theta$$

where D is the region bounded by the polar axis and the upper half of the cardioid $r = 1 + \cos\theta$.

Solution

We can describe the region D as $\{(r, \theta) \mid 0 \leq \theta \leq \pi,\ 0 \leq r \leq 1 + \cos\theta\}$ as shown in Figure 15.3.6.

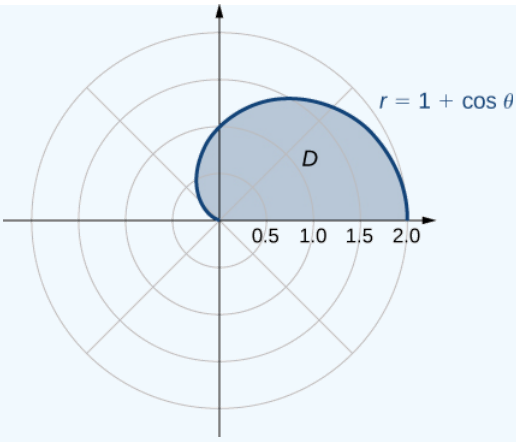


Figure 15.3.6: The region D is the top half of a cardioid.

Hence, we have

$$\begin{aligned}\iint_D r^2 \sin\,\theta\, r\,dr\,d\theta &= \int_{\theta=0}^{\theta=\pi}\int_{r=0}^{r=1+\cos\theta}(r^2\sin\,\theta)\,r\,dr\,d\theta \\ &= \frac{1}{4}\int_{\theta=0}^{\theta=\pi}\left[r^4\right]\Big|_{r=0}^{r=1+\cos\,\theta}\sin\,\theta\,d\theta \\ &= \frac{1}{4}\int_{\theta=0}^{\theta=\pi}(1+\cos\,\theta)^4\sin\,\theta\,d\theta \\ &= -\frac{1}{4}\left[\frac{(1+\cos\,\theta)^5}{5}\right]_0^{\pi} = \frac{8}{5}.\end{aligned}$$

Exercise 15.3.3

Evaluate the integral

$$\iint_D r^2\sin^2 2\theta\, r\,dr\,d\theta$$

where $D = \left\{(r,\theta)\,|\,0\le\theta\le\pi,\,0\le r\le 2\sqrt{\cos\,2\theta}\right\}$.

Hint

Graph the region and follow the steps in the previous example.

Answer

$\frac{\pi}{8}$

Polar Areas and Volumes

As in rectangular coordinates, if a solid S is bounded by the surface $z = f(r,\theta)$, as well as by the surfaces $r = a$, $r = b$, $\theta = \alpha$, and $\theta = \beta$, we can find the volume V of S by double integration, as

$$V = \iint_R f(r,\theta)\,r\,dr\,d\theta = \int_{\theta=\alpha}^{\theta=\beta}\int_{r=a}^{r=b} f(r,\theta)\,r\,dr\,d\theta.$$

If the base of the solid can be described as $D = \{(r,\theta)|\alpha\le\theta\le\beta,\ h_1(\theta)\le r\le h_2(\theta)\}$, then the double integral for the volume becomes

$$V = \iint_D f(r,\theta)\,r\,dr\,d\theta = \int_{\theta=\alpha}^{\theta=\beta}\int_{r=h_1(\theta)}^{r=h_2(\theta)} f(r,\theta)\,r\,dr\,d\theta.$$

We illustrate this idea with some examples.

Example 15.3.4A: Finding a Volume Using a Double Integral

Find the volume of the solid that lies under the paraboloid $z = 1 - x^2 - y^2$ and above the unit circle on the xy-plane (Figure 15.3.7).

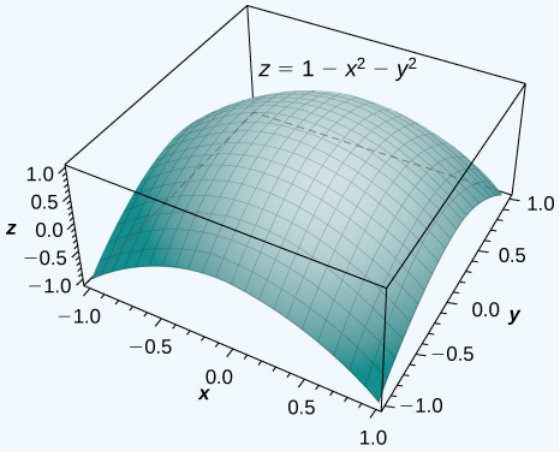


Figure 15.3.7: Finding the volume of a solid under a paraboloid and above the unit circle.

Solution

By the method of double integration, we can see that the volume is the iterated integral of the form

$$\iint_R (1 - x^2 - y^2)\, dA$$

where $R = \{(r, \theta) \mid 0 \le r \le 1,\ 0 \le \theta \le 2\pi\}$.

This integration was shown before in Example 15.3.2A, so the volume is $\frac{\pi}{2}$ cubic units.

Example 15.3.4B: Finding a Volume Using Double Integration

Find the volume of the solid that lies under the paraboloid $z = 4 - x^2 - y^2$ and above the disk $(x - 1)^2 + y^2 = 1$ on the xy-plane. See the paraboloid in Figure 15.3.8 intersecting the cylinder $(x - 1)^2 + y^2 = 1$ above the xy-plane.

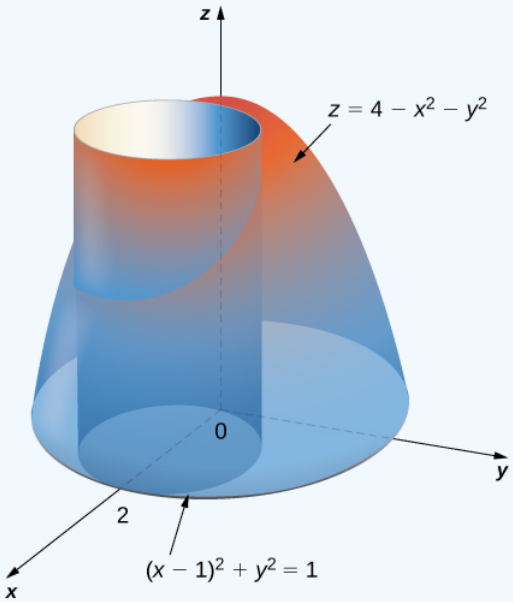


Figure 15.3.8: Finding the volume of a solid with a paraboloid cap and a circular base.

Solution

First change the disk $(x-1)^2+y^2=1$ to polar coordinates. Expanding the square term, we have $x^2-2x+1+y^2=1$. Then simplify to get $x^2+y^2=2x$, which in polar coordinates becomes $r^2=2r\cos\theta$ and then either $r=0$ or $r=2\cos\theta$. Similarly, the equation of the paraboloid changes to $z=4-r^2$. Therefore we can describe the disk $(x-1)^2+y^2=1$ on the xy -plane as the region

$$D=\{(r,\theta)\,|\,0\le\theta\le\pi,\ 0\le r\le 2\cos\theta\}.$$

Hence the volume of the solid bounded above by the paraboloid $z=4-x^2-y^2$ and below by $r=2\cos\theta$ is

$$\begin{aligned}V&=\iint_D f(r,\theta)\,r\,dr\,d\theta\\&=\int_{\theta=0}^{\theta=\pi}\int_{r=0}^{r=2\cos\theta}(4-r^2)\,r\,dr\,d\theta\\&=\int_{\theta=0}^{\theta=\pi}\left[4\frac{r^2}{2}-\frac{r^4}{4}\Big|_0^{2\cos\theta}\right]d\theta\\&=\int_0^{\pi}[8\cos^2\theta-4\cos^4\theta]\,d\theta\\&=\left[\frac{5}{2}\theta+\frac{5}{2}\sin\theta\cos\theta-\sin\theta\cos^3\theta\right]_0^{\pi}=\frac{5}{2}\pi\ \text{units}^3.\end{aligned}$$

Notice in the next example that integration is not always easy with polar coordinates. Complexity of integration depends on the function and also on the region over which we need to perform the integration. If the region has a more natural expression in polar coordinates or if f has a simpler antiderivative in polar coordinates, then the change in polar coordinates is appropriate; otherwise, use rectangular coordinates.

Example 15.3.5A: Finding a Volume Using a Double Integral

Find the volume of the region that lies under the paraboloid $z=x^2+y^2$ and above the triangle enclosed by the lines $y=x$, $x=0$, and $x+y=2$ in the xy-plane.

Solution

First examine the region over which we need to set up the double integral and the accompanying paraboloid.

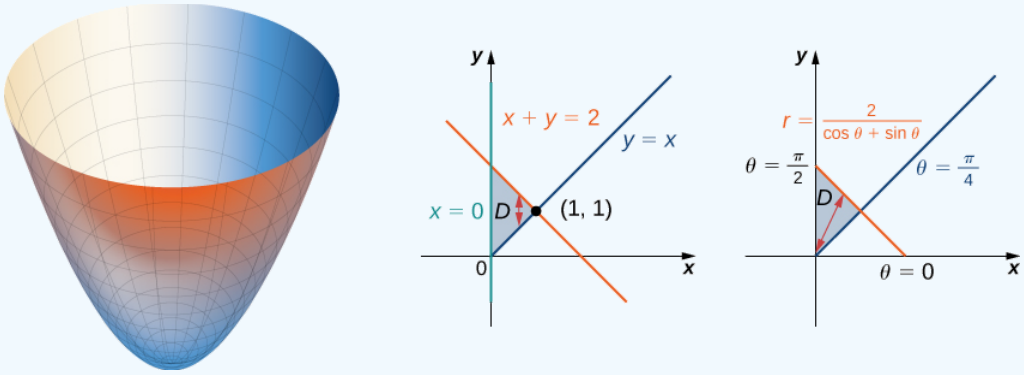


Figure 15.3.9: Finding the volume of a solid under a paraboloid and above a given triangle.

The region D is $\{(x,y)\,|\,0\le x\le 1,\ x\le y\le 2-x\}$. Converting the lines $y=x$, $x=0$, and $x+y=2$ in the xy-plane to functions of r and θ we have $\theta=\pi/4$, $\theta=\pi/2$, and $r=2/(\cos\theta+\sin\theta)$, respectively. Graphing the region on the xy- plane, we see that it looks like $D=\{(r,\theta)\,|\,\pi/4\le\theta\le\pi/2,\ 0\le r\le 2/(\cos\theta+\sin\theta)\}$.

Now converting the equation of the surface gives $z=x^2+y^2=r^2$. Therefore, the volume of the solid is given by the double integral

$$\begin{aligned} V &= \iint_D f(r,\theta)\, r\, dr\, d\theta \\ &= \int_{\theta=\pi/4}^{\theta=\pi/2} \int_{r=0}^{r=2/(\cos\,\theta+\sin\,\theta)} r^2 r\, dr d\theta \\ &= \int_{\pi/4}^{\pi/2} \left[\frac{r^4}{4}\right]_0^{2/(\cos\,\theta+\sin\,\theta)} d\theta \\ &= \frac{1}{4}\int_{\pi/4}^{\pi/2} \left(\frac{2}{\cos\,\theta+\sin\,\theta}\right)^4 d\theta \\ &= \frac{16}{4}\int_{\pi/4}^{\pi/2} \left(\frac{1}{\cos\,\theta+\sin\,\theta}\right)^4 d\theta \\ &= 4\int_{\pi/4}^{\pi/2} \left(\frac{1}{\cos\,\theta+\sin\,\theta}\right)^4 d\theta. \end{aligned}$$

As you can see, this integral is very complicated. So, we can instead evaluate this double integral in rectangular coordinates as

$$V = \int_0^1 \int_x^{2-x} (x^2+y^2)\, dy\, dx.$$

Evaluating gives

$$\begin{aligned} V &= \int_0^1 \int_x^{2-x} (x^2+y^2)\, dy\, dx \\ &= \int_0^1 \left[x^2 y + \frac{y^3}{3}\right]\Big|_x^{2-x} dx \\ &= \int_0^1 \frac{8}{3} - 4x + 4x^2 - \frac{8x^3}{3}\, dx \\ &= \left[\frac{8x}{3} - 2x^2 + \frac{4x^3}{3} - \frac{2x^4}{3}\right]\Big|_0^1 \\ &= \frac{4}{3}\ \text{units}^3. \end{aligned}$$

To answer the question of how the formulas for the volumes of different standard solids such as a sphere, a cone, or a cylinder are found, we want to demonstrate an example and find the volume of an arbitrary cone.

Example 15.3.5B: Finding a Volume Using a Double Integral

Use polar coordinates to find the volume inside the cone $z = 2 - \sqrt{x^2+y^2}$ and above the xy-plane.

Solution

The region D for the integration is the base of the cone, which appears to be a circle on the xy-plane (Figure 15.3.10).

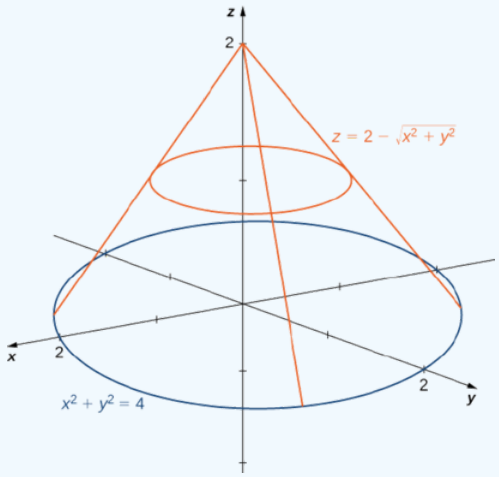


Figure 15.3.10: Finding the volume of a solid inside the cone and above the xy-plane.

We find the equation of the circle by setting $z = 0$:

$$\begin{aligned} 0 &= 2 - \sqrt{x^2 + y^2} \\ 2 &= \sqrt{x^2 + y^2} \\ x^2 + y^2 &= 4. \end{aligned}$$

This means the radius of the circle is 2 so for the integration we have $0 \le \theta \le 2\pi$ and $0 \le r \le 2$. Substituting $x = r\ \cos\theta$ and $y = r\ \sin\ \theta$ in the equation $z = 2 - \sqrt{x^2 + y^2}$ we have $z = 2 - r$. Therefore, the volume of the cone is

$$\int_{\theta=0}^{\theta=2\pi} \int_{r=0}^{r=2} (2 - r)\, r\, dr\, d\theta = 2\pi \frac{4}{3} = \frac{8\pi}{3} \text{ cubic units.}$$

Analysis

Note that if we were to find the volume of an arbitrary cone with radius α units and height h units, then the equation of the cone would be $z = h - \frac{h}{a}\sqrt{x^2 + y^2}$.

We can still use Figure 15.3.10 and set up the integral as

$$\int_{\theta=0}^{\theta=2\pi} \int_{r=0}^{r=a} \left(h - \frac{h}{a} r\right) r\, dr\, d\theta.$$

Evaluating the integral, we get $\frac{1}{3}\pi a^2 h$.

Exercise 15.3.5

Use polar coordinates to find an iterated integral for finding the volume of the solid enclosed by the paraboloids $z = x^2 + y^2$ and $z = 16 - x^2 - y^2$.

Hint

Sketching the graphs can help.

Answer

$$V = \int_0^{2\pi} \int_0^{2\sqrt{2}} (16 - 2r^2)\, r\, dr\, d\theta = 64\pi \text{ cubic units.}$$

As with rectangular coordinates, we can also use polar coordinates to find areas of certain regions using a double integral. As before, we need to understand the region whose area we want to compute. Sketching a graph and identifying the region can be helpful to realize the limits of integration. Generally, the area formula in double integration will look like

$$\text{Area of } A = \int_\alpha^\beta \int_{h_1(\theta)}^{h_2(\theta)} 1\, r\, dr\, d\theta.$$

Example 15.3.6A: Finding an Area Using a Double Integral in Polar Coordinates

Evaluate the area bounded by the curve $r = \cos\ 4\theta$.

Solution

Sketching the graph of the function $r = \cos\ 4\theta$ reveals that it is a polar rose with eight petals (see the following figure).

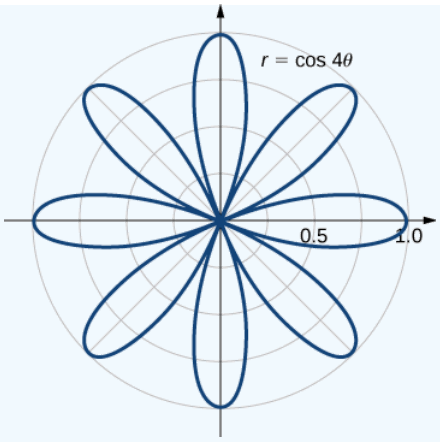


Figure 15.3.11: Finding the area of a polar rose with eight petals.

Using symmetry, we can see that we need to find the area of one petal and then multiply it by 8. Notice that the values of θ for which the graph passes through the origin are the zeros of the function $\cos\ 4\theta$, and these are odd multiples of $\pi/8$. Thus, one of the petals corresponds to the values of θ in the interval $[-\pi/8, \pi/8]$. Therefore, the area bounded by the curve $r = \cos\ 4\theta$ is

$$\begin{aligned} A &= 8\int_{\theta=-\pi/8}^{\theta=\pi/8}\int_{r=0}^{r=\cos\ 4\theta} 1\, r\, dr\, d\theta \\ &= 8\int_{\theta=-\pi/8}^{\theta=\pi/8}\left[\frac{1}{2}r^2\Big|_0^{\cos\ 4\theta}\right] d\theta \\ &= 8\int_{-\pi/8}^{\pi/8}\frac{1}{2}\cos^2 4\theta\, d\theta \\ &= 8\left[\frac{1}{4}\theta + \frac{1}{16}\sin\ 4\theta\ \cos\ 4\theta\Big|_{-\pi/8}^{\pi/8}\right] \\ &= 8\left[\frac{\pi}{16}\right] = \frac{\pi}{2}\ \text{units}^2. \end{aligned}$$

Example 15.3.6B: Finding Area Between Two Polar Curves

Find the area enclosed by the circle $r = 3\ \cos\ \theta$ and the cardioid $r = 1 + \cos\ \theta$.

Solution

First and foremost, sketch the graphs of the region (Figure 15.3.12).

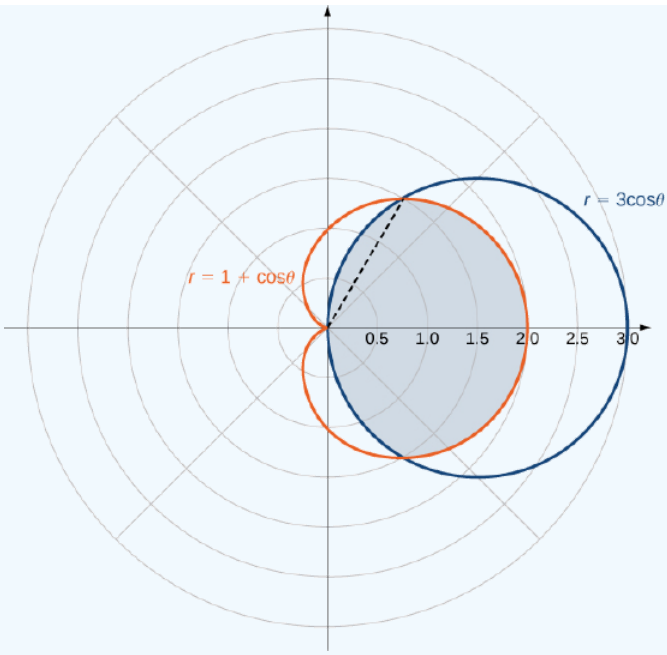


Figure 15.3.12: Finding the area enclosed by both a circle and a cardioid.

We can from see the symmetry of the graph that we need to find the points of intersection. Setting the two equations equal to each other gives

$$3\cos\theta = 1 + \cos\theta.$$

One of the points of intersection is $\theta = \pi/3$. The area above the polar axis consists of two parts, with one part defined by the cardioid from $\theta = 0$ to $\theta = \pi/3$ and the other part defined by the circle from $\theta = \pi/3$ to $\theta = \pi/2$. By symmetry, the total area is twice the area above the polar axis. Thus, we have

$$A = 2\left[\int_{\theta=0}^{\theta=\pi/3}\int_{r=0}^{r=1+\cos\theta} 1\,r\,dr\,d\theta + \int_{\theta=\pi/3}^{\theta=\pi/2}\int_{r=0}^{r=3\cos\theta} 1\,r\,dr\,d\theta\right].$$

Evaluating each piece separately, we find that the area is

$$A = 2\left(\frac{1}{4}\pi + \frac{9}{16}\sqrt{3} + \frac{3}{8}\pi - \frac{9}{16}\sqrt{3}\right) = 2\left(\frac{5}{8}\pi\right) = \frac{5}{4}\pi \text{ square units.}$$

? Exercise 15.3.6

Find the area enclosed inside the cardioid $r = 3 - 3\sin\theta$ and outside the cardioid $r = 1 + \sin\theta$.

Hint

Sketch the graph, and solve for the points of intersection.

Answer

$$A = 2\int_{-\pi/2}^{\pi/6}\int_{1+\sin\theta}^{3-3\sin\theta} r\,dr\,d\theta = \left(8\pi + 9\sqrt{3}\right) \text{ units}^2$$

✓ Example 15.3.7: Evaluating an Improper Double Integral in Polar Coordinates

Evaluate the integral

$$\iint_{R^2} e^{-10(x^2+y^2)}\,dx\,dy.$$

Solution

This is an improper integral because we are integrating over an unbounded region R^2. In polar coordinates, the entire plane R^2 can be seen as $0 \le \theta \le 2\pi$, $0 \le r \le \infty$.

Using the changes of variables from rectangular coordinates to polar coordinates, we have

$$\begin{aligned}\iint_{R^2} e^{-10(x^2+y^2)}\,dx\,dy &= \int_{\theta=0}^{\theta=2\pi}\int_{r=0}^{r=\infty} e^{-10r^2}\,r\,dr\,d\theta = \int_{\theta=0}^{\theta=2\pi}\left(\lim_{a\to\infty}\int_{r=0}^{r=a} e^{-10r^2}r\,dr\right)d\theta \\ &= \left(\int_{\theta=0}^{\theta=2\pi}\right)d\theta\left(\lim_{a\to\infty}\int_{r=0}^{r=a} e^{-10r^2}r\,dr\right) \\ &= 2\pi\left(\lim_{a\to\infty}\int_{r=0}^{r=a} e^{-10r^2}r\,dr\right) \\ &= 2\pi\lim_{a\to\infty}\left(-\frac{1}{20}\right)\left(e^{-10r^2}\Big|_0^a\right) \\ &= 2\pi\left(-\frac{1}{20}\right)\lim_{a\to\infty}\left(e^{-10a^2}-1\right) \\ &= \frac{\pi}{10}.\end{aligned}$$

? Exercise 15.3.7

Evaluate the integral

$$\iint_{R^2} e^{-4(x^2+y^2)}dx\,dy.$$

Hint

Convert to the polar coordinate system.

Answer

$\frac{\pi}{4}$

Key Concepts

- To apply a double integral to a situation with circular symmetry, it is often convenient to use a double integral in polar coordinates. We can apply these double integrals over a polar rectangular region or a general polar region, using an iterated integral similar to those used with rectangular double integrals.
- The area dA in polar coordinates becomes $r\,dr\,d\theta$.
- Use $x = r\cos\theta$, $y = r\sin\theta$, and $dA = r\,dr\,d\theta$ to convert an integral in rectangular coordinates to an integral in polar coordinates.
- Use $r^2 = x^2 + y^2$ and $\theta = tan^{-1}\left(\frac{y}{x}\right)$ to convert an integral in polar coordinates to an integral in rectangular coordinates, if needed.
- To find the volume in polar coordinates bounded above by a surface $z = f(r,\theta)$ over a region on the xy-plane, use a double integral in polar coordinates.

Key Equations

- **Double integral over a polar rectangular region** R

$$\iint_R f(r,\theta)dA = \lim_{m,n\to\infty}\sum_{i=1}^{m}\sum_{j=1}^{n} f(r_{ij}^*,\theta_{ij}^*)\Delta A = \lim_{m,n\to\infty}\sum_{i=1}^{m}\sum_{j=1}^{n} f(r_{ij}^*,\theta_{ij}^*)r_{ij}^*\Delta r\Delta\theta$$

- **Double integral over a general polar region**

$$\iint_D f(r,\theta)\,r\,dr\,d\theta = \int_{\theta=\alpha}^{\theta=\beta}\int_{r=h_1(\theta)}^{r_2(\theta)} f(r,\theta)\,r\,dr\,d\theta$$

Glossary

polar rectangle

the region enclosed between the circles $r = a$ and $r = b$ and the angles $\theta = \alpha$ and $\theta = \beta$; it is described as $R = \{(r, \theta) \mid a \leq r \leq b, \alpha \leq \theta \leq \beta\}$

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15.3E: EXERCISES FOR SECTION 15.3

TERMS AND CONCEPTS

1. When evaluating $\int\int_R f(x,y)\,dA$ using polar coordinates, $f(x,y)$ is replaced with _____ and dA is replaced with _____.

Answer

$f(x,y)$ is replaced with $f(r\cos\theta, r\sin\theta)$ and dA is replaced with $r\,dr\,d\theta$.

2. Why would one be interested in evaluating a double integral with polar coordinates?

DEFINING POLAR REGIONS

In exercises 3 - 6, express the region R in polar coordinates.

3) R is the region of the disk of radius 2 centered at the origin that lies in the first quadrant.

Answer

$R = \{(r,\theta)\,|\,0 \le r \le 2,\ 0 \le \theta \le \frac{\pi}{2}\}$

4) R is the region of the disk of radius 3 centered at the origin.

5) R is the region between the circles of radius 4 and radius 5 centered at the origin that lies in the second quadrant.

Answer

$R = \{(r,\theta)\,|\,4 \le r \le 5,\ \frac{\pi}{2} \le \theta \le \pi\}$

6) R is the region bounded by the y-axis and $x = \sqrt{1-y^2}$.

7) R is the region bounded by the x-axis and $y = \sqrt{2-x^2}$.

Answer

$R = \{(r,\theta)\,|\,0 \le r \le \sqrt{2},\ 0 \le \theta \le \pi\}$

8) $R = \{(x,y)\,|\,x^2+y^2 \le 4x\}$

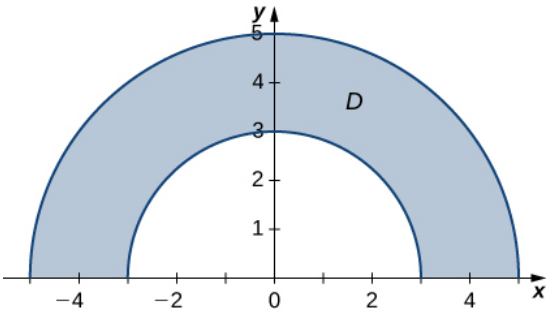
9) $R = \{(x,y)\,|\,x^2+y^2 \le 4y\}$

Answer

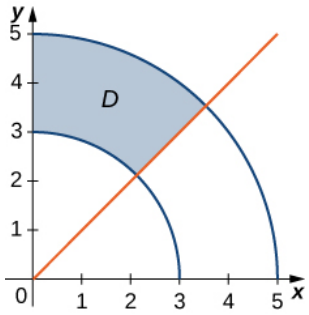
$R = \{(r,\theta)\,|\,0 \le r \le 4\sin\theta,\ 0 \le \theta \le \pi\}$

In exercises 10 - 15, the graph of the polar rectangular region D is given. Express D in polar coordinates. In exercises 10 - 13 the regions are bounded by circles centered at the origin, the coordinate axes, and/or lines of the form $y = \pm x$ (each of which should be able to be identified by inspection).

10)



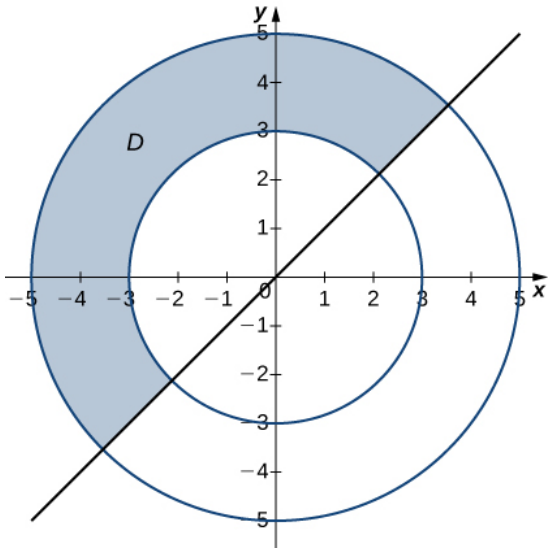
11)



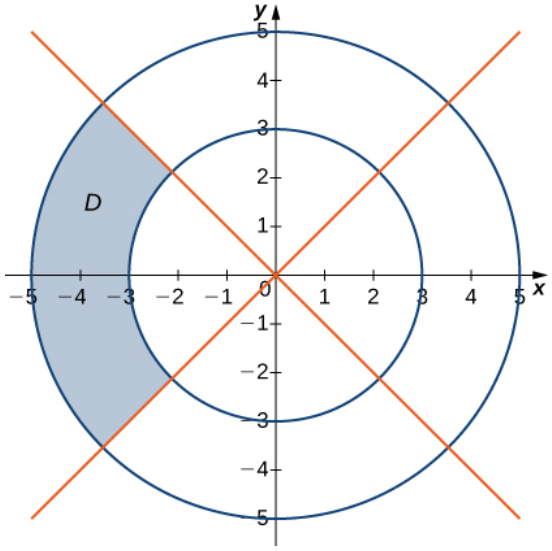
Answer

$D = \{(r, \theta) \mid 3 \leq r \leq 5,\ \frac{\pi}{4} \leq \theta \leq \frac{\pi}{2}\}$

12)



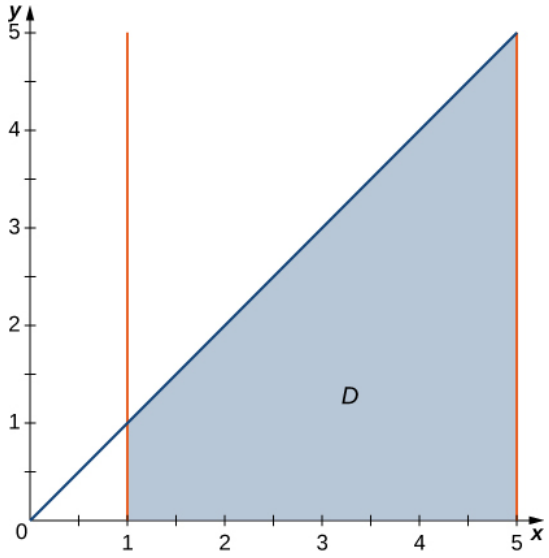
13)



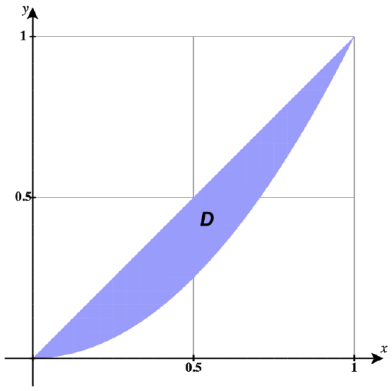
Answer

$D = \{(r, \theta) \mid 3 \leq r \leq 5,\ \frac{3\pi}{4} \leq \theta \leq \frac{5\pi}{4}\}$

14) In the following graph, the region D is situated below $y = x$ and is bounded by $x = 1,\ x = 5$, and $y = 0$.



15) In the following graph, the region D is bounded by $y = x$ and $y = x^2$.



Answer

$D = \{(r, \theta) \mid 0 \le r \le \tan\theta \sec\theta,\ 0 \le \theta \le \frac{\pi}{4}\}$

EVALUATING POLAR DOUBLE INTEGRALS

In exercises 16 - 25, evaluate the double integral $\iint_R f(x, y)\, dA$ over the polar rectangular region R.

16) $f(x, y) = x^2 + y^2$, $R = \{(r, \theta) \mid 3 \le r \le 5,\ 0 \le \theta \le 2\pi\}$

17) $f(x, y) = x + y$, $R = \{(r, \theta) \mid 3 \le r \le 5,\ 0 \le \theta \le 2\pi\}$

Answer

0

18) $f(x, y) = x^2 + xy$, $R = \{(r, \theta) \mid 1 \le r \le 2,\ \pi \le \theta \le 2\pi\}$

19) $f(x, y) = x^4 + y^4$, $R = \{(r, \theta) \mid 1 \le r \le 2,\ \frac{3\pi}{2} \le \theta \le 2\pi\}$

Answer

$\frac{63\pi}{16}$

20) $f(x, y) = \sqrt[3]{x^2 + y^2}$, where $R = \{(r, \theta) \mid 0 \le r \le 1,\ \frac{\pi}{2} \le \theta \le \pi\}$.

21) $f(x, y) = x^4 + 2x^2y^2 + y^4$, where $R = \{(r, \theta) \mid 3 \le r \le 4,\ \frac{\pi}{3} \le \theta \le \frac{2\pi}{3}\}$.

Answer

$\frac{3367\pi}{18}$

22) $f(x,y) = \sin(\arctan\frac{y}{x})$, where $R = \{(r,\theta) \mid 1 \le r \le 2,\ \frac{\pi}{6} \le \theta \le \frac{\pi}{3}\}$

23) $f(x,y) = \arctan\left(\frac{y}{x}\right)$, where $R = \{(r,\theta) \mid 2 \le r \le 3,\ \frac{\pi}{4} \le \theta \le \frac{\pi}{3}\}$

Answer

$\frac{35\pi^2}{576}$

24) $\iint_R e^{x^2+y^2}\left[1 + 2\ \arctan\left(\frac{y}{x}\right)\right] dA,\ R = \{(r,\theta) \mid 1 \le r \le 2,\ \frac{\pi}{6} \le \theta \frac{\pi}{3}\}$

25) $\iint_R \left(e^{x^2+y^2} + x^4 + 2x^2y^2 + y^4\right) \arctan\left(\frac{y}{x}\right) dA,\ R = \{(r,\theta) \mid 1 \le r \le 2,\ \frac{\pi}{4} \le \theta \le \frac{\pi}{3}\}$

Answer

$\frac{7}{576}\pi^2(21 - e + e^4)$

CONVERTING DOUBLE INTEGRALS TO POLAR FORM

In exercises 26 - 29, the integrals have been converted to polar coordinates. Verify that the identities are true and choose the easiest way to evaluate the integrals, in rectangular or polar coordinates.

26) $\int_1^2 \int_0^x (x^2 + y^2)\, dy\, dx = \int_0^{\frac{\pi}{4}} \int_{\sec\theta}^{2\sec\theta} r^3\, dr\, d\theta$

27) $\int_2^3 \int_0^x \frac{x}{\sqrt{x^2+y^2}}\, dy\, dx = \int_0^{\pi/4} \int_{2\sec\theta}^{3\sec\theta} r\ \cos\theta\, dr\, d\theta$

Answer

$\frac{5}{2}\ln(1+\sqrt{2})$

28) $\int_0^1 \int_{x^2}^x \frac{1}{\sqrt{x^2+y^2}}\, dy\, dx = \int_0^{\pi/4} \int_0^{\tan\theta\ \sec\theta} dr\, d\theta$

29) $\int_0^1 \int_{x^2}^x \frac{y}{\sqrt{x^2+y^2}}\, dy\, dx = \int_0^{\pi/4} \int_0^{\tan\theta\ \sec\theta} r\ \sin\theta\, dr\, d\theta$

Answer

$\frac{1}{6}(2-\sqrt{2})$

In exercises 30 - 37, draw the region of integration, R, labeling all limits of integration, convert the integrals to polar coordinates and evaluate them.

30) $\int_0^3 \int_0^{\sqrt{9-y^2}} (x^2 + y^2)\ dx\ dy$

31) $\int_0^2 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} (x^2 + y^2)^2\ dx\ dy$

Answer

$\int_0^{\pi} \int_0^2 r^5\, dr\, d\theta \quad = \quad \frac{32\pi}{3}$

32) $\int_0^1 \int_0^{\sqrt{1-x^2}} (x + y)\ dy\ dx$

33) $\int_0^4 \int_{-\sqrt{16-x^2}}^{\sqrt{16-x^2}} \sin(x^2 + y^2)\ dy\ dx$

Answer

$\int_{-\pi/2}^{\pi/2} \int_0^4 r\ \sin(r^2)\ dr\, d\theta \quad = \quad \pi\ \sin^2 8$

34) $\int_0^5 \int_{-\sqrt{25-x^2}}^{\sqrt{25-x^2}} \sqrt{x^2+y^2}\, dy\, dx$

35) $\int_{-4}^4 \int_{-\sqrt{16-y^2}}^0 (2y - x)\, dx\, dy$

Answer

$$\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \int_0^4 \left(2r\sin\theta - r\cos\theta\right) r\,dr\,d\theta \quad = \quad \frac{128}{3}$$

36) $\int_0^2 \int_y^{\sqrt{8-y^2}} (x+y)\,dx\,dy$

37) $\int_{-2}^{-1} \int_0^{\sqrt{4-x^2}} (x+5)\,dy\,dx + \int_{-1}^{1} \int_{\sqrt{1-x^2}}^{\sqrt{4-x^2}} (x+5)\,dy\,dx + \int_1^2 \int_0^{\sqrt{4-x^2}} (x+5)\,dy\,dx$

Answer

$$\int_0^{\pi} \int_1^2 \left(r\cos\theta + 5\right) r\,dr\,d\theta \quad = \quad \frac{15\pi}{2}$$

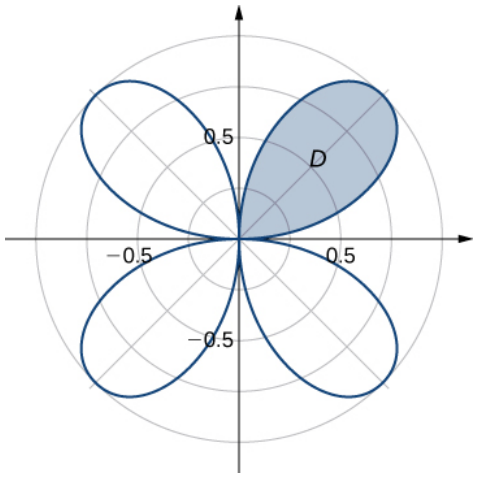
38) Evaluate the integral $\iint_D r\,dA$ where D is the region bounded by the polar axis and the upper half of the cardioid $r = 1 + \cos\theta$.

39) Find the area of the region D bounded by the polar axis and the upper half of the cardioid $r = 1 + \cos\theta$.

Answer

$\frac{3\pi}{4}$

40) Evaluate the integral $\iint_D r\,dA$, where D is the region bounded by the part of the four-leaved rose $r = \sin 2\theta$ situated in the first quadrant (see the following figure).



41) Find the total area of the region enclosed by the four-leaved rose $r = \sin 2\theta$ (see the figure in the previous exercise).

Answer

$\frac{\pi}{2}$

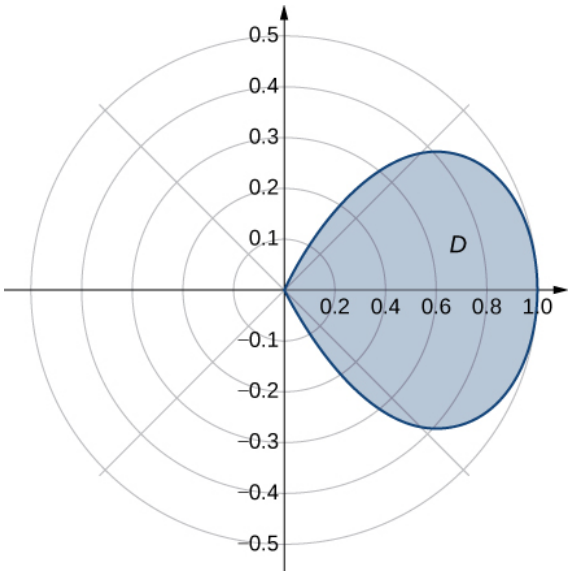
42) Find the area of the region D which is the region bounded by $y = \sqrt{4 - x^2}$, $x = \sqrt{3}$, $x = 2$, and $y = 0$.

43) Find the area of the region D, which is the region inside the disk $x^2 + y^2 \leq 4$ and to the right of the line $x = 1$.

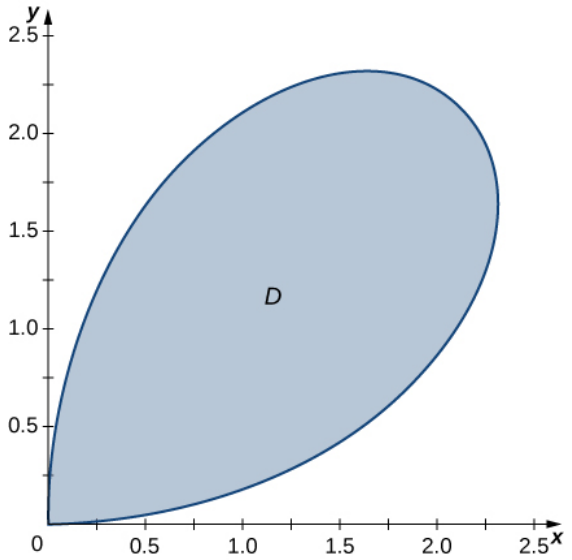
Answer

$\frac{1}{3}(4\pi - 3\sqrt{3})$

44) Determine the average value of the function $f(x, y) = x^2 + y^2$ over the region D bounded by the polar curve $r = \cos 2\theta$, where $-\frac{\pi}{4} \leq \theta \leq \frac{\pi}{4}$ (see the following graph).



45) Determine the average value of the function $f(x,y) = \sqrt{x^2+y^2}$ over the region D bounded by the polar curve $r = 3\sin 2\theta$, where $0 \le \theta \le \frac{\pi}{2}$ (see the following graph).



Answer

$\frac{16}{3\pi}$

46) Find the volume of the solid situated in the first octant and bounded by the paraboloid $z = 1 - 4x^2 - 4y^2$ and the planes $x = 0,\ y = 0$, and $z = 0$.

47) Find the volume of the solid bounded by the paraboloid $z = 2 - 9x^2 - 9y^2$ and the plane $z = 1$.

Answer

$\frac{\pi}{18}$

48)

1. Find the volume of the solid S_1 bounded by the cylinder $x^2 + y^2 = 1$ and the planes $z = 0$ and $z = 1$.
2. Find the volume of the solid S_2 outside the double cone $z^2 = x^2 + y^2$ inside the cylinder $x^2 + y^2 = 1$, and above the plane $z = 0$.
3. Find the volume of the solid inside the cone $z^2 = x^2 + y^2$ and below the plane $z = 1$ by subtracting the volumes of the solids S_1 and S_2.

49)

1. Find the volume of the solid S_1 inside the unit sphere $x^2 + y^2 + z^2 = 1$ and above the plane $z = 0$.

2. Find the volume of the solid S_2 inside the double cone $(z-1)^2 = x^2 + y^2$ and above the plane $z = 0$.
3. Find the volume of the solid outside the double cone $(z-1)^2 = x^2 + y^2$ and inside the sphere $x^2 + y^2 + z^2 = 1$.

Answer

a. $\frac{2\pi}{3}$; b. $\frac{\pi}{2}$; c. $\frac{\pi}{6}$

In Exercises 50-51, special double integrals are presented that are especially well suited for evaluation in polar coordinates.

50) The surface of a right circular cone with height h and base radius a can be described by the equation $f(x,y) = h - h\sqrt{\frac{x^2}{a^2} + \frac{y^2}{a^2}}$, where the tip of the cone lies at $(0,0,h)$ and the circular base lies in the xy-plane, centered at the origin.
Confirm that the volume of a right circular cone with height h and base radius a is $V = \frac{1}{3}\pi a^2 h$ by evaluating $\int\int_R f(x,y)\, dA$ in polar coordinates.

51) Consider $\int\int_R e^{-(x^2+y^2)}\, dA$.
(a) Why is this integral difficult to evaluate in rectangular coordinates, regardless of the region R?
(b) Let R be the region bounded by the circle of radius a centered at the origin. Evaluate the double integral using polar coordinates.
(c) Take the limit of your answer from (b), as $a \to \infty$. What does this imply about the volume under the surface of $e^{-(x^2+y^2)}$ over the entire xy -plane?

For the following two exercises, consider a spherical ring, which is a sphere with a cylindrical hole cut so that the axis of the cylinder passes through the center of the sphere (see the following figure).



52) If the sphere has radius 4 and the cylinder has radius 2 find the volume of the spherical ring.

53) A cylindrical hole of diameter 6 cm is bored through a sphere of radius 5 cm such that the axis of the cylinder passes through the center of the sphere. Find the volume of the resulting spherical ring.

Answer

$\frac{256\pi}{3}$ cm^3

54) Find the volume of the solid that lies under the double cone $z^2 = 4x^2 + 4y^2$, inside the cylinder $x^2 + y^2 = x$, and above the plane $z = 0$.

55) Find the volume of the solid that lies under the paraboloid $z = x^2 + y^2$, inside the cylinder $x^2 + y^2 = 1$ and above the plane $z = 0$.

Answer

$\frac{3\pi}{32}$

56) Find the volume of the solid that lies under the plane $x + y + z = 10$ and above the disk $x^2 + y^2 = 4x$.

57) Find the volume of the solid that lies under the plane $2x + y + 2z = 8$ and above the unit disk $x^2 + y^2 = 1$.

Answer

4π

58) A radial function f is a function whose value at each point depends only on the distance between that point and the origin of the system of coordinates; that is, $f(x,y) = g(r)$, where $r = \sqrt{x^2 + y^2}$. Show that if f is a continuous radial function, then

$$\iint_D f(x,y)dA = (\theta_2 - \theta_1)[G(R_2) - G(R_1)], \text{ where } G'(r) = rg(r)$$

and $(x,y) \in D = \{(r,\theta) \mid R_1 \le r \le R_2,\ 0 \le \theta \le 2\pi\}$, with $0 \le R_1 < R_2$ and $0 \le \theta_1 < \theta_2 \le 2\pi$.

59) Use the information from the preceding exercise to calculate the integral $\iint_D (x^2+y^2)^3 dA$, where D is the unit disk.

Answer

$\frac{\pi}{4}$

60) Let $f(x,y) = \dfrac{F'(r)}{r}$ be a continuous radial function defined on the annular region $D = \{(r,\theta) \mid R_1 \le r \le R_2,\ 0 \le \theta \le 2\pi\}$, where $r = \sqrt{x^2+y^2}$, $0 < R_1 < R_2$, and F is a differentiable function.

Show that $\iint_D f(x,y)\,dA = 2\pi[F(R_2) - F(R_1)]$.

61) Apply the preceding exercise to calculate the integral $\iint_D \dfrac{e^{\sqrt{x^2+y^2}}}{\sqrt{x^2+y^2}}\,dx\,dy$ where D is the annular region between the circles of radii 1 and 2 situated in the third quadrant.

Answer

$\frac{1}{2}\pi e(e-1)$

62) Let f be a continuous function that can be expressed in polar coordinates as a function of θ only; that is, $f(x,y) = h(\theta)$, where $(x,y) \in D = \{(r,\theta) \mid R_1 \le r \le R_2,\ \theta_1 \le \theta \le \theta_2\}$, with $0 \le R_1 < R_2$ and $0 \le \theta_1 < \theta_2 \le 2\pi$.

Show that $\iint_D f(x,y)\,dA = \dfrac{1}{2}(R_2^2 - R_1^2)[H(\theta_2) - H(\theta_1)]$, where H is an antiderivative of h.

63) Apply the preceding exercise to calculate the integral $\iint_D \dfrac{y^2}{x^2}\,dA$, where $D = \{(r,\theta) \mid 1 \le r \le 2,\ \frac{\pi}{6} \le \theta \le \frac{\pi}{3}\}$.

Answer

$\sqrt{3} - \frac{\pi}{4}$

64) Let f be a continuous function that can be expressed in polar coordinates as a function of θ only; that is $f(x,y) = g(r)h(\theta)$, where $(x,y) \in \{(r,\theta) \mid R_1 \le r \le R_2,\ \theta_1 \le \theta \le \theta_2\}$ with $0 \le R_1 < R_2$ and $0 \le \theta_1 < \theta_2 \le 2\pi$. Show that

$$\iint_D f(x,y)\,dA = [G(R_2) - G(R_1)]\ [H(\theta_2) - H(\theta_1)],$$

where G and H are antiderivatives of g and h, respectively.

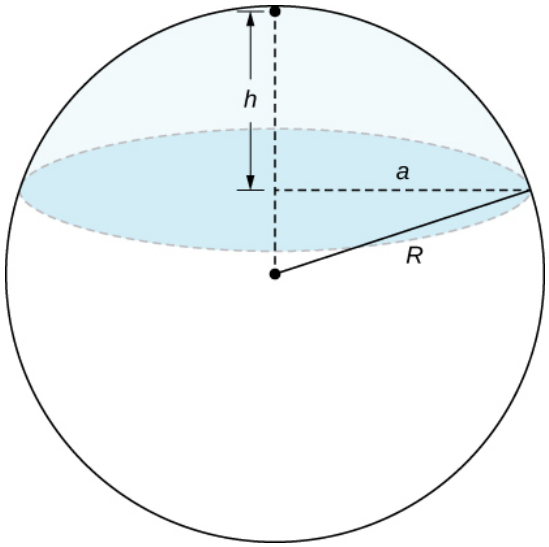
65) Evaluate $\iint_D \arctan\left(\dfrac{y}{x}\right)\sqrt{x^2+y^2}\,dA$, where $D = \{(r,\theta) \mid 2 \le r \le 3,\ \frac{\pi}{4} \le \theta \le \frac{\pi}{3}\}$.

Answer

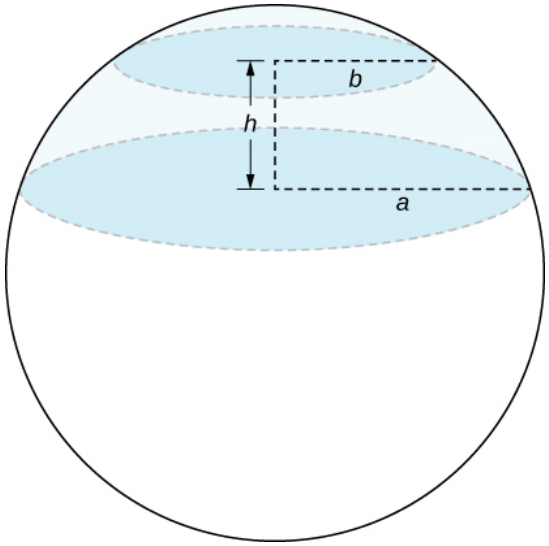
$\frac{133}{864}\pi^2$

66) A spherical cap is the region of a sphere that lies above or below a given plane.

a. Show that the volume of the spherical cap in the figure below is $\frac{1}{6}\pi h(3a^2 + h^2)$.



b. A spherical segment is the solid defined by intersecting a sphere with two parallel planes. If the distance between the planes is h show that the volume of the spherical segment in the figure below is $\frac{1}{6}\pi h(3a^2+3b^2+h^2)$.



67) In statistics, the joint density for two independent, normally distributed events with a mean $\mu=0$ and a standard distribution σ is defined by $p(x,y)=\frac{1}{2\pi\sigma^2}e^{-\frac{x^2+y^2}{2\sigma^2}}$. Consider (X,Y), the Cartesian coordinates of a ball in the resting position after it was released from a position on the z-axis toward the xy-plane. Assume that the coordinates of the ball are independently normally distributed with a mean $\mu=0$ and a standard deviation of σ (in feet). The probability that the ball will stop no more than a feet from the origin is given by

$$P[X^2+Y^2\le a^2]=\iint_D p(x,y)dy\,dx,$$

where D is the disk of radius a centered at the origin. Show that

$$P[X^2+Y^2\le a^2]=1-e^{-a^2/2\sigma^2}.$$

68) The double improper integral

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}e^{-x^2+y^2/2}\,dy\,dx$$

may be defined as the limit value of the double integrals $\iint_D e^{-x^2+y^2/2}\,dA$ over disks D_a of radii a centered at the origin, as a increases without bound; that is,

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}e^{-x^2+y^2/2}dy\,dx=\lim_{a\to\infty}\iint_{D_a}e^{-x^2+y^2/2}\,dA.$$

Use polar coordinates to show that $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-x^2+y^2/2}\,dy\,dx = 2\pi.$

69) Show that $\int_{-\infty}^{\infty} e^{-x^2/2}\,dx = \sqrt{2\pi}$ by using the relation

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-x^2+y^2/2}\,dy\,dx = \left(\int_{-\infty}^{\infty} e^{-x^2/2}dx\right)\left(\int_{-\infty}^{\infty} e^{-y^2/2}dy\right).$$

CONTRIBUTORS

- Gilbert Strang (MIT) and Edwin “Jed” Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.
- Problems 1, 2, 34 - 37 and 50 - 51 are from Apex Calculus, Chapter 13.3
- Edited by Paul Seeburger (Monroe Community College)

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15.4: Triple Integrals

Learning Objectives

- Recognize when a function of three variables is integrable over a rectangular box.
- Evaluate a triple integral by expressing it as an iterated integral.
- Recognize when a function of three variables is integrable over a closed and bounded region.
- Simplify a calculation by changing the order of integration of a triple integral.
- Calculate the average value of a function of three variables.

Previously, we discussed the double integral of a function $f(x, y)$ of two variables over a rectangular region in the plane. In this section we define the triple integral of a function $f(x, y, z)$ of three variables over a rectangular solid box in space, $\mathbb{R}^3$. Later in this section we extend the definition to more general regions in $\mathbb{R}^3$.

Integrable Functions of Three Variables

We can define a rectangular box B in $\mathbb{R}^3$ as

$$B = \{(x, y, z) \,|\, a \leq x \leq b,\, c \leq y \leq d,\, e \leq z \leq f\}.$$

We follow a similar procedure to what we did in previously. We divide the interval $[a, b]$ into l subintervals $[x_{i-1}, x_i]$ of equal length Δx with

$$\Delta x = \frac{x_i - x_{i-1}}{l},$$

divide the interval $[c, d]$ into m subintervals $[y_{i-1}, y_i]$ of equal length Δy with

$$\Delta y = \frac{y_j - y_{j-1}}{m},$$

and divide the interval $[e, f]$ into n subintervals $[z_{i-1}, z_i]$ of equal length Δz with

$$\Delta z = \frac{z_k - z_{k-1}}{n}$$

Then the rectangular box B is subdivided into lmn subboxes:

$$B_{ijk} = [x_{i-1}, x_i] \times [y_{i-1}, y_i] \times [z_{i-1}, z_i],$$

as shown in Figure 15.4.1.

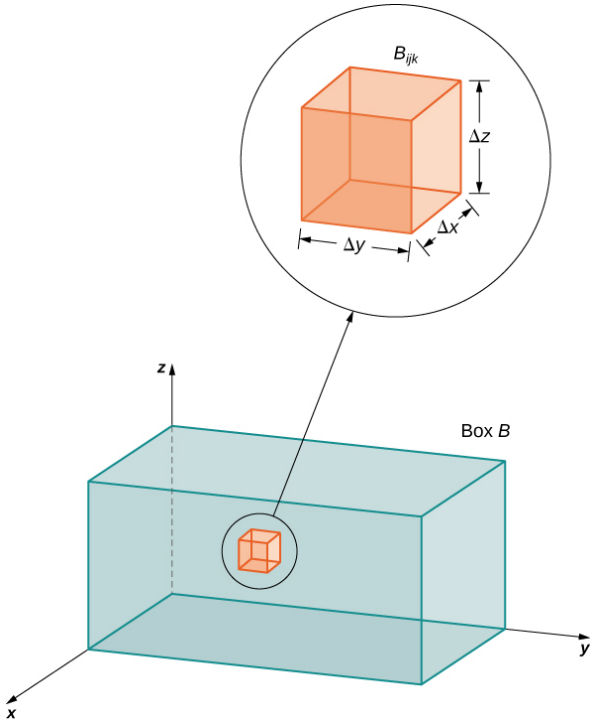


Figure 15.4.1: A rectangular box in $\mathbb{R}^3$ divided into subboxes by planes parallel to the coordinate planes.

For each i, j, and k, consider a sample point $(x^*_{ijk}, y^*_{ijk}, z^*_{ijk})$ in each sub-box B_{ijk}. We see that its volume is $\Delta V = \Delta x \Delta y \Delta z$. Form the triple Riemann sum

$$\sum_{i=1}^{l} \sum_{j=1}^{m} \sum_{k=1}^{n} f(x^*_{ijk}, y^*_{ijk}, z^*_{ijk})\, \Delta x \Delta y \Delta z.$$

We define the triple integral in terms of the limit of a triple Riemann sum, as we did for the double integral in terms of a double Riemann sum.

Definition: The triple integral

The triple integral of a function $f(x, y, z)$ over a rectangular box B is defined as

$$\lim_{l,m,n\to\infty} \sum_{i=1}^{l} \sum_{j=1}^{m} \sum_{k=1}^{n} f(x^*_{ijk}, y^*_{ijk}, z^*_{ijk})\, \Delta x \Delta y \Delta z = \iiint_B f(x, y, z)\, dV$$

if this limit exists.

When the triple integral exists on B the function $f(x, y, z)$ is said to be integrable on B. Also, the triple integral exists if $f(x, y, z)$ is continuous on B. Therefore, we will use continuous functions for our examples. However, continuity is sufficient but not necessary; in other words, f is bounded on B and continuous except possibly on the boundary of B. The sample point $(x^*_{ijk}, y^*_{ijk}, z^*_{ijk})$ can be any point in the rectangular sub-box B_{ijk} and all the properties of a double integral apply to a triple integral. Just as the double integral has many practical applications, the triple integral also has many applications, which we discuss in later sections.

Now that we have developed the concept of the triple integral, we need to know how to compute it. Just as in the case of the double integral, we can have an iterated triple integral, and consequently, a version of **Fubini's theorem** for triple integrals exists.

Fubini's Theorem for Triple Integrals

If $f(x, y, z)$ is continuous on a rectangular box $B = [a, b] \times [c, d] \times [e, f]$, then

$$\iiint_B f(x, y, z)\, dV = \int_e^f \int_c^d \int_a^b f(x, y, z)\, dx\, dy\, dz.$$

This integral is also equal to any of the other five possible orderings for the iterated triple integral.

For a, b, c, d, e and f real numbers, the iterated triple integral can be expressed in six different orderings:

$$\int_e^f \int_c^d \int_a^b f(x,y,z)\,dx\,dy\,dz = \int_e^f \left(\int_c^d \left(\int_a^b f(x,y,z)\,dx \right) dy \right) dz \tag{15.4.1}$$

$$= \int_c^d \left(\int_e^f \left(\int_a^b f(x,y,z)\,dx \right) dz \right) dy \tag{15.4.2}$$

$$= \int_a^b \left(\int_e^f \left(\int_c^d f(x,y,z)\,dy \right) dz \right) dx \tag{15.4.3}$$

$$= \int_e^f \left(\int_a^b \left(\int_c^d f(x,y,z)\,dy \right) dx \right) dz \tag{15.4.4}$$

$$= \int_c^d \left(\int_a^b \left(\int_c^d f(x,y,z)\,dz \right) dx \right) dy \tag{15.4.5}$$

$$= \int_a^b \left(\int_c^d \left(\int_e^f f(x,y,z)\,dz \right) dy \right) dx \tag{15.4.6}$$

For a rectangular box, the order of integration does not make any significant difference in the level of difficulty in computation. We compute triple integrals using Fubini's Theorem rather than using the Riemann sum definition. We follow the order of integration in the same way as we did for double integrals (that is, from inside to outside).

Example 15.4.1: Evaluating a Triple Integral

Evaluate the triple integral

$$\int_{z=0}^{z=1} \int_{y=2}^{y=4} \int_{x=-1}^{x=5} (x + yz^2)\,dx\,dy\,dz.$$

Solution

The order of integration is specified in the problem, so integrate with respect to x first, then y, and then z.

$$\begin{aligned}
&\int_{z=0}^{z=1} \int_{y=2}^{y=4} \int_{x=-1}^{x=5} (x + yz^2)\,dx\,dy\,dz \\
&= \int_{z=0}^{z=1} \int_{y=2}^{y=4} \left[\frac{x^2}{2} + xyz^2 \Big|_{x=-1}^{x=5} \right] dy\,dz && \text{Integrate with respect to } x. \\
&= \int_{z=0}^{z=1} \int_{y=2}^{y=4} \left[12 + 6yz^2 \right] dy\,dz && \text{Evaluate.} \\
&= \int_{z=0}^{z=1} \left[12y + 6\frac{y^2}{2}z^2 \Big|_{y=2}^{y=4} \right] dz && \text{Integrate with respect to } y. \\
&= \int_{z=0}^{z=1} \left[24 + 36z^2 \right] dz && \text{Evaluate.} \\
&= \left[24z + 36\frac{z^3}{3} \right]_{z=0}^{z=1} && \text{Integrate with respect to } z. \\
&= 36. && \text{Evaluate.}
\end{aligned}$$

Example 15.4.2: Evaluating a Triple Integral

Evaluate the triple integral

$$\iiint_B x^2yz\,dV$$

where $B = \{(x,y,z) \mid -2 \le x \le 1,\ 0 \le y \le 3,\ 1 \le z \le 5\}$ as shown in Figure 15.4.2.

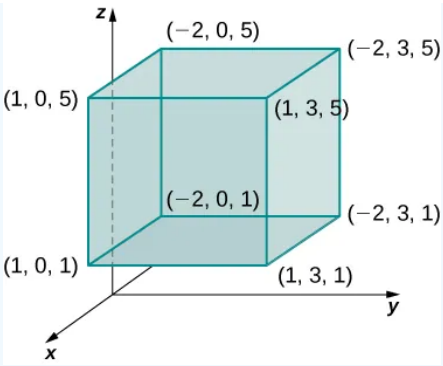


Figure 15.4.2: Evaluating a triple integral over a given rectangular box.

Solution

The order is not specified, but we can use the iterated integral in any order without changing the level of difficulty. Choose, say, to integrate y first, then x, and then z.

$$\begin{aligned}\iiint_B x^2yz\,dV &= \int_1^5\int_{-2}^1\int_0^3 [x^2yz]\,dy\,dx\,dz\\ &= \int_1^5\int_{-2}^1 \left[x^2\frac{y^2}{2}z\Big|_0^3\right]dx\,dz\\ &= \int_1^5\int_{-2}^1 \frac{9}{2}x^2z\,dx\,dz\\ &= \int_1^5 \left[\frac{9}{2}\frac{x^3}{3}z\Big|_{-2}^1\right]dz\\ &= \int_1^5 \frac{27}{2}z\,dz\\ &= \frac{27}{2}\frac{z^2}{2}\Big|_1^5 = \frac{27}{4}(25-1) = 162.\end{aligned}$$

Now try to integrate in a different order just to see that we get the same answer. Choose to integrate with respect to x first, then z, then y

$$\begin{aligned}\iiint_B x^2yz\,dV &= \int_0^3\int_1^5\int_{-2}^1 [x^2yz]\,dx\,dz\,dy\\ &= \int_0^3\int_1^5 \left[\frac{x^3}{3}yz\Big|_{-2}^1\right]dz\,dy\\ &= \int_0^3\int_1^5 3yz\,dz\,dy\\ &= \int_0^3 \left[3y\frac{z^2}{2}\Big|_1^5\right]dy\\ &= \int_0^3 36y\,dy\\ &= 36\frac{y^2}{2}\Big|_0^3 = 18(9-0) = 162.\end{aligned}$$

? Exercise 15.4.1

Evaluate the triple integral

$$\iiint_B z\sin x\cos y\,dV$$

where $B = \{(x,y,z)\,|\,0\le x\le\pi,\ \frac{3\pi}{2}\le y\le 2\pi,\ 1\le z\le 3\}$.

Hint

Follow the steps in the previous example.

Answer

$$\iiint_B z \sin x \cos y \, dV = 8$$

Triple Integrals over a General Bounded Region

We now expand the definition of the triple integral to compute a triple integral over a more **general bounded region** E in $\mathbb{R}^3$. The general bounded regions we will consider are of three types. First, let D be the bounded region that is a projection of E onto the xy-plane. Suppose the region E in $\mathbb{R}^3$ has the form

$$E = \{(x, y, z) \mid (x, y) \in D, u_1(x, y) \leq z \leq u_2(x, y)\}.$$

For two functions $z = u_1(x, y)$ and $u_2(x, y)$, such that $u_1(x, y) \leq u_2(x, y)$ for all (x, y) in D as shown in the following figure.

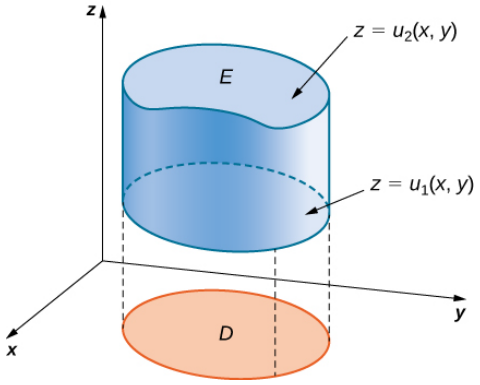


Figure 15.4.3: We can describe region E as the space between $u_1(x, y)$ and $u_2(x, y)$ above the projection D of E onto the xy-plane.

Triple Integral over a General Region

The triple integral of a continuous function $f(x, y, z)$ over a general three-dimensional region

$$E = \{(x, y, z) \mid (x, y) \in D, \, u_1(x, y) \leq z \leq u_2(x, y)\}$$

in $\mathbb{R}^3$, where D is the projection of E onto the xy-plane, is

$$\iiint_E f(x, y, z) \, dV = \iint_D \left[\int_{u_1(x,y)}^{u_2(x,y)} f(x, y, z) \, dz \right] dA.$$

Similarly, we can consider a general bounded region D in the xy-plane and two functions $y = u_1(x, z)$ and $y = u_2(x, z)$ such that $u_1(x, z) \leq u_2(x, z)$ for all (x, z) in D. Then we can describe the solid region E in $\mathbb{R}^3$ as

$$E = \{(x, y, z) \mid (x, z) \in D, \, u_1(x, z) \leq z \leq u_2(x, z)\}$$

where D is the projection of E onto the xy-plane and the triple integral is

$$\iiint_E f(x, y, z) \, dV = \iint_D \left[\int_{u_1(x,z)}^{u_2(x,z)} f(x, y, z) \, dy \right] dA.$$

Finally, if D is a general bounded region in the xy-plane and we have two functions $x = u_1(y, z)$ and $x = u_2(y, z)$ such that $u_1(y, z) \leq u_2(y, z)$ for all (y, z) in D, then the solid region E in $\mathbb{R}^3$ can be described as

$$E = \{(x, y, z) \mid (y, z) \in D, \, u_1(y, z) \leq z \leq u_2(y, z)\}$$

where D is the projection of E onto the xy-plane and the triple integral is

$$\iiint_E f(x, y, z) \, dV = \iint_D \left[\int_{u_1(y,z)}^{u_2(y,z)} f(x, y, z) \, dx \right] dA.$$

Note that the region D in any of the planes may be of Type I or Type II as described in previously. If D in the xy-plane is of Type I (Figure 15.4.4), then

$$E = \{(x, y, z) \mid a \le x \le b,\ g_1(x) \le y \le g_2(x),\ u_1(x, y) \le z \le u_2(x, y)\}.$$

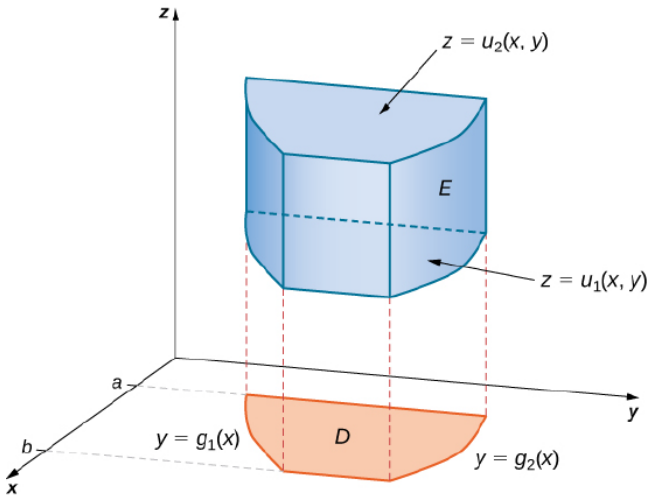


Figure 15.4.4: A box E where the projection D in the xy-plane is of Type I.

Then the triple integral becomes

$$\iiint_E f(x, y, z)\, dV = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{u_1(x,y)}^{u_2(x,y)} f(x, y, z)\, dz\, dy\, dx.$$

If D in the xy-plane is of Type II (Figure 15.4.5), then

$$E = \{(x, y, z) \mid c \le x \le d, h_1(x) \le y \le h_2(x),\ u_1(x, y) \le z \le u_2(x, y)\}.$$

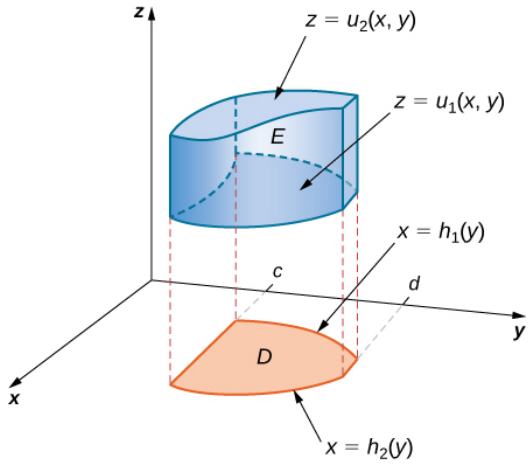


Figure 15.4.5: A box E where the projection D in the xy-plane is of Type II.

Then the triple integral becomes

$$\iiint_E f(x, y, z)\, dV = \int_{y=c}^{y=d} \int_{x=h_1(y)}^{x=h_2(y)} \int_{z=u_1(x,y)}^{z=u_2(x,y)} f(x, y, z)\, dz\, dx\, dy.$$

Example 15.4.3A: Evaluating a Triple Integral over a General Bounded Region

Evaluate the triple integral of the function $f(x, y, z) = 5x - 3y$ over the solid tetrahedron bounded by the planes $x = 0$, $y = 0$, $z = 0$, and $x + y + z = 1$.

Solution

Figure 15.4.6 shows the solid tetrahedron E and its projection D on the xy-plane.

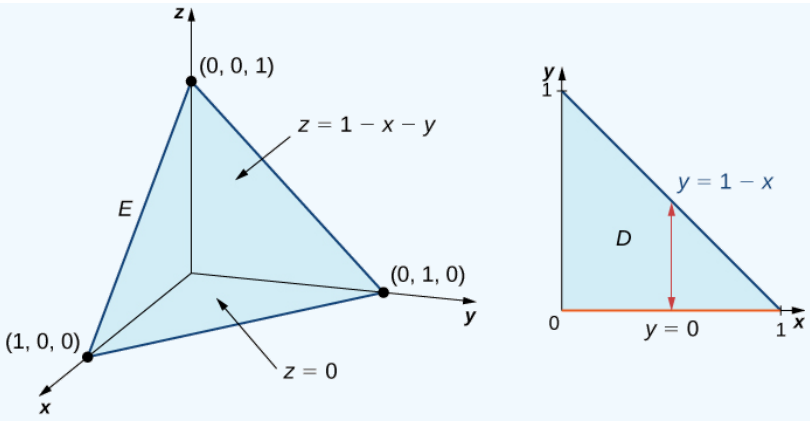


Figure 15.4.6: The solid E has a projection D on the xy-plane of Type I.

We can describe the solid region tetrahedron as

$$E = \{(x, y, z) \mid 0 \le x \le 1,\ 0 \le y \le 1 - x,\ 0 \le z \le 1 - x - y\}.$$

Hence, the triple integral is

$$\iiint_E f(x, y, z)\, dV = \int_{x=0}^{x=1} \int_{y=0}^{y=1-x} \int_{z=0}^{z=1-x-y} (5x - 3y)\, dz\, dy\, dx.$$

To simplify the calculation, first evaluate the integral $\int_{z=0}^{z=1-x-y} (5x - 3y)\, dz$. We have

$$\int_{z=0}^{z=1-x-y} (5x - 3y)\, dz = (5x - 3y)z \Big|_{z=0}^{z=1-x-y} = (5x - 3y)(1 - x - y).$$

Now evaluate the integral

$$\int_{y=0}^{y=1-x} (5x - 3y)(1 - x - y)\, dy,$$

obtaining

$$\int_{y=0}^{y=1-x} (5x - 3y)(1 - x - y)\, dy = \frac{1}{2}(x - 1)^2(6x - 1).$$

Finally evaluate

$$\int_{x=0}^{x=1} \frac{1}{2}(x - 1)^2(6x - 1)\, dx = \frac{1}{12}.$$

Putting it all together, we have

$$\iiint_E f(x, y, z)\, dV = \int_{x=0}^{x=1} \int_{y=0}^{y=1-x} \int_{z=0}^{z=1-x-y} (5x - 3y)\, dz\, dy\, dx = \frac{1}{12}.$$

Just as we used the double integral

$$\iint_D 1\, dA$$

to find the area of a general bounded region D we can use

$$\iiint_E 1\, dV$$

to find the volume of a general solid bounded region E. The next example illustrates the method.

Example 15.4.3B: Finding a Volume by Evaluating a Triple Integral

Find the volume of a right pyramid that has the square base in the xy-plane $[-1,1]\times[-1,1]$ and vertex at the point $(0,0,1)$ as shown in the following figure.

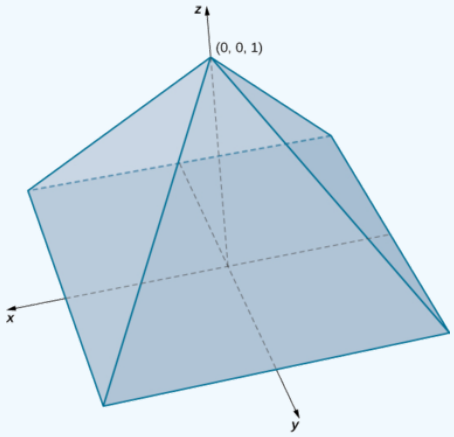


Figure 15.4.7: Finding the volume of a pyramid with a square base.

Solution

In this pyramid the value of z changes from 0 to 1 and at each height z the cross section of the pyramid for any value of z is the square

$$[-1+z,\,1-z]\times[-1+z,\,1-z].$$

Hence, the volume of the pyramid is

$$\iiint_E 1\,dV$$

where

$$E=\{(x,y,z)\,|\,0\le z\le 1,\,-1+z\le y\le 1-z,\,-1+z\le x\le 1-z\}.$$

Thus, we have

$$\begin{aligned}\iiint_E 1\,dV &= \int_{z=0}^{z=1}\int_{y=-1+z}^{y=1-z}\int_{x=-1+z}^{x=1-z} 1\,dx\,dy\,dz\\ &= \int_{z=0}^{z=1}\int_{y=-1+z}^{y=1-z}(2-2z)\,dy\,dz\\ &= \int_{z=0}^{z=1}(2-2z)^2\,dz=\frac{4}{3}.\end{aligned}$$

Hence, the volume of the pyramid is $\frac{4}{3}$ cubic units.

Exercise 15.4.3

Consider the solid sphere $E=\{(x,y,z)\,|\,x^2+y^2+z^2=9\}$. Write the triple integral

$$\iiint_E f(x,y,z)\,dV$$

for an arbitrary function f as an iterated integral. Then evaluate this triple integral with $f(x,y,z)=1$. Notice that this gives the volume of a sphere using a triple integral.

Hint

Follow the steps in the previous example. Use symmetry.

Answer

$$\iiint_E 1\,dV = 8\int_{x=-3}^{x=3}\int_{y=-\sqrt{9-z^2}}^{y=\sqrt{9-z^2}}\int_{z=-\sqrt{9-x^2-y^2}}^{z=\sqrt{9-x^2-y^2}} 1\,dz\,dy\,dx$$
$$= 36\pi \text{ cubic units.}$$

Changing the Order of Integration

As we have already seen in double integrals over general bounded regions, changing the order of the integration is done quite often to simplify the computation. With a triple integral over a rectangular box, the order of integration does not change the level of difficulty of the calculation. However, with a triple integral over a general bounded region, choosing an appropriate order of integration can simplify the computation quite a bit. Sometimes making the change to polar coordinates can also be very helpful. We demonstrate two examples here.

Example 15.4.4: Changing the Order of Integration

Consider the iterated integral

$$\int_{x=0}^{x=1}\int_{y=0}^{y=x^2}\int_{z=0}^{z=y} f(x,y,z)\,dz\,dy\,dx.$$

The order of integration here is first with respect to z, then y, and then x. Express this integral by changing the order of integration to be first with respect to x, then z, and then y. Verify that the value of the integral is the same if we let $f(x,y,z) = xyz$.

Solution

The best way to do this is to sketch the region E and its projections onto each of the three coordinate planes. Thus, let

$$E = \{(x,y,z)\,|\,0 \le x \le 1,\ 0 \le y \le x^2,\ 0 \le z \le y\}.$$

and

$$\int_{x=0}^{x=1}\int_{y=0}^{y=x^2}\int_{z=0}^{z=x^2} f(x,y,z)\,dz\,dy\,dx = \iiint_E f(x,y,z)\,dV.$$

We need to express this triple integral as

$$\int_{y=c}^{y=d}\int_{z=v_1(y)}^{z=v_2(y)}\int_{x=u_1(y,z)}^{x=u_2(y,z)} f(x,y,z)\,dx\,dz\,dy.$$

Knowing the region E we can draw the following projections (Figure 15.4.8):

on the xy-plane is $D_1 = \{(x,y)\,|\,0 \le x \le 1,\ 0 \le y \le x^2\} = \{(x,y)\,|\,0 \le y \le 1,\ \sqrt{y} \le x \le 1\}$,

on the yz-plane is $D_2 = \{(y,z)\,|\,0 \le y \le 1,\ 0 \le z \le y^2\}$, and

on the xz-plane is $D_3 = \{(x,z)\,|\,0 \le x \le 1,\ 0 \le z \le x^2\}$.

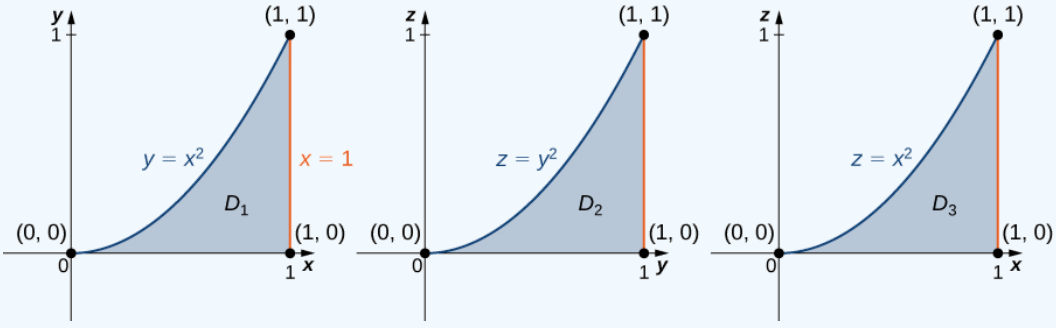


Figure 15.4.8. The three cross sections of E on the three coordinate planes.

Now we can describe the same region E as $\{(x,y,z)\,|\,0 \le y \le 1,\ 0 \le z \le y^2,\ \sqrt{y} \le x \le 1\}$, and consequently, the triple integral becomes

$$\int_{y=c}^{y=d}\int_{z=v_1(y)}^{z=v_2(y)}\int_{x=u_1(y,z)}^{x=u_2(y,z)} f(x,y,z)\,dx\,dz\,dy = \int_{y=0}^{y=1}\int_{z=0}^{z=x^2}\int_{x=\sqrt{y}}^{x=1} f(x,y,z)\,dx\,dz\,dy$$

Now assume that $f(x,y,z)=xyz$ in each of the integrals. Then we have

$$\begin{aligned}\int_{x=0}^{x=1}\int_{y=0}^{y=x^2}\int_{z=0}^{z=y^2} xyz\,dz\,dy\,dx &= \int_{x=0}^{x=1}\int_{y=0}^{y=x^2}\left[xy\frac{z^2}{2}\Big|_{z=0}^{z=y^2}\right]dy\,dx\\ &= \int_{x=0}^{x=1}\int_{y=0}^{y=x^2}\left(x\frac{y^5}{2}\right)dy\,dx\\ &= \int_{x=0}^{x=1}\left[x\frac{y^6}{12}\Big|_{y=0}^{y=x^2}\right]dx\\ &= \int_{x=0}^{x=1}\frac{x^{13}}{12}dx = \frac{x^{14}}{168}\Big|_{x=0}^{x=1}\\ &= \frac{1}{168},\end{aligned}$$

$$\begin{aligned}\int_{y=0}^{y=1}\int_{z=0}^{z=y^2}\int_{x=\sqrt{y}}^{x=1} xyz\,dx\,dz\,dy &= \int_{y=0}^{y=1}\int_{z=0}^{z=y^2}\left[yz\frac{x^2}{2}\Big|_{\sqrt{y}}^{1}\right]dz\,dy\\ &= \int_{y=0}^{y=1}\int_{z=0}^{z=y^2}\left(\frac{yz}{2}-\frac{y^2z}{2}\right)dz\,dy\\ &= \int_{y=0}^{y=1}\left[\frac{yz^2}{4}-\frac{y^2z^2}{4}\Big|_{z=0}^{z=y^2}\right]dy\\ &= \int_{y=0}^{y=1}\left(\frac{y^5}{4}-\frac{y^6}{4}\right)dy\\ &= \left(\frac{y^6}{24}-\frac{y^7}{28}\right)\Big|_{y=0}^{y=1}\\ &= \frac{1}{168}.\end{aligned}$$

The answers match.

Exercise 15.4.4

Write five different iterated integrals equal to the given integral

$$\int_{z=0}^{z=4}\int_{y=0}^{y=4-z}\int_{x=0}^{x=\sqrt{y}} f(x,y,z)\,dx\,dy\,dz.$$

Hint

Follow the steps in the previous example, using the region E as $\{(x,y,z)\mid 0\le z\le 4,\ 0\le y\le 4-z,\ 0\le x\le\sqrt{y}\}$, and describe and sketch the projections onto each of the three planes, five different times.

Answer

$$(i)\ \int_{z=0}^{z=4}\int_{x=0}^{x=\sqrt{4-z}}\int_{y=x^2}^{y=4-z} f(x,y,z)\,dy\,dx\,dz,\ (ii)\ \int_{y=0}^{y=4}\int_{z=0}^{z=4-y}\int_{x=0}^{x=\sqrt{y}} f(x,y,z)\,dx\,dz\,dy,\ (iii)\ \int_{y=0}^{y=4}\int_{x=0}^{x=\sqrt{y}}\int_{z=0}^{Z=4-y} f(x,y,z)\,dz\,dx\,dy,$$

$$(iv)\ \int_{x=0}^{x=2}\int_{y=x^2}^{y=4}\int_{z=0}^{z=4-y} f(x,y,z)\,dz\,dy\,dx,\ (v)\ \int_{x=0}^{x=2}\int_{z=0}^{z=4-x^2}\int_{y=x^2}^{y=4-z} f(x,y,z)\,dy\,dz\,dx$$

Example 15.4.5: Changing Integration Order and Coordinate Systems

Evaluate the triple integral

$$\iiint_E \sqrt{x^2+z^2}\,dV,$$

where E is the region bounded by the paraboloid $y=x^2+z^2$ (Figure 15.4.9) and the plane $y=4$.

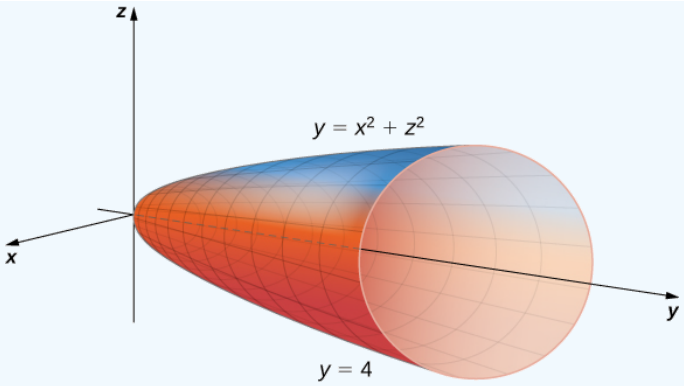


Figure 15.4.9. Integrating a triple integral over a paraboloid.

Solution

The projection of the solid region E onto the xy-plane is the region bounded above by $y = 4$ and below by the parabola $y = x^2$ as shown.

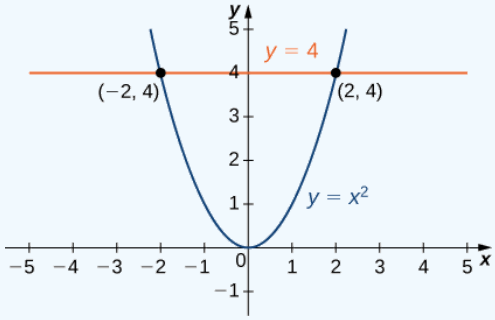


Figure 15.4.10. Cross section in the xy-plane of the paraboloid in Figure 15.4.9.

Thus, we have

$$E = \{(x, y, z) \mid -2 \le x \le 2,\ x^2 \le y \le 4,\ -\sqrt{y - x^2} \le z \le \sqrt{y - x^2}\}.$$

The triple integral becomes

$$\iiint_E \sqrt{x^2 + z^2}\, dV = \int_{x=-2}^{x=2} \int_{y=x^2}^{y=4} \int_{z=-\sqrt{y-x^2}}^{z=\sqrt{y-x^2}} \sqrt{x^2 + z^2}\, dz\, dy\, dx.$$

This expression is difficult to compute, so consider the projection of E onto the xz-plane. This is a circular disc $x^2 + z^2 \le 4$. So we obtain

$$\begin{aligned}\iiint_E \sqrt{x^2 + z^2}\, dV &= \int_{x=-2}^{x=2} \int_{y=x^2}^{y=4} \int_{z=-\sqrt{y-x^2}}^{z=\sqrt{y-x^2}} \sqrt{x^2 + z^2}\, dz\, dy\, dx \\ &= \int_{x=-2}^{x=2} \int_{z=-\sqrt{4-x^2}}^{z=\sqrt{4-x^2}} \int_{y=x^2+z^2}^{y=4} \sqrt{x^2 + z^2}\, dy\, dz\, dx.\end{aligned}$$

Here the order of integration changes from being first with respect to z then y and then x to being first with respect to y then to z and then to x. It will soon be clear how this change can be beneficial for computation. We have

$$\int_{x=-2}^{x=2} \int_{z=\sqrt{4-x^2}}^{z=\sqrt{4-x^2}} \int_{y=x^2+z^2}^{y=4} \sqrt{x^2 + z^2}\, dy\, dz\, dx = \int_{x=-2}^{x=2} \int_{z=-\sqrt{4-x^2}}^{z=\sqrt{4-x^2}} (4 - x^2 - z^2)\sqrt{x^2 + z^2}\, dz\, dx.$$

Now use the polar substitution $x = r\cos\theta$, $z = r\sin\theta$, and $dz\, dx = r\, dr\, d\theta$ in the xz-plane. This is essentially the same thing as when we used polar coordinates in the xy-plane, except we are replacing y by z. Consequently the limits of integration change and we have, by using $r^2 = x^2 + z^2$,

$$\int_{x=-2}^{x=2}\int_{z=-\sqrt{4-x^2}}^{z=\sqrt{4-x^2}}(4-x^2-z^2)\sqrt{x^2+z^2}\,dz\,dx=\int_{\theta=0}^{\theta=2\pi}\int_{r=0}^{r=2}(4-r^2)rr\,dr\,d\theta$$
$$=\int_0^{2\pi}\left[\frac{4r^3}{3}-\frac{r^5}{5}\Big|_0^2\right]d\theta=\int_0^{2\pi}\frac{64}{15}\,d\theta=\frac{128\pi}{15}$$

Average Value of a Function of Three Variables

Recall that we found the average value of a function of two variables by evaluating the double integral over a region on the plane and then dividing by the area of the region. Similarly, we can find the average value of a function in three variables by evaluating the triple integral over a solid region and then dividing by the volume of the solid.

Average Value of a Function of Three Variables

If $f(x,y,z)$ is integrable over a solid bounded region E with positive volume $V(E)$, then the average value of the function is

$$f_{ave}=\frac{1}{V(E)}\iiint_E f(x,y,z)\,dV.$$

Note that the volume is

$$V(E)=\iiint_E 1\,dV.$$

Example 15.4.6: Finding an Average Temperature

The temperature at a point (x,y,z) of a solid E bounded by the coordinate planes and the plane $x+y+z=1$ is $T(x,y,z)=(xy+8z+20)$ °C. Find the average temperature over the solid.

Solution

Use the theorem given above and the triple integral to find the numerator and the denominator. Then do the division. Notice that the plane $x+y+z=1$ has intercepts $(1,0,0)$, $(0,1,0)$, and $(0,0,1)$. The region E looks like

$$E=\{(x,y,z)\,|\,0\le x\le 1,\ 0\le y\le 1-x,\ 0\le z\le 1-x-y\}.$$

Hence the triple integral of the temperature is

$$\iiint_E f(x,y,z)\,dV=\int_{x=0}^{x=1}\int_{y=0}^{y=1-x}\int_{z=0}^{z=1-x-y}(xy+8z+20)\,dz\,dy\,dx=\frac{147}{40}.$$

The volume evaluation is

$$V(E)=\iiint_E 1\,dV=\int_{x=0}^{x=1}\int_{y=0}^{y=1-x}\int_{z=0}^{z=1-x-y}1\,dz\,dy\,dx=\frac{1}{6}.$$

Hence the average value is

$$T_{ave}=\frac{147/40}{1/6}=\frac{6(147)}{40}=\frac{441}{20}\ ^\circ\text{C}$$

.

Exercise 15.4.6

Find the average value of the function $f(x,y,z)=xyz$ over the cube with sides of length 4 units in the first octant with one vertex at the origin and edges parallel to the coordinate axes.

Hint

Follow the steps in the previous example.

Answer

$f_{ave} = 8$

Key Concepts

- To compute a triple integral we use Fubini's theorem, which states that if $f(x, y, z)$ is continuous on a rectangular box $B = [a, b] \times [c, d] \times [e, f]$, then

$$\iiint_B f(x, y, z)\, dV = \int_e^f \int_c^d \int_a^b f(x, y, z)\, dx\, dy\, dz$$

  and is also equal to any of the other five possible orderings for the iterated triple integral.
- To compute the volume of a general solid bounded region E we use the triple integral

$$V(E) = \iiint_E 1\, dV.$$

- Interchanging the order of the iterated integrals does not change the answer. As a matter of fact, interchanging the order of integration can help simplify the computation.
- To compute the average value of a function over a general three-dimensional region, we use

$$f_{ave} = \frac{1}{V(E)} \iiint_E f(x, y, z)\, dV.$$

Key Equations

- **Triple integral**

$$\lim_{l,m,n\to\infty} \sum_{i=1}^{l} \sum_{j=1}^{m} \sum_{k=1}^{n} f(x^*_{ijk}, y^*_{ijk}, z^*_{ijk})\, \Delta x \Delta y \Delta z = \iiint_B f(x, y, z)\, dV$$

Glossary

triple integral

the triple integral of a continuous function $f(x, y, z)$ over a rectangular solid box B is the limit of a Riemann sum for a function of three variables, if this limit exists

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15.4E: Exercises for Section 15.4

In exercises 1 - 4, evaluate the triple integrals over the rectangular solid box B.

1. $\iiint_B (2x + 3y^2 + 4z^3)\, dV$, where $B = \{(x, y, z) \mid 0 \le x \le 1,\ 0 \le y \le 2,\ 0 \le z \le 3\}$

   Answer

   192

2. $\iiint_B (xy + yz + xz)\, dV$, where $B = \{(x, y, z) \mid 1 \le x \le 2,\ 0 \le y \le 2,\ 1 \le z \le 3\}$
3. $\iiint_B (x \cos y + z)\, dV$, where $B = \{(x, y, z) \mid 0 \le x \le 1,\ 0 \le y \le \pi,\ -1 \le z \le 1\}$

   Answer

   0

4. $\iiint_B (z \sin x + y^2)\, dV$, where $B = \{(x, y, z) \mid 0 \le x \le \pi,\ 0 \le y \le 1,\ -1 \le z \le 2\}$

In exercises 5 - 8, change the order of integration by integrating first with respect to z, then x, then y.

5. $\int_0^1 \int_1^2 \int_2^3 (x^2 + \ln y + z)\, dx\, dy\, dz$

   Answer

   $$\int_0^1 \int_1^2 \int_2^3 (x^2 + \ln y + z)\, dx\, dy\, dz = \frac{35}{6} + 2 \ln 2$$

6. $\int_0^1 \int_{-1}^1 \int_0^3 (ze^x + 2y)\, dx\, dy\, dz$
7. $\int_{-1}^2 \int_1^3 \int_0^4 \left(x^2 z + \frac{1}{y}\right) dx\, dy\, dz$

   Answer

   $$\int_{-1}^2 \int_1^3 \int_0^4 \left(x^2 z + \frac{1}{y}\right) dx\, dy\, dz = 64 + 12 \ln 3$$

8. $\int_1^2 \int_{-2}^{-1} \int_0^1 \frac{x + y}{z}\, dx\, dy\, dz$

9. Let F, G, and H be continuous functions on $[a, b]$, $[c, d]$, and $[e, f]$, respectively, where a, b, c, d, e, and f are real numbers such that $a < b$, $c < d$, and $e < f$. Show that

$$\int_a^b \int_c^d \int_e^f F(x)\, G(y)\, H(z)\, dz\, dy\, dx = \left(\int_a^b F(x)\, dx\right) \left(\int_c^d G(y)\, dy\right) \left(\int_e^f H(z)\, dz\right).$$

10. Let F, G, and H be differential functions on $[a, b]$, $[c, d]$, and $[e, f]$, respectively, where a, b, c, d, e, and f are real numbers such that $a < b$, $c < d$, and $e < f$. Show that

$$\int_a^b \int_c^d \int_e^f F'(x)\, G'(y)\, H'(z)\, dz\, dy\, dx = [F(b) - F(a)]\, [G(d) - G(c)]\, H(f) - H(e)].$$

In exercises 11 - 14, evaluate the triple integrals over the bounded region

$E = \{(x, y, z) \mid a \le x \le b,\ h_1(x) \le y \le h_2(x),\ e \le z \le f\}$.

11. $\iiint_E (2x + 5y + 7z)\, dV$, where $E = \{(x, y, z) \mid 0 \le x \le 1,\ 0 \le y \le -x + 1,\ 1 \le z \le 2\}$

    Answer

    $\frac{77}{12}$

12. $\iiint_E (y \ln x + z)\, dV$, where $E = \{(x, y, z) \mid 1 \le x \le e,\ 0 \le y \ln x,\ 0 \le z \le 1\}$

13. $\iiint_E (\sin x + \sin y)\, dV$, where $E = \{(x, y, z) \mid 0 \le x \le \frac{\pi}{2},\ -\cos x \le y \cos x,\ -1 \le z \le 1\}$

Answer

2

14. $\iiint_E (xy + yz + xz) dV$ where $E = \{(x, y, z) \mid 0 \le x \le 1,\ -x^2 \le y \le x^2,\ 0 \le z \le 1\}$

In exercises 15 - 18, evaluate the triple integrals over the indicated bounded region E.

15. $\iiint_E (x + 2yz)\, dV$, where $E = \{(x, y, z) \mid 0 \le x \le 1,\ 0 \le y \le x,\ 0 \le z \le 5 - x - y\}$

Answer

$\frac{430}{120}$

16. $\iiint_E (x^3 + y^3 + z^3)\, dV$, where $E = \{(x, y, z) \mid 0 \le x \le 2,\ 0 \le y \le 2x,\ 0 \le z \le 4 - x - y\}$

17. $\iiint_E y\, dV$, where $E = \{(x, y, z) \mid -1 \le x \le 1,\ -\sqrt{1 - x^2} \le y \le \sqrt{1 - x^2},\ 0 \le z \le 1 - x^2 - y^2\}$

Answer

0

18. $\iiint_E x\, dV$, where $E = \{(x, y, z) \mid -2 \le x \le 2,\ -4\sqrt{1 - x^2} \le y \le \sqrt{4 - x^2},\ 0 \le z \le 4 - x^2 - y^2\}$

In exercises 19 - 22, evaluate the triple integrals over the bounded region E of the form

$E = \{(x, y, z) \mid g_1(y) \le x \le g_2(y),\ c \le y \le d,\ e \le z \le f\}$.

19. $\iiint_E x^2\, dV$, where $E = \{(x, y, z) \mid 1 - y^2 \le x \le y^2 - 1,\ -1 \le y \le 1,\ 1 \le z \le 2\}$

Answer

$-\frac{64}{105}$

20. $\iiint_E (y + \sin x)\, dV$, where $E = \{(x, y, z) \mid -y^4 \le x \le y^4,\ 0 \le y \le 2,\ 0 \le z \le 4\}$

21. $\iiint_E (x - yz)\, dV$, where $E = \{(x, y, z) \mid -y^6 \le x \le \sqrt{y},\ 0 \le y \le 1x,\ -1 \le z \le 1\}$

Answer

$\frac{11}{26}$

22. $\iiint_E z\, dV$, where $E = \{(x, y, z) \mid 2 - 2y \le x \le 2 + \sqrt{y},\ 0 \le y \le 1x,\ 2 \le z \le 3\}$

In exercises 23 - 26, evaluate the triple integrals over the bounded region

$E = \{(x, y, z) \mid g_1(y) \le x \le g_2(y),\ c \le y \le d,\ u_1(x, y) \le z \le u_2(x, y)\}$

23. $\iiint_E z\, dV$, where $E = \{(x, y, z) \mid -y \le x \le y,\ 0 \le y \le 1,\ 0 \le z \le 1 - x^4 - y^4\}$

Answer

$\frac{113}{450}$

24. $\iiint_E (xz + 1)\, dV$, where $E = \{(x, y, z) \mid 0 \le x \le \sqrt{y},\ 0 \le y \le 2,\ 0 \le z \le 1 - x^2 - y^2\}$

25. $\iiint_E (x - z)\, dV$, where $E = \{(x, y, z) \mid -\sqrt{1 - y^2} \le x \le y,\ 0 \le y \le \frac{1}{2}x,\ 0 \le z \le 1 - x^2 - y^2\}$

Answer

$\frac{1}{160}(6\sqrt{3}-41)$

26. $\iiint_E (x+y)\,dV$, where $E=\{(x,y,z)\,|\,0\le x\le\sqrt{1-y^2},\ 0\le y\le 1x,\ 0\le z\le 1-x\}$

In exercises 27 - 30, evaluate the triple integrals over the bounded region

$E=\{(x,y,z)\,|\,(x,y)\in D,\ u_1(x,y)x\le z\le u_2(x,y)\}$**, where** D **is the projection of** E **onto the** xy**-plane.**

27. $\iint_D \left(\int_1^2 (x+y)\,dz\right)\,dA$, where $D=\{(x,y)\,|\,x^2+y^2\le 1\}$

Answer

$\frac{3\pi}{2}$

28. $\iint_D \left(\int_1^3 x(z+1)\,dz\right)\,dA$, where $D=\{(x,y)\,|\,x^2-y^2\ge 1,\ x\le\sqrt{5}\}$

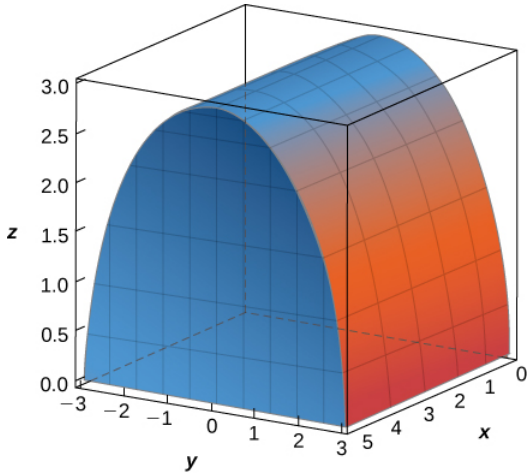
29. $\iint_D \left(\int_0^{10-x-y} (x+2z)\,dz\right)\,dA$, where $D=\{(x,y)\,|\,y\ge 0,\ x\ge 0,\ x+y\le 10\}$

Answer

1250

30. $\iint_D \left(\int_0^{4x^2+4y^2} y\,dz\right)\,dA$, where $D=\{(x,y)\,|\,x^2+y^2\le 4,\ y\ge 1,\ x\ge 0\}$

31. The solid E bounded by $y^2+z^2=9$, $z=0$, and $x=5$ is shown in the following figure. Evaluate the integral $\iiint_E z\,dV$ by integrating first with respect to z, then y, and then x.

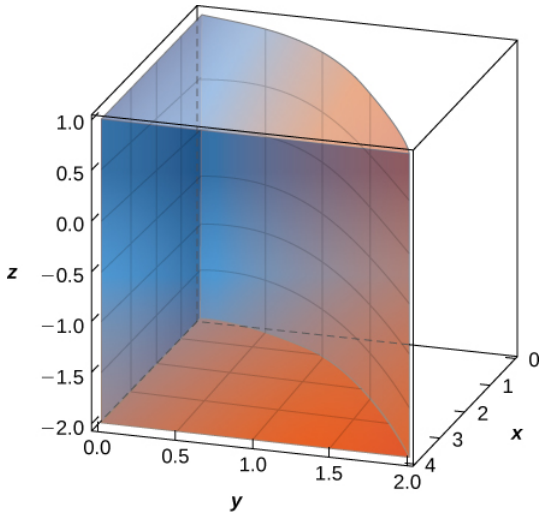


Answer

$\int_0^5\int_{-3}^3\int_0^{\sqrt{9-y^2}} z\,dz\,dy\,dx=90$

32. The solid E bounded by $y=\sqrt{x}$, $x=4$, $y=0$, and $z=1$ is given in the following figure. Evaluate the integral $\iiint_E xyz\,dV$ by integrating first with respect to x, then y, and then z.

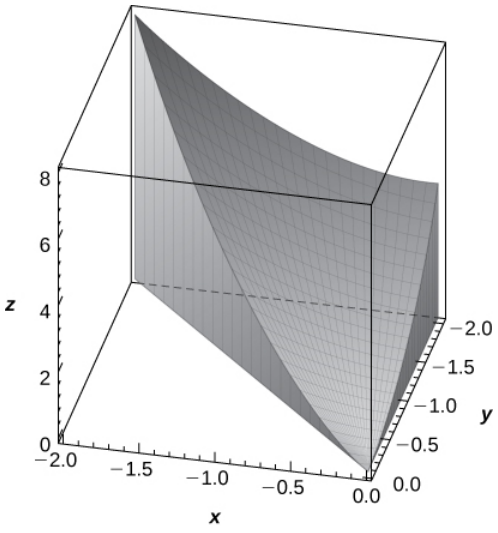
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33. [T] The volume of a solid E is given by the integral $\int_{-2}^{0}\int_{x}^{0}\int_{0}^{x^2+y^2} dz\,dy\,dx$. Use a computer algebra system (CAS) to graph E and find its volume. Round your answer to two decimal places.

Answer

$V \approx 5.33 \text{ units}^3$



34. [T] The volume of a solid E is given by the integral $\int_{-1}^{0}\int_{-x^3}^{0}\int_{0}^{1+\sqrt{x^2+y^2}} dz\,dy\,dx$. Use a CAS to graph E and find its volume V. Round your answer to two decimal places.

In exercises 35 - 38, use two circular permutations of the variables x, y, and z to write new integrals whose values equal the value of the original integral. A circular permutation of x, y, and z is the arrangement of the numbers in one of the following orders: y, z, and x or z, x, and y.

35. $\int_{0}^{1}\int_{1}^{3}\int_{2}^{4}(x^2z^2+1)\,dx\,dy\,dz$

Answer

$\int_{0}^{1}\int_{1}^{3}\int_{2}^{4}(y^2z^2+1)\,dz\,dx\,dy;$
$\int_{0}^{1}\int_{1}^{3}\int_{2}^{4}(x^2y^2+1)\,dy\,dz\,dx$

36. $\int_0^3 \int_0^1 \int_0^{-x+1} (2x+5y+7z)dy\,dx\,dz$

37. $\int_0^1 \int_{-y}^{y} \int_0^{1-x^4-y^4} \ln x \; dz\,dx\,dy$

38. $\int_{-1}^1 \int_0^1 \int_{-y^6}^{\sqrt{y}} (x+yz)\,dx\,dy\,dz$

39. Set up the integral that gives the volume of the solid E bounded by $y^2 = x^2 + z^2$ and $y = a^2$, where $a > 0$.

Answer

$$V = \int_{-a}^{a} \int_{-\sqrt{a^2-z^2}}^{\sqrt{a^2-z^2}} \int_{\sqrt{x^2+z^2}}^{a^2} dy\,dx\,dz$$

40. Set up the integral that gives the volume of the solid E bounded by $x = y^2 + z^2$ and $x = a^2$, where $a > 0$.

Average Value of a Function

41. Find the average value of the function $f(x, y, z) = x + y + z$ over the parallelepiped determined by $x + 0,\ x = 1,\ y = 0,\ y = 3,\ z = 0$, and $z = 5$.

Answer

$\frac{9}{2}$

42. Find the average value of the function $f(x, y, z) = xyz$ over the solid $E = [0, 1] \times [0, 1] \times [0, 1]$ situated in the first octant.

Finding Volumes using Triple Integrals

43. Find the volume of the solid E that lies under the plane $x + y + z = 9$ and whose projection onto the xy-plane is bounded by $x = sqrty - 1,\ x = 0$, and $x + y = 7$.

Answer

$\frac{156}{5}$ units^3

44. Find the volume of the solid E that lies under the plane $2x + y + z = 8$ and whose projection onto the xy-plane is bounded by $x = sin^{-1}y,\ y = 0$, and $x = \frac{\pi}{2}$.

45. Consider the pyramid with the base in the xy-plane of $[-2, 2] \times [-2, 2]$ and the vertex at the point $(0, 0, 8)$.

a. Show that the equations of the planes of the lateral faces of the pyramid are $4y + z = 8,\ 4y - z = -8,\ 4x + z = 8$, and $-4x + z = 8$.

b. Find the volume of the pyramid.

Answer

a. Answers may vary;

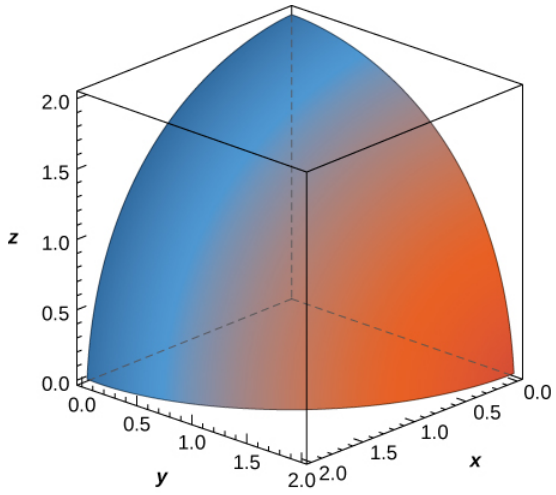
b. $\frac{128}{3}$ units^3

46. Consider the pyramid with the base in the xy-plane of $[-3, 3] \times [-3, 3]$ and the vertex at the point $(0, 0, 9)$.

a. Show that the equations of the planes of the side faces of the pyramid are $3y + z = 9,\ 3y + z = 9,\ y = 0$ and $x = 0$.

b. Find the volume of the pyramid.

47. The solid E bounded by the sphere of equation $x^2 + y^2 + z^2 = r^2$ with $r > 0$ and located in the first octant is represented in the following figure.



a. Write the triple integral that gives the volume of E by integrating first with respect to z, then with y, and then with x.

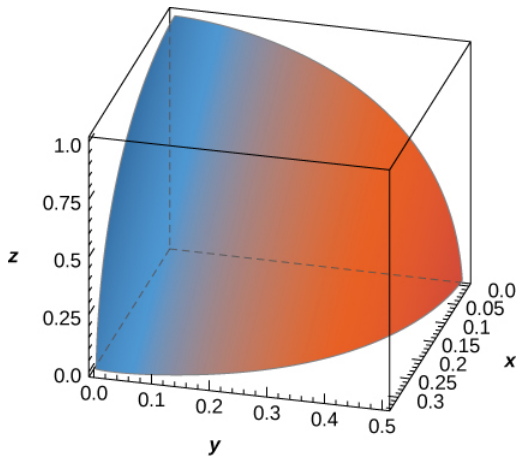
b. Rewrite the integral in part a. as an equivalent integral in five other orders.

Answer

a. $\int_0^4 \int_0^{\sqrt{r^2-x^2}} \int_0^{\sqrt{r^2-x^2-y^2}} dz\, dy\, dx$

b. $\int_0^2 \int_0^{\sqrt{r^2-x^2}} \int_0^{\sqrt{r^2-x^2-y^2}} dz\, dx\, dy$,
$\int_0^r \int_0^{\sqrt{r^2-x^2}} \int_0^{\sqrt{r^2-x^2-y^2}} dy\, dx\, dz$,
$\int_0^r \int_0^{\sqrt{r^2-x^2}} \int_0^{\sqrt{r^2-x^2-y^2}} dy\, dz\, dx$,
$\int_0^r \int_0^{\sqrt{r^2-x^2}} \int_0^{\sqrt{r^2-x^2-y^2}} dx\, dy\, dz$,
$\int_0^r \int_0^{\sqrt{r^2-x^2}} \int_0^{\sqrt{r^2-x^2-y^2}} dx\, dz\, dy$

48. The solid E bounded by the sphere of equation $9x^2 + 4y^2 + z^2 = 1$ and located in the first octant is represented in the following figure.



a. Write the triple integral that gives the volume of E by integrating first with respect to z then with y and then with x.

b. Rewrite the integral in part a. as an equivalent integral in five other orders.

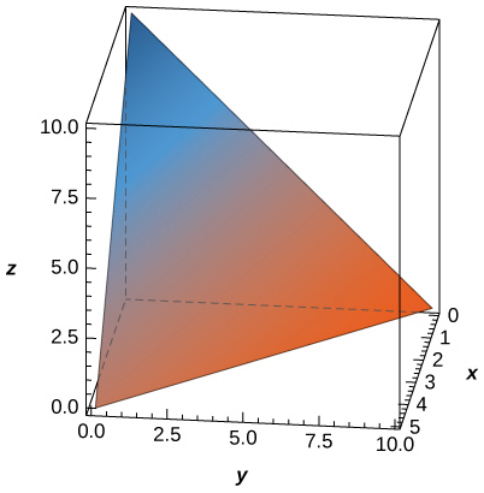
49. Find the volume of the prism with vertices $(0,0,0)$, $(2,0,0)$, $(2,3,0)$, $(0,3,0)$, $(0,0,1)$, and $(2,0,1)$.

Answer

3 units^3

50. Find the volume of the prism with vertices $(0,0,0)$, $(4,0,0)$, $(4,6,0)$, $(0,6,0)$, $(0,0,1)$, and $(4,0,1)$.

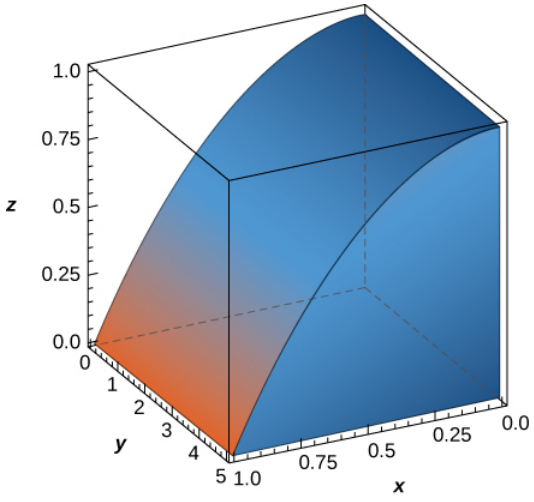
51. The solid E bounded by $z = 10 - 2x - y$ and situated in the first octant is given in the following figure. Find the volume of the solid.



Answer

$\frac{250}{3} \text{ units}^3$

52. The solid E bounded by $z = 1 - x^2$ and situated in the first octant is given in the following figure. Find the volume of the solid.



Approximating Triple Integrals

53. The midpoint rule for the triple integral $\iiint_B f(x,y,z)\,dV$ over the rectangular solid box B is a generalization of the midpoint rule for double integrals. The region B is divided into subboxes of equal sizes and the integral is approximated by the triple Riemann sum

$$\sum_{i=1}^{l}\sum_{j=1}^{m}\sum_{k=1}^{n} f(\bar{x}_i, \bar{y}_j, \bar{z}_k)\Delta V,$$

where $(\bar{x}_i, \bar{y}_j, \bar{z}_k)$ is the center of the box B_{ijk} and ΔV is the volume of each subbox. Apply the midpoint rule to approximate

$$\iiint_B x^2\,dV$$

over the solid $B = \{(x, y, z) \mid 0 \le x \le 1,\ 0 \le y \le 1,\ 0 \le z \le 1\}$ by using a partition of eight cubes of equal size. Round your answer to three decimal places.

Answer

$\iiint_B f(x, y, z)\,dV \quad \approx \quad \frac{5}{16} \approx 0.313 \text{ units}^3$

54. [T]

a. Apply the midpoint rule to approximate $\iiint_B e^{-x^2}\,dV$ over the solid $B = \{(x, y, z) \mid 0 \le x \le 1,\ 0 \le y \le 1,\ 0 \le z \le 1\}$ by using a partition of eight cubes of equal size. Round your answer to three decimal places.

b. Use a CAS to improve the above integral approximation in the case of a partition of n^3 cubes of equal size, where $n = 3, 4, \ldots, 10$.

Applications

55. Suppose that the temperature in degrees Celsius at a point (x, y, z) of a solid E bounded by the coordinate planes and the plane $x + y + z = 5$ is given by:

$$T(x, y, z) = xz + 5z + 10$$

Find the average temperature over the solid.

Answer

17.5° C

56. Suppose that the temperature in degrees Fahrenheit at a point (x, y, z) of a solid E bounded by the coordinate planes and the plane $x + y + z = 5$ is given by:

$$T(x, y, z) = x + y + xy$$

Find the average temperature over the solid.

57. Show that the volume of a right square pyramid of height h and side length a is $v = \dfrac{ha^2}{3}$ by using triple integrals.

58. Show that the volume of a regular right hexagonal prism of edge length a is $\dfrac{3a^3\sqrt{3}}{2}$ by using triple integrals.

59. Show that the volume of a regular right hexagonal pyramid of edge length a is $\dfrac{a^3\sqrt{3}}{2}$ by using triple integrals.

60. If the charge density at an arbitrary point (x, y, z) of a solid E is given by the function $\rho(x, y, z)$, then the total charge inside the solid is defined as the triple integral $\iiint_E \rho(x, y, z)\,dV$. Assume that the charge density of the solid E enclosed by the paraboloids $x = 5 - y^2 - z^2$ and $x = y^2 + z^2 - 5$ is equal to the distance from an arbitrary point of E to the origin. Set up the integral that gives the total charge inside the solid E.

Answer

Total Charge inside the Solid $E \quad = \quad \int_{-\sqrt{5}}^{\sqrt{5}} \int_{-\sqrt{5-y^2}}^{\sqrt{5-y^2}} \int_{y^2+z^2-5}^{5-y^2-z^2} \sqrt{x^2 + y^2 + z^2}\,dx\,dz\,dy$

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15.5: Triple Integrals in Cylindrical and Spherical Coordinates

Learning Objectives

- Evaluate a triple integral by changing to cylindrical coordinates.
- Evaluate a triple integral by changing to spherical coordinates.

Earlier in this chapter we showed how to convert a double integral in rectangular coordinates into a double integral in polar coordinates in order to deal more conveniently with problems involving circular symmetry. A similar situation occurs with triple integrals, but here we need to distinguish between cylindrical symmetry and spherical symmetry. In this section we convert triple integrals in rectangular coordinates into a triple integral in either cylindrical or spherical coordinates.

Also recall the chapter prelude, which showed the opera house l'Hemisphèric in Valencia, Spain. It has four sections with one of the sections being a theater in a five-story-high sphere (ball) under an oval roof as long as a football field. Inside is an IMAX screen that changes the sphere into a planetarium with a sky full of 9000 twinkling stars. Using triple integrals in spherical coordinates, we can find the volumes of different geometric shapes like these.

Review of Cylindrical Coordinates

As we have seen earlier, in two-dimensional space $\mathbb{R}^2$ a point with rectangular coordinates (x, y) can be identified with (r, θ) in polar coordinates and vice versa, where $x = r\cos\theta$, $y = r\sin\theta$, $r^2 = x^2 + y^2$ and $\tan\theta = \frac{y}{x}$ are the relationships between the variables.

In three-dimensional space $\mathbb{R}^3$ a point with rectangular coordinates (x, y, z) can be identified with cylindrical coordinates (r, θ, z) and vice versa. We can use these same conversion relationships, adding z as the vertical distance to the point from the $(xy$-plane as shown in $15.5.1$.

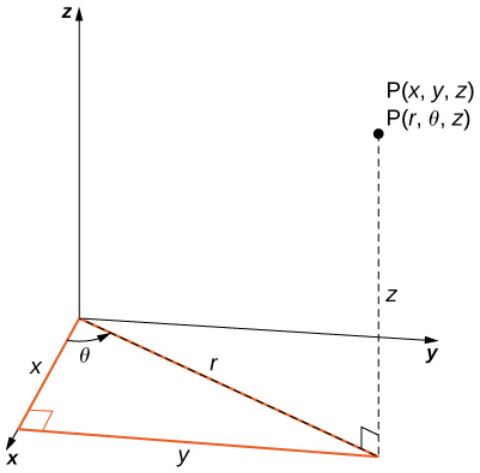


Figure $15.5.1$: Cylindrical coordinates are similar to polar coordinates with a vertical z coordinate added.

To convert from rectangular to cylindrical coordinates, we use the conversion

- $x = r\cos\theta$
- $y = r\sin\theta$
- $z = z$

To convert from cylindrical to rectangular coordinates, we use

- $r^2 = x^2 + y^2$ and
- $\theta = \tan^{-1}\left(\frac{y}{x}\right)$
- $z = z$

Note that that z-coordinate remains the same in both cases.

In the two-dimensional plane with a rectangular coordinate system, when we say $x = k$ (constant) we mean an unbounded vertical line parallel to the y-axis and when $y = l$ (constant) we mean an unbounded horizontal line parallel to the x-axis. With the polar coordinate system, when we say $r = c$ (constant), we mean a circle of radius c units and when $\theta = \alpha$ (constant) we mean an infinite ray making an angle α with the positive x-axis.

Similarly, in three-dimensional space with rectangular coordinates (x, y, z) the equations $x = k,\ y = l$ and $z = m$ where $k,\ l$ and m are constants, represent unbounded planes parallel to the yz-plane, xz-plane and xy-plane, respectively. With cylindrical coordinates (r, θ, z), by $r = c,\ \theta = \alpha$, and $z = m$, where c, α, and m are constants, we mean an unbounded vertical cylinder with the z-axis as its radial axis; a plane making a constant angle α with the xy-plane; and an unbounded horizontal plane parallel to the xy-plane, respectively. This means that the circular cylinder $x^2 + y^2 = c^2$ in rectangular coordinates can be represented simply as $r = c$ in cylindrical coordinates. (Refer to Cylindrical and Spherical Coordinates for more review.)

Integration in Cylindrical Coordinates

Triple integrals can often be more readily evaluated by using cylindrical coordinates instead of rectangular coordinates. Some common equations of surfaces in rectangular coordinates along with corresponding equations in cylindrical coordinates are listed in Table 15.5.1. These equations will become handy as we proceed with solving problems using triple integrals.

Table 15.5.1: Equations of Some Common Shapes

	Circular cylinder	**Circular cone**	**Sphere**	**Paraboloid**
Rectangular	$x^2 + y^2 = c^2$	$z^2 = c^2(x^2 + y^2)$	$x^2 + y^2 + z^2 = c^2$	$z = c(x^2 + y^2)$
Cylindrical	$r = c$	$z = cr$	$r^2 + z^2 = c^2$	$z = cr^2$

As before, we start with the simplest bounded region B in $\mathbb{R}^3$ to describe in cylindrical coordinates, in the form of a cylindrical box, $B = \{(r, \theta, z) | a \le r \le b,\ \alpha \le \theta \le \beta,\ c \le z \le d\}$ (Figure 15.5.2). Suppose we divide each interval into l, m, and n subdivisions such that $\Delta r = \frac{b \cdot a}{l}$, $\Delta\theta = \frac{\beta \cdot \alpha}{m}$, and $\Delta z = \frac{d \cdot c}{n}$. Then we can state the following definition for a triple integral in cylindrical coordinates.

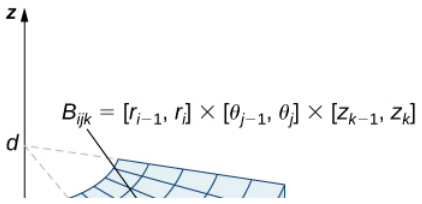


Figure 15.5.2: A cylindrical box B described by cylindrical coordinates.

DEFINITION: triple integral in cylindrical coordinates

Consider the cylindrical box (expressed in cylindrical coordinates)

$$B = \{(r, \theta, z) | a \le r \le b,\ \alpha \le \theta \le \beta,\ c \le z \le d\}.$$

If the function $f(r, \theta, z)$ is continuous on B and if $(r^*_{ijk}, \theta^*_{ijk}, z^*_{ijk})$ is any sample point in the cylindrical subbox $B_{ijk} = |r_{i-1}, r_i| \times |\theta_{j-1}, \theta_j| \times |z_{k-1}, k_i|$ (Figure 15.5.2), then we can define the *triple integral in cylindrical coordinates* as the limit of a triple Riemann sum, provided the following limit exists:

$$\lim_{l,m,n\to\infty} \sum_{i=1}^{l} \sum_{j=1}^{m} \sum_{k=1}^{n} f(r^*_{ijk}, \theta^*_{ijk}, z^*_{ijk})\, \Delta r \Delta\theta \Delta z.$$

Note that if $g(x, y, z)$ is the function in rectangular coordinates and the box B is expressed in rectangular coordinates, then the triple integral

$$\iiint_B g(x, y, z)\, dV$$

is equal to the triple integral

$$\iiint_B g(r\, \cos\theta,\, r\, \sin\, \theta,\, z) r\, dr\, d\theta\, dz$$

and we have

$$\iiint_B g(x, y, z)\, dV = \iiint_B g(r\, \cos\theta,\, r\, \sin\, \theta,\, z)\, r\, dr\, d\theta\, dz = \iiint_B f(r, \theta\, z)\, r\, dr\, d\theta\, dz.$$

As mentioned in the preceding section, all the properties of a double integral work well in triple integrals, whether in rectangular coordinates or cylindrical coordinates. They also hold for iterated integrals. To reiterate, in cylindrical coordinates, Fubini's theorem takes the following form:

Theorem: Fubini's Theorem in Cylindrical Coordinates

Suppose that $g(x, y, z)$ is continuous on a rectangular box B which when described in cylindrical coordinates looks like $B = \{(r, \theta, z) | a \le r \le b,\, \alpha \le \theta \le \beta,\, c \le z \le d\}$.

Then $g(x, y, z) = g(r\, \cos\theta, r\, \sin\, \theta, z) = f(r, \theta, z)$ and

$$\iiint_B g(x, y, z)\, dV = \int_c^d \int_\beta^\alpha \int_a^b f(r, \theta, z)\, r\, dr\, d\theta\, dz.$$

The iterated integral may be replaced equivalently by any one of the other five iterated integrals obtained by integrating with respect to the three variables in other orders.

Cylindrical coordinate systems work well for solids that are symmetric around an axis, such as cylinders and cones. Let us look at some examples before we define the triple integral in cylindrical coordinates on general cylindrical regions.

Example 15.5.1: Evaluating a Triple Integral over a Cylindrical Box

Evaluate the triple integral

$$\iiint_B (zr\, \sin\, \theta) r\, dr\, d\theta\, dz$$

where the cylindrical box B is $B = \{(r, \theta, z) | 0 \le r \le 2,\, 0 \le \theta \le \pi/2,\, 0, \le z \le 4\}$.

Solution

As stated in Fubini's theorem, we can write the triple integral as the iterated integral

$$\iiint_B (zr\, \sin\, \theta) r\, dr\, d\theta\, dz = \int_{\theta=0}^{\theta=\pi/2} \int_{r=0}^{r=2} \int_{z=0}^{z=4} (zr\, \sin\, \theta)\, r\, dz\, dr\, d\theta.$$

The evaluation of the iterated integral is straightforward. Each variable in the integral is independent of the others, so we can integrate each variable separately and multiply the results together. This makes the computation much easier:

$$\int_{\theta=0}^{\theta=\pi/2}\int_{r=0}^{r=2}\int_{z=0}^{z=4}(zr\,\sin\,\theta)\,r\,dz\,dr\,d\theta=\left(\int_0^{\pi/2}\sin\,\theta\,d\theta\right)\left(\int_0^2 r^2dr\right)\left(\int_0^4 z\,dz\right)$$
$$=\left(-\cos\theta|_0^{\pi/2}\right)\left(\frac{r^3}{3}\Big|_0^2\right)\left(\frac{z^2}{2}\Big|_0^4\right)=\frac{64}{3}.$$

? Exercise 15.5.1:

Evaluate the triple integral

$$\int_{\theta=0}^{\theta=\pi}\int_{r=0}^{r=1}\int_{z=0}^{z=4}(rz\,\sin\,\theta)\,r\,dz\,dr\,d\theta.$$

Hint

Follow the same steps as in the previous example.

Answer

8

If the cylindrical region over which we have to integrate is a general solid, we look at the projections onto the coordinate planes. Hence the triple integral of a continuous function $f(r,\theta,z)$ over a general solid region $E=\{(r,\theta,z)|(r,\theta)\in D, u_1(r,\theta)\le z\le u_2(r,\theta)\}$ in $\mathbb{R}^3$ where D is the projection of E onto the $r\theta$-plane, is

$$\iiint_E f(r,\theta,z)\,r\,dr\,d\theta\,dz=\iint_D\left[\int_{u_1(r,\theta)}^{u_2(r,\theta)}f(r,\theta,z)dz\right]r\,dr\,d\theta.$$

In particular, if $D=\{(r,\theta)|g_1(\theta)\le r\le g_2(\theta),\alpha\le\theta\le\beta\}$, then we have

$$\iiint_E f(r,\theta,z)\,r\,dr\,d\theta=\int_{\theta=\alpha}^{\theta=\beta}\int_{r=g_1(\theta)}^{r=g_2(\theta)}\int_{z=u_1(r,\theta)}^{z=u_2(r,\theta)}f(r,\theta,z)\,r\,dz\,dr\,d\theta.$$

Similar formulas exist for projections onto the other coordinate planes. We can use polar coordinates in those planes if necessary.

✓ Example 15.5.2: Setting up a Triple Integral in Cylindrical Coordinates over a General Region

Consider the region E inside the right circular cylinder with equation $r=2\,\sin\,\theta$, bounded below by the $r\theta$-plane and bounded above by the sphere with radius 4 centered at the origin (Figure 15.5.3). Set up a triple integral over this region with a function $f(r,\theta,z)$ in cylindrical coordinates.

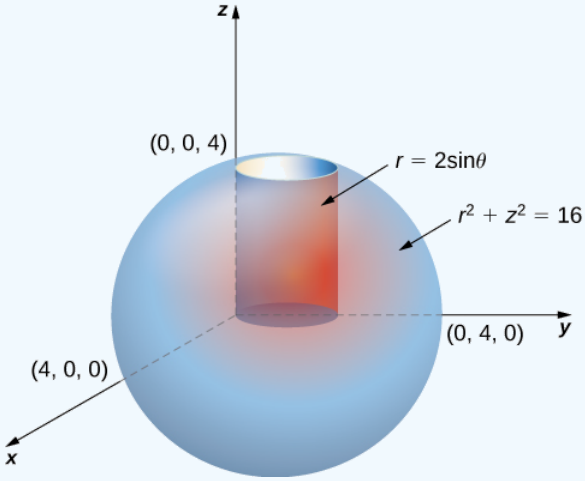


Figure 15.5.3: Setting up a triple integral in cylindrical coordinates over a cylindrical region.

Solution

First, identify that the equation for the sphere is $r^2 + z^2 = 16$. We can see that the limits for z are from 0 to $z = \sqrt{16 - r^2}$. Then the limits for r are from 0 to $r = 2 \sin \theta$. Finally, the limits for θ are from 0 to π. Hence the region is $E = \{(r, \theta, z) | 0 \leq \theta \leq \pi, \ 0 \leq r \leq 2 \sin \theta, \ 0 \leq z \leq \sqrt{16 - r^2}\}$. Therefore, the triple integral is

$$\iiint_E f(r, \theta, z) \, r \, dz \, dr \, d\theta = \int_{\theta=0}^{\theta=\pi} \int_{r=0}^{r=2 \sin \theta} \int_{z=0}^{z=\sqrt{16-r^2}} f(r, \theta, z) \, r \, dz \, dr \, d\theta.$$

? Exercise 15.5.2:

Consider the region inside the right circular cylinder with equation $r = 2 \sin \theta$ bounded below by the $r\theta$-plane and bounded above by $z = 4 - y$. Set up a triple integral with a function $f(r, \theta, z)$ in cylindrical coordinates.

Hint

Analyze the region, and draw a sketch.

Answer

$$\iiint_E f(r, \theta, z) r \, dz \, dr \, d\theta = \int_{\theta=0}^{\theta=\pi} \int_{r=0}^{r=2 \sin \theta} \int_{z=0}^{z=4-r \sin \theta} f(r, \theta, z) r \, dz \, dr \, d\theta.$$

✓ Example 15.5.3: Setting up a Triple Integral in Two Ways

Let E be the region bounded below by the cone $z = \sqrt{x^2 + y^2}$ and above by the paraboloid $z = 2 - x^2 - y^2$. (Figure 15.5.4). Set up a triple integral in cylindrical coordinates to find the volume of the region, using the following orders of integration:

a. $dz \, dr \, d\theta$

b. $dr \, dz \, d\theta$

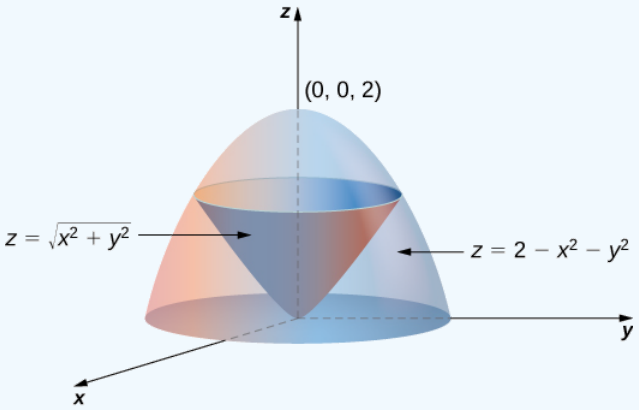


Figure 15.5.4: Setting up a triple integral in cylindrical coordinates over a conical region.

Solution

a. The cone is of radius 1 where it meets the paraboloid. Since $z = 2 - x^2 - y^2 = 2 - r^2$ and $z = \sqrt{x^2 + y^2} = r^2$ (assuming r is nonnegative), we have $2 - r^2 = r$. Solving, we have $r^2 + r - 2 = (r + 2)(r - 1) = 0$. Since $r \geq 0$, we have $r = 1$. Therefore $z = 1$. So the intersection of these two surfaces is a circle of radius 1 in the plane $z = 1$. The cone is the lower bound for z and the paraboloid is the upper bound. The projection of the region onto the xy-plane is the circle of radius 1 centered at the origin.

Thus, we can describe the region as $E = \{(r, \theta, z) | 0 \leq \theta \leq 2\pi, \ 0 \leq r \leq 1, \ r \leq z \leq 2 - r^2\}$.

Hence the integral for the volume is

$$V = \int_{\theta=0}^{\theta=2\pi} \int_{r=0}^{r=1} \int_{z=r}^{z=2-r^2} r \, dz \, dr \, d\theta.$$

b. We can also write the cone surface as $r = z$ and the paraboloid as $r^2 = 2 - z$. The lower bound for r is zero, but the upper bound is sometimes the cone and the other times it is the paraboloid. The plane $z = 1$ divides the region into two regions. Then the region can be described as

$$E = \{(r, \theta, z) | 0 \leq \theta \leq 2\pi,\ 0 \leq z \leq 1,\ 0 \leq r \leq z\}$$
$$\cup \{(r, \theta, z) | 0 \leq \theta \leq 2\pi,\ 1 \leq z \leq 2,\ 0 \leq r \leq \sqrt{2 - z}\}.$$

Now the integral for the volume becomes

$$V = \int_{\theta=0}^{\theta=2\pi} \int_{z=0}^{z=1} \int_{r=0}^{r=z} r\, dr\, dz\, d\theta + \int_{\theta=0}^{\theta=2\pi} \int_{z=1}^{z=2} \int_{r=0}^{r=\sqrt{2-z}} r\, dr\, dz\, d\theta.$$

Exercise 15.5.3:

Redo the previous example with the order of integration $d\theta\, dz\, dr$.

Hint

Note that θ is independent of r and z.

Answer

$E = \{(r, \theta, z) | 0 \leq \theta \leq 2\pi,\ 0 \leq z \leq 1,\ 0 \leq r \leq 2 - z^2\}$ and

$$V = \int_{r=0}^{r=1} \int_{z=0}^{z=2-r^2} \int_{\theta=0}^{\theta=2\pi} r\, d\theta\, dz\, dr.$$

Example 15.5.4: Finding a Volume with Triple Integrals in Two Ways

Let E be the region bounded below by the $r\theta$-plane, above by the sphere $x^2 + y^2 + z^2 = 4$, and on the sides by the cylinder $x^2 + y^2 = 1$ (Figure 15.5.5). Set up a triple integral in cylindrical coordinates to find the volume of the region using the following orders of integration, and in each case find the volume and check that the answers are the same:

a. $dz\, dr\, d\theta$
b. $dr\, dz\, d\theta$.

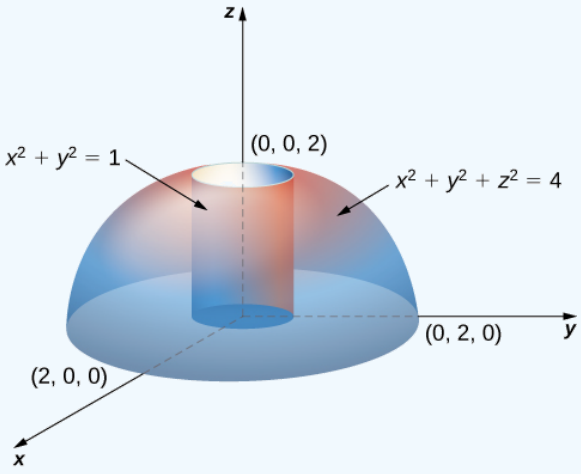


Figure 15.5.5: Finding a cylindrical volume with a triple integral in cylindrical coordinates.

Solution

a. Note that the equation for the sphere is

$$x^2 + y^2 + z^2 = 4 \text{ or } r^2 + z^2 = 4$$

and the equation for the cylinder is

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$$x^2 + y^2 = 1 \text{ or } r^2 = 1.$$

Thus, we have for the region E

$$E = \{(r, \theta, z) | 0 \leq z \leq \sqrt{4 - r^2},\ 0 \leq r \leq 1,\ 0 \leq \theta \leq 2\pi\}$$

Hence the integral for the volume is

$$\begin{aligned} V(E) &= \int_{\theta=0}^{\theta=2\pi} \int_{r=0}^{r=1} \int_{z=0}^{z=\sqrt{4-r^2}} r\, dz\, dr\, d\theta \\ &= \int_{\theta=0}^{\theta=2\pi} \int_{r=0}^{r=1} \left[rz \Big|_{z=0}^{z=\sqrt{4-r^2}} \right] dr\, d\theta \\ &= \int_{\theta=0}^{\theta=2\pi} \int_{r=0}^{r=1} \left(r\sqrt{4 - r^2} \right) dr\, d\theta \\ &= \int_0^{2\pi} \left(\frac{8}{3} - \sqrt{3} \right) d\theta = 2\pi \left(\frac{8}{3} - \sqrt{3} \right) \text{ cubic units.} \end{aligned}$$

b. Since the sphere is $x^2 + y^2 + z^2 = 4$, which is $r^2 + z^2 = 4$, and the cylinder is $x^2 + y^2 = 1$, which is $r^2 = 1$, we have $1 + z^2 = 4$, that is, $z^2 = 3$. Thus we have two regions, since the sphere and the cylinder intersect at $(1, \sqrt{3})$ in the rz-plane

$$E_1 = \{(r, \theta, z) | 0 \leq r \leq \sqrt{4 - r^2},\ \sqrt{3} \leq z \leq 2,\ 0 \leq \theta \leq 2\pi\}$$

and

$$E_2 = \{(r, \theta, z) | 0 \leq r \leq 1,\ 0 \leq z \leq \sqrt{3},\ 0 \leq \theta \leq 2\pi\}.$$

Hence the integral for the volume is

$$\begin{aligned} V(E) &= \int_{\theta=0}^{\theta=2\pi} \int_{z=\sqrt{3}}^{z=2} \int_{r=0}^{r=\sqrt{4-r^2}} r\, dr\, dz\, d\theta + \int_{\theta=0}^{\theta=2\pi} \int_{z=0}^{z=\sqrt{3}} \int_{r=0}^{r=1} r\, dr\, dz\, d\theta \\ &= \sqrt{3}\pi + \left(\frac{16}{3} - 3\sqrt{3} \right) \pi \\ &= 2\pi \left(\frac{8}{3} - \sqrt{3} \right) \text{ cubic units.} \end{aligned}$$

? Exercise 15.5.4

Redo the previous example with the order of integration $d\theta\, dz\, dr$.

Hint

A figure can be helpful. Note that θ is independent of r and z.

Answer

$E_2 = \{(r, \theta, z) | 0 \leq \theta \leq 2\pi,\ 0 \leq r \leq 1,\ r \leq z \leq \sqrt{4 - r^2}\}$ and

$$V = \int_{r=0}^{r=1} \int_{z=r}^{z=\sqrt{4-r^2}} \int_{\theta=0}^{\theta=2\pi} r\, d\theta\, dz\, dr.$$

Review of Spherical Coordinates

In three-dimensional space $\mathbb{R}^3$ in the spherical coordinate system, we specify a point P by its distance ρ from the origin, the polar angle θ from the positive x-axis (same as in the cylindrical coordinate system), and the angle φ from the positive z-axis and the line OP (Figure 15.5.6). Note that $\rho > 0$ and $0 \leq \varphi \leq \pi$. (Refer to Cylindrical and Spherical Coordinates for a review.) Spherical coordinates are useful for triple integrals over regions that are symmetric with respect to the origin.

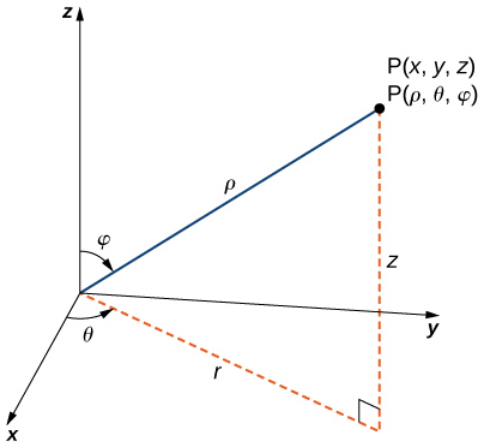


Figure 15.5.6: The spherical coordinate system locates points with two angles and a distance from the origin.

Recall the relationships that connect rectangular coordinates with spherical coordinates.

From spherical coordinates to rectangular coordinates:

$$x = \rho \sin \varphi \cos\theta,\ y = \rho \sin \varphi \sin \theta,\ and\, z = \rho \cos \varphi.$$

From rectangular coordinates to spherical coordinates:

$$\rho^2 = x^2 + y^2 + z^2,\ \tan \theta = \frac{y}{x},\ \varphi = \arccos\left(\frac{z}{\sqrt{x^2 + y^2 + z^2}}\right).$$

Other relationships that are important to know for conversions are

- $r = \rho \sin \varphi$
- $\theta = \theta$ These equations are used to convert from spherical coordinates to cylindrical coordinates.
- $z = \rho \cos \varphi$

and

- $\rho = \sqrt{r^2 + z^2}$
- $\theta = \theta$ These equations are used to convert from cylindrical coordinates to spherical coordinates.
- $\varphi = \arccos\left(\frac{z}{\sqrt{r^2+z^2}}\right)$

15.5.7 shows a few solid regions that are convenient to express in spherical coordinates.

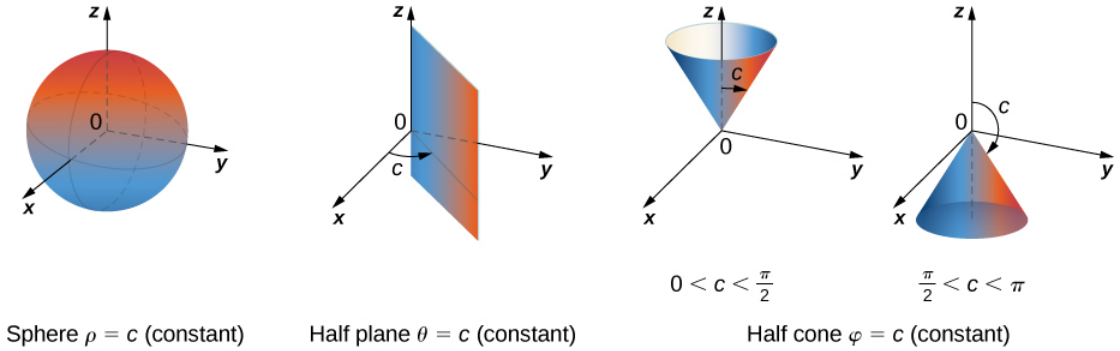


Figure 15.5.7: Spherical coordinates are especially convenient for working with solids bounded by these types of surfaces. (The letter c indicates a constant.)

Integration in Spherical Coordinates

We now establish a triple integral in the spherical coordinate system, as we did before in the cylindrical coordinate system. Let the function $f(\rho, \theta, \varphi)$ be continuous in a bounded spherical box, $B = \{(\rho, \theta, \varphi) | a \le \rho \le b,\ \alpha \le \theta \le \beta,\ \gamma \le \varphi \le \psi\}$. We then divide each interval into l, m, n and n subdivisions such that $\Delta\rho = \frac{b-a}{l}$, $\Delta\theta = \frac{\beta-\alpha}{m}$. $\Delta\varphi = \frac{\psi-\gamma}{n}$. Now we can illustrate the following theorem for

triple integrals in spherical coordinates with $(\rho^*_{ijk}, \theta^*_{ijk}, \varphi^*_{ijk})$ being any sample point in the spherical subbox B_{ijk}. For the volume element of the subbox ΔV in spherical coordinates, we have $\Delta V = (\Delta\rho)\,(\rho\Delta\varphi)\,(\rho\,\sin\,\varphi\,\Delta\theta)$, as shown in the following figure.

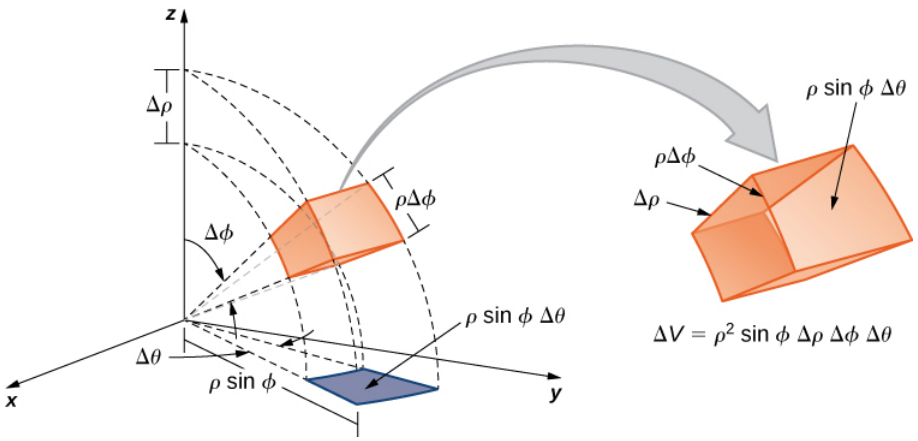


Figure 15.5.8: The volume element of a box in spherical coordinates.

Definition: triple integral in spherical coordinates

The **triple integral in spherical coordinates** is the limit of a triple Riemann sum,

$$\lim_{l,m,n\to\infty} \sum_{i=1}^{l}\sum_{j=1}^{m}\sum_{k=1}^{n} f(\rho^*_{ijk}, \theta^*_{ijk}, \varphi^*_{ijk})(\rho^*_{ijk})^2 \sin\,\varphi\Delta\rho\,\Delta\theta\Delta\varphi$$

provided the limit exists.

As with the other multiple integrals we have examined, all the properties work similarly for a triple integral in the spherical coordinate system, and so do the iterated integrals. Fubini's theorem takes the following form.

Theorem: Fubini's Theorem for Spherical Coordinates

If $f(\rho, \theta, \varphi)$ is continuous on a spherical solid box $B = [a, b] \times [\alpha, \beta] \times [\gamma, \psi]$, then

$$\iiint_B f(\rho, \theta, \varphi)\,\rho^2 \sin\,\varphi\,d\rho\,d\varphi\,d\theta = \int_{\varphi=\gamma}^{\varphi=\psi}\int_{\theta=\alpha}^{\theta=\beta}\int_{\rho=a}^{\rho=b} f(\rho, \theta, \varphi)\,\rho^2 \sin\,\varphi\,d\rho\,d\varphi\,d\theta.$$

This iterated integral may be replaced by other iterated integrals by integrating with respect to the three variables in other orders.

As stated before, spherical coordinate systems work well for solids that are symmetric around a point, such as spheres and cones. Let us look at some examples before we consider triple integrals in spherical coordinates on general spherical regions.

Example 15.5.5: Evaluating a Triple Integral in Spherical Coordinates

Evaluate the iterated triple integral

$$\int_{\theta=0}^{\theta=2\pi}\int_{\varphi=0}^{\varphi=\pi/2}\int_{\rho=0}^{\rho=1} \rho^2 \sin\,\varphi\,d\rho\,d\varphi\,d\theta.$$

Solution

As before, in this case the variables in the iterated integral are actually independent of each other and hence we can integrate each piece and multiply:

$$\int_0^{2\pi}\int_0^{\pi/2}\int_0^1 \rho^2 \sin\,\varphi\,d\rho\,d\varphi\,d\theta = \int_0^{2\pi} d\theta \int_0^{\pi/2} \sin\,\varphi\,d\varphi \int_0^1 \rho^2 d\rho = (2\pi)\,(1)\left(\frac{1}{3}\right) = \frac{2\pi}{3}$$

The concept of triple integration in spherical coordinates can be extended to integration over a general solid, using the projections onto the coordinate planes. Note that dV and dA mean the increments in volume and area, respectively. The variables V and A are used as the variables for integration to express the integrals.

The triple integral of a continuous function $f(\rho, \theta, \varphi)$ over a general solid region

$$E = \{(\rho, \theta, \varphi) | (\rho, \theta) \in D, u_1(\rho, \theta) \leq \varphi \leq u_2(\rho, \theta)\}$$

in $\mathbb{R}^3$, where D is the projection of E onto the $\rho\theta$-plane, is

$$\iiint_E f(\rho, \theta, \varphi)\, dV = \iint_D \left[\int_{u_1(\rho,\theta)}^{u_2(\rho,\theta)} f(\rho, \theta, \varphi)\, d\varphi \right] dA.$$

In particular, if $D = \{(\rho, \theta) | g_1(\theta) \leq \rho \leq g_2(\theta),\ \alpha \leq \theta \leq \beta\}$, the we have

$$\iiint_E f(\rho, \theta, \varphi)\, dV = \int_\alpha^\beta \int_{g_1(\theta)}^{g_2(\theta)} \int_{u_1(\rho,\theta)}^{u_2(\rho,\theta)} f(\rho, \theta, \varphi) \rho^2 \sin\, \varphi\, d\varphi\, d\rho\, d\theta.$$

Similar formulas occur for projections onto the other coordinate planes.

Example 15.5.6: Setting up a Triple Integral in Spherical Coordinates

Set up an integral for the volume of the region bounded by the cone $z = \sqrt{3(x^2 + y^2)}$ and the hemisphere $z = \sqrt{4 - x^2 - y^2}$ (see the figure below).

Figure 15.5.9: A region bounded below by a cone and above by a hemisphere.

Solution

Using the conversion formulas from rectangular coordinates to spherical coordinates, we have:

For the cone: $z = \sqrt{3(x^2 + y^2)}$ or $\rho \cos\, \varphi = \sqrt{3}\rho \sin\, \varphi$ or $\tan\, \varphi = \frac{1}{\sqrt{3}}$ or $\varphi = \frac{\pi}{6}$.

For the sphere: $z = \sqrt{4 - x^2 - y^2}$ or $z^2 + x^2 + y^2 = 4$ or $\rho^2 = 4$ or $\rho = 2$.

Thus, the triple integral for the volume is

$$V(E) = \int_{\theta=0}^{\theta=2\pi} \int_{\varphi=0}^{\varphi=\pi/6} \int_{\rho=0}^{\rho=2} \rho^2 \sin\, \varphi\, d\rho\, d\varphi\, d\theta.$$

Exercise 15.5.5

Set up a triple integral for the volume of the solid region bounded above by the sphere $\rho = 2$ and bounded below by the cone $\varphi = \pi/3$.

Hint

Follow the steps of the previous example.

Answer

$$V(E) = \int_{\theta=0}^{\theta=2\pi} \int_{\varphi=0}^{\varphi=\pi/3} \int_{\rho=0}^{\rho=2} \rho^2 \sin\, \varphi\, d\rho\, d\varphi\, d\theta$$

Example 15.5.7: Interchanging Order of Integration in Spherical Coordinates

Let E be the region bounded below by the cone $z = \sqrt{x^2 + y^2}$ and above by the sphere $z = x^2 + y^2 + z^2$ (Figure 15.5.10). Set up a triple integral in spherical coordinates and find the volume of the region using the following orders of integration:

a. $d\rho\, d\phi\, d\theta$
b. $d\varphi\, d\rho\, d\theta$

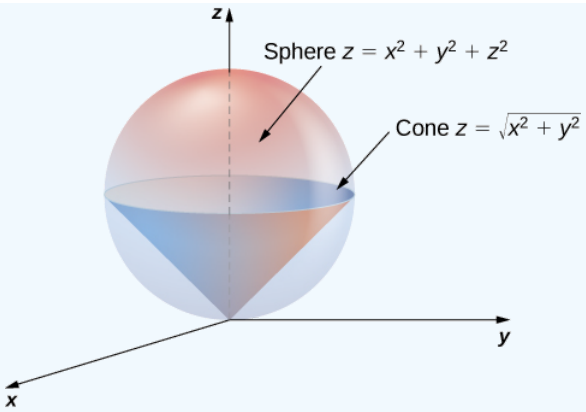


Figure 15.5.10:. A region bounded below by a cone and above by a sphere.

Solution

a. Use the conversion formulas to write the equations of the sphere and cone in spherical coordinates.

For the sphere:

$$\begin{aligned} x^2 + y^2 + z^2 &= z \\ \rho^2 &= \rho \cos \varphi \\ \rho &= \cos \varphi. \end{aligned}$$

For the cone:

$$\begin{aligned} z &= \sqrt{x^2 + y^2} \\ \rho \cos \varphi &= \sqrt{\rho^2 \sin^2 \varphi \cos^2 \theta + \rho^2 \sin^2 \varphi \sin^2 \theta} \\ \rho \cos \varphi &= \sqrt{\rho^2 \sin^2 \varphi (\cos^2 \theta + \sin^2 \theta)} \\ \rho \cos \varphi &= \rho \sin \varphi \\ \cos \varphi &= \sin \varphi \\ \varphi &= \pi/4. \end{aligned}$$

Hence the integral for the volume of the solid region E becomes

$$V(E) = \int_{\theta=0}^{\theta=2\pi} \int_{\varphi=0}^{\varphi=\pi/4} \int_{\rho=0}^{\rho=\cos \varphi} \rho^2 \sin \varphi \, d\rho \, d\varphi \, d\theta.$$

b. Consider the $\varphi\rho$-plane. Note that the ranges for φ and ρ (from part a.) are

$$\begin{aligned} 0 \leq \varphi \leq \pi/4 \\ 0 \leq \rho \leq \cos \varphi \end{aligned}$$

The curve $\rho = \cos \varphi$ meets the line $\varphi = \pi/4$ at the point $(\pi/4, \sqrt{2}/2)$. Thus, to change the order of integration, we need to use two pieces:

$$0 \leq \rho \leq \sqrt{2}/2, \, 0 \leq \varphi \leq \pi/4$$

and

$$\sqrt{2}/2 \leq \rho \leq 1, \, 0 \leq \varphi \leq \cos^{-1} \rho.$$

Hence the integral for the volume of the solid region E becomes

$$V(E) = \int_{\theta=0}^{\theta=2\pi} \int_{\rho=0}^{\rho=\sqrt{2}/2} \int_{\varphi=0}^{\varphi=\pi/4} \rho^2 \sin \varphi \, d\varphi \, d\rho \, d\theta + \int_{\theta=0}^{\theta=2\pi} \int_{\rho=\sqrt{2}/2}^{\rho=1} \int_{\varphi=0}^{\varphi=\cos^{-1} \rho} \rho^2 \sin \varphi \, d\varphi \, d\rho \, d\theta$$

In each case, the integration results in $V(E) = \frac{\pi}{8}$.

Before we end this section, we present a couple of examples that can illustrate the conversion from rectangular coordinates to cylindrical coordinates and from rectangular coordinates to spherical coordinates.

Example 15.5.8: Converting from Rectangular Coordinates to Cylindrical Coordinates

Convert the following integral into cylindrical coordinates:

$$\int_{y=-1}^{y=1}\int_{x=0}^{x=\sqrt{1-y^2}}\int_{z=x^2+y^2}^{z=\sqrt{x^2+y^2}} xyz\,dz\,dx\,dy.$$

Solution

The ranges of the variables are

$$\begin{aligned} -1 &\le y \le y \\ 0 &\le x \le \sqrt{1-y^2} \\ x^2+y^2 &\le z \le \sqrt{x^2+y^2}. \end{aligned}$$

The first two inequalities describe the right half of a circle of radius 1. Therefore, the ranges for θ and r are

$$-\frac{\pi}{2} \le \theta \le \frac{\pi}{2} \text{ and } 0 \le r \le 1.$$

The limits of z are $r^2 \le z \le r$, hence

$$\int_{y=-1}^{y=1}\int_{x=0}^{x=\sqrt{1-y^2}}\int_{z=x^2+y^2}^{z=\sqrt{x^2+y^2}} xyz\,dz\,dx\,dy = \int_{\theta=-\pi/2}^{\theta=\pi/2}\int_{r=0}^{r=1}\int_{z=r^2}^{z=r} r(r\cos\theta)\,(r\sin\theta)\,z\,dz\,dr\,d\theta.$$

Example 15.5.9: Converting from Rectangular Coordinates to Spherical Coordinates

Convert the following integral into spherical coordinates:

$$\int_{y=0}^{y=3}\int_{x=0}^{x=\sqrt{9-y^2}}\int_{z=\sqrt{x^2+y^2}}^{z=\sqrt{18-x^2-y^2}} (x^2+y^2+z^2)\,dz\,dx\,dy.$$

Solution

The ranges of the variables are

$$\begin{aligned} 0 &\le y \le 3 \\ 0 &\le x \le \sqrt{9-y^2} \\ \sqrt{x^2+y^2} &\le z \le \sqrt{18-x^2-y^2}. \end{aligned}$$

The first two ranges of variables describe a quarter disk in the first quadrant of the xy-plane. Hence the range for θ is $0 \le \theta \le \frac{\pi}{2}$.

The lower bound $z = \sqrt{x^2+y^2}$ is the upper half of a cone and the upper bound $z = \sqrt{18-x^2-y^2}$ is the upper half of a sphere. Therefore, we have $0 \le \rho \le \sqrt{18}$, which is $0 \le \rho \le 3\sqrt{2}$.

For the ranges of φ we need to find where the cone and the sphere intersect, so solve the equation

$$\begin{aligned} r^2+z^2 &= 18 \\ \left(\sqrt{x^2+y^2}\right)^2+z^2 &= 18 \\ z^2+z^2 &= 18 \\ 2z^2 &= 18 \\ z^2 &= 9 \\ z &= 3. \end{aligned}$$

This gives

$$3\sqrt{2}\cos\varphi = 3$$
$$\cos\varphi = \frac{1}{\sqrt{2}}$$
$$\varphi = \frac{\pi}{4}.$$

Putting this together, we obtain

$$\int_{y=0}^{y=3}\int_{x=0}^{x=\sqrt{9-y^2}}\int_{z=\sqrt{x^2+y^2}}^{z=\sqrt{18-x^2-y^2}}(x^2+y^2+z^2)\,dz\,dx\,dy = \int_{\varphi=0}^{\varphi=\pi/4}\int_{\theta=0}^{\theta=\pi/2}\int_{\rho=0}^{\rho=3\sqrt{2}}\rho^4\sin\varphi\,d\rho\,d\theta\,d\varphi.$$

? Exercise 15.5.6:

Use rectangular, cylindrical, and spherical coordinates to set up triple integrals for finding the volume of the region inside the sphere $x^2+y^2+z^2=4$ but outside the cylinder $x^2+y^2=1$.

Answer: Rectangular

$$\int_{x=-2}^{x=2}\int_{y=-\sqrt{4-x^2}}^{y=\sqrt{4-x^2}}\int_{z=-\sqrt{4-x^2-y^2}}^{z=\sqrt{4-x^2-y^2}}dz\,dy\,dx - \int_{x=-1}^{x=1}\int_{y=-\sqrt{1-x^2}}^{y=\sqrt{1-x^2}}\int_{z=-\sqrt{4-x^2-y^2}}^{z=\sqrt{4-x^2-y^2}}dz\,dy\,dx.$$

Answer: Cylindrical

$$\int_{\theta=0}^{\theta=2\pi}\int_{r=1}^{r=2}\int_{z=-\sqrt{4-r^2}}^{z=\sqrt{4-r^2}}r\,dz\,dr\,d\theta.$$

Answer: Spherical

$$\int_{\varphi=\pi/6}^{\varphi=5\pi/6}\int_{\theta=0}^{\theta=2\pi}\int_{\rho=\csc\varphi}^{\rho=2}\rho^2\sin\varphi\,d\rho\,d\theta\,d\varphi.$$

Now that we are familiar with the spherical coordinate system, let's find the volume of some known geometric figures, such as spheres and ellipsoids.

✓ Example 15.5.10: Chapter Opener: Finding the Volume of l'Hemisphèric

Find the volume of the spherical planetarium in l'Hemisphèric in Valencia, Spain, which is five stories tall and has a radius of approximately 50 ft, using the equation $x^2+y^2+z^2=r^2$.

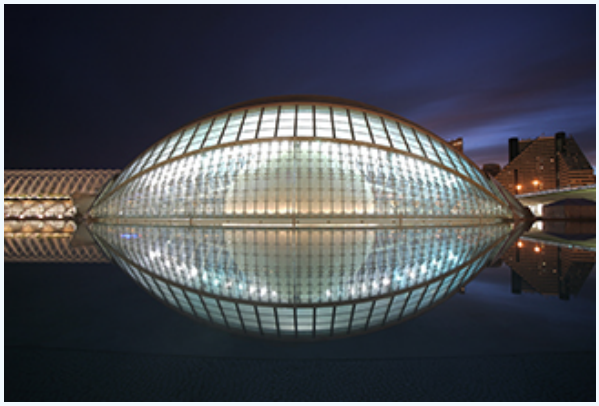


Figure 15.5.11: (credit: modification of work by Javier Yaya Tur, Wikimedia Commons)

Solution

We calculate the volume of the ball in the first octant, where $x \le 0$, $y \le 0$, and $z \le 0$, using spherical coordinates, and then multiply the result by 8 for symmetry. Since we consider the region D as the first octant in the integral, the ranges of the variables are

$$0 \le \varphi \le \frac{\pi}{2},\ 0 \le \rho \le r,\ 0 \le \theta \le \frac{\pi}{2}.$$

Therefore,

$$\begin{aligned} V = \iiint_D dx\,dy\,dz &= 8\int_{\theta=0}^{\theta=\pi/2}\int_{\rho=0}^{\rho=\pi}\int_{\varphi=0}^{\varphi=\pi/2} \rho^2 \sin\,\theta\,d\varphi\,d\rho\,d\varphi \\ &= 8\int_{\varphi=0}^{\varphi=\pi/2} d\varphi \int_{\rho=0}^{\rho=r} \rho^2 d\rho \int_{\theta=0}^{\theta=\pi/2} \sin\,\theta\,d\theta \\ &= 8\left(\frac{\pi}{2}\right)\left(\frac{r^3}{3}\right)(1) \\ &= \frac{4}{3}\pi r^3. \end{aligned}$$

This exactly matches with what we knew. So for a sphere with a radius of approximately 50 ft, the volume is $\frac{4}{3}\pi(50)^3 \approx 523,600\ ft^3$.

For the next example we find the volume of an ellipsoid.

Example 15.5.11: Finding the Volume of an Ellipsoid

Find the volume of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.

Solution

We again use symmetry and evaluate the volume of the ellipsoid using spherical coordinates. As before, we use the first octant $x \le 0$, $y \le 0$, and $z \le 0$ and then multiply the result by 8.

In this case the ranges of the variables are

$$0 \le \varphi \le \frac{\pi}{2}\ 0 \le \rho \le 1,\ \text{and}\ 0 \le \theta \le \frac{\pi}{2}.$$

Also, we need to change the rectangular to spherical coordinates in this way:

$$x = a\rho\,\cos\,\varphi\,\sin\,\theta,\ y = b\rho\,\sin\,\varphi\,\sin\,\theta,\ \text{and}\ z = cp\,\cos\,\theta.$$

Then the volume of the ellipsoid becomes

$$\begin{aligned} V &= \iiint_D dx\,dy\,dz \\ &= 8\int_{\theta=0}^{\theta=\pi/2}\int_{\rho=0}^{\rho=1}\int_{\varphi=0}^{\varphi=\pi/2} abc\,\rho^2 \sin\,\theta\,d\varphi\,d\rho\,d\theta \\ &= 8abc\int_{\varphi=0}^{\varphi=\pi/2} d\varphi \int_{\rho=0}^{\rho=1} \rho^2 d\rho \int_{\theta=0}^{\theta=\pi/2} \sin\,\theta\,d\theta \\ &= 8abc\left(\frac{\pi}{2}\right)\left(\frac{1}{3}\right)(1) \\ &= \frac{4}{3}\pi abc. \end{aligned}$$

Example 15.5.12: Finding the Volume of the Space Inside an Ellipsoid and Outside a Sphere

Find the volume of the space inside the ellipsoid $\frac{x^2}{75^2} + \frac{y^2}{80^2} + \frac{z^2}{90^2} = 1$ and outside the sphere $x^2 + y^2 + z^2 = 50^2$.

Solution

This problem is directly related to the l'Hemisphèric structure. The volume of space inside the ellipsoid and outside the sphere might be useful to find the expense of heating or cooling that space. We can use the preceding two examples for the volume of the sphere and ellipsoid and then substract.

First we find the volume of the ellipsoid using $a = 75$ ft, $b = 80$ ft, and $c = 90$ ft in the result from Example. Hence the volume of the ellipsoid is

$$V_{ellipsoid} = \frac{4}{3}\pi(75)(80)(90) \approx 2,262,000\ ft^3.$$

From Example, the volume of the sphere is

$$V_{sphere} \approx 523,600\ ft^3.$$

Therefore, the volume of the space inside the ellipsoid $\frac{x^2}{75^2} + \frac{y^2}{80^2} + \frac{z^2}{90^2} = 1$ and outside the sphere $x^2 + y^2 + z^2 = 50^2$ is approximately

$$V_{Hemispheric} = V_{ellipsoid} - V_{sphere} = 1,738,400\ ft^3.$$

Student Project: Hot air balloons

Hot air ballooning is a relaxing, peaceful pastime that many people enjoy. Many balloonist gatherings take place around the world, such as the Albuquerque International Balloon Fiesta. The Albuquerque event is the largest hot air balloon festival in the world, with over 500 balloons participating each year.

Figure 15.5.12: Balloons lift off at the 2001 Albuquerque International Balloon Fiesta. (credit: David Herrera, Flickr)

As the name implies, hot air balloons use hot air to generate lift. (Hot air is less dense than cooler air, so the balloon floats as long as the hot air stays hot.) The heat is generated by a propane burner suspended below the opening of the basket. Once the balloon takes off, the pilot controls the altitude of the balloon, either by using the burner to heat the air and ascend or by using a vent near the top of the balloon to release heated air and descend. The pilot has very little control over where the balloon goes, however—balloons are at the mercy of the winds. The uncertainty over where we will end up is one of the reasons balloonists are attracted to the sport.

In this project we use triple integrals to learn more about hot air balloons. We model the balloon in two pieces. The top of the balloon is modeled by a half sphere of radius 28

feet. The bottom of the balloon is modeled by a frustum of a cone (think of an ice cream cone with the pointy end cut off). The radius of the large end of the frustum is 28 feet and the radius of the small end of the frustum is 28 feet. A graph of our balloon model and a cross-sectional diagram showing the dimensions are shown in the following figure.

Figure 15.5.13: (a) Use a half sphere to model the top part of the balloon and a frustum of a cone to model the bottom part of the balloon. (b) A cross section of the balloon showing its dimensions.

We first want to find the volume of the balloon. If we look at the top part and the bottom part of the balloon separately, we see that they are geometric solids with known volume formulas. However, it is still worthwhile to set up and evaluate the integrals we would need to find the volume. If we calculate the volume using integration, we can use the known volume formulas to check our answers. This will help ensure that we have the integrals set up correctly for the later, more complicated stages of the project.

1. Find the volume of the balloon in two ways.

a. Use triple integrals to calculate the volume. Consider each part of the balloon separately. (Consider using spherical coordinates for the top part and cylindrical coordinates for the bottom part.)

b. Verify the answer using the formulas for the volume of a sphere, $V = \frac{4}{3}\pi r^3$, and for the volume of a cone, $V = \frac{1}{3}\pi r^2 h$.

In reality, calculating the temperature at a point inside the balloon is a tremendously complicated endeavor. In fact, an entire branch of physics (thermodynamics) is devoted to studying heat and temperature. For the purposes of this project, however, we

are going to make some simplifying assumptions about how temperature varies from point to point within the balloon. Assume that just prior to liftoff, the temperature (in degrees Fahrenheit) of the air inside the balloon varies according to the function

$$T_0(r,\theta,z) = \frac{z-r}{10} + 210.$$

2. What is the average temperature of the air in the balloon just prior to liftoff? (Again, look at each part of the balloon separately, and do not forget to convert the function into spherical coordinates when looking at the top part of the balloon.)

Now the pilot activates the burner for 10 seconds. This action affects the temperature in a 12-foot-wide column 20 feet high, directly above the burner. A cross section of the balloon depicting this column in shown in the following figure

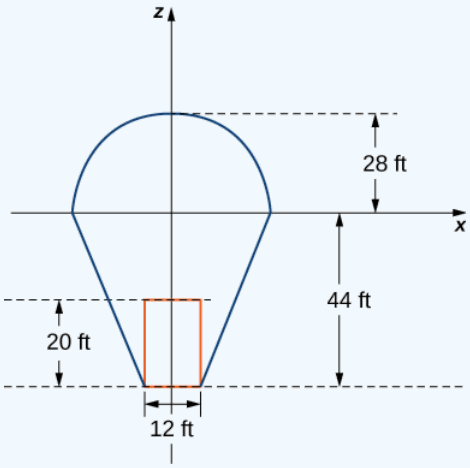


Figure $15.5.14$: Activating the burner heats the air in a 20-foot-high, 12-foot-wide column directly above the burner.

Assume that after the pilot activates the burner for 10 seconds, the temperature of the air in the column described above *increases* according to the formula

$$H(r,\theta,z) = -2z - 48.$$

Then the temperature of the air in the column is given by

$$T_1(r,\theta,z) = \frac{z-r}{10} + 210 + (-2z - 48),$$

while the temperature in the remainder of the balloon is still given by

$$T_0(r,\theta,z) = \frac{z-r}{10} + 210.$$

3. Find the average temperature of the air in the balloon after the pilot has activated the burner for 10 seconds.

Key Concepts

- To evaluate a triple integral in cylindrical coordinates, use the iterated integral

$$\int_{\theta=\alpha}^{\theta=\beta}\int_{r=g_1(\theta)}^{r=g_2(\theta)}\int_{z=u_1(r,\theta)}^{u_2(r,\theta)} f(r,\theta,z)r\,dz\,dr\,d\theta.$$

- To evaluate a triple integral in spherical coordinates, use the iterated integral

$$\int_{\theta=\alpha}^{\theta=\beta}\int_{\rho=g_1(\theta)}^{\rho=g_2(\theta)}\int_{\varphi=u_1(r,\theta)}^{u_2(r,\theta)} f(\rho,\theta,\varphi)\,\rho^2\sin\varphi\,d\varphi\,d\rho\,d\theta.$$

Key Equations

- **Triple integral in cylindrical coordinates**

$$\iiint_B g(s,y,z)dV = \iiint_B g(r\cos\theta,\ r\sin\theta,\ z)r\,dr\,d\theta\,dz = \iiint_B f(r,\theta,z)r\,dr\,d\theta\,dz$$

- **Triple integral in spherical coordinates**

$$\iiint_B f(\rho,\theta,\varphi)\rho^2\sin\varphi\,d\rho\,d\varphi\,d\theta = \int_{\varphi=\gamma}^{\varphi=\psi}\int_{\theta=\alpha}^{\theta=\beta}\int_{\rho=a}^{\rho=b} f(\rho,\theta,\varphi)\rho^2\sin\,\varphi\,d\rho\,d\varphi\,d\theta$$

Glossary

triple integral in cylindrical coordinates

the limit of a triple Riemann sum, provided the following limit exists:

$$lim_{l,m,n\to\infty}\sum_{i=1}^{l}\sum_{j=1}^{m}\sum_{k=1}^{n} f(r^*_{ijk},\theta^*_{ijk},s^*_{ijk})r^*_{ijk}\Delta r\Delta\theta\Delta z$$

triple integral in spherical coordinates

the limit of a triple Riemann sum, provided the following limit exists:

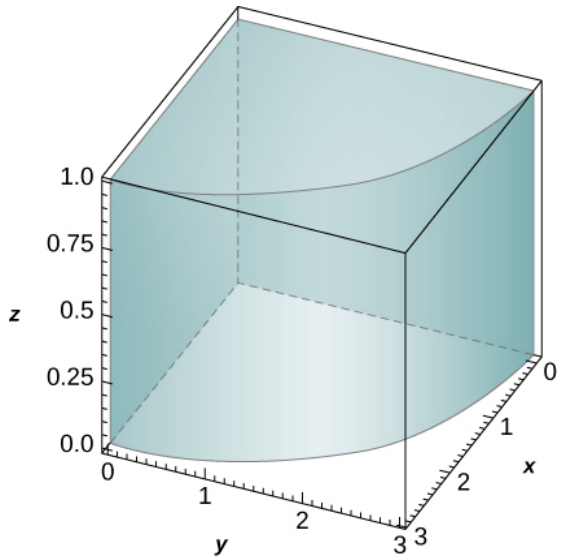
$$lim_{l,m,n\to\infty}\sum_{i=1}^{l}\sum_{j=1}^{m}\sum_{k=1}^{n} f(\rho^*_{ijk},\theta^*_{ijk},\varphi^*_{ijk})(\rho^*_{ijk})^2\sin\,\varphi\Delta\rho\Delta\theta\Delta\varphi$$

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15.5E: EXERCISES FOR SECTION 15.5

In exercises 1 - 8, evaluate the triple integrals $\iiint_E f(x,y,z)\,dV$ over the solid E.

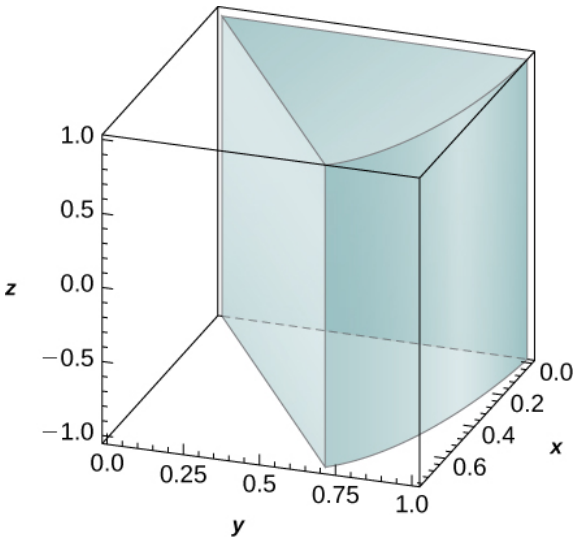
1. $f(x,y,z)=z,\quad B=\{(x,y,z)\,|\,x^2+y^2\le 9,\quad x\le 0,\quad y\le 0,\quad 0\le z\le 1\}$



Answer

$\frac{9\pi}{8}$

2. $f(x,y,z)=xz^2,\ B=\{(x,y,z)\,|\,x^2+y^2\le 16,\ x\ge 0,\ y\le 0,\ -1\le z\le 1\}$
3. $f(x,y,z)=xy,\ B=\{(x,y,z)\,|\,x^2+y^2\le 1,\ x\ge 0,\ x\ge y,\ -1\le z\le 1\}$



Answer

$\frac{1}{8}$

4. $f(x,y,z)=x^2+y^2,\ B=\{(x,y,z)\,|\,x^2+y^2\le 4,\ x\ge 0,\ x\le y,\ 0\le z\le 3\}$
5. $f(x,y,z)=e^{\sqrt{x^2+y^2}},\ B=\{(x,y,z)\,|\,1\le x^2+y^2\le 4,\ y\le 0,\ x\le y\sqrt{3},\ 2\le z\le 3\}$

Answer

$\frac{\pi e^2}{6}$

https://math.libretexts.org/@go/page/66956

6. $f(x,y,z)=\sqrt{x^2+y^2},\ B=\{(x,y,z)\,|\,1\le x^2+y^2\le 9,\ y\le 0,\ 0\le z\le 1\}$

7. a. Let B be a cylindrical shell with inner radius a outer radius b, and height c where $0<a<b$ and $c>0$. Assume that a function F defined on B can be expressed in cylindrical coordinates as $F(x,y,z)=f(r)+h(z)$, where f and h are differentiable functions. If $\int_a^b \bar{f}(r)\,dr=0$ and $\bar{h}(0)=0$, where $\bar{f}$ and $\bar{h}$ are antiderivatives of f and h, respectively, show that $\iiint_B F(x,y,z)\,dV=2\pi c(b\bar{f}(b)-a\bar{f}(a))+\pi(b^2-a^2)\bar{h}(c)$.

b. Use the previous result to show that $\iiint_B \left(z+\sin\sqrt{x^2+y^2}\right)\,dx\,dy\,dz=6\pi^2(\pi-2)$, where B is a cylindrical shell with inner radius π outer radius 2π, and height 2.

8. a. Let B be a cylindrical shell with inner radius a outer radius b and height c where $0<a<b$ and $c>0$. Assume that a function F defined on B can be expressed in cylindrical coordinates as $F(x,y,z)=f(r)g(\theta)f(z)$, where $f,\ g$, and h are differentiable functions. If $\int_a^b \tilde{f}(r)\,dr=0$, where $\tilde{f}$ is an antiderivative of f, show that $\iiint_B F(x,y,z)\,dV=[b\tilde{f}(b)-a\tilde{f}(a)][\tilde{g}(2\pi)-\tilde{g}(0)][\tilde{h}(c)-\tilde{h}(0)]$, where $\tilde{g}$ and $\tilde{h}$ are antiderivatives of g and h, respectively.

b. Use the previous result to show that $\iiint_B z\sin\sqrt{x^2+y^2}\,dx\,dy\,dz=-12\pi^2$, where B is a cylindrical shell with inner radius π outer radius 2π, and height 2.

In exercises 9 - 12, the boundaries of the solid E are given in cylindrical coordinates.

a. Express the region E in cylindrical coordinates.

b. Convert the integral $\iiint_E f(x,y,z)\,dV$ to cylindrical coordinates.

9. E is bounded by the right circular cylinder $r=4\sin\theta$, the $r\theta$-plane, and the sphere $r^2+z^2=16$.

Answer

a. $E=\{(r,\theta,z)\,|\,0\le\theta\le\pi,\ 0\le r\le 4\sin\theta,\ 0\le z\le\sqrt{16-r^2}\}$

b. $\int_0^{\pi}\int_0^{4\sin\theta}\int_0^{\sqrt{16-r^2}} f(r,\theta,z)r\,dz\,dr\,d\theta$

10. E is bounded by the right circular cylinder $r=\cos\theta$, the $r\theta$-plane, and the sphere $r^2+z^2=9$.

11. E is located in the first octant and is bounded by the circular paraboloid $z=9-3r^2$, the cylinder $r=\sqrt{3}$, and the plane $r(\cos\theta+\sin\theta)=20-z$.

Answer

a. $E=\{(r,\theta,z)\,|\,0\le\theta\le\frac{\pi}{2},\ 0\le r\le\sqrt{3},\ 9-r^2\le z\le 10-r(\cos\theta+\sin\theta)\}$

b. $\int_0^{\pi/2}\int_0^{\sqrt{3}}\int_{9-r^2}^{10-r(\cos\theta+\sin\theta)} f(r,\theta,z)r\,dz\,dr\,d\theta$

12. E is located in the first octant outside the circular paraboloid $z=10-2r^2$ and inside the cylinder $r=\sqrt{5}$ and is bounded also by the planes $z=20$ and $\theta=\frac{\pi}{4}$.

In exercises 13 - 16, the function f and region E are given.

a. Express the region E and the function f in cylindrical coordinates.

b. Convert the integral $\iiint_B f(x,y,z)\,dV$ into cylindrical coordinates and evaluate it.

13. $f(x,y,z)=x^2+y^2,\ E=\{(x,y,z)\,|\,0\le x^2+y^2\le 9,\ x\ge 0,\ y\ge 0,\ 0\le z\le x+3\}$

Answer

a. $E=\{(r,\theta,z)\,|\,0\le r\le 3,\ 0\le\theta\le\frac{\pi}{2},\ 0\le z\le r\ \cos\theta+3\}$,
$f(r,\theta,z)=\frac{1}{r\ \cos\theta+3}$

b. $\int_0^3\int_0^{\pi/2}\int_0^{r\ \cos\theta+3}\frac{r}{r\ \cos\theta+3}\,dz\,d\theta\,dr=\frac{9\pi}{4}$

14. $f(x,y,z)=x^2+y^2,\ E=\{(x,y,z)|0\le x^2+y^2\le 4,\ y\ge 0,\ 0\le z\le 3-x\}$

15. $f(x,y,z)=x,\ E=\{(x,y,z)\,|\,1\le y^2+z^2\le 9,\ 0\le x\le 1-y^2-z^2\}$

Answer

a. $y = r\ \cos\theta,\ z = r\ \sin\theta,\ x = z,\ E = \{(r,\theta,z)\,|\,1 \le r \le 3,\ 0 \le \theta \le 2\pi,\ 0 \le z \le 1 - r^2\},\ f(r,\theta,z) = z;$

b. $\int_1^3 \int_0^{2\pi} \int_0^{1-r^2} zr\ dz\ d\theta\ dr = \dfrac{356\pi}{3}$

16. $f(x,y,z) = y,\ E = \{(x,y,z)\,|\,1 \le x^2 + z^2 \le 9,\ 0 \le y \le 1 - x^2 - z^2\}$

In exercises 17 - 24, find the volume of the solid E whose boundaries are given in rectangular coordinates.

17. E is above the xy-plane, inside the cylinder $x^2 + y^2 = 1$, and below the plane $z = 1$.

Answer

π

18. E is below the plane $z = 1$ and inside the paraboloid $z = x^2 + y^2$.

19. E is bounded by the circular cone $z = \sqrt{x^2 + y^2}$ and $z = 1$.

Answer

$\frac{\pi}{3}$

20. E is located above the xy-plane, below $z = 1$, outside the one-sheeted hyperboloid $x^2 + y^2 - z^2 = 1$, and inside the cylinder $x^2 + y^2 = 2$.

21. E is located inside the cylinder $x^2 + y^2 = 1$ and between the circular paraboloids $z = 1 - x^2 - y^2$ and $z = x^2 + y^2$.

Answer

π

22. E is located inside the sphere $x^2 + y^2 + z^2 = 1$, above the xy-plane, and inside the circular cone $z = \sqrt{x^2 + y^2}$.

23. E is located outside the circular cone $x^2 + y^2 = (z-1)^2$ and between the planes $z = 0$ and $z = 2$.

Answer

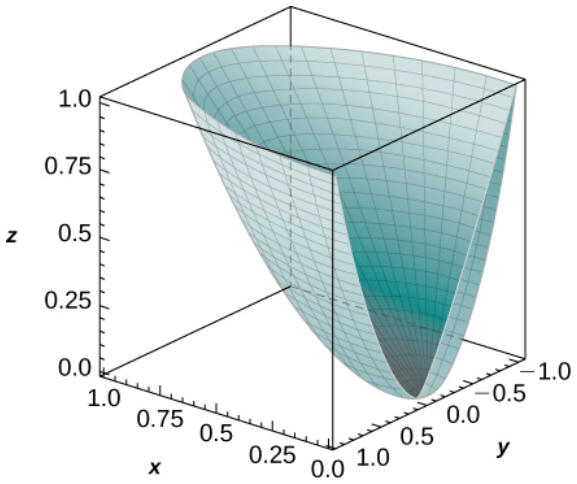
$\frac{4\pi}{3}$

24. E is located outside the circular cone $z = 1 - \sqrt{x^2 + y^2}$, above the xy-plane, below the circular paraboloid, and between the planes $z = 0$ and $z = 2$.

25. [T] Use a computer algebra system (CAS) to graph the solid whose volume is given by the iterated integral in cylindrical coordinates $\int_{-\pi/2}^{\pi/2} \int_0^1 \int_{r^2}^{r} r\ dz\ dr\ d\theta$. Find the volume V of the solid. Round your answer to four decimal places.

Answer

$V = \frac{pi}{12} \approx 0.2618$



26. [T] Use a CAS to graph the solid whose volume is given by the iterated integral in cylindrical coordinates $\int_0^{\pi/2} \int_0^1 \int_{r^4}^{r} r\ dz\ dr\ d\theta$. Find the volume E of the solid. Round your answer to four decimal places.

27. Convert the integral $\int_0^1 \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} \int_{x^2+y^2}^{\sqrt{x^2+y^2}} xz\ dz\ dx\ dy$ into an integral in cylindrical coordinates.

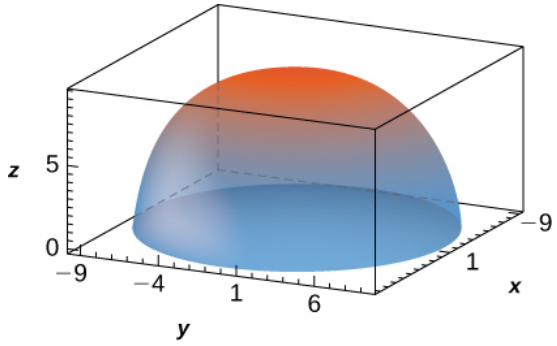
Answer

$$\int_0^1 \int_0^{\pi} \int_{r^2}^{r} zr^2\ \cos\theta\, dz\ d\theta\ dr$$

28. Convert the integral $\int_0^2 \int_0^y \int_0^1 (xy+z)\, dz\ dx\ dy$ into an integral in cylindrical coordinates.

In exercises 29 - 32, evaluate the triple integral $\iiint_B f(x,y,z)\, dV$ over the solid B.

29. $f(x,y,z)=1,\ B=\{(x,y,z)\,|\,x^2+y^2+z^2\le 90,\ z\ge 0\}$

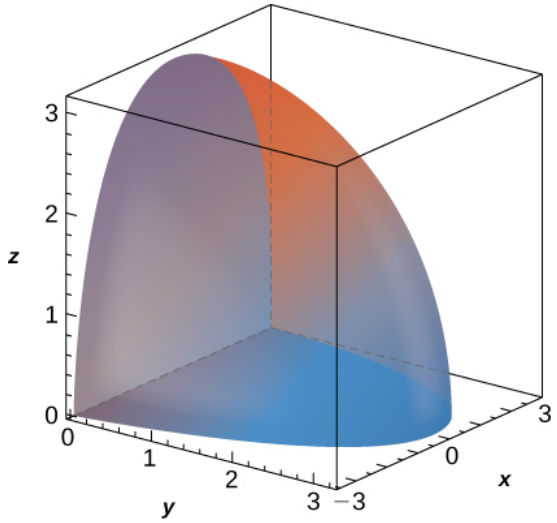


[Hide Solution]

Answer

$180\pi\sqrt{10}$

30. $f(x,y,z)=1-\sqrt{x^2+y^2+z^2},\ B=\{(x,y,z)\,|\,x^2+y^2+z^2\le 9,\ y\ge 0,\ z\ge 0\}$



31. $f(x,y,z)=\sqrt{x^2+y^2},\ B$ is bounded above by the half-sphere $x^2+y^2+z^2=9$ with $z\ge 0$ and below by the cone $2z^2=x^2+y^2$.

Answer

$\frac{81\pi(\pi-2)}{16}$

32. $f(x,y,z)=\sqrt{x^2+y^2},\ B$ is bounded above by the half-sphere $x^2+y^2+z^2=16$ with $z\ge 0$ and below by the cone $2z^2=x^2+y^2$.

33. Show that if $F(\rho,\theta,\varphi)=f(\rho)g(\theta)h(\varphi)$ is a continuous function on the spherical box $B=\{(\rho,\theta,\varphi)\,|\,a\le\rho\le b,\ \alpha\le\theta\le\beta,\ \gamma\le\varphi\le\psi\}$, then $\iiint_B F\ dV=\left(\int_a^b \rho^2 f(\rho)\ dr\right)\left(\int_\alpha^\beta g(\theta)\ d\theta\right)\left(\int_\gamma^\psi h(\varphi)\ \sin\varphi\ d\varphi\right)$.

34. A function F is said to have spherical symmetry if it depends on the distance to the origin only, that is, it can be expressed in spherical coordinates as $F(x,y,z) = f(\rho)$, where $\rho = \sqrt{x^2+y^2+z^2}$. Show that $\iiint_B F(x,y,z)\,dV = 2\pi \int_a^b \rho^2 f(\rho)\,d\rho$, where B is the region between the upper concentric hemispheres of radii a and b centered at the origin, with $0 < a < b$ and F a spherical function defined on B.

Use the previous result to show that $\iiint_B (x^2+y^2+z^2)\sqrt{x^2+y^2+z^2}\,dV = 21\pi$, where $B = \{(x,y,z) \mid 1 \le x^2+y^2+z^2 \le 2,\ z \ge 0\}$.

35. Let B be the region between the upper concentric hemispheres of radii a and b centered at the origin and situated in the first octant, where $0 < a < b$. Consider F a function defined on B whose form in spherical coordinates (ρ,θ,φ) is $F(x,y,z) = f(\rho)\cos\varphi$. Show that if $g(a) = g(b) = 0$ and $\int_a^b h(\rho)\,d\rho = 0$, then $\iiint_B F(x,y,z)\,dV = \frac{\pi^2}{4}[ah(a) - bh(b)]$, where g is an antiderivative of f and h is an antiderivative of g.

Use the previous result to show that $\iiint_B = \frac{z\cos\sqrt{x^2+y^2+z^2}}{\sqrt{x^2+y^2+z^2}}\,dV = \frac{3\pi^2}{2}$, where B is the region between the upper concentric hemispheres of radii π and 2π centered at the origin and situated in the first octant.

In exercises 36 - 39, the function f and region E are given.

a. Express the region E and function f in cylindrical coordinates.

b. Convert the integral $\iiint_B f(x,y,z)\,dV$ into cylindrical coordinates and evaluate it.

36. $f(x,y,z) = z;\ E = \{(x,y,z) \mid 0 \le x^2+y^2+z^2 \le 1,\ z \ge 0\}$

37. $f(x,y,z) = x+y;\ E = \{(x,y,z) \mid 1 \le x^2+y^2+z^2 \le 2,\ z \ge 0,\ y \ge 0\}$

Answer

a. $f(\rho,\theta,\varphi) = \rho\ \sin\varphi\ (\cos\theta + \sin\theta),\ E = \{(\rho,\theta,\varphi) \mid 1 \le \rho \le 2,\ 0 \le \theta \le \pi,\ 0 \le \varphi \le \frac{\pi}{2}\}$;

b. $\int_0^{\pi}\int_0^{\pi/2}\int_1^2 \rho^3 \cos\varphi\ \sin\varphi\ d\rho\,d\varphi\,d\theta = \frac{15\pi}{8}$

38. $f(x,y,z) = 2xy;\ E = \{(x,y,z) \mid \sqrt{x^2+y^2} \le z \le \sqrt{1-x^2-y^2},\ x \ge 0,\ y \ge 0\}$

39. $f(x,y,z) = z;\ E = \{(x,y,z) \mid x^2+y^2+z^2-2x \le 0,\ \sqrt{x^2+y^2} \le z\}$

Answer

a. $f(\rho,\theta,\varphi) = \rho\ \cos\varphi;\ E = \{(\rho,\theta,\varphi) \mid 0 \le \rho \le 2\ \cos\varphi,\ 0 \le \theta \le \frac{\pi}{2},\ 0 \le \varphi \le \frac{\pi}{4}\}$;

b. $\int_0^{\pi/2}\int_0^{\pi/4}\int_0^{2\ \cos\varphi} \rho^3 \sin\varphi\ \cos\varphi\ d\rho\,d\varphi\,d\theta = \frac{7\pi}{24}$

In exercises 40 - 41, find the volume of the solid E whose boundaries are given in rectangular coordinates.

40. $E = \{(x,y,z) \mid \sqrt{x^2+y^2} \le z \le \sqrt{16-x^2-y^2},\ x \ge 0,\ y \ge 0\}$

41. $E = \{(x,y,z) \mid x^2+y^2+z^2-2z \le 0,\ \sqrt{x^2+y^2} \le z\}$

Answer

$\frac{\pi}{4}$

42. Use spherical coordinates to find the volume of the solid situated outside the sphere $\rho = 1$ and inside the sphere $\rho = \cos\varphi$, with $\varphi \in [0, \frac{\pi}{2}]$.

43. Use spherical coordinates to find the volume of the ball $\rho \le 3$ that is situated between the cones $\varphi = \frac{\pi}{4}$ and $\varphi = \frac{\pi}{3}$.

Answer

$9\pi(\sqrt{2}-1)$

44. Convert the integral $\int_{-4}^{4}\int_{-\sqrt{16-y^2}}^{\sqrt{16-y^2}}\int_{-\sqrt{16-x^2-y^2}}^{\sqrt{16-x^2-y^2}} (x^2+y^2+z^2)\,dz\,dx\,dy$ into an integral in spherical coordinates.

45. Convert the integral $\int_0^4\int_0^{\sqrt{16-x^2}}\int_{-\sqrt{16-x^2-y^2}}^{\sqrt{16-x^2-y^2}} (x^2+y^2+z^2)^2\,dz\,dy\,dx$ into an integral in spherical coordinates.

Answer

$\int_0^{\pi/2}\int_0^{\pi/2}\int_0^4 \rho^6 \sin\varphi\,d\rho\,d\phi\,d\theta$